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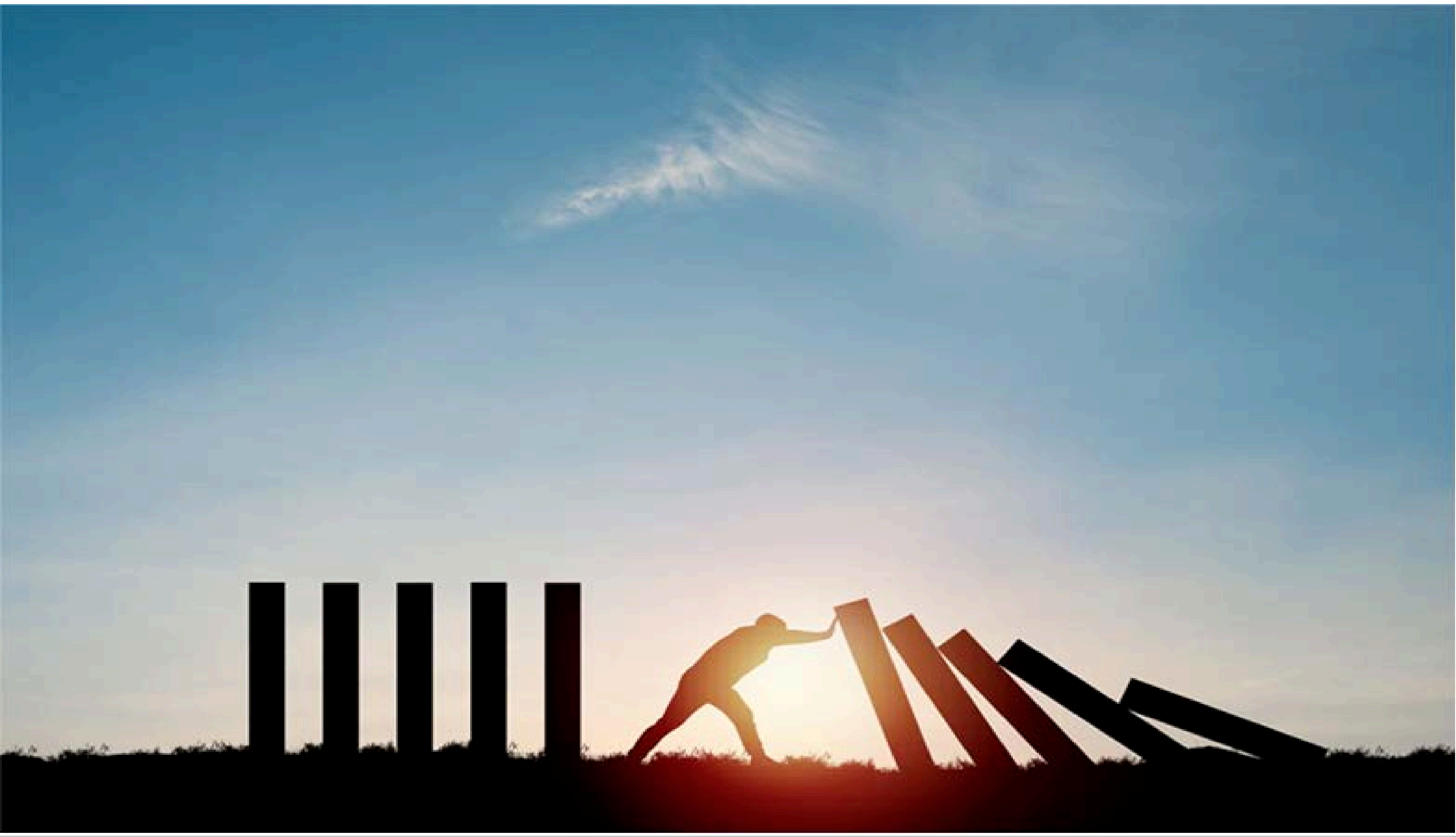
ANCHOR REPORT

Global Markets Research

12 April 2024



Greater China Analog IC

Expectations reset, risk-reward improves

The share prices of Greater China analog IC companies significantly underperformed the domestic and global semiconductor sector during the downturn of 2022-23, owing to slow tech end-demand and Texas Instruments' (TI; TXN US, Not rated) aggressive pricing. Although short-term visibility remains low and the competitive backdrop is still tough for these companies, investment risk-reward has improved, following severe inventory digestion and share price correction. We do not think TI can be as aggressive on pricing in 2024 as it was in 2023. Also, we see Greater China analog IC vendors' cost improving in 2024F, either on the migration to 12" fabs or the end of high-priced LTAs with foundries. We thus expect a profitability recovery from 2024F. Our analysis of Taiwan analog IC in the late-2010s suggests that the China analog IC localization trend is ongoing. We initiate coverage of Silergy (6415 TT) and 3Peak (688536 CH) at Buy, and Novosense (688052 CH) at Neutral. Silergy is one of the top Asia analog fabless companies, while the other two are exposed to industrial applications.

Key themes and analysis in this Anchor Report:

- A deep dive into TI's cost structure, capacity plan, pricing, and UTR dynamics.
- A case study of Taiwan analog IC industry dynamics in the late-2010s to assess the mature localization rate facing China's analog IC vendors.
- A detailed analysis of China analog IC vendors in terms of their cycles, mix, and foundry suppliers.

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Production Complete: 2024-04-12 09:35 UTC

See Appendix A-1 for analyst certification, important disclosures and the status of non-US analysts.

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EQUITY: TECHNOLOGY

Expectations reset, risk-reward improves

Worst appears behind despite still-low visibility; picking quality companies facing challenging times

Initiating coverage of Silergy and 3Peak at Buy, and Novosense at Neutral

Following our analog IC sector initiation (*link*) in Oct-21, we provide an update on the dynamics in the industry in this report, as well as initiate coverage on three of Greater China's leading analog IC vendors. The industry backdrop has undergone significant change since two-and-a-half years ago. The semiconductor cycle peaked in 1H22 (*Cycle peak is here*, Apr-2022), and then came a downturn that lasted (at least for one-and-a-half year) until late-2023 (*More signs of cycle bottom*, Oct-2023). For the Greater China analog IC sector, the downturn has been even more severe given TI's aggressive pricing over the same period. However, investment risk-reward should have also improved following the sector's severe two years of underperformance. Our stock selection favors quality companies that are facing challenges amid a severe downturn in the industry. We initiate coverage of Silergy (6415 TT) with a Buy rating in view of its product expansion plan in Greater China, cost strategy, and its own fab production know-how. We also initiate coverage of 3Peak (688536 CH) with a Buy rating, and Novosense (688052 CH) with a Neutral rating – both of which are industrial/auto-related analog IC fabless companies with little consumer exposure. Although the industrial and auto supply chains are still around their cycle trough levels (unlike some consumer applications, which have seen rush orders since 2023), we believe China local analog vendors have long-term value with industrial/auto exposure (where local competition is much less than that in consumer applications). The US/China PMI data indicate somewhat positive signals for industrial applications, too. We prefer 3Peak over Novosense in the China industrial analog IC sector, given its stronger profitability, better inventory control, and more resilient margin.

TI, the market leader, won't be able to price as aggressively in 2024 as it did in 2023, in our view

We assume Ti's further price cuts in 2024 would be more moderate as: (1) TI set a long-term sales CAGR target of 10% in Feb-23; (2) pricing and fab UTR are two key factors affecting TI's sales and margin trends; and (3) TI's GPM came down to 55-60% in 1Q24 from the peak of 70% in 1Q22.

Current China analog IC localization rate of 15-20% is well below Taiwan analog IC localization rate of 40-50% (i.e., our defined mature level) in late 2010s

The current China analog IC localization rate is likely at 15-20% (likely USD4-5bn in combined sales for China analog IC vendors sales of c.USD27bn China TAM, based on WSTS data), in our view, far below our estimated mature analog IC localization rate of 40-50% recorded by Taiwan analog IC vendors vs their consumer electronics-related analog IC TAM by the mid-to-late 2010s (2006-10).

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Fig. 1: Stocks for action

Company	Ticker	Rating	Market Cap (USD mn)	Nomura analyst	Target Price	Avg. TO (USD mn)	Price (Local CCY)	Upside (%)
Silergy	6415 TT	Buy*	3,946	Aaron Jeng, CFA	450	36.8	328	37%
3Peak	688536 CH	Buy*	1,764	Aaron Jeng, CFA	120	19.5	96.2	25%
Awinic	688798 CH	Buy	1,702	Donnie Teng	100	11.9	53.0	89%
SG Micro	300661 CH	Neutral	3,999	Donnie Teng	93	32.7	61.9	50%
Novosense	688052 CH	Neutral*	1,860	Aaron Jeng, CFA	105	29.9	94.4	11%

Note: Prices as of 9 Apr 2024; * initiating coverage
Source: Bloomberg Finance L.P., Nomura research

Executive summary

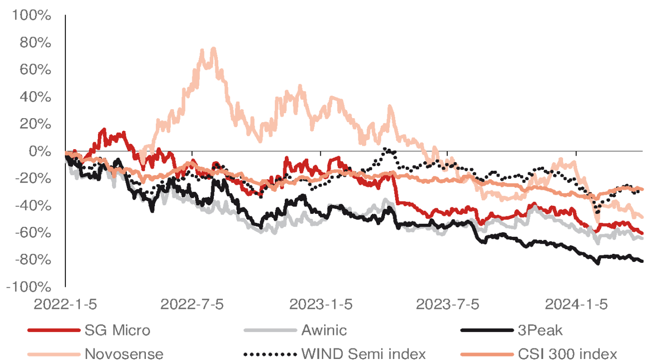
We initiate coverage of Silergy and 3 Peak at Buy, and Novosense at Neutral. The inventory correction for consumer applications is coming to an end, while the US/China PMI data indicate somewhat positive signals for industrial applications. Investment risk-reward should have also improved following the sector's severe two years of underperformance.

Following our analog IC sector initiation report (*link*) in Oct-21 (when the semiconductor sector was still in an upcycle, and the severe shortage drove TI to abandon its focus on China business), we provide an update on the industry in this report, as well as initiate coverage of Silergy, 3Peak , and Novosense. The industry environment has changed dramatically since and two-and-a-half years ago. The semi cycle peaked in 1H22 (*C ycle peak is here,* Apr-2022), and then entered an at least one-and-a-half-year downturn that lasted until late-2023 (*M ore signs of cycle bottom* , Oct-2023). In our view, we are currently at the semi cycle's trough but with limited long-term visibility.

For the Greater China analog IC sector, the downturn has been even more severe than other types of chips, given TI's 12" fab capacity expansions and aggressive pricing in this downturn, as well as the slow end-demand recovery. Analog IC companies have also delivered worse-than-peers share price performance during this downturn (*Fig. 2* - *Fig. 3*). While competition (either with global or China local vendors) is tough in this slow demand and low visibility environment, "long-term" risk-reward appears to be turning more positive now following the significant share price underperformance, six quarters of inventory correction, GPM downtrend (down by 12-13pp until 4Q23 from the peak of 56%/41% for China signal chain/PMIC companies) (*Fig. 6*), and inventory write-offs. Our stock selection approach focuses on quality companies that have exhibited weakness amid the cyclical downturn in the industry.

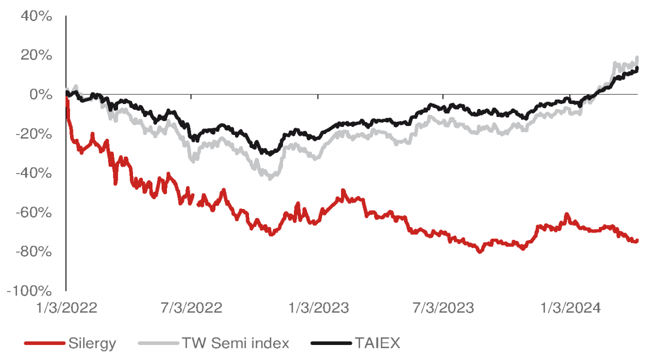
We initiate coverage of Silergy with a Buy rating based on its impressive product expansion plan in Greater China, cost strategy, and own fab production know-how. We also initiate coverage of 3Peak with a Buy rating, and Novosense with a Neutral rating – both of which are industrial/auto-related analog IC fabless with little consumer exposure. Although industrial and auto supply chains are still around their cycle troughs (unlike some consumer applications which are seeing rush orders from 2023), we believe the China local analog vendors have long-term value with industrial/auto exposure (where local competition is much less than consumer applications). The US PMI is a good indicator of industrial applications (*Fig. 7*), which appear to be bottoming out. The China PMI recently broke the 50-level, after remaining below 50 in 2023, which also seems to be a positive sign (*Fig. 8*).

Fig. 2: China A-share analog vendors' share price performance vs index



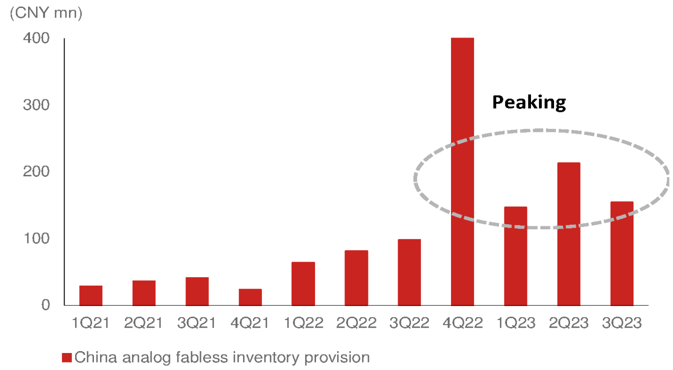
Source: Company data, Nomura research

Fig. 3: Silergy share price performance vs index



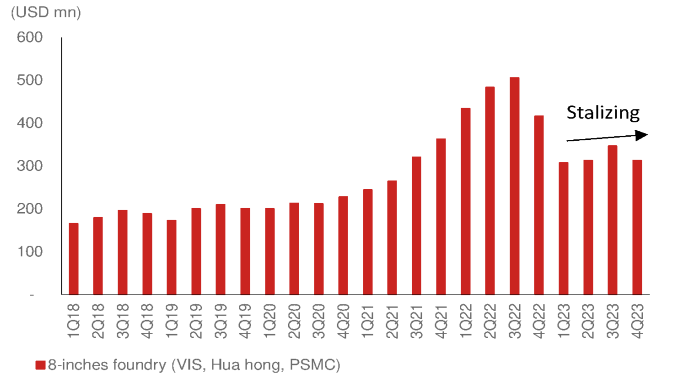
Source: Company data, Bloomberg Finance L.P., Nomura research

Fig. 4: China analog fabless inventory provision



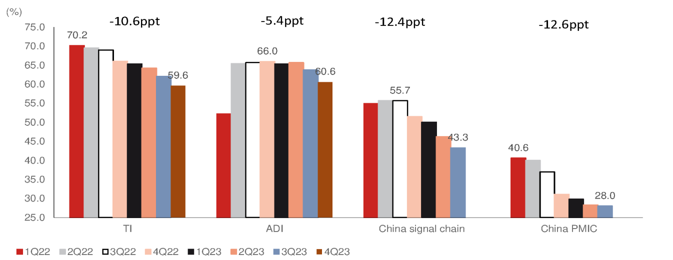
Source: Company data, Nomura research

Fig. 5: Greater China tier-2 foundries' analog/PMIC sales



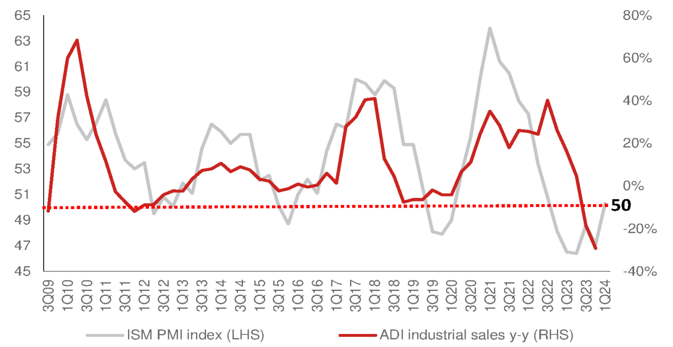
Note: Companies include Vanguard, Huahong, PSMC
Source: Company data, Nomura research

Fig. 6: GPM comparison over 1Q22-4Q23



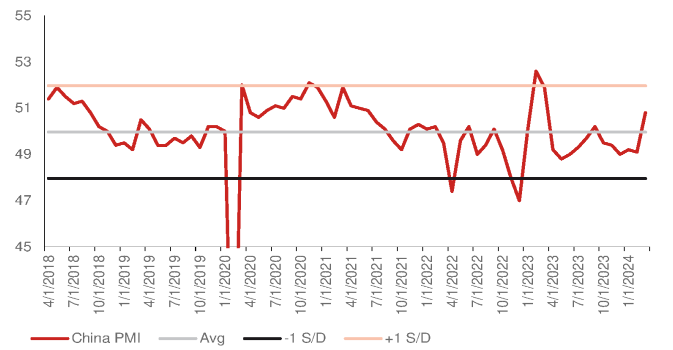
Source: Company data, Nomura research

Fig. 7: ADI industrial sales y-y vs ISM PMI index



Source: Company data, Nomura research

Fig. 8: China manufacturing PMI

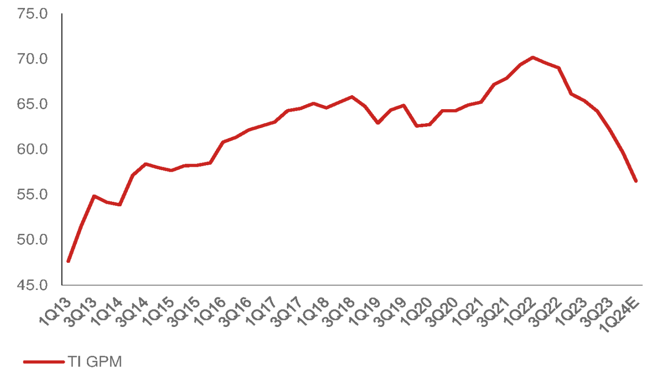


Source: Nomura research

TI, the market leader, will not be able to price as aggressively as it did in 2023 in 2024F, according to our simulation

TI is a major topic of discussion in this report. In response to the severe IC shortage in 2021, TI announced an aggressive capacity expansion plan in Feb-2022 and decided to further hike long-term capex in Feb-2023 due to its expectation of long-term demand, particularly from auto and industrial applications. However, short-term end-demand probably came in lower and took longer than TI had expected (starting with consumer applications). As a result, TI initiated aggressive pricing from late-2022/early-2023 in order to regain market share to sustain its fab UTR. However, **we assume any further price cuts from TI in 2024 would be much more moderate**: 1) TI set a long-term sales CAGR target of 10% for Feb-23; 2) pricing and fab UTR are two key factors affecting TI's sales and margin trend; (*Fig. 44* - *Fig. 45*); and 3) TI's GPM came down to below-60% in 1Q24 from 70% peak in 1Q22 (*Fig. 9*). We will need to monitor if TI reviews its capex plan if end-demand continues to be slow.

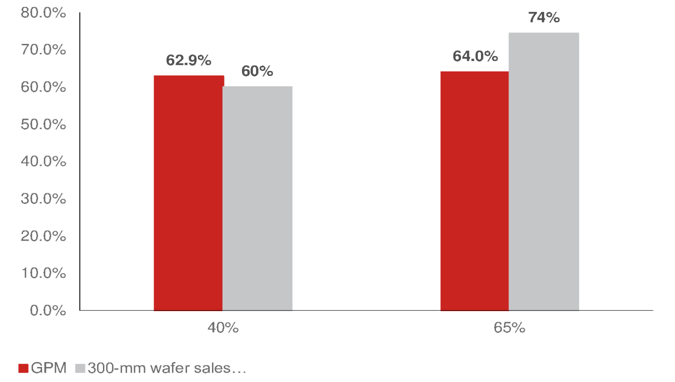
Fig. 9: TI: Quarterly GPM



Note: 1Q24 estimates by Bloomberg consensus
Source: Company data, Nomura research

Fig. 10: 300mm wafer GPM comparison

300mm wafer mix of 65% would only improve GPM by 1pp vs current mix of 40%

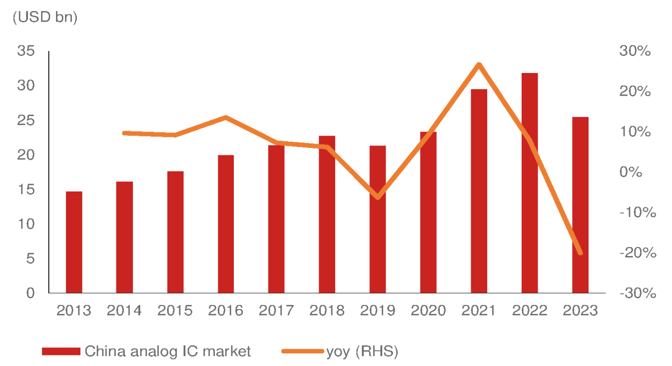


Source: Company data, Nomura estimates

Current China analog IC localization rate of 15-20% is well below Taiwan analog IC localization rate of 40-50% (i.e., our defined "mature" level) in the late 2010s – indicating share gain potential

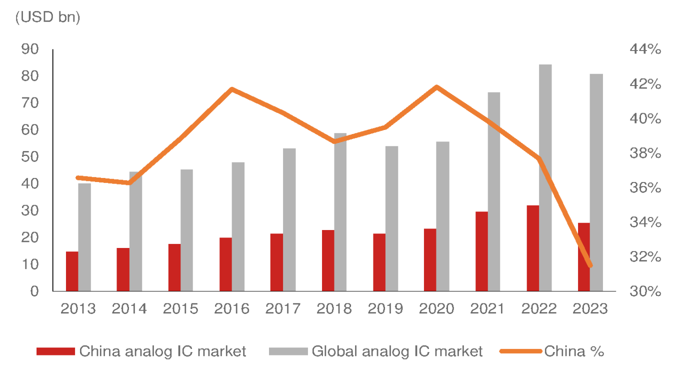
Structurally, we see ongoing share gain potential for China analog IC players. **The current analog IC localization rate is likely at 15-20%** (likely USD4-5bn combined sales for China analog IC vendors sales out of c.USD27bn China TAM from WSTS data; *Fig. 11* - *Fig. 12*) **which is far below our perceived mature analog IC localization rate of 40-50% recorded by Taiwan analog IC vendors vs their consumer electronics-related analog IC TAM during mid-to-late 2010s (2006-10)**. Of note, Richtek [unlisted], one of the top analog IC companies in Taiwan, delivered speedy growth before 2010 but stopped growing after 2010 once all Taiwan analog IC companies combined had captured 40-50% of their TAM (and resumed growth only after being acquired by MediaTek [2454 TT, Neutral] in 2016; *Fig. 15*).

Fig. 11: China analog IC market



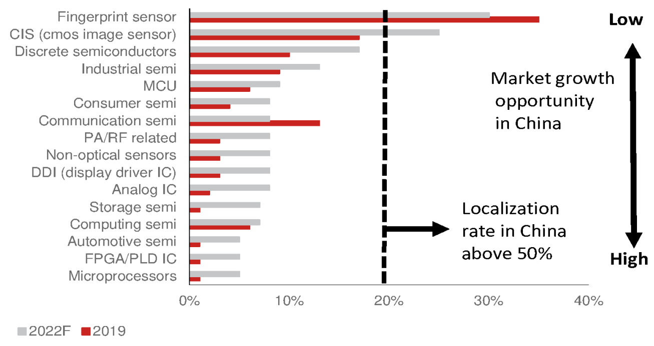
Source: WSTS, Nomura research

Fig. 12: Global analog IC market vs China analog IC market



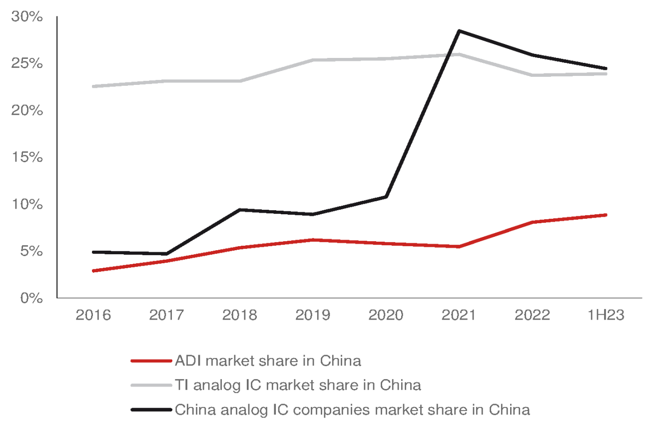
Source: WSTS, Nomura research

Fig. 13: China local vendors' market share globally



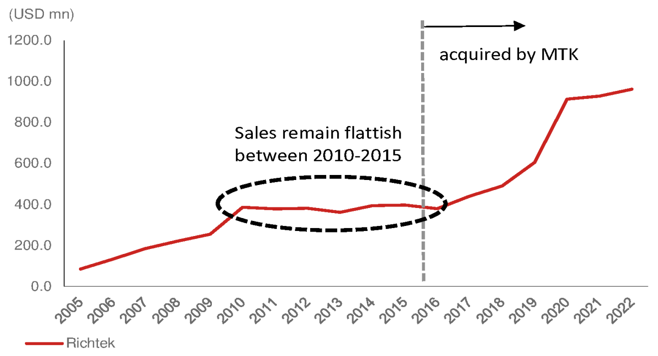
Source: Nomura estimates

Fig. 14: TI, ADI, China analog IC companies market share in China



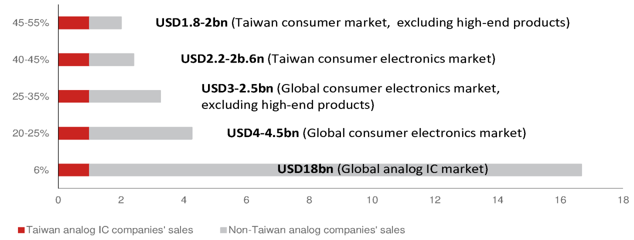
Source: WSTS, CSIA, Nomura estimates

Fig. 15: Richtek: annual sales between 2005-22



Source: Company data, Nomura research

Fig. 16: Taiwan analog companies' market share by different TAM in 2007



Source: Nomura estimates

Key observations of China analog IC market and vendors

- There were many China analog IC IPOs in 2022 (*Fig. 51*).
- Their product portfolio expansions have been aggressive (*Fig. 81*, *Fig. 82*).
- Inventory provisions have surged from 4Q22 (*Fig. 63*).
- Major analog IC (and discrete) foundries in Greater China saw relevant sales peaks in 3Q22 and consolidation through 2023 (*Fig. 62*).
- By application, consumer applications were the first to correct (peaked from 1H22) and the first to recover (bottomed from 1Q23, particularly for the smartphone supply chain *Fig. 67*, *Fig. 68*); auto applications have been on an upward trend despite some minor quarterly corrections – likely because they just started from scratch in 2020 *Fig. 69*, *Fig. 70*); industrial applications peaked out from 1H22 and consolidated at a low level through 2023 (*Fig. 73*, *Fig. 74*).
- For industrial applications, the businesses of global industrial analog IC companies like TI and ADI [ADI US, Not rated] are highly correlated with the US ISM manufacturing PMI cycle (*Fig. 7*). The China PMI has been mostly below 50 through 2023 (*Fig. 8*). Both the US and China PMIs recently bottomed out, which may augur well for industrial application analog IC vendors (*Fig. 78*).
- SMIC is the most commonly used foundry for China local analog IC vendors, followed by TSMC and DB Hitek (*Fig. 83*).
- By sales exposure, consumer application analog ICs are very crowded. 3Peak, Novosense and Belling each have 50% sales exposure to industrial applications, and Novosense has the highest exposure to auto among China analog vendors (*Fig. 84*).

- The following figures detail sales and GPM trends of major China, Taiwan and global analog IC companies over 2021-23 (*Fig. 88* , *Fig. 89* , *Fig. 90*). China vendors saw their business peak out much earlier than global IDMs.

Fig. 17: Great China analog vendors' sales application (%)

	Industrial	Consumer	Auto	Telecom/ Networking / Comm.	Computing/ Server	Others	Product details
Silergy	34%	38%	8%	7%	13%		PMIC:85% Signal chain and others:15%
Awinic	2%	96%				2%	
SGMicro	30%	50%	5%	5%		10%	PMIC 68.29%, Signal chain 31.67%
Will Semi	10%	85%				5%	
3 Peak	47%	9%	12%	32%			Signal chain: 70%, PMIC: 30%
Belling	50%					50%	PMIC 27.25%, General Analog 37.5%, NVM 2.5%, Discrete 5.6%, distributor 27.2%
Novosense	55%	12%	33%			0%	Signal chain: 63%, PMIC: 30.5%, Sensor: 7%
Chipown	21%	24%				55%	Power: 34.35%, others 3.37%
Kiwi instrument		87%				13%	PMIC: 78% (LED 65%, General power 22%, Home Appliance/IoT 13%)
Wuxi Etek		95%				5%	PMIC 85.5%, Others 14.5%
Injoinic		76%	16%			8%	PMIC:73% (Portable battery 52%, auto 16.4%, wireless charging 16%, TWS 8%, others 8%)
Fine Made		51%				49%	PMIC:39%, LED driver:42.3%, Mosfet 4%, others 14.7%
Bright Power		90%				10%	
Joulwatt	15%	30%	2%	40%	10%	3%	PMIC:95%, Signal chain:2%
Southchip	3%	95%	1%			1%	
Dioo	12%	65%		5%		18%	
Cellwise		95%			5%		

Source: Company data, Nomura estimates

Fig. 18: Greater China analog vendors' sales peaked out earlier than global peers

Also bottomed out earlier than global peers

Sales (USD mn)	1Q21	2Q21	3Q21	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23
Silergy	148.8	188.1	212.3	221.2	214.9	**230.9**	197.2	150.0	**113.0**	117.1	129.9
SG Micro	60.8	80.8	95.8	110.1	122.2	**132.4**	111.1	109.0	**75.0**	90.5	101.1
AWINIC	76.4	88.4	91.9	104.2	93.8	**106.4**	**54.2**	59.0	56.2	89.0	106.9
Southchip	25.9	25.9	48.9	53.8	**66.3**	53.6	53.6	53.6	**41.7**	53.4	75.3
Joulwatt	28.9	28.9	46.4	**59.1**	53.1	54.3	47.7	58.5	**44.0**	49.5	49.0
Injoinic	26.5	29.9	33.1	32.9	**33.3**	30.1	**29.3**	36.0	32.2	42.1	45.2
Kiwi instrument	30.0	30.0	**42.7**	36.4	26.5	22.0	**12.7**	17.6	19.3	24.1	16.7
Fine Made	41.0	**90.6**	51.2	29.4	43.4	26.6	22.1	23.7	20.0	27.5	26.6
Cellwise	13.0	13.0	12.8	**14.5**	8.4	8.4	5.2	7.8	**4.2**	8.0	9.2
Bpsemi	62.8	101.9	**117.3**	74.8	47.6	43.8	**31.4**	38.4	38.7	49.9	41.5
Chipown	22.0	28.5	32.3	**34.1**	29.2	28.8	**22.2**	27.0	27.3	28.1	27.0
Wuxi ETEK	25.6	31.5	31.7	31.2	**41.7**	31.3	20.9	21.4	26.1	27.8	36.1
Belling	66.2	91.4	76.5	79.8	70.3	73.8	71.9	**86.8**	57.6	68.2	76.9
3Peak	25.8	49.2	62.8	68.1	69.7	**84.0**	68.7	44.3	44.9	43.4	27.8
Novosense	21.2	31.4	36.9	44.3	53.4	68.7	**70.5**	55.5	68.8	36.0	38.3
Dioo	17.7	17.7	21.3	23.0	**25.3**	20.0	20.0	14.1	11.0	15.0	15.9
China total	692.6	927.1	1013.8	**1016.8**	999.1	1015.1	838.7	802.7	**680.2**	769.6	823.3
Richtek	269.6	314.1	327.7	321.6	**404.9**	368.3	327.2	241.2	**220.3**	224.1	244.7
GMT	77.7	85.1	90.3	83.9	**90.7**	80.8	58.8	**54.7**	58.5	67.8	70.8
Anpec	53.8	57.6	65.8	65.0	69.2	**69.6**	45.2	**32.5**	36.5	42.9	46.7
uPI	41.6	41.6	60.0	60.1	61.0	**63.5**	49.8	31.7	23.5	25.9	22.8
TW total	442.7	498.5	543.8	530.5	**625.7**	582.2	481.0	360.1	**338.7**	360.7	385.0
Greater China	1135.3	1425.5	1557.7	1547.3	**1624.8**	1597.3	1319.7	1162.8	**1018.8**	1130.3	1208.3
Global											
TI	4289	4580	4643	4832	4905	5212	**5241**	4670	4379	4531	4532
ADI	2178	2290	2424	3059	2684	2972	3110	3248	3250	**3263**	3076
MPS	254	293	324	337	378	461	**495**	460	451	441	475
Infineon	3137	3253	3281	3544	3612	3700	3851	4170	4037	4421	**4452**
STM	3016	2992	3197	3556	3546	3837	4321	4424	4247	4326	**4431**
ON	1482	1670	1742	1846	1945	2085	**2193**	2104	1960	2094	2181
Global total	14356	15078	15610	17174	17070	18267	**19211**	19076	18324	19076	19147

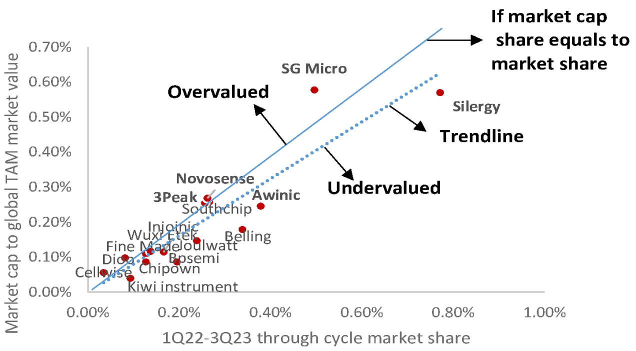
Peak **Bottom**

Source: Company data, Nomura research

Valuation methodology

We have mainly used the P/E multiple as our valuation approach (except for Novosense, where we have used P/B, as it might not make money by 2025F). However, we have also noticed that **the global market share (in terms of sales) of a particular analog IC company is positively correlated with its market cap share** – a concept of price-to-sales, which puts the global companies at the same benchmark. We use TI and ADI as a reference to derive the global analog IC sector market cap (*Fig. 104* , *Fig. 105*), and compare individual China analog IC companies' global sales share with its market cap share. To eliminate the impact of cyclical swings, we use sales from 1Q22 (cycle peak) to 3Q23 (cycle trough) for the comparison. From this approach, within our coverage, Silergy (Buy) and Awinic (Buy) are undervalued, while SG Micro (Neutral) appears overvalued. 3Peak (Buy) and Novosense (Neutral) appear fairly valued in our market share vs. market cap share analysis. *Fig. 104* , *Fig. 105* , *and Fig. 106* elaborate this with numbers.

Fig. 19: China analog through cycle sales' market share vs. market cap share



Source: Company data, Nomura estimates

Fig. 20: China major analog vendors' valuaiton analysis

	Market cap (USD mn)	1Q22-3Q23 sales (USD mn)	1Q22-3Q23 market share	Market cap to global TAM market value
Silergy	3,946	1,152.9	0.77%	0.57%
SG Micro	3,999	741.4	0.50%	0.58%
AWINIC	1,702	565.3	0.38%	0.25%
Southchip	1,782	397.6	0.27%	0.26%
Joulwatt	1,019	356.3	0.24%	0.15%
Injoinic	795	248.2	0.17%	0.11%
Kiwi instrument	274	139.0	0.09%	0.04%
Fine Made	755	189.9	0.13%	0.11%
Cellwise	388	51.1	0.03%	0.06%
Bpsemi	598	291.3	0.19%	0.09%
Chipown	600	189.6	0.13%	0.09%
Wuxi ETEK	808	205.4	0.14%	0.12%
Belling	1,240	505.6	0.34%	0.18%
3Peak	1,764	382.8	0.26%	0.25%
Novosense	1,860	391.1	0.26%	0.27%
Dioo	678	121.3	0.08%	0.10%
Total	22,209		4.0%	3.2%

Source: Bloomberg Finance L.P., Nomura research

TI's pricing along with capacity expansion

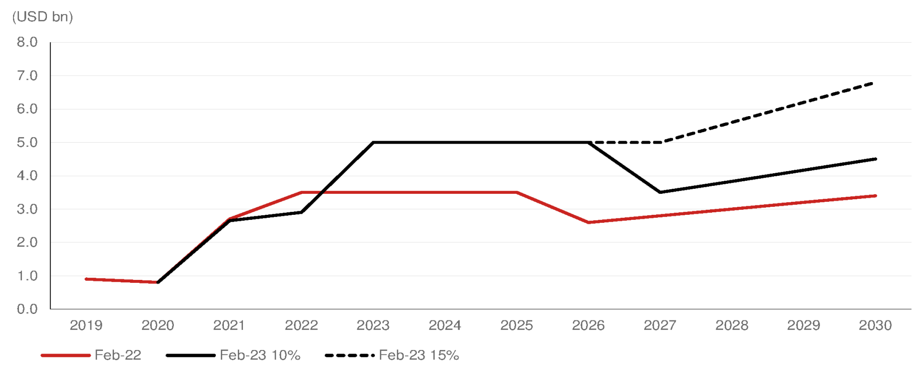
Aggressive pricing cut to compete in greater China market

Texas Instruments (TI) is the world's largest analog vendor, and owns three 150mm (6-inch) fabs, five 200mm (8-inch) fabs, and four 300mm (12-inch) fabs. 300mm accounted for 40% of internal wafer usage in 2022 and 2023. In Feb-22, TI announced an aggressive capacity expansion plan and further raised its capex amount in 2023 (*Fig. 21*). TI decided to build seven new 300mm fabs starting from 2022. In 2022, TI had two old 300mm fabs in the US, one in Dallas (DMOS6), and another in Richardson (R-FAB-1), which ramped up production in 2009 and 2014, respectively. Therefore, this was the first time that TI had to build new 300mm fabs since 2014. The first two 300mm fabs had already ramped up production in 2H22, the fab in Richardson (RFAB-2) had an annual sales run-rate of USD5bn, and we estimate the acquired fab Lehi (L-Fab) has an annual sales run-rate of USD3bn. TI plans to build four other 300mm fabs in Sherman, Texas, with the first phase of two fabs to start production in late-2025, while another fab in Lehi (L-FAB2) is expected to start mass production in 2026E.

TI's internal 300mm wafer goal of >65% in 2026E, and >80% in 2030E

With its two new 300mm fabs ramp up in 2022, TI recorded its 300mm fab of 40% internal wafer contribution in 2022 (implying 200mm and below contributed the other 60%). TI believes its 300mm new fab can extend its cost competitive advantage and expects 300mm wafer to achieve >65% wafer contribution in 2026E, and >80% in 2030E along with capacity expansion in Lehi (L-FAB2) and Sherman (SM1). On the other hand, TI expects its total capacity can support sales of USD30/45bn in 2026E/2030E and use 85%/90% of its internal wafers (*Fig. 22*).

Fig. 21: TI capex plan



Note: 10%/15% means TI's capex expenditure comes to 10%/15% of its revenue each year in 2027 and beyond..
Source: Company data, Nomura research

Fig. 22: TI capex plan key metrics and revision

	2022	2026		2030	
Key metrics		2022 version	2023 version	2022 version	2023 version
Sales supported	~USD20bn	~USD24bn	**~USD30bn**	~USD34bn	**~USD45bn**
% wafer internal	80%	85%	>85%	90%	>90%
% of internal wafer 300-mm	40%	65%	>65%	75%	**>80%**
% of Assembly internal	60%	75%	>75%	85%	**>90%**

Source: Company data, Nomura research

Fig. 23: TI 300-mm fabs expansion

TI fabs		2009	2015	2017	2020	2021	2022	2023E	2024E	2025E	2026E	2030E
300-mm												
1	Dallas (DMOS6)		2000	3000	3000	3000	3000	3000	3000	3000	3000	3000
2	Richardson (RFAB-1)	2000	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000
3	**Richardson (R-FAB-2)**						1500	2000	5000	5000	5000	5000
4	**Lehi (L-FAB)**						500	1500	3500	3500	3500	3500
5	Lehi (L-FAB2)									1000	2500	5000
6	Sherman-1(SM1)									500	3000	5000
7	Sherman-2(SM2)									500	3000	5000
8	Sherman-3 (build after 2025)											4000
9	Sherman-4 (build after 2025)											4000
	300-mm capacity sales value(USD bn)	2000	7000	8000	8000	8000	10000	11500	16500	18500	25000	39500

Source: Company data, Nomura research

Street cautious on 300mm fab beneficiary

Analog IC has been mostly made in 200mm fab due to its small volume and variety of characteristics, which makes it cost advantageous compared with 300mm, due to its better wafer size, for analog vendors. On the other hand, TI is the largest vendor globally that makes analog IC in 300mm fab, because TI has sufficient volume to cover the high upfront cost of 300mm fab, and can also enjoy higher wafer area utilization. According to TI's estimates, the 300mm fab can reduce chip manufacturing cost by 40%, which results in 8% GPM improvement (*Fig. 24*). Thus, TI's 300mm fab expansion may create pressure on other analog vendors' pricing strategy when it expands its 300mm fab aggressively.

Fig. 24: TI estimates on 300-mm beneficiary

300-mm can reduce chip cost through better area utilization

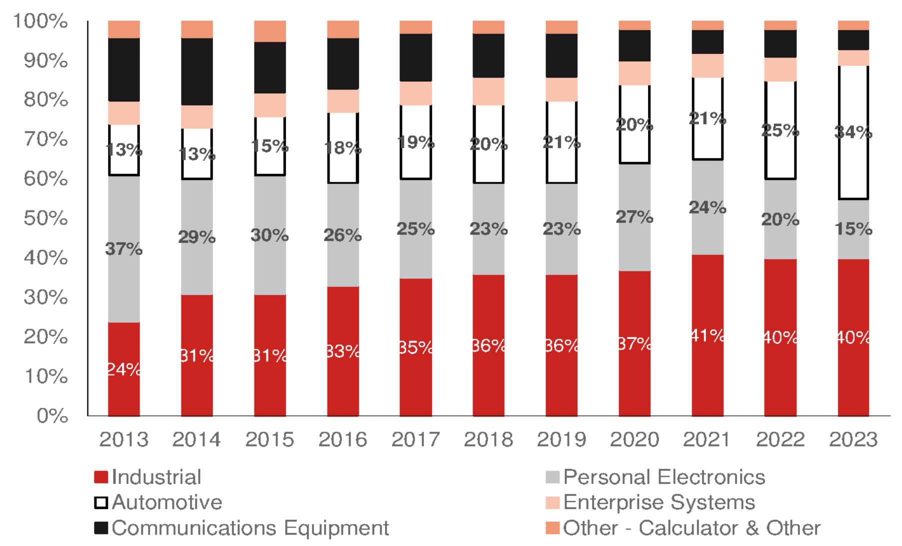
TI		Build on 200-mm wafer	Build on 300-mm wafer
Sales price of example part		1	1
Cost of goods:	**Chip cost**	**0.2**	**0.12**
	Asembly, test, others	0.2	0.2
	Total	0.4	0.32
GPM (%)		**60%**	**68%**

Source: Company data, Nomura research

Why didn't TI expand capacity aggressively in 2015-20?

From our observation, TI's overall UTR remained only at 60-80%(*Fig. 43*) during 2015-19. Thus, it was not necessary for TI to expand its new 300mm fab. According to TI's forecast, it will invest a higher R&D resource in industrial and automotive applications (74% of total sales in 2023), which also have a high growth opportunity, while others' R&D resource will remain flattish or slightly up.

Fig. 25: TI: Product mix (%)



Source: Company data, Nomura research

Fig. 26: TI's R&D investment allocation plan

Market segment	R&D investments	% of TI revenue 2013	2019	2023
Industrial	Up broadly	30%	36%	40%
Automotive	Up broadly	12%	21%	34%
Personal Electronics	Slightly up, continue to be selective	32%	23%	15%
Enterprise Systems	Steady	15%	11%	4%
Communications Equipment	Slightly up	6%	6%	5%
Other - Calculator & Other	Flat, at low levels	5%	3%	2%

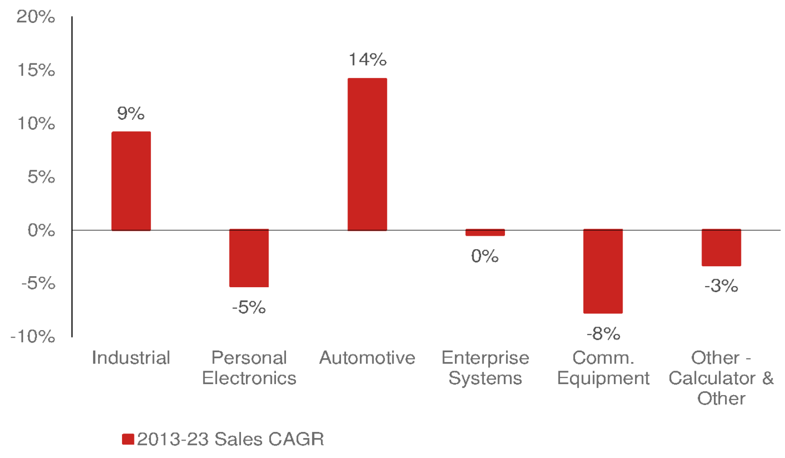
Source: Company data, Nomura research

TI's planned capacity in 2026E/30E implied industrial and auto to record 15% CAGR over 2026-30E

TI's capacity supported sales of USD30bn/USD45bn in 2026E/30E suggest industrial and auto combined sales would record a c.15% CAGR over 2026-30E, and other applications will remain at the same level as 2022.

TI's industrial and auto recorded a respective 9% and 14% sales CAGR between 2013 and 2023 (*Fig. 27*).

Fig. 27: TI application sales CAGR during 2013-23



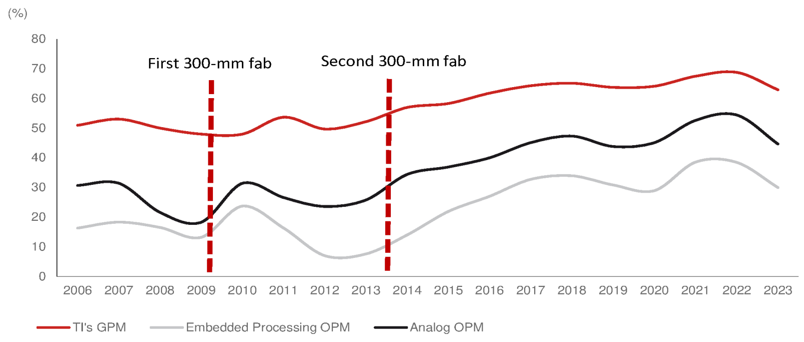
Source: Company data, Nomura research

Previous major 300mm fabs expansion in 2009 and 2013

TI built its first 300mm fab in 2009 and the second one in 2013, and TI's GPM remained at a similar level after first 300mm ramp up in 2009 to 2013, but continued to improve (*Fig. 28*) during 2014-22. *Fig. 29*

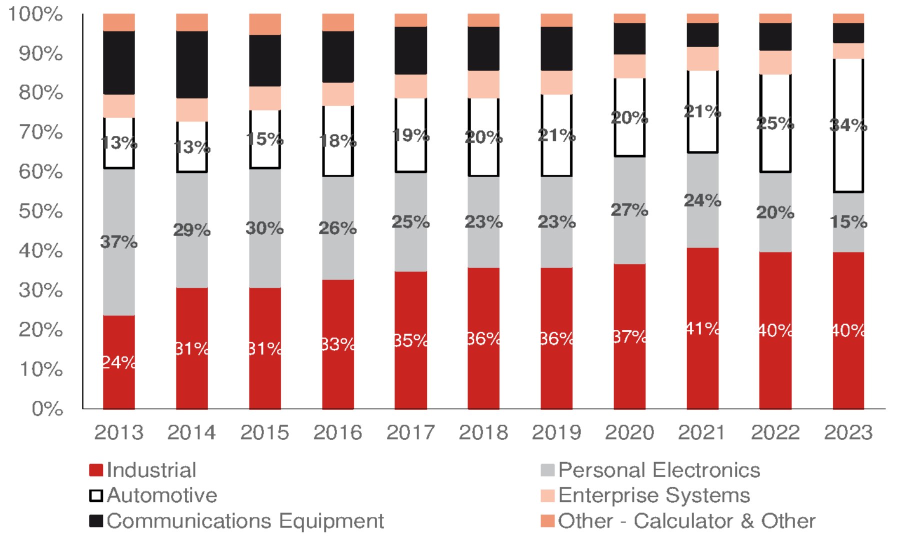
Fig. 28: TI's GPM, analog/embedded processing OPM trend

Analog OPM remained at similar level in 2010-13 before the first 300mm ramp up in 2006-07.



Source: Bloomberg Finance L.P., Nomura research

Fig. 29: TI: Product mix (%)



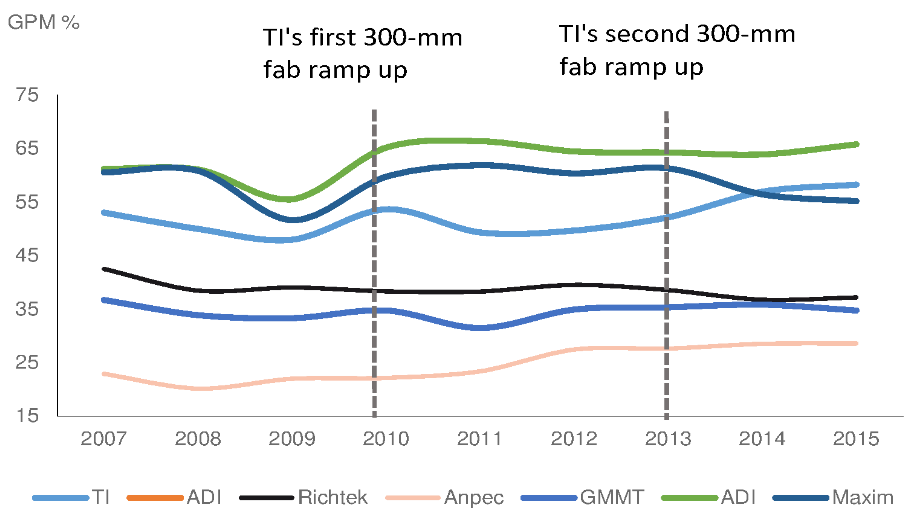
Source: Company data, Nomura research

Previous 300mm fabs impact on Greater China analog peers

We also look at other Asia analog vendors to gauge whether they were also affected by TI's new 300mm fab in the past.

During the 2010s, due to limited China analog vendors' data in that period, Taiwan analog fabless vendors were the major players in the Greater China market, and Richtek (unlisted) faced pressure in its GPM, which declined from a peak level of 42% in 2009 to 37% in 2015. GMMT's (8081 TT, Not rated) and Anpec's (6138 TT, Not rated) GPM remained stable or slightly improved within TI's 300mm fab expansion period.

Fig. 30: Global and Taiwan analog peers' GPM over 2007-15



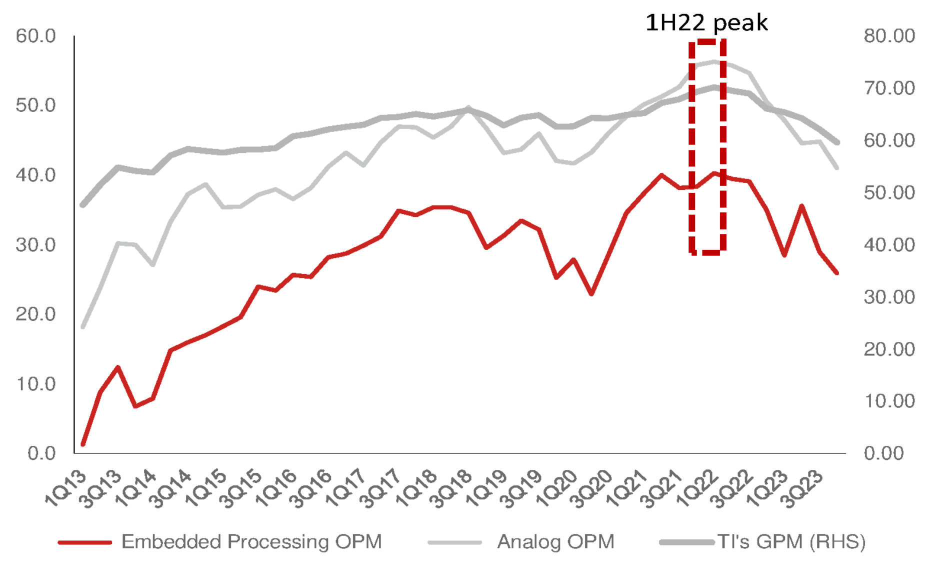
Source: Bloomberg Finance L.P., Nomura research

TI's return with aggressive pricing

However, after the shortage eased, TI decided to return to the China analog market and followed an aggressive pricing strategy through its strong 300mm fab ramp-up in 2H22. Through our industry channel checks, we noticed TI cut prices by 10-20% in 4Q22 and another 10-20% in 1H23 in the Greater China market for consumer application.

Considering TI's OPM trend, its analog product OPM declined further to 41% in 4Q23, from the 1Q22 peak of 56.3%, which is the lowest level since 2016. Embedded process OPM declined to 26% in 4Q23 from the 1Q22 peak level of 40%. The OPM decline was relatively low in embedded process than in analog.

Fig. 31: TI's quarterly GPM and analog/embedded processing OPM



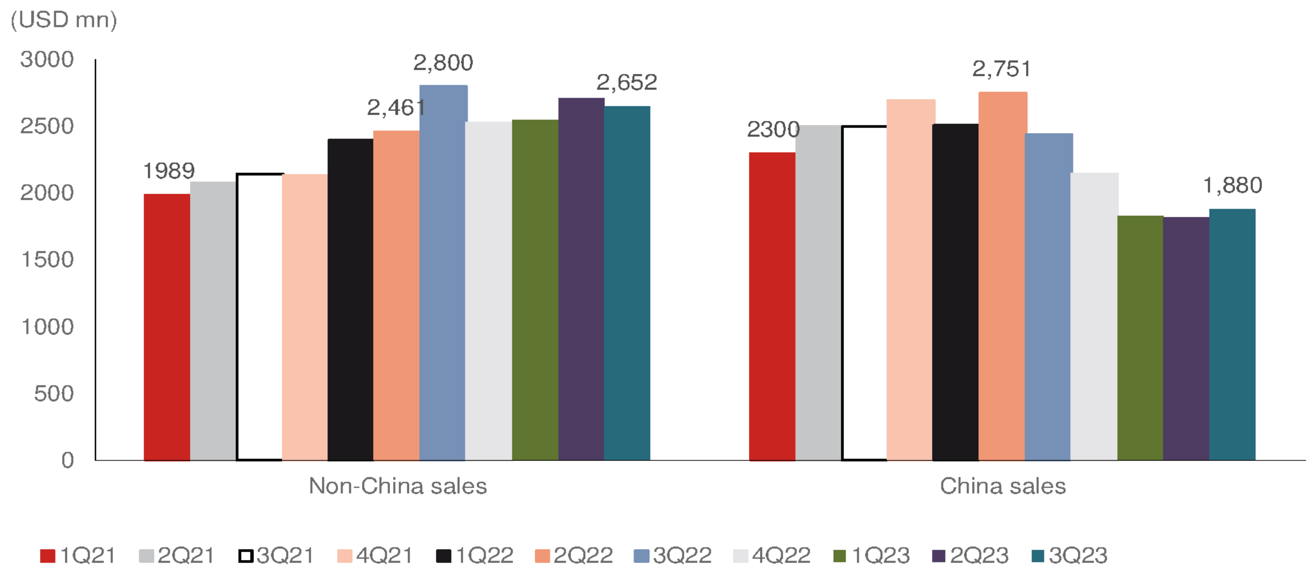
Source: Company data, Nomura research

TI's China market strategy

Facing global IC shortages in 2020-22, TI prioritized the capacity for high-end products and applications, and we think it does provide a substitution opportunity for China local vendors, which we emphasized in our previous analog IC report in 2021 (*link*). After capacity shortage eased in 2H22, TI returned to the China analog market with an aggressive pricing strategy to compete with local vendors to offset weak end-demand impact. Therefore, TI's China sales mix fell from 55.3% in 2020 to 49.2% in 2022, and further declined to mid-to-low 40s% in 1H23, but stabilized in 3Q23. TI's China sales declined to USD1.8-1.9bn in 1Q-3Q23 from its quarterly peak level of USD2.4-2.7bn in 2022, but stabilized in 3Q23. China sales declined at a higher pace of mid-20s% vs non-China sales which declined 5-10% from its peak level in 1H23, owing to China customers adjusting inventory earlier than non-China customers leading to loss of market share in 2022.

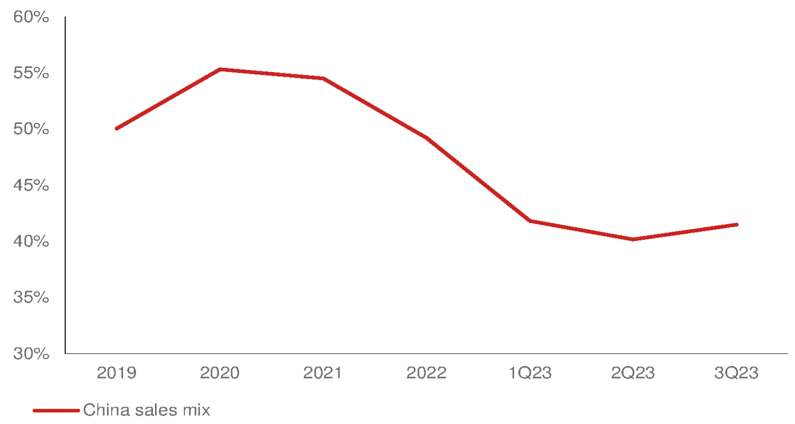
Fig. 32: TI's China and non-China sales comparison

The sales decline is higher in China than non-China region



Source: Bloomberg Finance L.P., Nomura research

Fig. 33: TI: China sales mix (%) trend



Source: Company data, Nomura research

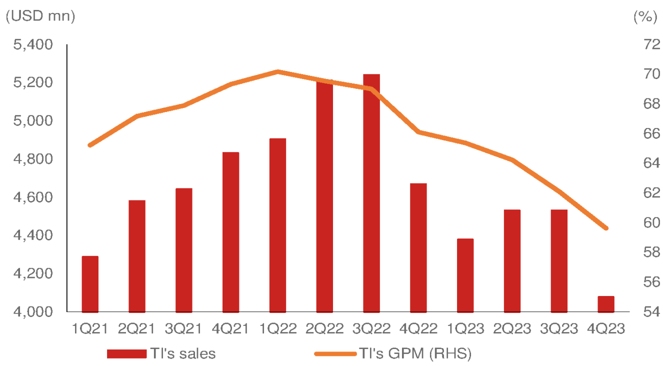
TI's pricing strategy

Its aggressive pricing strategy was a result of: 1) the Analog IC industry downturn, high inventory pressure and weakening sales performance; 2) high GPM due to depreciated capacity and benefit from 300mm fabs; and 3) its focus on acquiring market share.

1) Industry downturn:

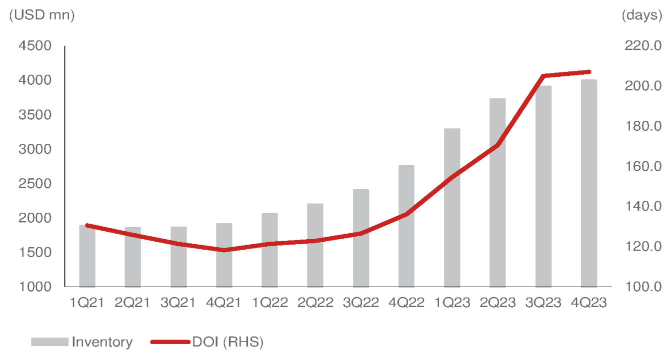
In 1H22, TI's sales were affected by weak consumer demand, and the semi industry was also entering a downcycle. TI's inventory and days were also under pressure.
Inventory began to increase in 1H22, and inventory days also increased from 2021's 120 days to 4Q23's 200+ days (*Fig. 35*).

Fig. 34: TI: quarterly sales and GPM



Source: Company data, Nomura research

Fig. 35: TI: inventory and DOI



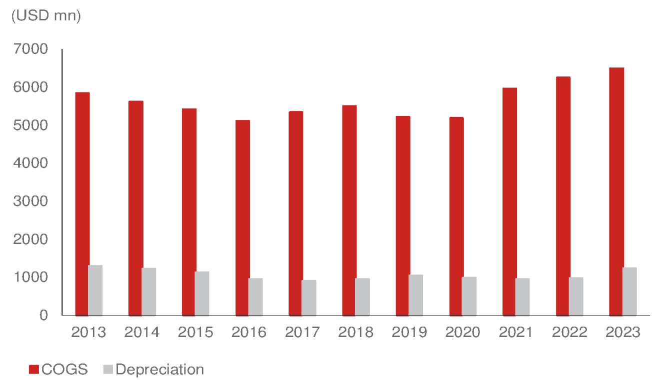
Source: Company data, Nomura research

2) High GPM:

TI's analog IC products mainly operate through the IDM model, and TI's high GPM also provides it competitive advantage in pricing. We list below several reasons behind TI's high GPM.

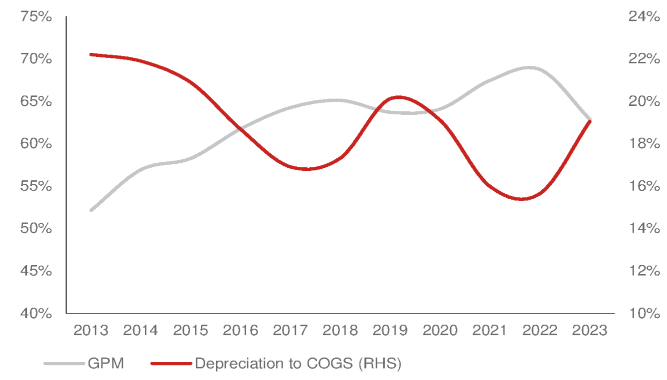
a) 300mm wafer and low depreciation cost: Most of TI's fabs have been largely depreciated. TI's latest 300mm fab before 2022 was the Dallas fab (DMOS6) in 2013-14. Except during lower sales in the 2018-19 cycle downturn, TI has maintained stable depreciation costs, and the ratio of depreciation cost to COGS has declined over the past 10 years (*Fig. 37*), even during the 2021-22 chip shortage.

Fig. 36: TI: COGS and depreciation



Source: Company data, Nomura research

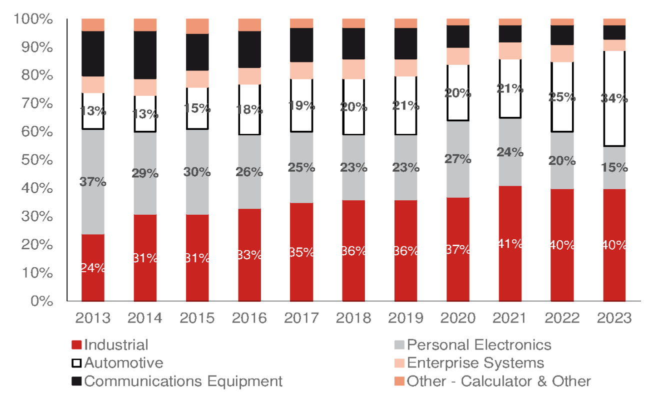
Fig. 37: TI: GPM and depreciation cost to COGS (%)



Note: Depreciation to COGS increase in 2019-20 owing to lower sales scale
Source: Company data, Nomura research

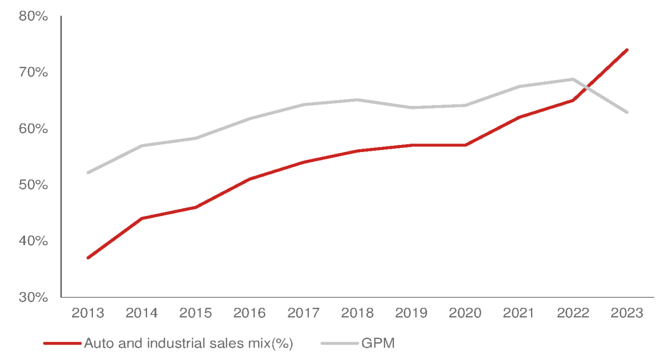
b) Better product mix: TI's industrial and auto application sales have significantly increased over the past 10 years (*Fig. 38*). Industrial and auto products have high entry barriers and gross profit.

Fig. 38: TI: Product mix (%)



Source: Company data, Nomura research

Fig. 39: TI's auto and industrial sales mix vs GPM



Source: Company data, Nomura research

c) High direct sales mix: TI announced its TI.com platform to provide customers with a way to order product directly to improve sales efficiency and quality, and also to lower its channel and distributor cost; thus, the direct sales mix increased from 2019's 35% to 2022's 70%.

3) Focus on acquiring market share

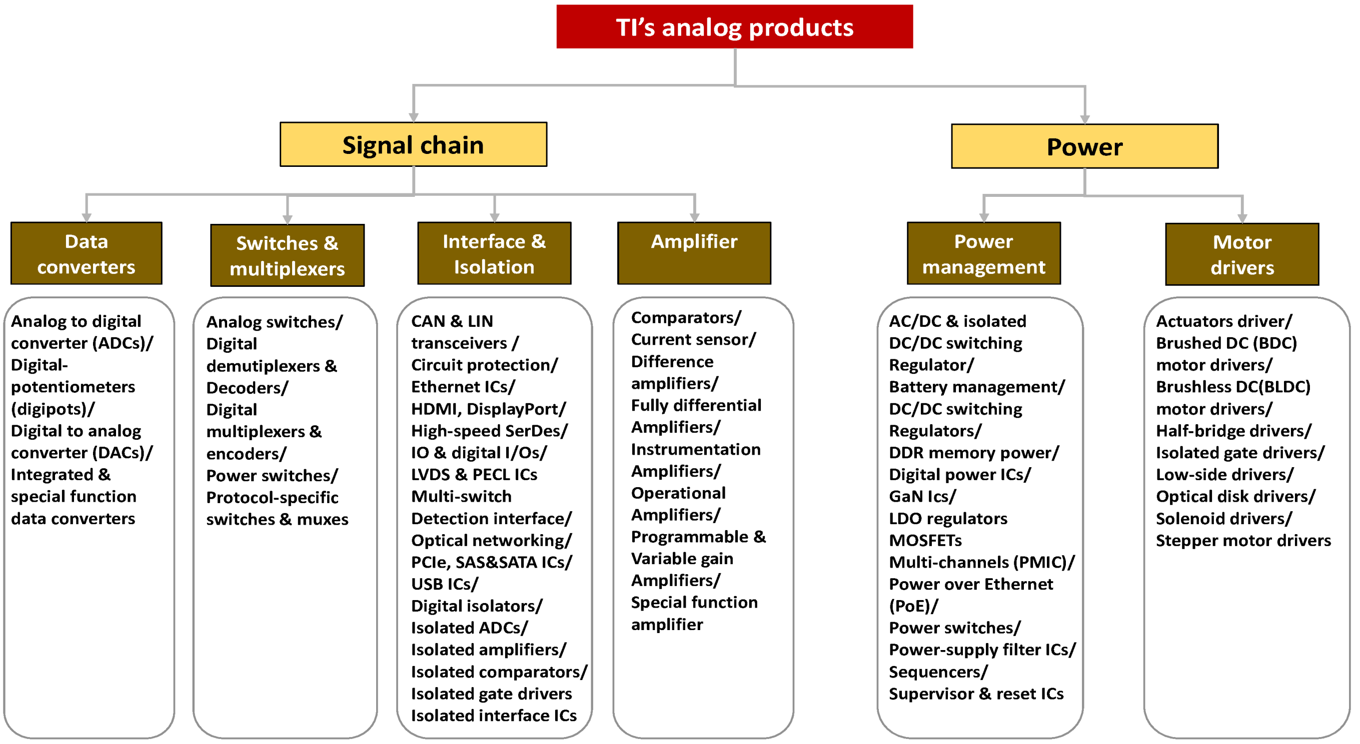
During TI's 2023 earnings call, the company reiterated that it would continue to monitor its product pricing and always focus on acquiring market share. That said, TI uses pricing as one of its advantage to satisfy customers' demand for low-cost products.

TI's aggressive pricing significantly impacts China consumer market

Within TI's product portfolio, the general-purpose power products for consumer applications is the major product impacted by pricing. These analog products have lower entry barriers, and hence, large local consumer electronics clients have an incentive to adopt local vendors' products with the goal of localization in China. We also notice China local analog vendors' products are also entering some high-end power products.

Due to the general purpose products being mostly pin-to-pin products, supplier replacement is not difficult, in our view. As a result, local vendors are naturally impacted by TI's pricing pressure and market share focus. Although industrial and auto products' entry barriers are higher due to their long verification cycle, high safety requirement, and high customer loyalty, we have also seen that some vendors' pricing and profitability have also been impacted by TI's pricing strategy.

Fig. 40: TI: Analog product portfolio



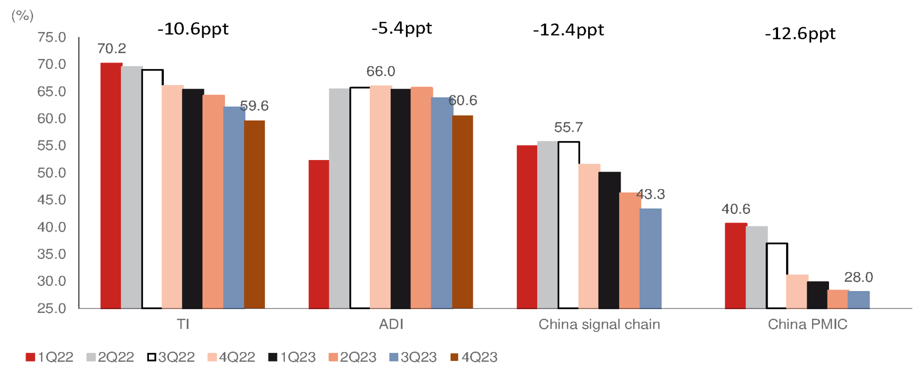
Source: Company data, Nomura research

Significant y-y decline in GPM for China vendors

China consumer analog vendors, which originally had pricing and cost advantages, have been severely impacted by margin pressure since 2H22. In addition to competition from TI, China analog vendors are also facing high competition from local peers.

In both PMIC or signal chain markets, most China local vendors witnessed significant sales and GPM decline in 2H22 and 1H23. Hence, this shows that TI's strategy not only intensified competition in the entire analog chip industry, but also caused many local analog vendors to be passively involved in price wars, causing them to suffer heavy losses.

Fig. 41: GPM comparison over 1Q22-4Q23



Source: Company data, Nomura research

Smaller room for ASP downside from current level

TI targeting sales CAGR of 10% in 2022-26E

At TI's 2023 capital day, management raised its long-term sales growth CAGR through 2030 from **7%** to **10%.** TI management stated three major reasons behind the sales CAGR goal.

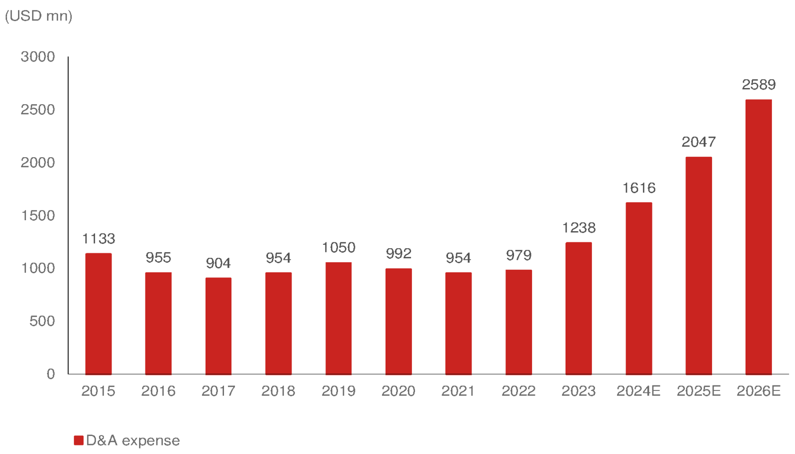
1) Semi content growth: Particularly in auto and industrial, which will witness structural content growth through faster EV adoption and factory automation in the next 10 years.

2) Market position: TI management believes the adoption of its products will increase due to its strong product portfolio. However, management doesn't expect that the market share gain will happen quickly in both analog and embedded processing products. Market share gain of 0.3-0.4% annually is normal for TI.

3) Supply chain security: TI thinks the supply chain will become more cautious on supply chain security now after the pandemic and geopolitical tensions in 2020-22. Customers are looking to understand resiliency of their supply chains, so they find TI highly attractive.

According to management, TI's depreciation cost will significantly increase through 2026F (*Fig. 42*) due to its aggressive 300mm capacity expansion. At TI's capital day, management stated that TI changed its pricing behavior when capital intensity increased. We have seen TI's aggressive pricing behavior since 2H22.

Fig. 42: TI's depreciation cost

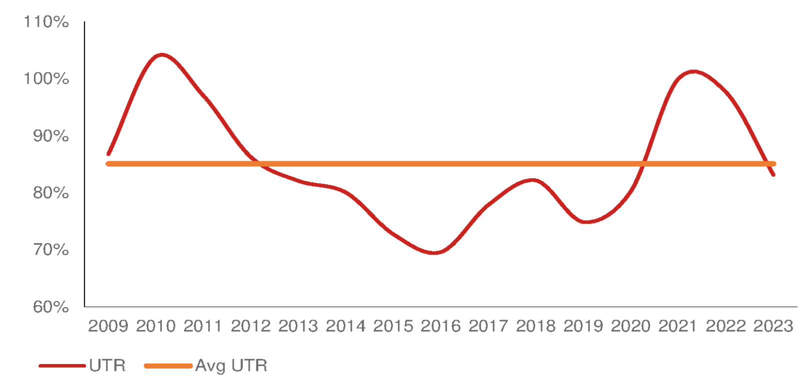


Source: Company data, Bloomberg Financials L.P., Nomura research

Sales CAGR and GPM target analysis through ASP and UTR

By our observation, TI's long-term UTR would reach between 80% and 90% (*Fig. 43*). We think in order to achieve TI's sales CAGR target of **10%**, the ASP and UTR will be major factors. Therefore, we can analyze the ASP and UTR difference to better understand TI's limited capacity to lower ASPs. TI's 2023 UTR of 83% and 2023 GPM of 62.9% imply a blended ASP decline of 5% vs 2022 level.

Fig. 43: TI's UTR



Source: Company data, Nomura research

ASP and UTR analysis suggests limited ASP downside from now

In *Fig. 44* - *Fig. 45* , we show that if TI cuts prices by 10% (implying 10% less than 2022's ASP), it requires full capacity utilization of 90%+ to 100% throughout the year to achieve its 10% CAGR goal. However, we think a UTR of 80-90% is more reasonable for TI's IDM model, based on its historical average of 85%. That said, if TI's UTR remains at an average of 85%, we think ASPs cannot decline much from now in order to achieve the above sales CAGR goal in 2026.

Fig. 44: 2023-24F TI GPM and sales CAGR sensitivity analysis

GPM	2023			
UTR / ASP	100%	90%	80%	70%
100%	65.4%	64.8%	64.0%	62.9%
95%	63.6%	**62.9%**	62.1%	61.0%
90%	61.6%	60.9%	60.0%	58.8%
85%	59.3%	58.6%	57.6%	56.4%
80%	56.8%	56.0%	54.9%	53.6%

GPM	2024			
UTR / ASP	100%	90%	80%	70%
100%	65.3%	64.6%	63.8%	62.7%
95%	63.5%	62.8%	61.9%	60.8%
90%	61.5%	60.7%	59.8%	58.6%
85%	59.2%	58.4%	57.4%	56.2%
80%	56.6%	55.8%	54.8%	53.4%

Annual sales run-rate	2023			
UTR / ASP	100%	90%	80%	70%
100%	21,069	18,962	16,855	14,749
95%	20,016	18,014	16,013	14,011
90%	18,962	17,066	15,170	13,274
85%	17,909	16,118	14,327	12,536
80%	16,855	15,170	13,484	11,799

Annual sales run-rate	2024			
UTR / ASP	100%	90%	80%	70%
100%	26,952	24,257	21,561	18,866
95%	25,604	23,044	20,483	17,923
90%	24,257	21,831	19,405	16,980
85%	22,909	20,618	18,327	16,036
80%	21,561	19,405	17,249	15,093

Sales CAGR	2023			
UTR / ASP	100%	90%	80%	70%
100%	5%	-5%	-16%	-26%
95%	0%	-10%	-20%	-30%
90%	-5%	-15%	-24%	-34%
85%	-11%	-20%	-28%	-37%
80%	-16%	-24%	-33%	-41%

Sales CAGR	2024			
UTR / ASP	100%	90%	80%	70%
100%	16%	10%	4%	-3%
95%	13%	7%	1%	-5%
90%	10%	4%	-2%	-8%
85%	7%	1%	-4%	-11%
80%	4%	-2%	-7%	-13%

Source: Nomura estimates

Fig. 45: 2025-26F TI GPM and sales CAGR sensitivity analysis

GPM	2025			
ASP \ UTR	100%	90%	80%	70%
100%	64.3%	63.5%	62.6%	61.3%
95%	62.4%	61.6%	60.6%	59.3%
90%	60.4%	59.5%	58.4%	57.0%
85%	58.0%	57.1%	56.0%	54.5%
80%	55.4%	54.4%	53.2%	51.7%

GPM	2026			
ASP \ UTR	100%	90%	80%	70%
100%	63.9%	63.1%	62.1%	60.8%
95%	62.0%	61.2%	60.1%	58.7%
90%	59.9%	59.0%	57.9%	56.4%
85%	57.6%	56.6%	55.4%	53.9%
80%	54.9%	53.9%	52.6%	51.0%

Annual sales run-rate	2025			
ASP \ UTR	100%	90%	80%	70%
100%	29,305	26,374	23,444	20,513
95%	27,839	25,055	22,272	19,488
90%	26,374	23,737	21,099	18,462
85%	24,909	22,418	19,927	17,436
80%	23,444	21,099	18,755	16,411

Annual sales run-rate	2026			
ASP \ UTR	100%	90%	80%	70%
100%	35,187	31,668	28,150	24,631
95%	33,428	30,085	26,742	23,399
90%	31,668	28,501	25,335	22,168
85%	29,909	26,918	23,927	20,936
80%	28,150	25,335	22,520	19,705

Sales CAGR	2025			
ASP \ UTR	100%	90%	80%	70%
100%	14%	10%	5%	1%
95%	12%	8%	4%	-1%
90%	10%	6%	2%	-3%
85%	8%	4%	0%	-5%
80%	5%	2%	-2%	-6%

Sales CAGR	2026			
ASP \ UTR	100%	90%	80%	70%
100%	15%	12%	9%	5%
95%	14%	11%	7%	4%
90%	12%	9%	6%	3%
85%	11%	8%	5%	1%
80%	9%	6%	3%	0%

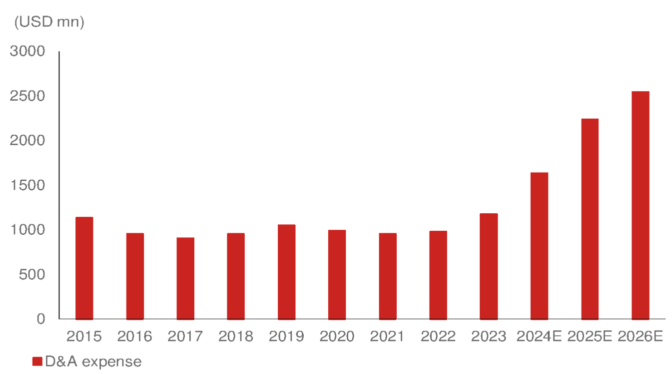
Source: Nomura estimates

TI's 70% GPM goal set in 2022

On the other hand, at its Feb-2022 Capital Day, TI's management mentioned its GPM target of 70-75% is based on its old capex plan of USD3.5bn each year between 2022-25. TI raised its capex plan in Feb-2023 and didn't mention about the GPM range of 70-75%.

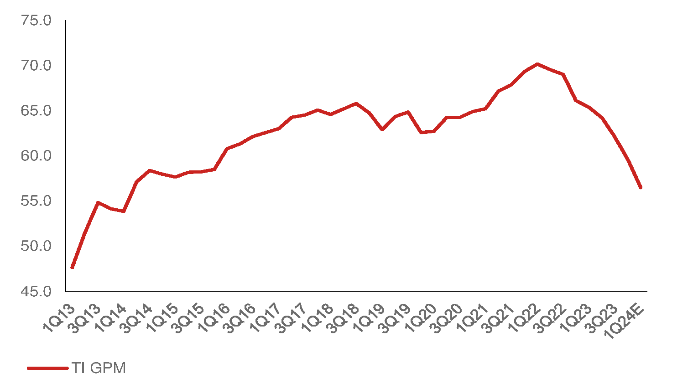
TI's depreciation cost will increase significantly after its significant 300mm fab capex from 2022 (*Fig. 46*), which is different from TI's previously low depreciation cost advantages. Therefore, if TI tries to maintain its GPM at above 60%, the ASP can't decline higher than 5% from the current level when UTR is at 80-90%.

Fig. 46: TI: Depreciation cost and consensus estimates



Source: Bloomberg Financials L.P., Nomura research

Fig. 47: TI: Quarterly GPM



Note: 1Q24 estimates are Bloomberg consensus
Source: Company data, Nomura research

300mm wafer advantage, product mix, and direct sales mix

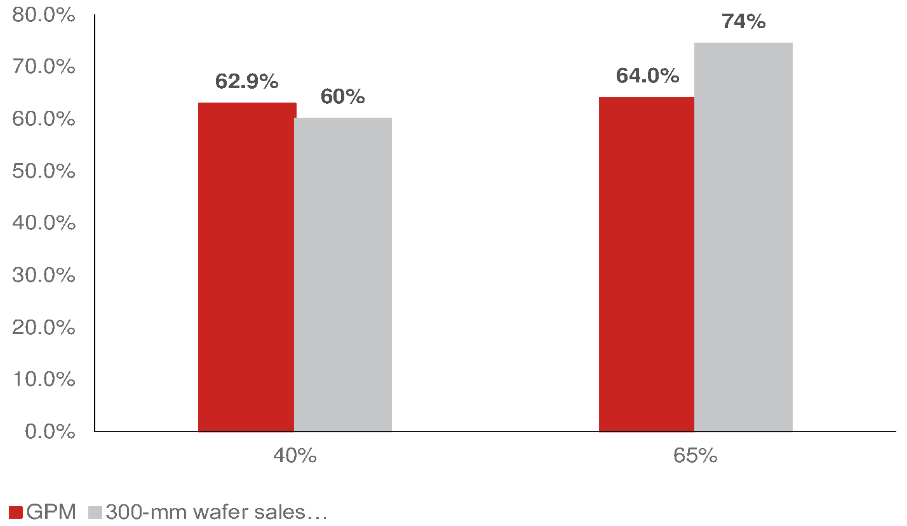
According to management, TI will continue to strengthen its GPM through its high 300mm internal wafer mix, better product mix, and higher direct sales mix.

GPM improves by 1pp if 300mm internal wafer mix goes to 65% from the current 40%

In 2023, 40% of TI's internal wafers came from the 300mm fab. 300mm wafer area is 2.25x larger than 200mm wafer, and thus, 300mm may have accounted for 60% of internal wafer sales. Also, 300mm wafer has 40% lower chip process cost than 200mm wafer. Based on its 2023 financials, if we assume TI consumes 65% (TI's goal for 2026E) of its wafers from 300mm, this would suggest it accounts for 74% of total sales, or 14% of its COGS can be reduced through the 300mm wafer advantage. Our sensitivity analysis suggests GPM improves by c.1pp (*Fig. 48*) when the 300mm wafer mix increases from 40% to 65%.

Fig. 48: 300mm wafer GPM comparison

300mm wafer mix of 65% can only improve GPM by 1pp vs current mix of 40%

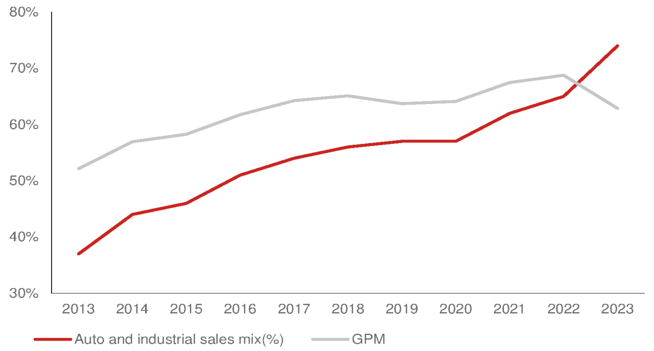


Source: Company data, Nomura estimates

Narrowing GPM benefit from product mix and TI.com direct sales

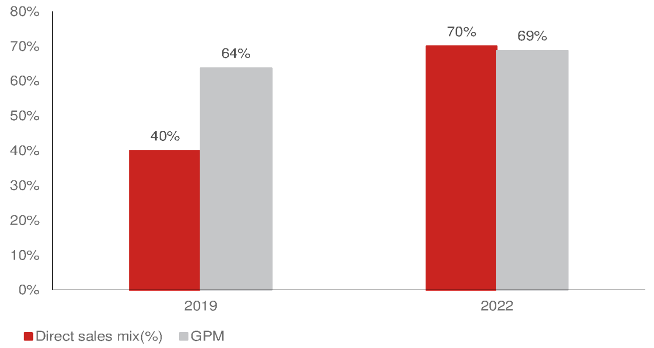
On the other hand, according to TI management, TI's further GPM improvement from better product mix and direct sales through TI.com would be challenging. We assume that TI's product strategy will continue to focus on industrial and auto applications, while TI's auto and industrial sales mix already accounted for 74% of its 2023 sales. Also, sales through TI.com account for 70% of its total sales. Due to limited growth opportunity from product mix and direct sales mix for TI, management thinks the benefit to GPM could gradually decrease.

Fig. 49: TI's auto and industrial sales mix vs GPM



Source: Company data, Nomura research

Fig. 50: Direct sales mix vs GPM



Source: Company data, Nomura research

China analog market – localization trend unchanged

China local vendors still have significant opportunity to grow in China analog market

We provided a basic introduction to analog ICs in our previous analog IC Anchor Report (*link*), where we stated that analog ICs have a broad end-applications due to which normal analog IC companies have more products than other MCU, memory, or logic IC companies.

Due to analog products' long lifecycle, lower market share of leading global vendors (compared to memory, smartphone AP, GPU, CPU, etc.), and broad applications (used in all electronics such as communications, industrial, consumer, and auto), we believe China analog fabless can leverage enough Greater China foundry partners to compete with global peers. Thus, analog ICs will become a growth market for new vendors, in our view.

Most analog fabless IPO through STAR market since 2019

Since the launch of the Sci-Tech Innovation Board (STAR market, 科創版) on the Shanghai exchange in June 2019, the STAR market has become the major stock exchange for start-ups. Most China analog companies are listed on the STAR market (*Fig. 51*), which has accelerated its IPO review and fund-raising process.

Fig. 51: China major analog companies' IPO market/year

Most China analog companies are listed through Shanghai STAR market

Company	Chinese name	Ticker list	Stock Exchange	Year
Silergy	硅力杰	**6415 TT**	TAIEX (Taipei)	2013
SG Micro	圣邦	**300661 CH**	Shanghai A-share	2017
Fullhan	富瀚微	**300613 CH**	ChiNext (Shenzhen)	2017
Fine Made	富满微	**300671 CH**	ChiNext (Shenzhen)	2017
Bpsemi	晶丰明源	**688368 CH**	STAR (Shanghai)	2019
3Peak	思瑞浦	**688536 CH**	STAR (Shanghai)	2020
AWINIC	艾为电子	**688798 CH**	STAR (Shanghai)	2021
Joulwatt	杰华特	**688141 CH**	STAR (Shanghai)	2022
Injoinic	英集芯	**688209 CH**	STAR (Shanghai)	2022
Kiwi instrument	必易微	**688045 CH**	STAR (Shanghai)	2022
Cellwise	赛微微电	**688325 CH**	STAR (Shanghai)	2022
Novosense	纳芯微	**688052 CH**	STAR (Shanghai)	2022
Dioo	帝奥	**688381 CH**	STAR (Shanghai)	2022
halomicro	希荻微	**688173 CH**	STAR (Shanghai)	2022
Southchip	南芯	**688484 CH**	STAR (Shanghai)	2023

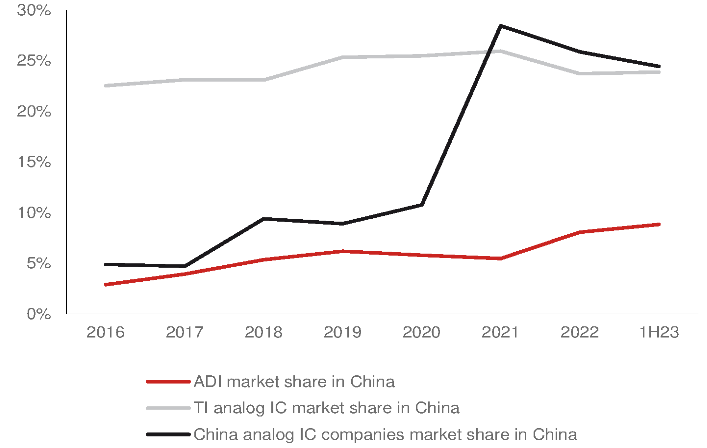
Source: Company data, Nomura research

Two global IDMs gained share in China analog market in 1H23

Analog ICs have a large variety of product applications, and China local vendors started developing products in the low-end pin-to-pin market at an early stage (before 2019). We believe most low-end consumer analog products are highly localized now, which is supported by our previous analog anchor report where we stated that there's an opportunity for local vendors to replace TI in the China market. However, except pin-to-pin products or consumer applications, we believe China analog IC vendors are needed to compete with global vendors.

The China Semiconductor Industry Association (CSIA) forecasts China analog sales of CNY55.5bn in 2022, which we think accounts for 26% market share of the China analog market. In 1H23, global analog vendors gained market share due to several reasons (mentioned below), while local vendors market share growth plateaued, as per our estimates. **1) Pricing advantages:** TI cut pricing aggressively to gain share from local vendors in 1H23. **2) Consumer application weakness:** China local vendors' end-applications are focusing more on consumer products, which are impacted most from globally weak macro, while global IDM sales have more exposure to industrial or automotive applications, where demand is more resilient than consumer products. **3) Leading position:** In an industry downturn, customers will prefer to place more orders with leading vendors (TI or ADI) with a richer product portfolio, larger capacity, and better pricing.

Fig. 52: TI, ADI, China analog IC companies market share in China



Source: WSTS, CSIA, Nomura estimates

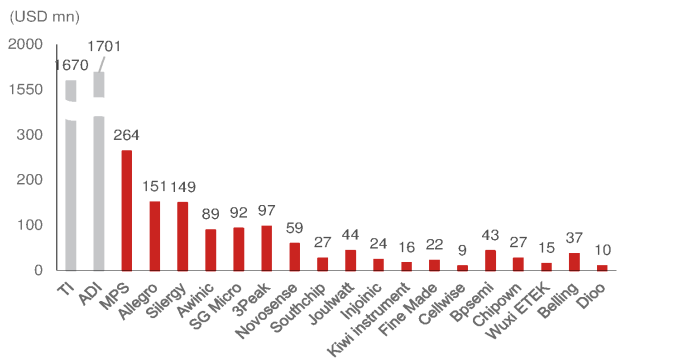
China analog market localization opportunity unchanged

We believe that China analog fabless will continue to regain market share from global vendors through their **richer product portfolios**, **competitive foundry support**, and **localization requirements** to expand local market share.

China local vendors’ number of products increase significantly

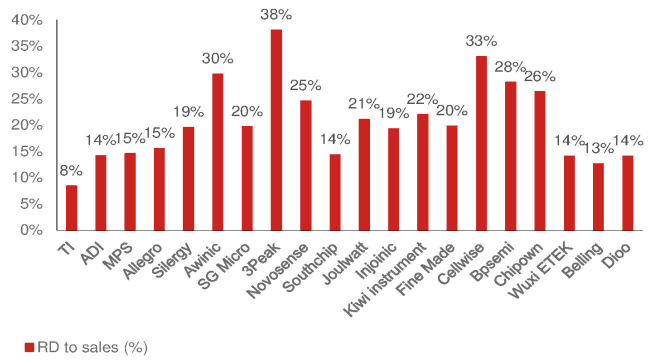
According to our analysis, major analog companies' product portfolio have grown significantly (*Fig. 55*). Compared to 2021, most China analog vendors’ number of products grew more than twice within two years. For customers, a complete analog product portfolio can save their time and cost; thus, we saw that China analog vendors’ product development speed has significantly accelerated from 2021’s 200-300 new products each year to 2023’s 500-1,000.

Fig. 53: Global and China analog IC companies' RD expense in 2022



Source: Company data, Nomura research

Fig. 54: Global and China analog companies' RD expense to sales (%)



Source: Company data, Nomura research

Fig. 55: China analog vendors' number of products

Number of products		
Company	2021	End-2023
Silergy	2000	5000
SG Micro	1700	5000
AWINIC	470	1100
Joulwatt		1000
Injoinic		3000
Kiwi instrument		700
3Peak	470	2400
Novosense	800	1500
Dioo		1400
Chipown	770	1500

Source: Company data, Nomura research

Case study: Taiwan analog IC companies delivered fast growth before they reached 40-50% share of their TAM by 2010

One of the most attractive parts of the China analog IC market is its large addressable market that Silergy estimates at CNY200bn (similar to WSTS's forecast of USD26bn in 2023). According to CSIA, China analog IC companies had combined sales of CNY50bn in 2023, while for Silergy it is estimated to be CNY30bn.

Given our estimate that the combined analog IC sales from listed China analog IC companies were c.CNY20bn (USD2.6bn) in 2022 excluding Silergy (*Fig. 56*), or c.CNY23bn (USD3.1bn) including Silergy's sales to China domestic customers (i.e., Silergy reported c.USD800m 2022 sales, with half of the contribution coming from China local customers), it looks to us that Silergy's estimate might be more realistic than the CSIA data. We can accordingly conclude that the current localization rate of China analog IC market is likely at 15-20% level.

Fig. 56: China analog IC companies' analog IC sales in 2022

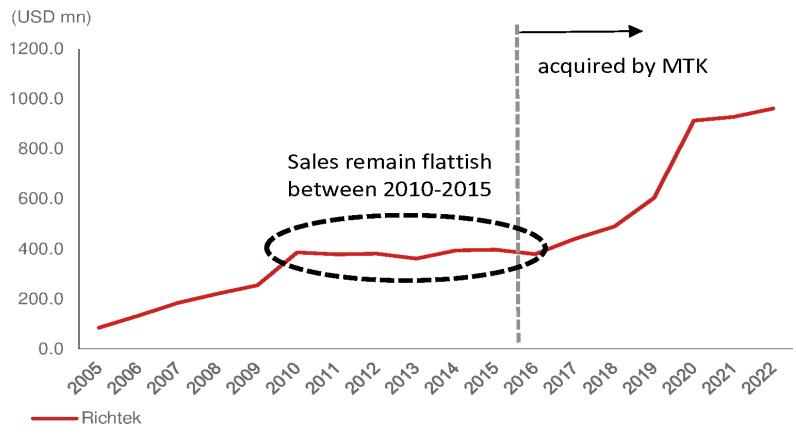
Companies	Analog IC sales (USD mn)
Silergy	793.0
SG Micro	474.7
AWINIC	313.3
Southchip	227.2
Joulwatt	213.6
Injoinic	96.5
Kiwi instrument	61.5
Fine Made	35.9
Cellwise	29.8
Bpsemi	145.0
Chipown	107.2
Wuxi ETEK	115.4
Belling	257.4
3Peak	266.7
Novosense	248.0
Dioo	79.3
Total	3068.1
Total ex-Silergy	2671.6

Source: Company data, Nomura research

Richtek's sales remained flattish over 2010-15

What does the 15-20% localization level imply to us? The case study on Taiwan analog IC industry sales vs its TAM back in mid-to-late-2000s (2006-09) could be a good reference. This is because, Richtek was the most outstanding analog IC company in Asia during 2000-10 but it entered 5-6 years of nearly no growth in 2010-16 (before being acquired by Mediatek [2454 TT, Neutral] in 2016; *Fig. 57*).

Fig. 57: Richtek: Annual sales between 2005-22



Source: Company data, Nomura research

Taiwan analog beneficiary of consumer analog substitution since early-2000s

We think the Taiwan analog market growth can be a reference to assess the growth opportunity in the China analog market. Taiwan analog fabless companies were beneficiaries of import substitution as local ODM/OEM/EMS companies started replacing global analog vendors' products between 2003-07. However, we think the addressable market for Taiwan analog companies should be restricted largely to major consumer electronics, rather than the whole analog market.

Taiwan analog companies recorded 40-45% global consumer analog market share

On our breakdown of addressable market for Taiwan analog IC industry and Richtek in 2007 (*Fig. 58*, *Fig. 59*), even Richtek likely hit the growth ceiling when Taiwan analog IC companies captured 20-30% of global (or 40-50% of Taiwan/China production) consumer electronics-related analog IC content by 2010.

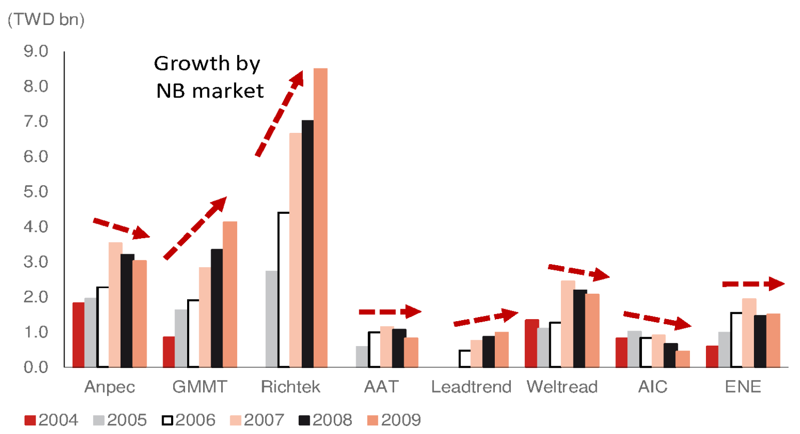
We believe this is why Silergy (and other ambitious China analog ICcompanies) are expanding their analog IC product portfolios and applications rapidly. Their addressable markets in China today are much bigger than they were in the late-2000s in Taiwan given a significantly larger domestic market and application expansion beyond consumer electronics products (such as auto and industrial). If China analog IC companies are only capturing 15-20% of their TAM now, they should still have decent growth upside, in our view.

Separately, if we compare the localization rate of analog IC vs other semiconductor chips in China, we note that the analog IC localization rate of 15-20% now is likely well below others given TI's strong competitiveness and leadership (such as fingerprint, RFFE, CIS etc.; *Fig. 61*).

Fig. 58: Taiwan analog companies' market share by different TAM in 2007

Source: Nomura estimates

Fig. 59: Taiwan analog companies' sales



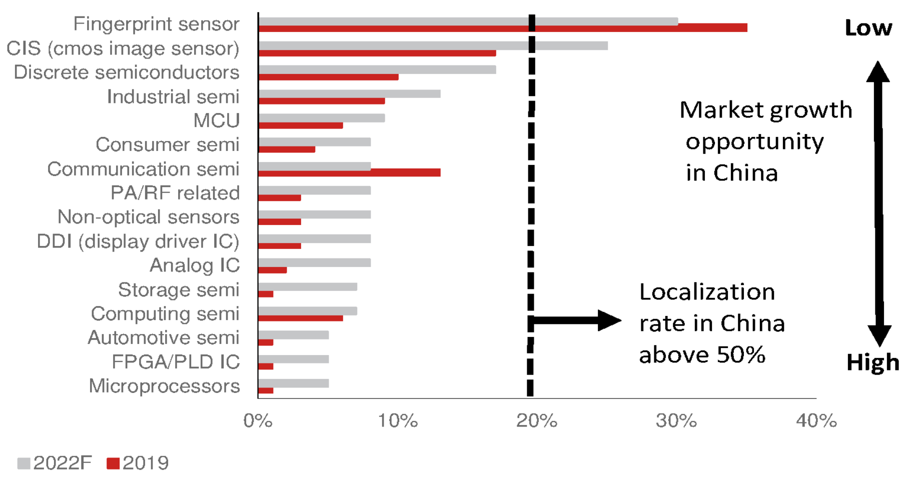
Source: Company data, Nomura research

Fig. 60: Analog IC content of major consumer electronics products and Taiwan analog IC vendors' market share in 2007

		Shipment (mn)		Value (USDmn)	~ manufacturered by TW+CN (2007)			Taiwan vendors' sales	TAM: ww 3C products	TAM: TW 3C products
Application	ASP/pcs (US$)	2007	09-13 CAGR	2007	%	shipments (mn)	Value (USD)		Taiwan's mkt shr	Taiwan's mkt shr
MB -analog IC -MOS	3.0-3.5 1.5-2.0 1.5-2.0	138	3%	500-550 250-300 250-300	95%	131.1	450-500 200-250 200-250	203	45%	47%
NB -analog IC -MOS	6.5-7.0 3.0-3.5 3.5	108	22%	700-750 c.350 350-400	80%	86.4	550-600 250-300 300-350	79	11%	13%
VGA card -analog IC -MOS	0.6 0.5 0.1	85	2%	c.50 c.40 c.10	95%	80.75	c.50 c.40 c.10	25	49%	52%
Mobile -analog IC -MOS	0.8-1.3 0.7-1.2 0.1	1150	9%	1300-1500 1150-1350 100-150	26%	300	350-400 300-350 <50	45	3%	12%
DSC -analog IC -MOS	1 1 <0.1	105	8%	100-120 100-110 <10	45%	47.25	c.50 45-50 <5	33	29%	65%
Networking -analog IC -MOS	0.5 0.5 0	290	26%	100-150 100-150 0	80%	232	80-130 80-130 0	35	27%	34%
Large panel -analog IC -MOS	TV:0.6-0.65; IT:0.5 TV:0.6-0.65; IT:0.5 0	400	14%	200-250 200-250	TV:45%, MNT:70%, NB:40%	222.85	100-130 100-130	53	25%	45%
CCFL inverter -analog IC -MOS	TV:1.3; IT:0.3 TV:0.3; IT:0.15 TV:1; IT:0.15	400	14%	200-250 70-80 120-150	TV:45%, MNT:70%, NB:40%	222.85	100-120 30-50 60-70	21	10%	20%
Control board -analog IC -MOS	TV:0.6-0.7; MNT:0.3-0.4 TV:0.6; MNT:0.3 TV: <0.1; MNT: <0.1	233	14%	100-140 90-110 10-30	TV:20%, MNT:80%	139	50-70 40-60 c.10	46	44%	88%
Handheld -analog IC -MOS	0.9-1.0 0.9-1.0 <0.1	200	21%	180-210 170-190 10-20	70%	140	130-150 120-140 c.10	37	19%	27%
SPS (AC-DC) -analog IC -MOS	0.6-1.2 0.4-0.7 0.2-0.5	1290	9%	1000-1200 600-800 300-500	18%	232.2	150-250 100-150 50-100	0	0%	0%
Others								421		
DC-DC Total				**4000-4500**			**2200-2600**	**0**	**20-25%**	**40-45%**
DC-DC Total (excluding high-entry barrier products, such as NB Vcore and charger IC, mobile phone MPU and CCFL				**3000-3500**			**1800-2200**	**0**	**25-35%**	**45-55%**

Note: Others include: 1) sales of listed companies that couldn't be categorized (e.g., shipping via distributors); 2) sales by unlisted companies in Taiwan
Source: Nomura estimates

Fig. 61: China local vendors' market share globally



Source: Nomura estimates

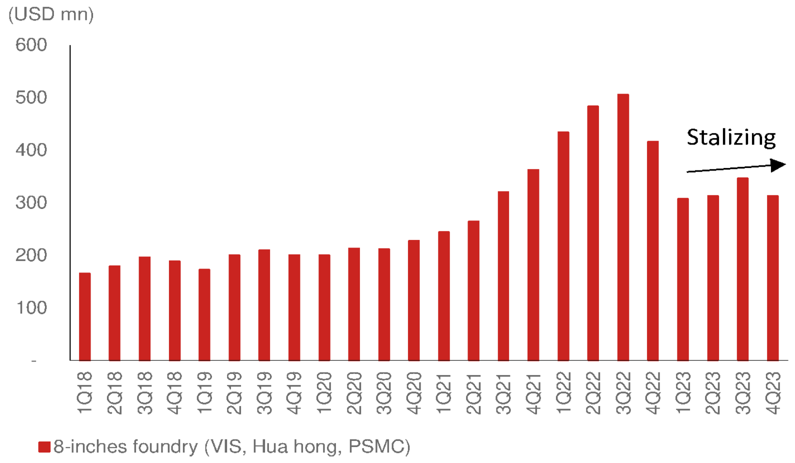
China analog shows signs of bottoming in inventory and foundry orders

Eased China analog downside risk through foundry sales level

In our Aug-23 *report*, we highlighted that leading foundries have started offering select customers 5-10% lower prices. On the other hand, we estimate that China foundries including SMIC (981 HK, Neutral) and Hua Hong (1347 HK, Neutral) and tier-2 foundries in Taiwan such as UMC (2303 TT, Buy) and Vanguard (5347 TT, Neutral) have lowered prices by 10-30%.

As most China analog fabless companies have listed after 2021, we do not have enough data to compare the sales performance with the pre-pandemic period. However, we have tracked tier-2 foundry partner sales to assess analog fabless order dynamics. Tier-2 foundry analog/PMIC sales declined significantly since mid-2022 peak (*Fig. 62*), and stabilized in 2H23, suggesting that downside risk has eased, in our view. Once the semi cycle starts turning upward, we believe foundry partners will provide more pricing incentives to fabless customers.

Fig. 62: Greater China tier-2 foundries' analog/PMIC sales

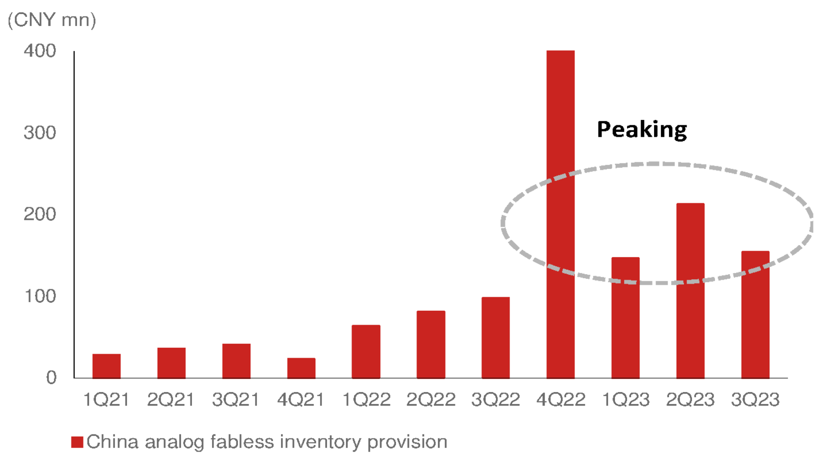


Note: Companies include Vanguard, Huahong, and PSMC
Source: Company data, Nomura estimates

Provision cost peak passed; expects GPM to recover gradually

In 2021, most major China analog fabless benefited from price hikes, leading to lower inventory provision cost. After facing inventory adjustment in 2022, the provision cost increased and GPM declined significantly, especially due to TI's two significant price cuts in 4Q22 and 2Q23. We expect the provision cost weakness to sustain into 2H23. However, we think the most challenging period has passed, and the provision cost impact on vendors will gradually ease in 2H23F, and GPM will gradually recover in 2024F.

Fig. 63: China analog fabless inventory provision



Source: Company data, Nomura estimates

Application sales performance of global and Greater China vendors

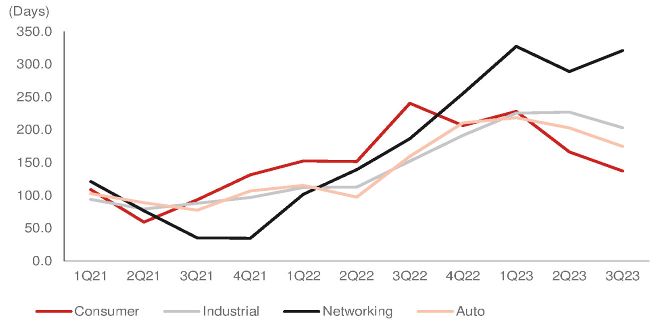
Applications' recovery pattern in China analog

We have been conservative about the China analog IC sector since June-22 (*report*) due to easing supply, and product lead time has also shortened with inventory overhang being witnessed across local vendors. The headwinds will continue to weigh on local vendors to end-2Q23, while some niche analog IC companies are turning less conservative and face reduced pricing pressure along with lower wafer cost, in our view. However, we think pricing pressure eased in end-2023, as TI became less aggressive in lowering prices in the China analog market.

In terms of China analog end-application recovery, we think consumer sales will be the first to recover as it turned soft the earliest in 2H21, followed by industrial and auto that bottomed in mid-2023, and we expect networking/commmunication to bottom in 1H24F.

As per Gartner's estimate, the industrial and auto applications accounted for 51% (*Fig. 66*) of global general purpose analog sales. Thus, we think the two applications are critical for the analog IC market recovery.

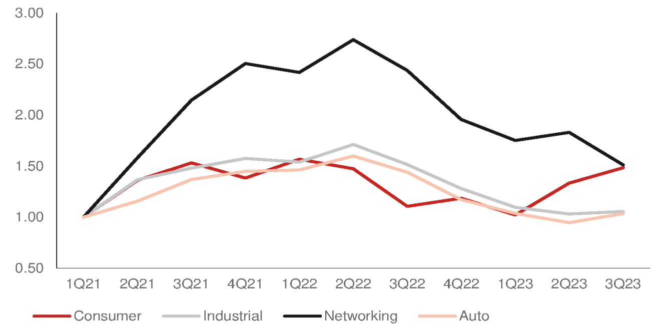
Fig. 64: China analog DOI by application



Source: Company data, Nomura research

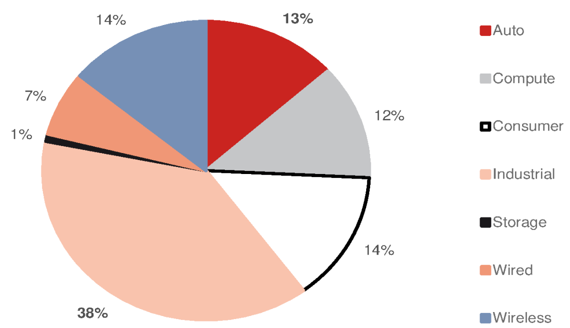
Fig. 65: China analog sales performance

We have set 1Q21 as 1.0x



Source: Company data, Nomura estimates

Fig. 66: General purpose analog semi sales mix by applications in 2022



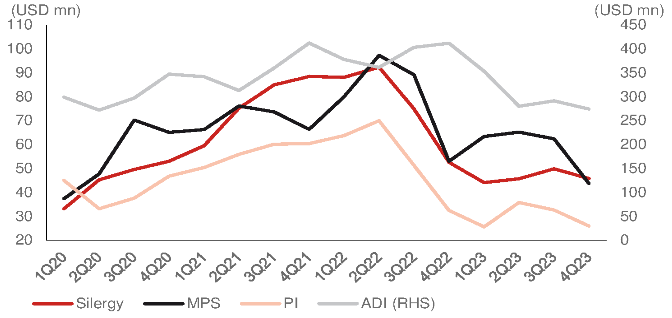
Source: Gartner, Nomura research

Consumer

The China consumer application-focused vendors were the first group in the analog market to enter inventory correction in 3Q21, while global peers faced lagging pressure in 2Q-3Q22. Further, during the 2022-23 industry downturn, we think China local consumer analog vendors' market share declined due to global IDMs' aggressive pricing strategy. On the other hand, the consumer products' end-demand was also deteriorating in 1H23. Thus, we saw most analog vendors' sales decline q-q in 1Q23-2Q23, with GPM also falling in the same period due to TI's price cuts.

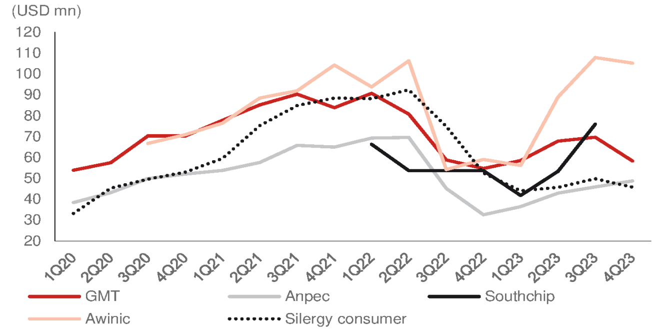
However, we think China consumer analog IC demand has stabilized and have started to recover (*Fig. 68*), according to our industry survey. In terms of pricing, we think TI's pricing impact will weigh until end-2023, but we think GPM of local China vendors will gradually recover because of foundry pricing support and eased pricing pressure into 2024.

Fig. 67: Global analog IC vendors' consumer sales



Source: Company data, Nomura research

Fig. 68: Greater China consumer analog IC companies sales



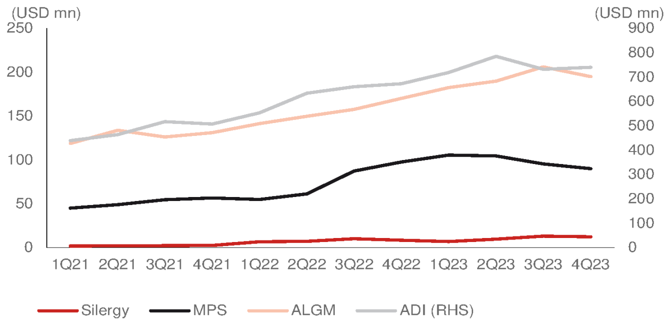
Source: Company data, Nomura research

Auto

We think the auto analog market was the least impacted by the semi downcycle in 2022, as major analog vendors' auto application sales strength was sustained in 2023 (*Fig. 69*). However, we saw the China auto market starting inventory adjustments (*Fig. 70*) in late-2022 to early-2023, which impacted auto analog vendors' orders (1Q-2Q23 sales declined q-q). However, we think this China auto inventory correction period would be short (vs the consumer application inventory correction which lasted for more than four quarters). We think auto analog vendors' demand will soon recover as China auto demand has now stabilized in 2H23. Forn global auto analog IC peers, the sales started to turn soft in 2H23, and we think the global auto analog IC vendors' sales will remain slow in 1H24F.

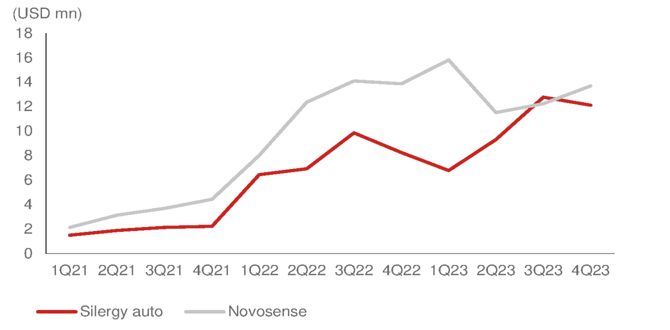
We believe auto has one of the lowest localization rates in the China analog market (as it requires higher quality and longer verification process) for each application, which should accelerate the domestic supply ratio from local vendors.

Fig. 69: Global analog IC vendors' auto sales



Source: Company data, Nomura research

Fig. 70: Greater China auto analog IC companies sales



Source: Company data, Nomura research

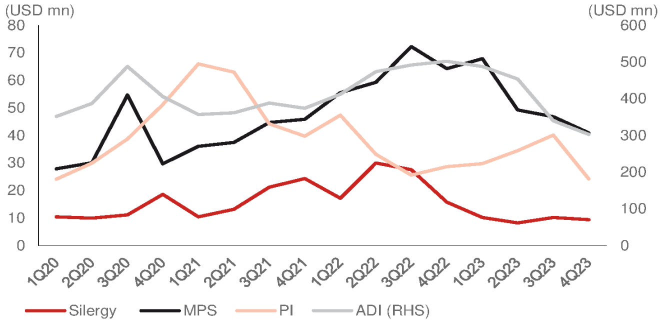
Networking/communication

Due to the maturation of 5G infrastructure in the global and China markets in 2022-23, and the global telecom equipment market turning soft in 2H22 after the pandemic, we saw networking end-market demand weakening in 2H22. China networking analog vendors' sales remained weak in 1H23, and we think vendors are also facing pricing pressure.

The networking market is forecasted to record steady growth of 6.6% by Gartner from 2022 to 2027E, and we think that China networking analog vendors can continue to gain share from global vendors.

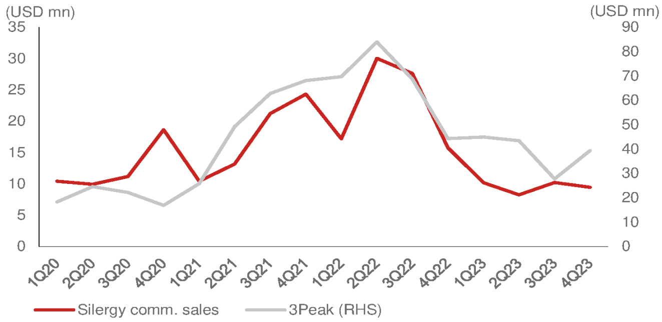
In terms of quarterly sales trend, we think China comm. analog IC vendors' sales bottomed in 4Q23, while global vendors' declining sales haven't stabilized yet. We think China comm. analog IC vendors' sales can gradually improve as customers' inventories return to healthy levels in mid-24F, while current global comm. vendors' order momentum is unclear in 2024F.

Fig. 71: Global analog IC vendors' communication sales



Source: Company data, Nomura research

Fig. 72: Greater China communication analog IC companies sales



Source: Company data, Nomura research

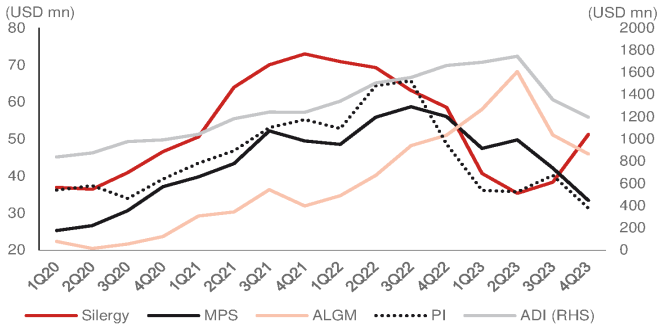
Industrial

Industrial end-demand turned soft in 2H22, and we are yet to see the demand recover in 2H23 (*Fig. 73*). We think it is because some industrial applications have stocked up early to prevent the shortage of 2021-22. However, the industrial analog vendors' sales declined significantly over 2023, and not only were shipments impacted by customers' excess inventory owing to weak macroeconomy, but the vendors also faced pricing pressure from competitors.

In the global industrial market, we believe the industrial analog IC demand is largely affected by macro environment. The global industrial analog IC leader ADI's industrial sales y-y are highly correlated with the global ISM PMI index (*Fig. 75*). We think the current PMI index has decreased to a historical low level at -1 S/D in late-23, suggesting low downside risk from here. China and global PMI have both strongly rebounded to above 50 in March at 50.8 and 50.3, respectively (*Fig. 77*-*Fig. 78*).

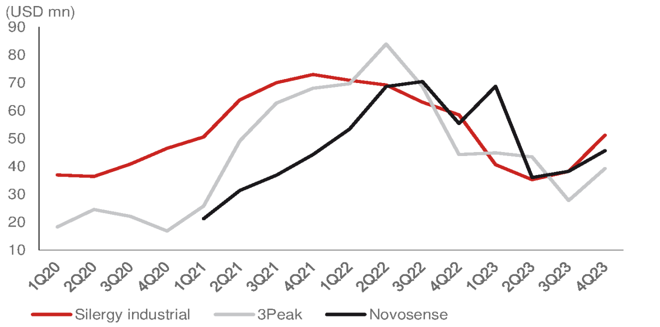
We think China local vendors were able to only satisfy some mid to low-end industrial analog product demand in 2021. However, we think more China local analog vendors' products are meeting more mid to high-end industrial products' requirements. On the other hand, the industrial analog market growth is forecast to grow above the normal analog average of 7%, as per TI's forecast. We think the China industrial analog market still has significant opportunity for local vendors due to its high CAGR and low localization rate.

Fig. 73: Global analog IC vendors' industrial sales



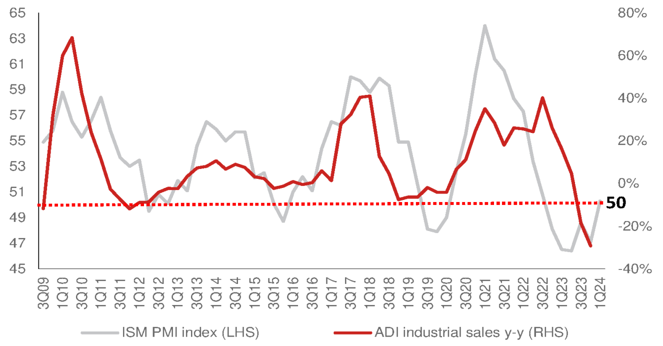
Source: Company data, Nomura research

Fig. 74: Greater China industrial analog IC companies sales



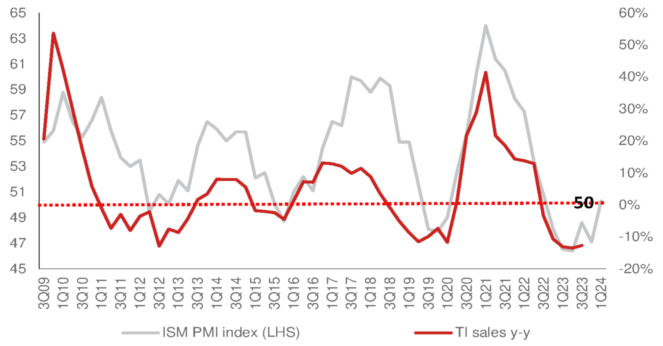
Note: 3Peak and Novosense are total sales, and Silergy shows industrial sales only.
Source: Company data, Nomura research

Fig. 75: ADI industrial sales y-y vs. ISM PMI index



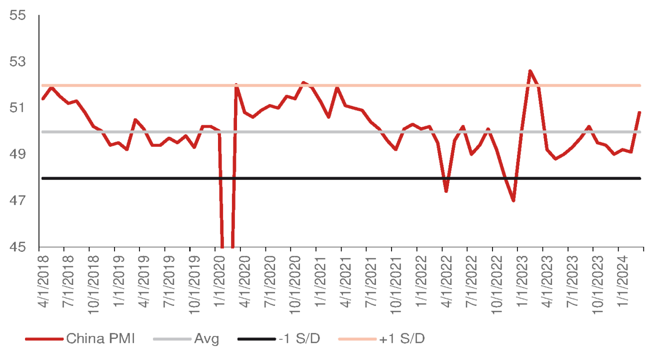
Source: Company data, Nomura research

Fig. 76: TI sales y-y vs. ISM PMI index



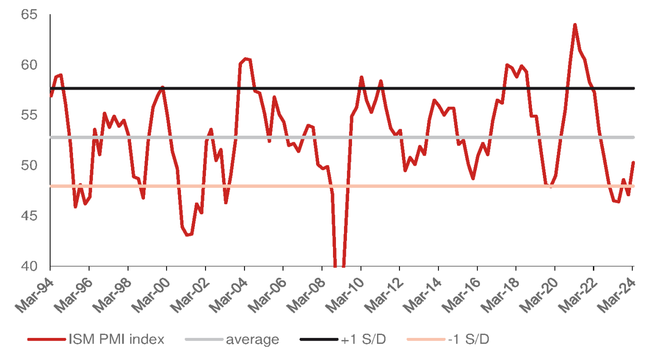
Source: Company data, Nomura research

Fig. 77: China manufacturing PMI



Source: Nomura research

Fig. 78: ISM PMI index



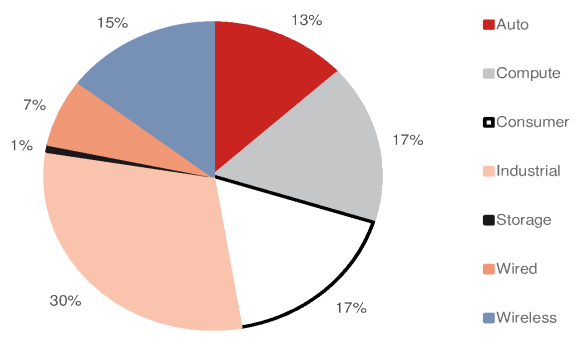
Source: Nomura research

China analog IC vendors

Localization trend turning to high-end and premium products, as well as non-consumer applications

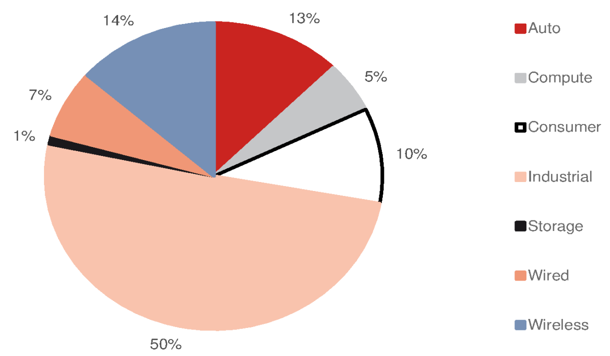
China analog vendors are usually involved in either PMIC or signal chain, or both, and these two major analog products require different technologies. Compared to signal chain, some analog vendors are still mainly focused on PMIC products primarily due to the following: **1)** the China PMIC market is large enough for multiple local vendors to survive; **2)** the technology requirements are low; **3)** China's local foundries have mature foundry capabilities to support fabless customers; and **4)** the global PMIC market continues to grow due to electrification. The signal chain products' end-applications are more fragmented and niche (*Fig. 80*), mostly in the industrial/automotive market, which is not a large volume and require custom-made products. Also, signal chain products' development is more difficult than that of PMIC, and local OEM customers are not mature enough, so there are fewer vendors or products involved.

Fig. 79: General purpose PMIC end-applications mix (%)



Source: Gartner, Nomura research

Fig. 80: General purpose signal chain end-application mix (%)



Source: Gartner, Nomura research

In the PMIC industry, DC-DC accounts for the largest market size (13% in 2022), followed by LDO (9%), Gate driver (6%), AC-DC (3%), and most of the other PMICs are an extension of the integration of AC-DC or DC-DC products. Therefore, China local PMIC vendors use DC-DC and AC-DC products as a starting product and then further expand their product portfolio. Compared to our previous data in 2021 (*2021 Anchor Report Fig. 18-19*), we think China local vendors' products have become more complete.

In the general signal chain market, according to IC insight, converters (ADC or DAC products) have the largest market size and account for 39%, interface accounts for 24%, linear products such as amplifiers and comparators and other products account for 37%. Currently, most leading China analog vendors have their own signal chain product lines, which can also strengthen their product portfolio and customers' demand.

Echoing our China analog *report* in Nov-2023, we think China analog vendors with higher sales exposure to industrial/auto are more likely to compete with TI in the China market due to higher pricing pressure and later inventory adjustment than other consumer applications peers that recovered earlier in mid-2023.

Fig. 81: Major China analog IC companies' PMIC product portfolios

	Driver			Non-driver							
							Battery management				
	LED	MOSFET	Motor	LDO	DC-DC	AC-DC	Charge management	Power Monitoring	Overvoltage protection	Load Switch	Supervisors/ Reset
Sliergy	√	√	√	√	√	√	√	√	√	√	√
SG Micro	√	√		√	√		√	√	√	√	√
Awinic	√		√	√	√		√	√	√	√	
3Peak		√	√	√	√	ICM-semi	ICM-semi	ICM-semi	ICM-semi		√
Novosense	√	√	√	√			√		√	√	
Southchip					√	√	√		√	√	
Joulwatt				√	√	√	√		√	√	
Injoinic					√	√	√				
Chipown	√	√		√	√	√					√
Wuxi ETEK	√			√	√		√		√	√	
Dioo	√	√	√	√	√	√	√			√	√
Cellwise			√	√	√		√	√	√	√	
Fine Made	√			√	√	√	√				
Belling	√	√	√	√	√	√	√			√	√
Kiwi instrument	√	√	√	√	√	√	√	√		√	
Bpsemi	√		√		√	√					

Source: Company data, Nomura research

Fig. 82: Major China analog IC companies' signal chain product portfolios

	Amplifier			Converter			Driver		Analog Switch	Interface circuits	Isolator
	Operational Amp	Audio/ Video Amp	Comparator	ADC	DAC	AFE	Audio	Video			
Sliergy	√	√			√				√		√
SG Micro	√	√	√	√			√	√	√	√	√
Awinic		√					√	√	√	√	√
3Peak	√	√	√	√	√	√	√		√	√	√
Novosense		KTmicro		KTmicro			KTmicro			√	√
Southchip											
Joulwatt						√				√	
Injoinic											
Chipown											
Wuxi ETEK		√		√					√		
Dioo	√	√	√				√	√	√		
Cellwise											
Fine Made											
Belling	√	√		√	√				√	√	√
Kiwi instrument						√					
Bpsemi											

Source: Company data, Nomura research

Fig. 83: Global analog vendors' foundry allocation

Companies	TSMC	UMC	SMIC	Vanguard	Global Foundry	Hua Hong	PSMC	Nexchip	DB Hitek	Tower Jazz	Others
China companies											
Silergy	<5%	60%	<10%	<5%		<10%	<10%				<10%
Awinic	70%		5%-10%		2%	10%-20%				3%	
SGMicro	80%+	5%	5%						5%	5%	
Will semi			50%						50%		
3 Peak			20%							50%+	
Belling											100%(GTA or in-house)
Novosense	5%		55%						40%		
Chipone		5%	25%	35%				35%			
Injoinic	50%				50%						
Southchip			60%			25%			15%		
Joulwatt			90%	10%							
Wuxi ETEK									50%+		CR Micro 30%+
Kiwi instrument			20%								CR Micro 50%+
Dioo		30%							60%		10%
Taiwan companies											
Richtek (MediaTek)	60%	5%	5%								30%
Anpec	20%	20%	10%								30%
GMMT	70%	30%									
Weltrend	85%			15%							
uPI		10%		10%			80%				
Global companies											
Texas instrument	<5%	<5%									Mostly in house foundry
Analog devices	50%										50% in house foundry
STMicro											Mostly in house foundry
NXP											In house foundry
Infineon											Mostly in house foundry
Renesas											50% in house foundry
Microchip											Mostly in house foundry
Maxim (under ADI)											<50% in house foundry

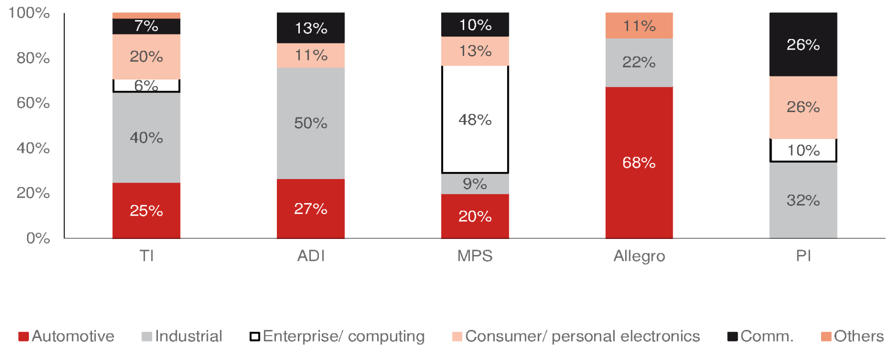
Source: Company data, Nomura estimates

Fig. 84: Great China analog vendors' sales application (%)

	Industrial	Consumer	Auto	Telecom/ Networking / Comm.	Computing/ Server	Others	Product details
Silergy	34%	38%	8%	7%	13%		PMIC:85% Signal chain and others:15%
Awinic	2%	96%				2%	
SGMicro	30%	50%	5%	5%		10%	PMIC 68.29%, Signal chain 31.67%
Will Semi	10%	85%				5%	
3 Peak	47%	9%	12%	32%			Signal chain: 70%, PMIC: 30%
Belling	50%					50%	PMIC 27.25%, General Analog 37.5%, NVM 2.5%, Discrete 5.6%, distributor 27.2%
Novosense	55%	12%	33%			0%	Signal chain: 63%, PMIC: 30.5%, Sensor: 7%
Chipown	21%	24%				55%	Power: 34.35%, others 3.37%
Kiwi instrument		87%				13%	PMIC: 78% (LED 65%, General power 22%, Home Appliance/IoT 13%)
Wuxi Etek		95%				5%	PMIC 85.5%, Others 14.5%
Injoinic		76%	16%			8%	PMIC:73% (Portable battery 52%, auto 16.4%, wireless charging 16%, TWS 8%, others 8%)
Fine Made		51%				49%	PMIC:39%, LED driver:42.3%, Mosfet 4%, others 14.7%
Bright Power		90%				10%	
Joulwatt	15%	30%	2%	40%	10%	3%	PMIC:95%, Signal chain:2%
Southchip	3%	95%	1%			1%	
Dioo	12%	65%		5%		18%	
Cellwise		95%			5%		

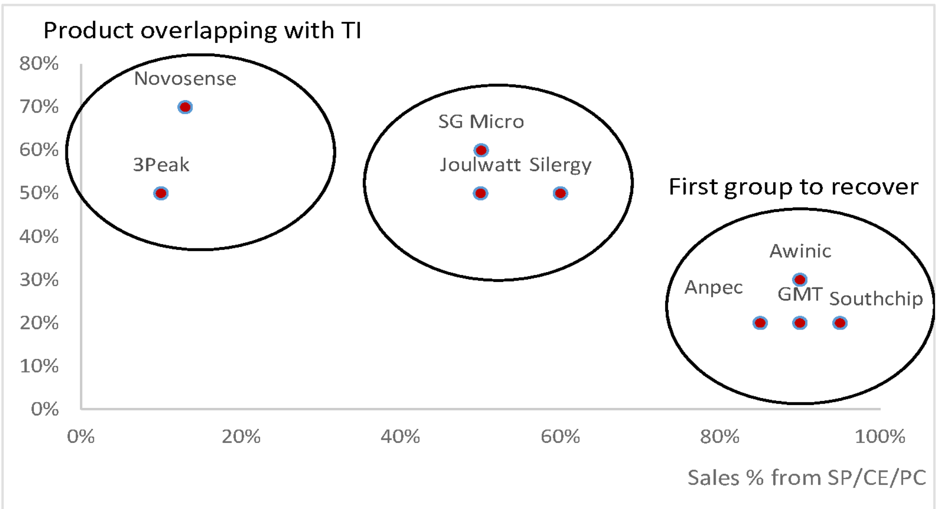
Source: Company data, Nomura estimates

Fig. 85: Global analog vendors: application sales mix %



Source: Company data, Nomura research

Fig. 86: Greater China analog IC companies: Product overlap with TI vs sales % from SP/CE/PC



Source: Company data, Nomura estimates

Fig. 87: Global and China analog IC companies – employment metrics

Analog IC companies	Business Model	2022 sales (USD mn)	Number of employees	Number of R&D employee	Number of products	R&D employee percentage	R&D spending	R&D spending %	R&D employee salary	Annual salary per R&D employee (CNY k)
TI	IDM	20028.0	33,000		100,000		1670.0	8%		
ADI	IDM	12014.0	26,000	11,000.0	80,000	42.3%	1700.5	14%		
MPS	Virtual IDM	1794.1	3,300	1,000.0	3,500	30.3%	263.6	15%		
Allegro	Fabless	973.7	4,000				150.9	15%		
Silergy	Virtual IDM	767.0	1,700	1,000.0	5,000	58.8%	149.4	19%		
Awinic	Fabless	300.0	1,186	766.0	1,100	64.6%	89.0	30%	354,735	463.1
SG Micro	Fabless	457.7	1,243	896.0	5,000	72.1%	92.5	20%		
3Peak	Fabless	256.1	714	508.0	2,400	71.2%	97.3	38%	290,932	572.7
Novosense	Fabless	240.0	645	326.0	1,500	50.5%	58.9	25%	191,134	586.3

Source: Company data, Nomura research

Fig. 88: Greater China analog vendors' sales peaked out earlier than global peers'

Also bottomed out earlier than global peers

Sales (USD mn)	1Q21	2Q21	3Q21	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23
Silergy	148.8	188.1	212.3	221.2	214.9	**230.9**	197.2	150.0	**113.0**	117.1	129.9
SG Micro	60.8	80.8	95.8	110.1	122.2	**132.4**	111.1	109.0	**75.0**	90.5	101.1
AWINIC	76.4	88.4	91.9	104.2	93.8	**106.4**	**54.2**	59.0	56.2	89.0	106.9
Southchip	25.9	25.9	48.9	53.8	**66.3**	53.6	53.6	53.6	**41.7**	53.4	75.3
Joulwatt	28.9	28.9	46.4	**59.1**	53.1	54.3	47.7	58.5	**44.0**	49.5	49.0
Injoinic	26.5	29.9	33.1	32.9	**33.3**	30.1	**29.3**	36.0	32.2	42.1	45.2
Kiwi instrument	30.0	30.0	**42.7**	36.4	26.5	22.0	**12.7**	17.6	19.3	24.1	16.7
Fine Made	41.0	**90.6**	51.2	29.4	43.4	26.6	22.1	23.7	20.0	27.5	26.6
Cellwise	13.0	13.0	12.8	**14.5**	8.4	8.4	5.2	7.8	**4.2**	8.0	9.2
Bpsemi	62.8	101.9	**117.3**	74.8	47.6	43.8	**31.4**	38.4	38.7	49.9	41.5
Chipown	22.0	28.5	32.3	**34.1**	29.2	28.8	**22.2**	27.0	27.3	28.1	27.0
Wuxi ETEK	25.6	31.5	31.7	31.2	**41.7**	31.3	20.9	21.4	26.1	27.8	36.1
Belling	66.2	91.4	76.5	79.8	70.3	73.8	71.9	**86.8**	57.6	68.2	76.9
3Peak	25.8	49.2	62.8	68.1	69.7	**84.0**	68.7	44.3	44.9	43.4	27.8
Novosense	21.2	31.4	36.9	44.3	53.4	68.7	**70.5**	55.5	68.8	36.0	38.3
Dioo	17.7	17.7	21.3	23.0	**25.3**	20.0	20.0	14.1	11.0	15.0	15.9
China total	692.6	927.1	1013.8	**1016.8**	999.1	1015.1	838.7	802.7	**680.2**	769.6	823.3
Richtek	269.6	314.1	327.7	321.6	**404.9**	368.3	327.2	241.2	**220.3**	224.1	244.7
GMT	77.7	85.1	90.3	83.9	**90.7**	80.8	58.8	**54.7**	58.5	67.8	70.8
Anpec	53.8	57.6	65.8	65.0	69.2	**69.6**	45.2	**32.5**	36.5	42.9	46.7
uPI	41.6	41.6	60.0	60.1	61.0	**63.5**	49.8	31.7	23.5	25.9	22.8
TW total	442.7	498.5	543.8	530.5	**625.7**	582.2	481.0	360.1	**338.7**	360.7	385.0
Greater China	1135.3	1425.5	1557.7	1547.3	**1624.8**	1597.3	1319.7	1162.8	**1018.8**	1130.3	1208.3
Global											
TI	4289	4580	4643	4832	4905	5212	**5241**	4670	4379	4531	4532
ADI	2178	2290	2424	3059	2684	2972	3110	3248	3250	**3263**	3076
MPS	254	293	324	337	378	461	**495**	460	451	441	475
Infineon	3137	3253	3281	3544	3612	3700	3851	4170	4037	4421	**4452**
STM	3016	2992	3197	3556	3546	3837	4321	4424	4247	4326	**4431**
ON	1482	1670	1742	1846	1945	2085	**2193**	2104	1960	2094	2181
Global total	14356	15078	15610	17174	17070	18267	**19211**	19076	18324	19076	19147

Peak **Bottom**

Source: Company data, Nomura research

Fig. 89: Greater China analog vendors' GPM peaked out earlier than global peers'

But the bottomed out pattern is unclear

GPM (%)	1Q21	2Q21	3Q21	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23
Silergy	47.7	53.4	**56.3**	54.0	53.6	53.5	51.8	50.9	44.5	**40.7**	43.1
SG Micro	47.9	53.8	59.9	57.2	60.6	59.0	**60.7**	55.6	52.7	50.6	49.0
AWINIC	32.7	41.9	42.1	43.4	**46.5**	41.6	35.0	**22.9**	28.8	26.4	22.1
Southchip					**44.5**	43.1	43.1	43.1	**41.2**	41.5	43.1
Joulwatt					**44.5**	40.0			32.8	30.6	23.4
Injoinic	35.8		**49.1**	48.8	45.8	43.9	37.7	35.8	31.1	30.3	29.9
Kiwi instrument			**49.7**	41.1	37.4	32.9	19.2	**15.6**	22.4	25.0	23.5
Fine Made	36.9	58.8	**71.8**	31.4	35.7	31.6	17.8	-18.9	11.6	3.3	7.7
Cellwise			**63.7**	60.4	58.2	59.4	55.2	56.2	55.4	**54.8**	55.1
Bpsemi	37.4	**52.6**	51.8	44.4	29.8	28.1	**0.7**	6.2	23.2	25.8	23.9
Chipown	40.5	42.2	**44.9**	43.4	41.5	41.4	41.4	40.3	39.1	38.8	38.1
Wuxi ETEK	31.0	38.6	42.7	42.3	44.7	**47.1**	43.6	42.1	42.1	41.4	44.2
Belling	30.7	34.1	**36.7**	34.5	36.2	35.9	34.2	31.0	30.8	29.2	31.0
3Peak	58.1	60.8	**62.3**	59.6	57.0	59.4	58.9	59.1	57.9	51.6	50.0
Novosense	52.3	**55.5**	54.0	52.3	51.2	50.4	51.9	46.3	45.3	38.2	36.5
Dioo				**58.3**	57.5	58.2	58.2	48.4	48.2	50.1	48.2
GMT	41.8	47.7	**52.0**	51.2	48.8	48.6	44.2	**35.2**	38.4	39.9	41.0
Anpec	31.0	36.1	40.6	41.1	41.6	**43.3**	41.1	36.1	**30.3**	31.3	32.4
uPI	38.1	38.1	45.0	42.3	44.7	**45.0**	40.4	27.8	**22.7**	26.4	26.1
TI	65.2	67.2	67.9	69.3	**70.2**	69.6	69.0	66.1	65.4	64.2	62.1
ADI	67.1	68.4	**69.4**	47.9	52.2	65.4	65.7	66.0	65.4	65.7	63.8
MPS	55.4	56.0	57.6	57.6	57.9	**58.8**	58.7	58.2	57.4	56.1	55.5
Infineon	37.4	36.0	39.1	41.2	41.5	42.9	43.2	44.4	**47.2**	46.6	44.5
STM	39.0	40.5	41.6	45.2	46.7	47.4	47.7	47.5	**49.7**	49.0	47.6
ON	35.2	38.3	41.4	45.1	49.4	**49.7**	48.3	48.5	46.8	47.4	47.3

Source: Company data, Nomura research

Fig. 90: Greater China analog vendors' sales y-y peaked out at similar period as global peers in 1H21

Greater China analog vendors' sales y-y bottomed out earlier than global peers in 2H22-1H23

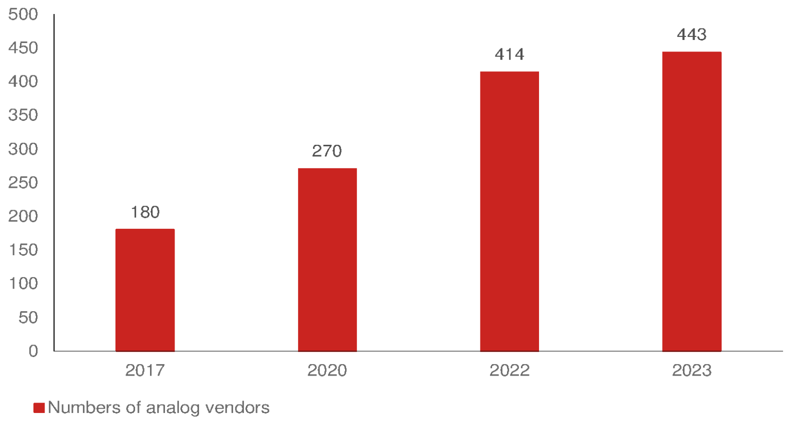
Sales y-y (%)	1Q21	2Q21	3Q21	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23
Silergy	57%	70%	**71%**	54%	44%	23%	-7%	-32%	-47%	**-49%**	-34%
SG Micro	**120%**	110%	67%	118%	101%	64%	16%	-1%	**-39%**	-32%	-9%
AWINIC			38%	**47%**	23%	20%	-41%	**-43%**	-40%	-16%	97%
Southchip					156%	107%	10%	0%	**-37%**	0%	40%
Joulwatt					84%	88%	3%	-1%	**-17%**	-9%	3%
Injoinic					26%	1%	**-12%**	9%	-3%	40%	54%
Kiwi instrument					-12%	-27%	**-70%**	-52%	-27%	10%	31%
Fine Made	185%	**328%**	35%	-40%	6%	**-71%**	-57%	-19%	-54%	3%	20%
Cellwise					-35%	-35%	**-59%**	-47%	-50%	-5%	76%
Bpsemi	141%	**256%**	152%	25%	-24%	-57%	**-73%**	-49%	-19%	14%	32%
Chipown	**143%**	117%	81%	50%	33%	1%	**-31%**	-21%	-6%	-2%	22%
Wuxi ETEK			16%	59%	**63%**	-1%	-34%	-31%	**-38%**	-11%	73%
Belling	96%	**109%**	50%	21%	6%	**-19%**	-6%	9%	-18%	-8%	7%
3Peak	41%	100%	184%	**303%**	170%	71%	9%	-35%	-36%	-48%	-60%
Novosense					**151%**	119%	91%	25%	29%	**-48%**	-46%
Dioo					**43%**	13%	-6%	-39%	**-56%**	-25%	-20%
China average					44%	9%	-17%	-21%	**-32%**	-24%	-2%
MTK (Richtek)	**57%**	53%	32%	26%	50%	17%	0%	-25%	**-46%**	-39%	-25%
GMT	44%	**48%**	29%	19%	17%	-5%	**-35%**	-35%	-35%	-16%	20%
Anpec	**40%**	33%	32%	25%	29%	21%	-31%	**-50%**	-47%	-38%	3%
uPI	51%	51%	26%	51%	47%	**53%**	-17%	-47%	**-61%**	-59%	-54%
TW average	**270%**	289%	224%	227%	41%	17%	-12%	-32%	**-46%**	-38%	-20%
Greater China					43%	12%	-15%	-25%	**-37%**	-29%	-8%
Global											
TI	29%	**41%**	22%	19%	14%	14%	13%	-3%	-11%	-13%	-14%
ADI	19%	23%	20%	**48%**	23%	30%	28%	6%	21%	10%	-1%
MPS	53%	**58%**	25%	44%	48%	57%	53%	37%	19%	-4%	-4%
Infineon	48%	**49%**	37%	22%	15%	14%	17%	18%	12%	19%	16%
STM	35%	**43%**	20%	10%	18%	28%	35%	24%	20%	13%	3%
ON	16%	**38%**	32%	28%	31%	25%	26%	14%	1%	0%	-1%
Global average	31%	**40%**	25%	23%	19%	21%	23%	11%	7%	4%	0%

Source: Company data, Nomura research

Major China analog vendors

According to CSIA data, there were 180 analog companies registered in China in 2017, which grew to 414 in 2022 and further to 443 in 2023 (*Fig. 91*). Most analog vendors benefited during 2020-21 chip shortages and localization requirement of local customers. We think each China local vendor has an advantage in terms of applications, product portfolio, performance, or pricing to continue to grow in China local analog market.

Fig. 91: Number of China analog vendors



Source: CSIA, Nomura research

Silergy, 3Peak, and Novosense are major beneficiaries in China analog IC market's further localization

Silergy, 3Peak, and Novosense are the major beneficiary companies of further recovery of industrial analog IC application as PMI in the US/China turned positive to above 50 in Mar-24. We expect the three companies are also major beneficiaries of further China local analog localization with high industrial/auto sales exposure. We think consumer application in China has higher localization with stronger competition for local vendors or global peers.

Fig. 92: Silergy, 3Peak,and Novosense's business

Company	Business introduction
Silergy	Silergy is a largest China analog IC fabless company that mainly provides PMIC products. Silergy's end-to-end analog, power, and signal chain solutions are widely used in automotive, industrial, consumer, computing, and telecom equipment.
3Peak	3peak is a China leading industrial analog fabless IC company, which ranks fourth largest fabless in China listed analog vendors. The company is focused on high-performance, high-quality, and highly reliable IC products, including signal chains, power management IC (PMIC), mixed-signal front-ends, and embedded processors to provide customers with all-around solutions. The applications of the products cover many markets such as communication, industrial, consumer, and automotive industry.
Novosense	Novosense is one of China's leading analog IC vendors, which ranked fifth largest fabless vendors in China listed analog vendors. It specializes in industrial, new energy, and automotive products. Novosense focused on signal chain, power management IC (PMIC) products, and one of few vendor has sensor products to provide customers total solution in analog products

Source: Nomura research

Silergy (6415 TT, Buy)

Silergy is one of the largest analog vendors in the China analog market, and has more than 5,000 products, covering most PMICs. In 2022, Silergy recorded annual sales of USD793mn and 85% was from power products. Silergy has a virtual IDM business model, which means it collaborates with foundry partners to design its own manufacturing technology and capacity, but doesn't need to incur high capex. Silergy's sales by application was: industrial 33%, consumer 39%, computing 13%, communication 11%, and automotive 4%.

SG Micro (300661 CH, Neutral)

SG Micro is the leading analog fabless company in China, covering PMIC and signal chain, with more than 3,800 products in 2023. Among China local vendors, SG Micro has one of the most complete product portfolios in PMIC. In 2022, the company recorded annual sales of CNY3.2bn, and 62% of sales were from PMIC, and 37% from signal chain. For more details, please refer to our *initiation report* in 2021.

Awinic (688798 CH, Buy)

Awinic focuses on smartphone applications. In 2022, Awinic recorded annual sales of CNY2.1bn, and 97% sales were from smartphone. Besides PMIC, Awinic has also expanded its product portfolio to signal chain, which accounted for 8.3%/13.7% of sales in 2022/1H23. For more details, please refer to our *initiation report* in 2021.

3Peak (688536 CH, Buy)

3Peak is one of the leading China signal chain vendors, and has also expanded its product portfolio to PMIC. In 2023, 3Peak had more than 1,400 products, including amplifiers, analog switches, interface, isolators, and motor drivers. 3Peak has high application sales exposure to industrial (42%), and networking/communication (41%). Other sales applications include auto (9%), consumer (7%), and others (1%). In 2022, 3Peak recorded annual sales of CNY1.8bn, and by product mix, 70% of sales were from signal chain and 29% from PMIC. 3Peak announced to acquire ICM-semi in 1H23. ICM-semi is an analog fabless in Shenzhen, China, focusing on low-power consumption, and high-density PMIC, with annual sales of CNY200mn/CNY182mn, and net profit of CNY55mn/4.4mn in 2021/2022.

Novosense (688052 CH, Neutral)

Novosense is one of the leading digital isolator analog vendors in the China industrial market. Isolators and sampling are Novosense's major products, and the company has further expanded its PMIC product portfolio, and its product mix has increased from 20% to 30% during 2022-1H23. In 2022, Novosense recorded annual sales of CNY1.67bn, and has high sales exposure to industrial (70%), auto (23%), and other (7%) in terms of consumer applications. Novosense announced to acquire KTmicro in July-2023. KTmicro is an analog and mix-signal fabless company based in Beijing, and focuses on audio SoC for the consumer market, signal chain product of ADC/ DAC and integrated ADC & DAC ICs for industrial and communication market. KTmicro recorded annual sales of CNY315mn/CNY305mn in 2021/22.

Southchip (688484 CH, Not rated)

Southchip is a leading analog fabless company manufacturing charging pump products in China. The company is one of the few vendors in China offering a comprehensive bouquet of solutions ranging from charging components to device components, catering to global customers such as Huawei (unlisted), Xiaomi (1810 HK, Neutral), Anker (300866 CH, Not rated), DJI (unlisted), Honor (unlisted) and Samsung (005930 KS). In terms of product mix, 90% of sales are attributable to charging applications, and 70% of which came from charge pump products. In terms of end-applications, we estimate a majority of Southchip's sales came from consumer products (>90%). In 2022, Southchip recorded CNY1.3bn of annual sales.

Joulwatt (688141 CH, Not rated)

Similar to Silergy, Joulwatt is another virtual IDM fabless company in China. It develops AC/DC PMIC products in their early stage, and has expanded to signal chain as well (2% sales contribution in 2022). Joulwatt now has more than 1,000 products, including AC/DC, DC/DC, linear power, DrMOS, and battery management IC products. Currently, Joulwatt's application mix focuses on consumer applications. Sales contribution by segment in 2022: 52% from DC/DC, 18% from consumer AC/DC, 24% from linear power, 1% from battery management IC, and the rest from technology service and signal chain IC products.

Injoinic (688029 CH, Not rated)

Founded in 2014, Injoinic focuses on mixed-signal analog product, including fast charging interface, low-power PMIC, ADC, high power/voltage, and gauge products. The applications include portable battery, auto charging, true-wireless stereo (TWS), and wireless charging. Injoinic is the major supplier of fast charging protocol IC in consumer electronics, and it also has exposure to audio processing, home appliances, IoT, and automotive (10% of sales in 1H21). Injoinic recorded annual sales of CNY867mn in 2022, and PMIC accounted for 73% of total sales, and 27% from fast-charging IC sales.

Dioo (688381 CH, Not rated)

Dioo started with analog switches, and then expanded to amplifiers and converters. It now has a balanced product portfolio of analog ICs, and generates 54% of sales from PMICs, and 46% from signal chain. In terms of application mix in 2022, Dioo has more sales exposure to consumer applications, which comprise 66% of total sales. Its customers include Xiaomi, OPPO, VIVO, Wingtech, etc. Sales from other applications in 2021 were: 17% from LED drivers, 12% from industrial/ surveillance, 5% from communication equipment, and 0.6% from medical equipment. Dioo recorded annual sales of CNY501mn in 2022.

Cellwise (688325 CH, Not rated)

Founded in 2009, Cellwise focuses on battery management and peripherals, such as battery protection, battery charging controllers, and voltage gauge products. In 2022, Cellwise recorded annual sales of CNY200mn. In terms of application mix, Cellwise is more exposed to consumer applications, including NBs, tablets, smartphones, and others such as drones and home appliances. Cellwise will also accelerate its battery management IC exposure to new energy and storage customers.

Comparison between 3Peak and Novosense

3Peak and Novosense are the two major industrial/auto application-focused analog IC companies in China, with high industrial application sales exposure that we estimate at 47% and 55% in 2023, respectively. The two companies had similar sales growth patterns between 2018 and 2023 (*Fig. 94*), as both benefited from analog IC localization.

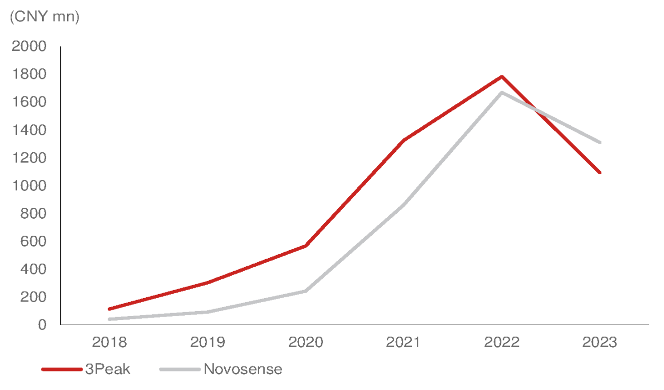
However, we think 3Peak managed inventory better by decreasing its inventory level faster and reducing inventory days, thereby protecting profitability better than Novosense despite substantial price cuts by global and local peers since 4Q22 (Fig. 96, Fig. 97). We believe Novosense suffered higher pricing pressure, which resulted in a lower GPM from China auto customers given its high auto sales exposure (30%+ vs. 3Peak at 10-20%).

Fig. 93: Industrial/auto analog vendor comparison

	3Peak	Novosense
Founded	2012	2013
2023 Sales (CNYmn)	1,094	1,311
2023 GPM (%)	52%	40%
Product SKU	2,400	1,500
2022 Employee	714	645
RD employee (%)	71.2%	50.5%
2023 RD expense (CNYmn)	554	650
2023 RD to sales (%)	51%	50%
1Q-3Q23 Application sales mix (%)		
Auto	12%	33%
Industrial	47%	55%
Consumer	9%	12%
Product sales mix (%)		
Signal chain	79%	65%
PMIC	20%	27%
Supply chian		
Key foundry partner	Tower, SMIC	SMIC, DB Hitek

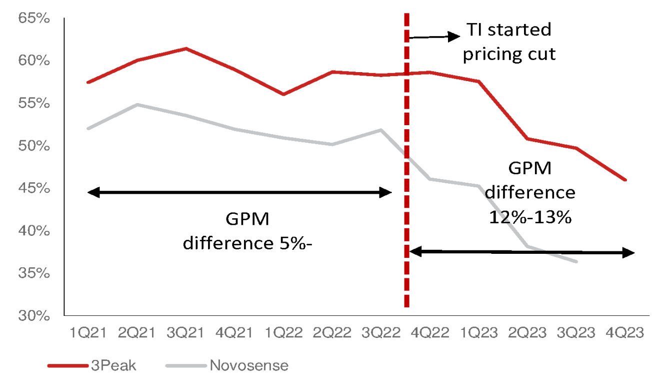
Source: Nomura research

Fig. 94: 3Peak and Novosense sales



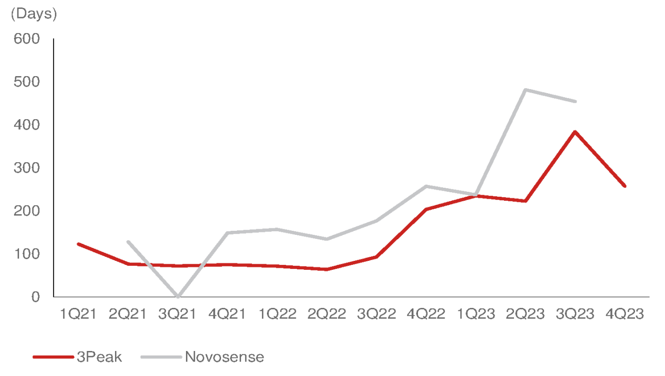
Source: Company data, Nomura research

Fig. 95: 3Peak and Novosense GPM



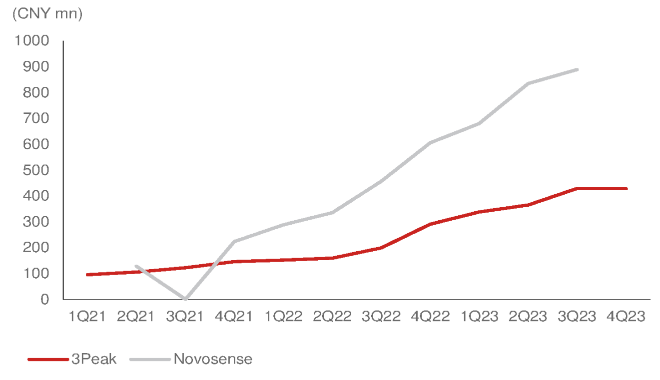
Source: Company data, Nomura research

Fig. 96: 3Peak and Novosense inventory days



Source: Company data, Nomura research

Fig. 97: 3Peak and Novosense inventories



Source: Company data, Nomura research

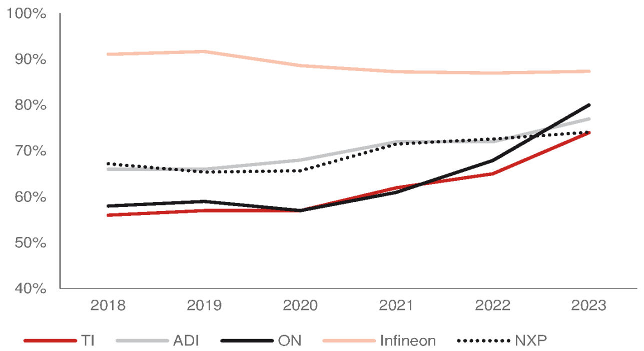
China PMIC and signal chain market trends

Trend toward high-voltage/current in auto and industrial market to drive analog market

All global IDMs aiming at auto and industrial market growth opportunity

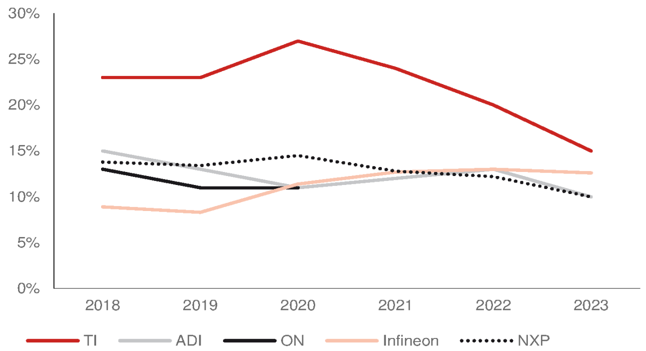
Based on TI forecasts, auto and industrial applications will outperform the global analog market average sales CAGR of 7% over 2023-2030. Global analog IDMs such as TI and ADI are both positive on these two end-segments due to: **1) Stable demand:** auto and industrial demand is less volatile and it can balance IDM's capacity operation plan and lower capacity risk compared to consumer applications. **2) Higher GPM:** auto and industrial applications require higher product life and quality than consumer products, and global IDMs are more willing to get involved in these two segments. **3) Strong growth opportunity:** electrification and intelligence are the two major growth drivers of auto and industrial applications, and will increase analog IC consumption in line with the market megatrend.

Fig. 98: Revenue contribution from industrial and automotive segments for top analog suppliers



Source: Company data, Nomura research

Fig. 99: Revenue contribution from personal electronics segment for top analog suppliers



Source: Company data, Nomura research

Fig. 100: TI's R&D investment allocation plan

Market segment	R&D investments	% of TI revenue 2013	2019	2023
Industrial	Up broadly	30%	36%	40%
Automotive	Up broadly	12%	21%	34%
Personal Electronics	Slightly up, continue to be selective	32%	23%	15%
Enterprise Systems	Steady	15%	11%	4%
Communications Equipment	Slightly up	6%	6%	5%
Other - Calculator & Other	Flat, at low levels	5%	3%	2%

Source: Company data, Nomura research

Fig. 101: NXP's 3-year growth guidance comparison by end market

NXP revised up auto and industrial 3-yr CAGR vs. 2020 version

End Market	Percentage of 2020 revenue	Percentage of 2023 revenue	2020 version Normalized 3-yr.CAGR	Nov-2021 version Normalized 3-yr.CAGR
Automotive	44%	56%	Up 7 to 10%	Up 9 to 14%
Industrial & IoT	21%	18%	Up 8 to 11%	Up 9 to 14%
Mobile	14%	10%	4 to 6%	8 to 10%
Comm.Infra.& Other	20%	16%	0 to 2%	2 to 6%
Total			Up 5 to 7%	Up 8 to 12%

Source: NXP, Nomura research

High voltage/current to require better power technology

Auto: High voltage requires better BCD technology

Electric vehicles (EVs) is still the major focus segment in the auto analog market. EV OEMs are raising the power voltage solution in powertrain to 800V (vs current mainstream of 400V) to accelerate charging time (using more SiC and GaN device). Therefore, we think the coming auto analog product will need higher voltage BCD technology to meet 800V requirement and better isolation technology to prevent current interference situation in the EV systems.

On the other hand, the latest Tesla (TSLA US, Not rated) Cybertruck upgraded its electric system into **48V** from the current EV mainstream of 12V. The upgrade has several advantages. **1) Demand requirement**: higher voltage can meet the requirement of increasing demand in the vehicle control system. **2) Power consumption:** the 4x increase makes the current pass-through smaller, which can reduce the loss during energy transmission, and can reduce the temperature. **3) Cost:** It can reduce the wire thickness in vehicle and save cost.

Industrial: Automation and energy applications seeing strong growth

On the other hand, industrial applications are seeing strong growth in automation, (including robotics, inverters, serves, human-machine interface, etc.) to create significant demand for PMIC and signal chain products. For example, the robot in edge AI is equipped by high computing power and control precision to require better motor drivers and higher current ampere performance in analog products.

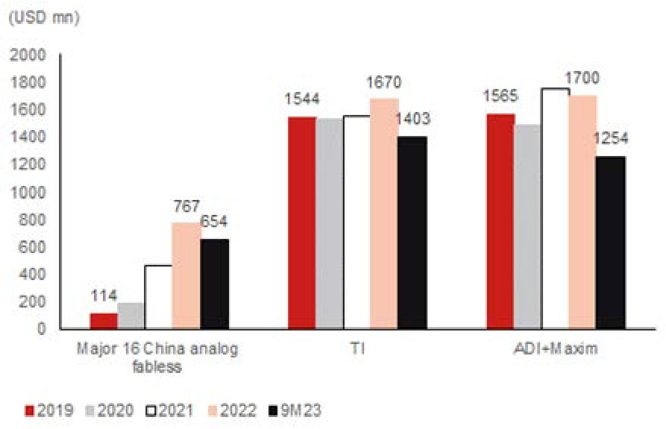
Higher integration in analog products

We believe analog products will become more integrated. The single function PMIC is being replaced by multi-function PMIC products. Furthermore, we think the integration of PMIC and signal chain will become more common in the future, as customers prefer more functions in single analog IC. Thus, we are seeing more China analog vendors expanding their product portfolio into both PMIC and signal chain.

Significant R&D investment growth among China analog vendors

Global analog vendors have maintained their strong position in the premium and high-end market, such as higher than 30A. However, China local vendors are able to only supply below 20A products. China's analog fabless vendors have significantly increased their R&D by 4x over the past four years to compete with global analog IDMs (*Fig. 80* , *Fig. 81*).

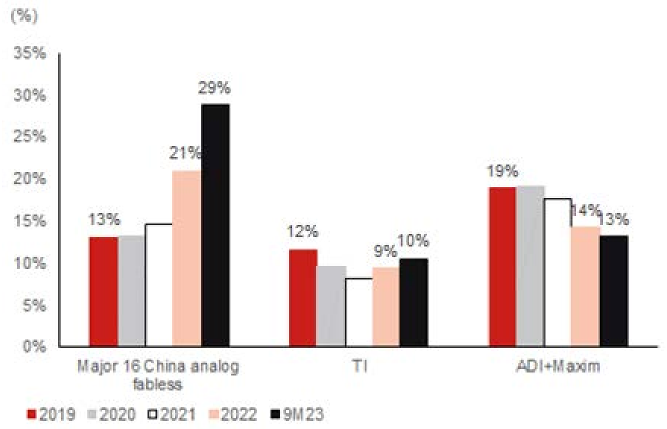
Fig. 102: China analog vendors, TI, ADI RD expense during 2019-22



Source: Company data, Nomura research

Fig. 103: China analog vendors, TI, ADI RD expense % during 2019-9M22

China analog vendors are aggressively expanding RD expense



Source: Company data, Nomura research

The long-term China localization story

Local analog vendors can strongly compete and survive in the China market, but they are still weak while competing with global peers

Most China local analog vendors have advantages in:

1) Pricing: Local analog vendors can only gain market share by sacrificing ASP and GPM in the China low-end analog product; and hence, their GPMs remain at 30-40%.

2) Cost control: China analog vendors are mostly fabless, where managing supply chain cost is a key issue. If they can acquire enough capacity and lower their costs by partnering local foundry and OSAT partners, it will provide them with better opportunity to compete with other analog vendors.

3) Service: Support capability is usually a critical issue when customers make their supplier decisions. Analog products require constant adjustment in the end-product. China analog vendors are relatively small, and they may not have enough personnel or offices near every region to provide service instantly.

4) Products. With a better product portfolio, the analog vendors can save more time and cost for customers. These advantages provide them with an opportunity to grow in the China analog market, and ultimately list on the stock market.

However, these advantages are not as powerful as global peers that global IDM have lower product price, owned capacity, strong service in China, as well as better product portfolio than China analog vendors. We think local vendors need to make significant breakthrough in: **1) product performance**, **2) reliability**, **3) product portfolio**, and **4) self-developed technology**. After overcoming these weaknesses, we believe the local vendors will be able to compete with global peers in more high-end markets.

Analog products in auto or industrial may need a long development cycle, auto certifications such as AEC-Q100, or stringent performance verification for high-end functions. It’s not easy to replace global vendors quickly due to technology, time, and development costs. Therefore, although we think localization in the China auto or industrial analog market will continue to grow, it will be a long-term stable growth instead of a sudden sharp one.

Valuation analysis

Sales share positively correlated with market cap share

We have used different approaches to value China analog vendors.

We analyze the China analog vendors from different perspectives and valuation approaches to better determine their reasonable valuations compared with global peers and China fabless companies.

1. Global analog TAM and peer comparison (P/S)
2. China analog peers P/B vs ROE comparison
3. China analog peers P/E vs sales CAGR comparison
4. China fabless and global analog peers P/E comparison

We use TI and ADI as global analog market valuation samples. TI and ADI are the top two global analog suppliers, while other global analog peers Qualcomm (QCOM US, Not Rated), Renesas (6723 JP, Buy), Mediatek (2454 TT, Neutral), STMicro (STM US, Not rated), and Infineon (IFX GY, Not rated) have diverse product portfolios, which mainly consist of other logic ICs. These companies' analog product sales are relatively small, while TI and ADI are considered analog-focused suppliers, and their product lines overlap with that of China's leading analog companies; hence, we use the latter as our global sample indicators.

TI's and ADI's 1Q22-3Q23 combined sales of USD55bn accounted for around 30-35% of the global analog market of USD149.5bn in the same period (based on WSTS estimates). We compare TI's and ADI's combined sales with the global analog market cap to arrive at a market valuation (P/S) of about **4.6x**. Based on WSTS's estimate of the global analog market of USD89bn in 2024E, the global analog market participants' combined market cap comes to USD692bn. The combined market cap of 16 major China analog vendors is around USD22.2bn, which is 3.2% of our global analog market value estimate.

Based on our estimates, the 16 major local vendors currently account for 4% of global market share, which is significantly lower than their market cap to global TAM market share. Therefore, based on this approach (using TI and ADI as a valuation comparison base), China analog companies' market share vs. market cap of 4.0% and 3.2%, respectively, appear that the combined market cap is undervalued at current share price levels, in our view.

Fig. 104: TI and ADI: Analog TAM and market valuation

	Market cap (USD mn)	1Q22-3Q23 sales (USD mn)	1Q22-3Q23 NI	P/S	P/E
TI	157,461	33,470	13888	4.70	11.34
ADI	97,505	21,603	5783	4.51	16.86
TI+ADI	254,967	55,073		4.63	

Source: Bloomberg Finance L.P., Nomura research

Fig. 105: Global analog TAM and vendors – P/S valuation analysis

	2022E	2023E	2024E
Global analog market value (USD mn)	88,983	83,907	88,902
Global TAM to TI+ADI combines sales (P/S)			4.6
Global analog market participants market cap combines(USD mn)			**692,081**

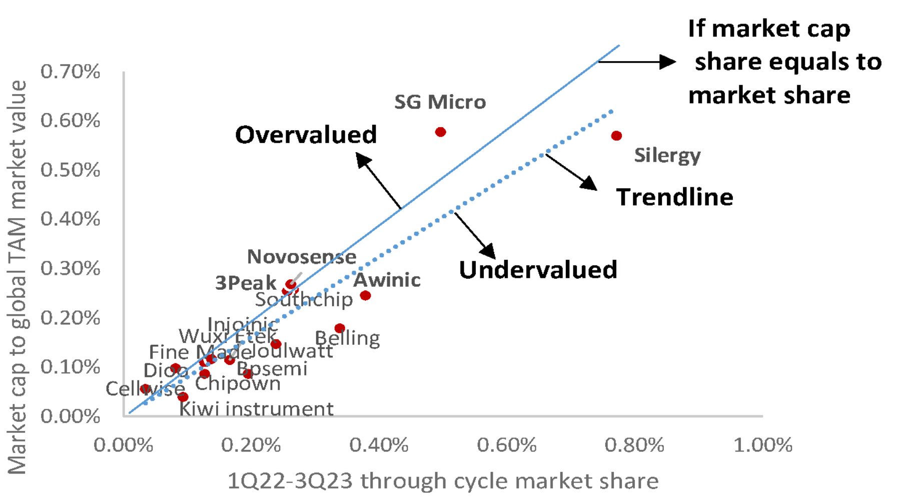
Source: WSTS, Nomura estimates

Fig. 106: China major analog vendors – valuation analysis

	Market cap (USD mn)	1Q22-3Q23 sales (USD mn)	1Q22-3Q23 market share	Market cap to global TAM market value
Silergy	3,946	1,152.9	0.77%	0.57%
SG Micro	3,999	741.4	0.50%	0.58%
AWINIC	1,702	565.3	0.38%	0.25%
Southchip	1,782	397.6	0.27%	0.26%
Joulwatt	1,019	356.3	0.24%	0.15%
Injoinic	795	248.2	0.17%	0.11%
Kiwi instrument	274	139.0	0.09%	0.04%
Fine Made	755	189.9	0.13%	0.11%
Cellwise	388	51.1	0.03%	0.06%
Bpsemi	598	291.3	0.19%	0.09%
Chipown	600	189.6	0.13%	0.09%
Wuxi ETEK	808	205.4	0.14%	0.12%
Belling	1,240	505.6	0.34%	0.18%
3Peak	1,764	382.8	0.26%	0.25%
Novosense	1,860	391.1	0.26%	0.27%
Dioo	678	121.3	0.08%	0.10%
Total	22,209		4.0%	3.2%

Source: Bloomberg Finance L.P., Nomura research

Fig. 107: China analog through cycle sales' market share vs market cap share



Source: Company data, Nomura estimates

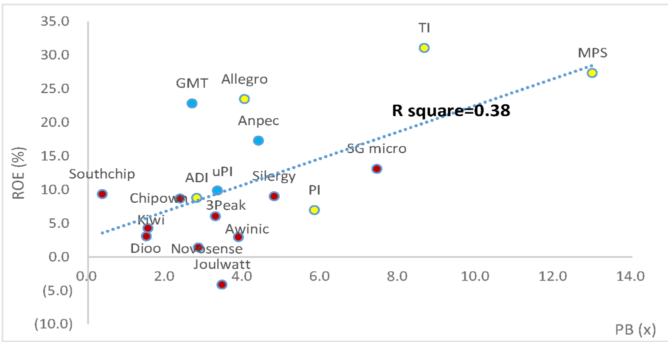
Global and China analog peer valuation overview

P/B-ROE relationship between global peers and China fabless

We use the P/B-ROE chart to show the trend of global peers and Greater China fabless companies. We exclude companies with negative ROE or extreme ROE, which, in our view, are not representable. The P/B-ROE valuation of China or global analog companies doesn't differ significantly from the trend line (see chart below).

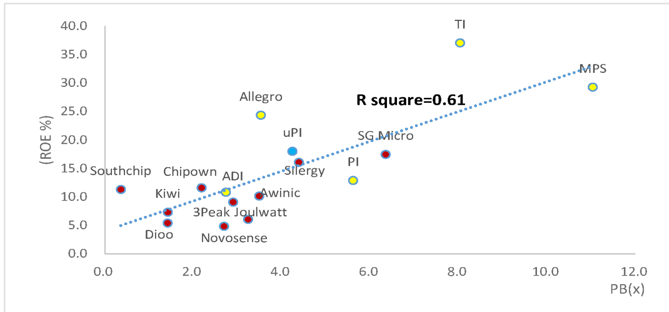
We think that China analog companies' P/B-ROE is correlated with their ROE outlook (R square is 0.61 in 2025F), while China fabless companies' R-square relationship doesn't have high correlation. Thus, a 2025F ROE of 10-20% would imply a target P/B range of 3-6x, suggesting Silergy, 3Peak, and Novosense share prices should range between TWD325-650, CNY107-214, and CNY129-258, respectively.

Fig. 108: Global analog companies: 2024E ROE vs P/B



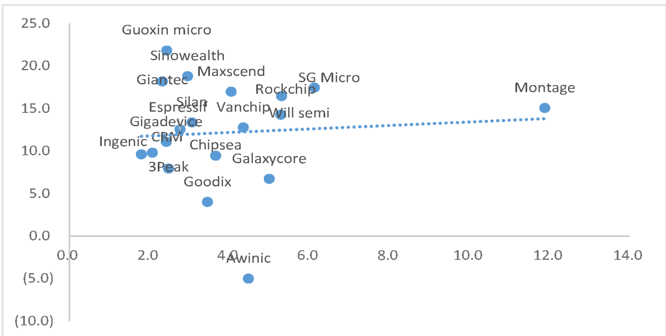
Source: Bloomberg Finance L.P., Nomura research

Fig. 109: Global analog companies: 2025E ROE vs P/B



Source: Bloomberg Finance L.P., Nomura research

Fig. 110: China fabless companies' 2025E P/B vs ROE

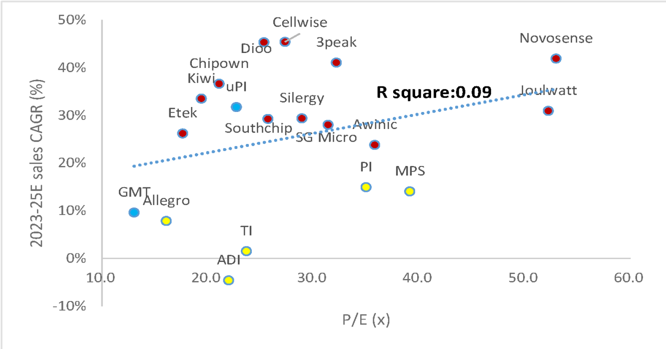


Source: Bloomberg Finance L.P., Nomura research

Sales CAGR vs P/E of global analog and China fabless peers

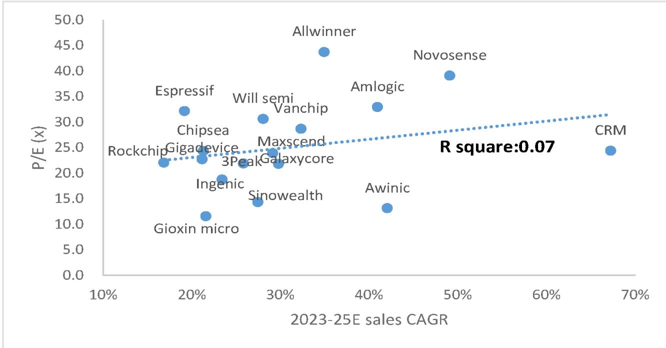
Based on the P/E valuation and sales CAGRs of global and China's analog companies, we think China analog companies' valuations are higher than global peers, while the P/E-sales CAGR relationship is not significant; the China A-share fabless P/E- sales CAGR relationship is not significant as well.

Fig. 111: Global analog semi companies' 2025 P/E vs. 2023-25E sales CAGR



Source: Bloomberg Finance L.P., Nomura estimates for covered companies

Fig. 112: China A-share fabless companies: 2025F P/E vs 2023-25E sales CAGR



Source: WIND, Nomura estimates for covered companies

Analog and China fabless peers' valuation comparison

Analog peer valuation

Global analog peers are trading at 29x 2025E EPS, and Taiwan analog peers are trading at 22x 2025F EPS. Global power semi companies are trading at 16x 2025E EPS.

Fig. 113: Analog vendors' valuation comparison

Ticker	Company	Nomura Analyst	Nomura Rating	Price(Local CCY) 4/9/2024	Mkt cap USDmn	P/E (x)		P/S(x)		P/B (x)		ROE (%)		EPS (Local CCY)		Div Yield (%)	
						FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F
Global analog																	
TXN US	TI		Not Rated	173.5	152,306	31.4	24.9	8.7	9.7	8.9	8.3	33.4	30.6	5.3	6.7	3.3	3.5
ADI US	ADI		Not Rated	204.1	96,583	32.5	25.3	7.9	10.6	2.8	2.7	11.4	13.7	6.0	7.7	2.0	2.2
MPWR US	MPS		Not Rated	682.2	31,839	51.1	40.5	17.5	15.6	13.7	11.6	28.9	29.8	12.8	16.1	0.8	0.9
ALGM US	Allegro		Not Rated	28.0	5,091	29.8	19.6	4.9	5.4	4.0	3.4	13.5	17.7	0.9	1.3	n.a.	n.a.
POWI US	Power Integration		Not Rated	71.7	3,917	57.5	33.2	8.8	8.7	5.1	4.7	10.3	16.8	1.2	2.1	1.2	1.2
Global Avg						40.5	28.7	9.5	10.0	6.9	6.2	19.5	21.7			1.8	2.0
Taiwan analog																	
2454 TT	MediaTek	Aaron Jeng	Neutral	1,160.0	57,896	23.4	21.0	3.7	3.3	4.6	4.4	19.2	21.1	49.6	55.2	3.6	4.0
6415 TT	Silergy	Aaron Jeng	Buy	328.0	3,946	51.0	25.5	6.7	5.4	3.7	3.3	7.6	13.8	6.4	12.9	0.2	0.6
8081 TT equity	GMT		Not Rated	271.5	728	14.8	14.9	2.8	2.6	2.6	n.a	18.1	n.a.	20.4	18.0	n.a.	n.a.
6719 TT equity	uPI		Not Rated	288.5	717	42.8	26.7	7.6	5.8	3.1	3.1	11.6	n.a.	6.0	11.8	1.8	2.9
TW Avg						33.0	22.0	5.2	4.3	3.5	3.6	14.1	17.4			1.8	2.5
A-share analog																	
688052 CH	Novosense	Aaron Jeng	Neutral	94.4	1,859	(0.0)	0.0	8.9	6.9	2.1	2.2	(2.5)	(3.0)	(1.4)	0.0	0.0	0.0
688536 CH	3Peak	Aaron Jeng	Buy	96.2	1,763	64.4	36.0	8.4	6.4	2.4	2.3	6.5	7.6	1.5	2.7	0.2	0.2
688798 CH	Awinic	Donnie Teng	Buy	53.0	1,699	142.4	38.9	3.8	3.2	3.4	3.2	2.3	8.1	0.4	1.4	0.1	0.1
300661 CH	SG Micro	Donnie Teng	Neutral	61.9	3,999	67.6	40.0	8.8	7.2	7.2	6.3	10.9	16.5	0.9	1.6	0.4	0.4
688484 CH	Southchip		Not Rated	30.0	1,689	35.5	26.0	7.4	5.5	3.4	0.4	10.9	12.5	0.8	1.2	0.8	1.5
688141 CH	Joulwatt		Not Rated	16.0	934	(50.0)	50.8	5.2	4.0	2.8	2.7	(5.7)	5.0	(0.3)	0.3	n.a.	n.a.
A share Avg						43.3	32.0	10.0	7.5	3.6	2.8	3.8	7.8			0.3	0.4

Source: Bloomberg Financials L.P., Nomura estimates forn coverage companies

Fig. 114: Global power semi peer valuation comparison

Ticker	Company	Nomura Analyst	Nomura Rating	Price(Local CCY) 4/9/2024	Mkt cap USDmn	P/E (x)		P/S(x)		P/B (x)		ROE (%)		EPS (Local CCY)		Div Yield (%)	
						FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F
Global Power semi																	
NXPI US	NXP		Not Rated	252	61,827	18.7	16.4	4.9	4.9	7.3	6.7	35.8	33.7	13.5	15.4	1.8	2.0
STM US	STM		Not Rated	44	38,178	13.8	10.8	2.5	2.3	2.4	2.1	19.1	18.6	3.1	4.0	0.7	0.6
IFNNY US	Infineon		Not Rated	37	47,849	17.9	14.2	3.4	2.7	2.8	2.6	17.5	18.6	2.0	2.6	1.2	1.3
6723 JP	Renesas	Masaya Yamasaki	Buy	2,761	34,048	15.4	13.1	3.0	3.5	2.5	2.2	15.5	15.6	183.0	215.0	0.0	0.0
ON US	ON Semi		Not Rated	71	19,758	16.7	13.8	3.7	3.7	3.8	3.2	22.7	25.5	4.2	5.2	-	-
TXN US	TI		Not Rated	173	153,596	32.5	25.8	7.9	9.0	9.4	9.2	34.6	31.7	5.3	6.7	3.2	3.4
ADI US	ADI		Not Rated	204	97,361	34.1	26.5	8.5	8.2	2.9	2.9	12.0	14.4	6.0	7.7	1.9	2.1
VSH US	Vishay		Not Rated	23	2,990	17.7	11.5	0.9	0.9	n.a.	n.a.	n.a.	n.a.	1.3	2.0	1.9	n.a.
DIOD US	Diode		Not Rated	72	3,210	31.8	18.9	1.7	2.0	1.9	1.8	n.a.	n.a.	2.3	3.8	n.a.	n.a.
Average						22.1	16.8	4.0	4.1	4.1	3.8	22.4	22.6			1.3	1.4

Source: Bloomberg Finance L.P., Nomura estimates on coverage companies

China Fabless peers P/E comparison

China fabless peers trade at 30x 2025F P/E on average, based on WIND consensus estimates (36x/30x 2024F/25F EPS).

Our target P/E multiple of 35x 2025F P/E for Silergy, and 50x for 3Peak is above the average of the China fabless group. We believe Silergy deserves a higher-than-average multiple given its market position and low localization in the China analog IC market.

Fig. 115: China fabless peers valuation comparison

Ticker	Company Chinese	Company English	Mkt cap CNYmn	P/E (x)		P/B (x)		P/S (x)		ROE (%)		Div Yield (%)	
				FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F
002049 CH	紫光国微	GUOXIN MICRO	54,027	15.8	12.5	3.4	2.7	5.3	4.2	22.6	22.1	0.4%	0.3%
300613 CH	富瀚微	FULLHAN	7,614	17.3	14.1	2.5	2.2	3.3	2.8	14.4	15.1	0.5%	0.5%
688099 CH	晶晨股份	AMLOGIC	19,682	15.8	13.7	2.7		2.8	2.3	13.5	17.2	0.3%	0.3%
300327 CH	中颖电子	SINO WEALTH	7,200	19.3	15.0	3.4	2.6	4.2	3.4	13.8	17.3	0.0%	0.0%
688396 CH	华润微	CRM	51,895	22.4		2.0		4.4	3.8	8.0	8.9	0.3%	0.5%
688018 CH	乐鑫科技	ESPRESSIF	7,653	29.2	21.4	3.3	2.9	4.2	3.3	9.0	11.5	0.8%	1.1%
688123 CH	聚辰股份	GIANTEC SEMICONDUCTOR	8,491	17.4	12.5	3.1	2.5	7.8	6.0	15.0	18.1	0.9%	1.2%
300223 CH	北京君正	INGENIC	30,156	35.9	25.3	2.4	2.2	5.1	4.2	6.8	8.9	0.0%	0.0%
300782 CH	卓胜微	MAXSCEND	53,131	26.7		4.1		9.6	7.9	14.6	15.5	0.9%	0.8%
300672 CH	国科微	GOKE MICROELECTRONICS	10,866	42.1	25.1	2.5	2.3	2.4	2.0	5.9	9.0	0.2%	0.3%
603986 CH	兆易创新	GIGADEVICE	47,757	37.2	25.2	2.9	2.6	6.3	5.1	7.9	10.5	0.6%	0.9%
300458 CH	全志科技	ALLWINNERTECH	12,224	37.9	27.6	3.7	3.3	5.5	4.4	5.4	10.0	0.6%	1.0%
603501 CH	韦尔股份	WILLSEMI	117,894	43.8	30.9	5.5	4.8	4.5	3.8	12.8	15.6	0.8%	1.0%
688008 CH	澜起科技	MONTAGE TECHNOLOGY	51,674	23.3		3.8		12.0	8.2	12.4	16.8	0.7%	0.7%
300661 CH	圣邦股份	SGMICRO	29,401	58.7	37.4	7.1	6.1	9.1	7.3	12.3	16.7	0.2%	0.3%
688536 CH	思瑞浦	3PEAK	13,216	36.5	25.1	2.4	2.0	8.0	5.8	4.0	6.7	0.2%	0.3%
688153 CH	唯捷创芯	VANCHIP	22,443	35.6		4.5		5.9	4.7	9.0	12.8	0.3%	0.5%
600460 CH	士兰微	SILAN	32,682	53.7	41.1	3.8	3.5	2.9	2.5	7.0	8.1	0.5%	0.9%
688728 CH	格科微	GALAXYCORE	44,548	83.2		5.0		6.9	5.6	3.7	6.1	0.0%	0.0%
603160 CH	汇顶科技	GOODIX	27,219	52.1	40.3	3.2	3.0	5.3	4.5	6.3	7.5	0.4%	0.6%
603068 CH	博通集成	BEKEN	3,299		106.4	2.0	2.0	4.0	3.5	(3.2)	1.9	0.0%	0.0%
688798 CH	艾为电子	AWINIC	12,709	42.2		3.2		4.1	3.3	2.8	7.6	0.3%	0.7%
688595 CH	芯海科技	CHIPSEA	4,301	51.4	45.3	4.0	4.5	6.3	4.4	1.0	7.5	0.0%	0.1%
Avg				**36.3**	**30.5**	**3.5**	**3.1**	**5.6**	**4.5**	**8.9**	**11.8**	**0.4%**	**0.5%**

Source: WIND, Nomura research

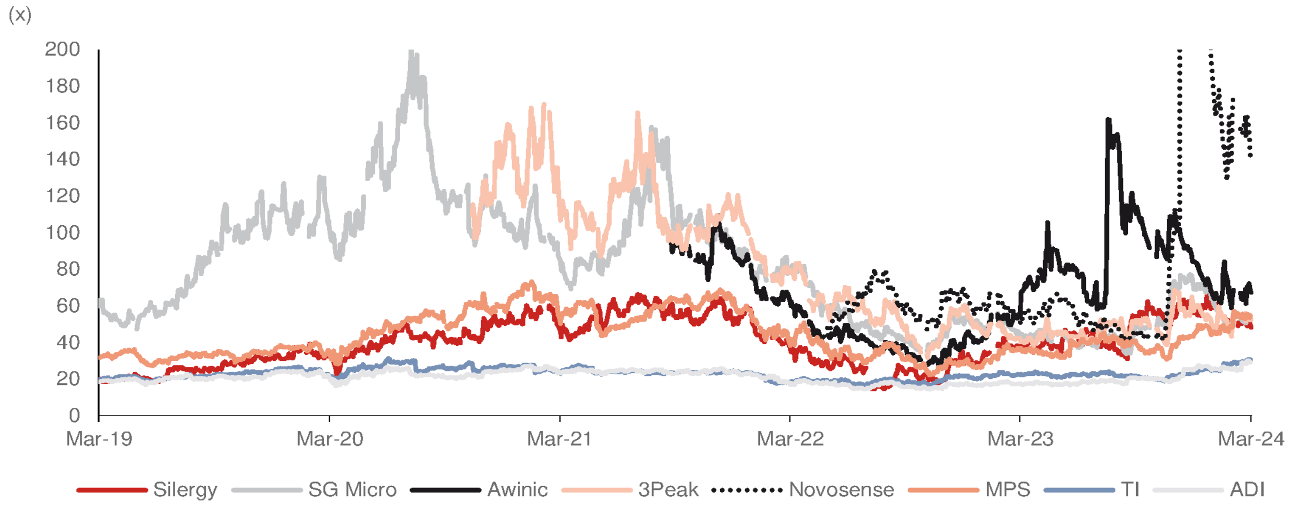
Valuation comparison between global and China analog IC companies

We further analyzed the valuation comparison in our 2021 Anchor Report (*report*), and found that China analog companies' valuations in P/E, P/B, and P/S are significantly lower than the 2021 level, except for MPS, which is the only analog vendor currently enjoying an AI application valuation re-rating.

- P/E: In 2020-21, SG Micro and 3Peak were trading at higher than 100x P/E, and Awinic, Silergy, and MPS were also trading in the 50-100x range, while large analog vendors such as TI and ADI were trading in the 20-30x P/E range. In 2024, Novosense, which is at near-breakeven level based on one-year forward 2025F PE, has reached a significanlty high P/E valuation. Other China analog vendors and MPS are trading in the 40-60x P/E range, and TI and ADI are trading at the 30x P/E level (Bloomberg consensus).
- P/B: In 2020-21, SG Micro was trading at higher than 20x P/B, while other China and global peers were in the 10-20x P/B range. In 2024, China and global analog vendors P/B valuations have declined to below the 10x level (Bloomberg consensus), except for MPS.
- P/S: In 2020-21, SG Micro and 3Peak were trading at higher than 40x P/S, Silergy and MPS were at around 10-20x P/S, while Awinic, TI, and ADI were in the 5-10x P/S range. In 2024, however, China and global analog vendors' P/S valuations have declined to 5-10x (Bloomberg consensus), except for MPS.

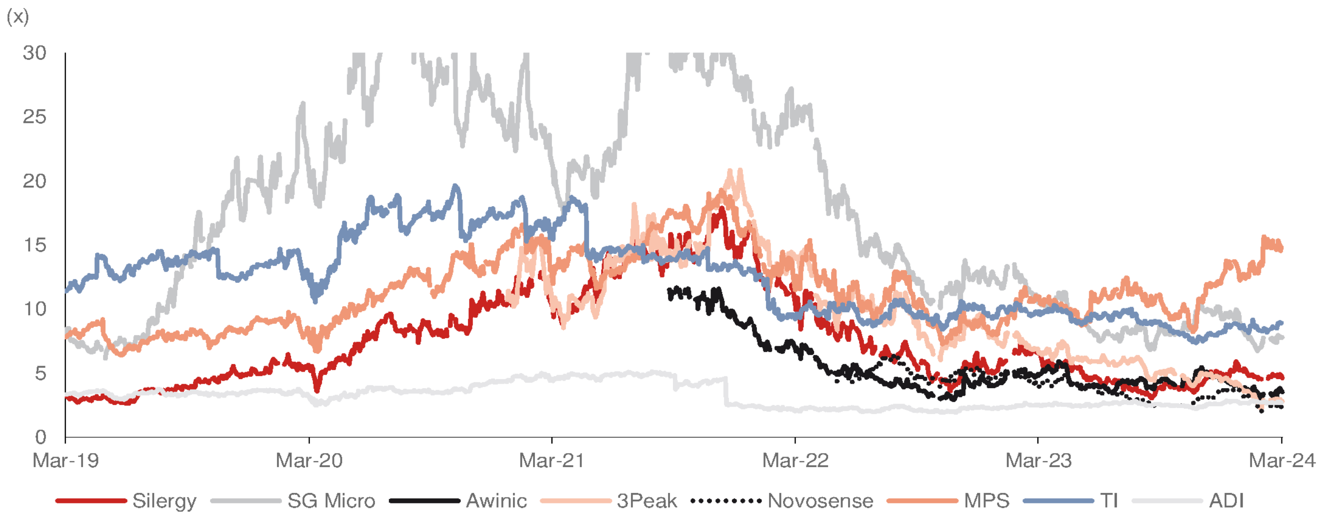
We think China analog vendors' valuations are still slightly higher than those of global peers. However, these China vendors' valuations are not as high as during 2020-21, and are getting closer to their global analog peers now.

Fig. 116: Global and China analog IC companies' one-year-forward P/E



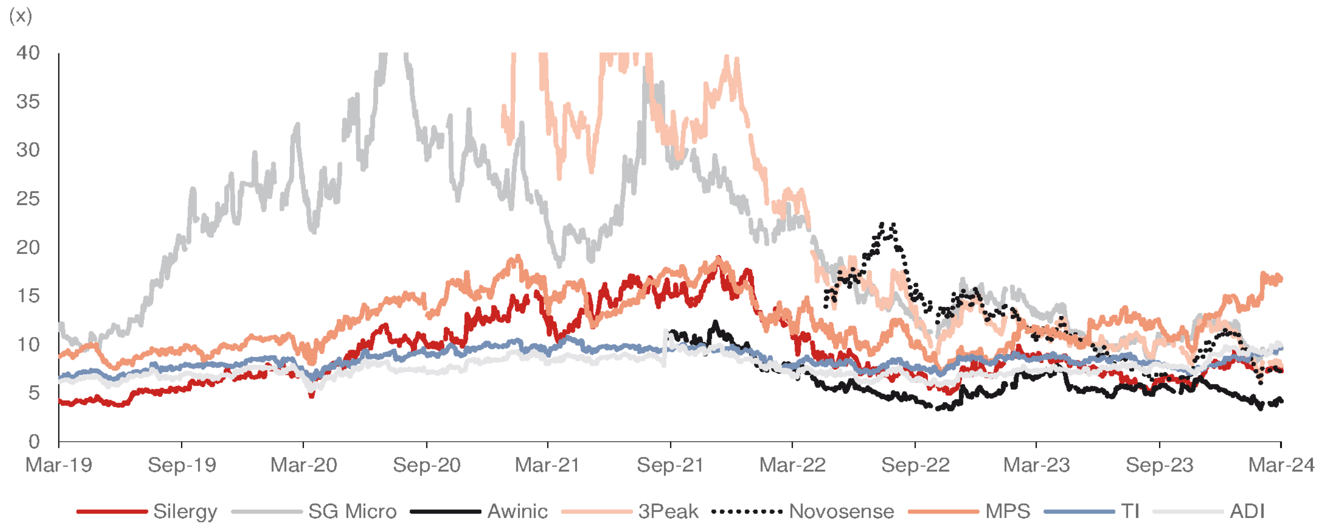
Source: Bloomberg Finance L.P., Nomura research

Fig. 117: Global and China analog IC companies' one-year-forward P/B



Source: Bloomberg Finance L.P., Nomura research

Fig. 118: Global and China analog IC companies' one-year-forward P/S



Source: Bloomberg Finance L.P., Nomura research

Appendix

Share price comparison in the China semi market

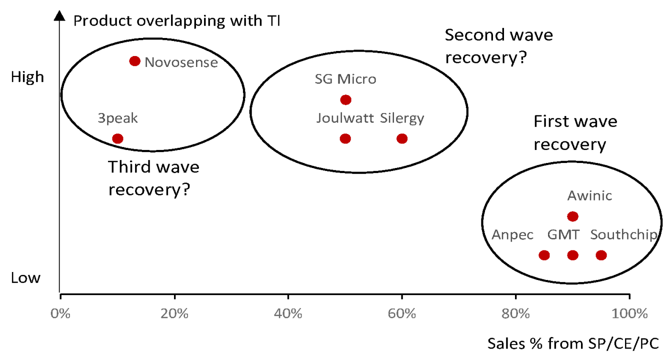
China analog vendors' share prices have been the worst performers among China A-share semi companies since Jan-22. We think this is due to: 1) some vendors' end-application exposure being more in industrial and auto; 2) analog companies losing market share to global vendors, while other semi vendors have continued to gain share from global vendors; and 3) higher pricing pressure in 2023 than in other products.

First, we expect China local analog vendors to regain market share. Second, we think TI's pricing pressure on vendors is easing. Last, we expect industrial and auto application end-demand can gradually improve in China. Therefore, we expect China analog vendors' share performance to gradually recover.

Share price performance comparison of China analog vendors

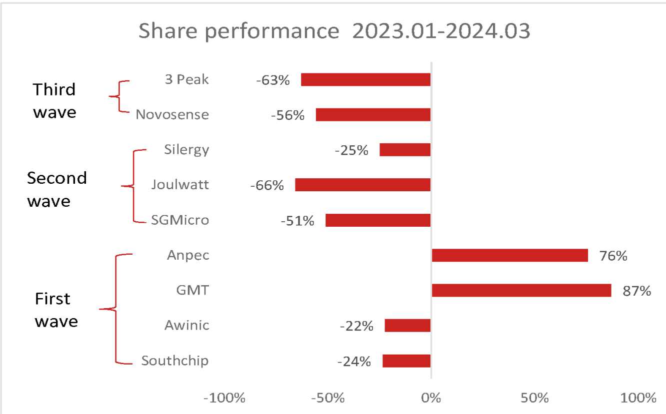
We think the China analog vendors' share price performance is related to their sales recovery pattern. We have highlighted that Greater China analog IC vendors have different product exposure to TI in our Nov-23 report (*link*). The percentage of product exposure to TI will also impact the sales recovery timing, in our view, and companies with less sales exposure to TI will recover earlier. Therefore, we see that the share prices of companies' with higher SP/CE/PC sales exposure have performed better than other analog vendors since 2023 (***Fig. 120***).

Fig. 119: Greater China analog IC companies: product overlapping with TI vs sales % from SP/CE/PC



Source: Company data, Nomura estimates

Fig. 120: Greater China analog vendors' share performance



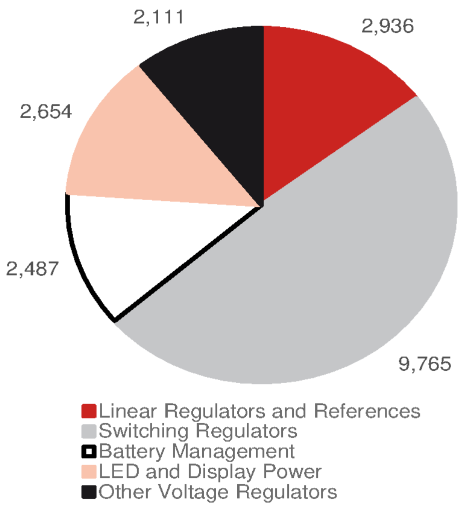
Source: Company data, Nomura research

General purpose analog product market share by vendor

In this Anchor Report, we have discussed mainly about general purpose analog IC, which includes power IC and signal chain IC, with market sizes of USD20bn and USD6.5bn, respectively, in 2022, based on Gartner estimates. General power IC has four major products (***Fig. 121***) – linear, switching, battery management, LED power and display power, with shares of 15%/49%/12%/13% respectively. On the other hand, the general purpose signal chain has two major products (***Fig. 122***), ADC and DAC, with shares of 65% and 21%, respectively.

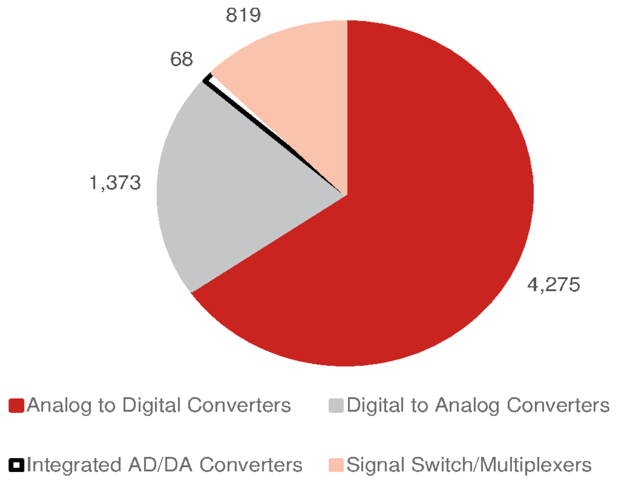
The analog IC market leaders are mainly global vendors, while some China analog IC vendors have gained meaningful market share, including Silergy in general purpose power IC, Chipone [unlisted]/ Bpsemi [688368 CH, Not rated]/ Silan [600460 CH, Not rated] in LED/display power IC, and 3Peak in the signal chain ADC and DAC markets.

Fig. 121: General power IC product sales in 2022



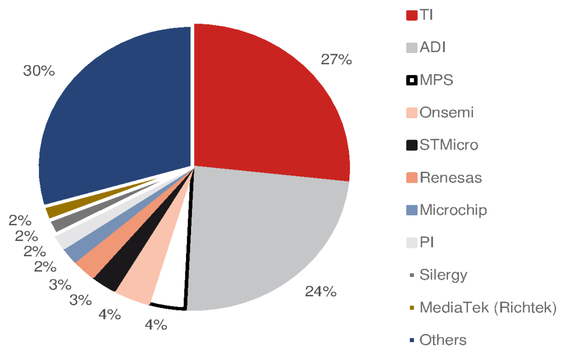
Source: Gartner, Nomura research

Fig. 122: General signal chain product sales in 2022



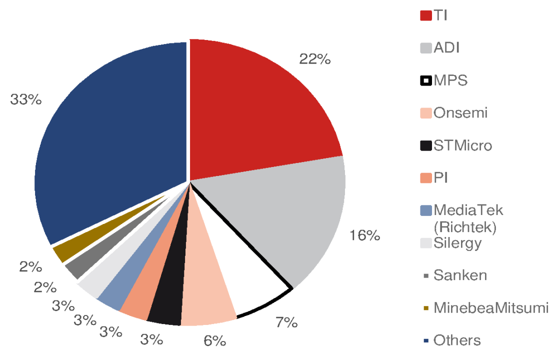
Source: Gartner, Nomura research

Fig. 123: General purpose analog semi market share by vendors' sales in 2022



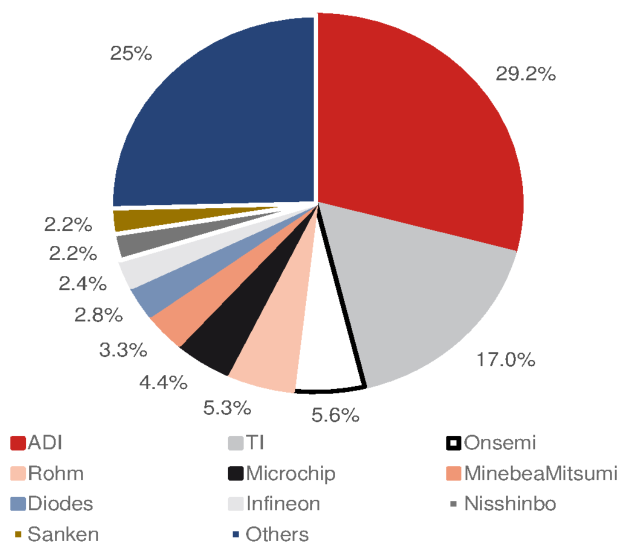
Source: Gartner, Nomura research

Fig. 124: Voltage regulator power IC market share by vendors' sales in 2022



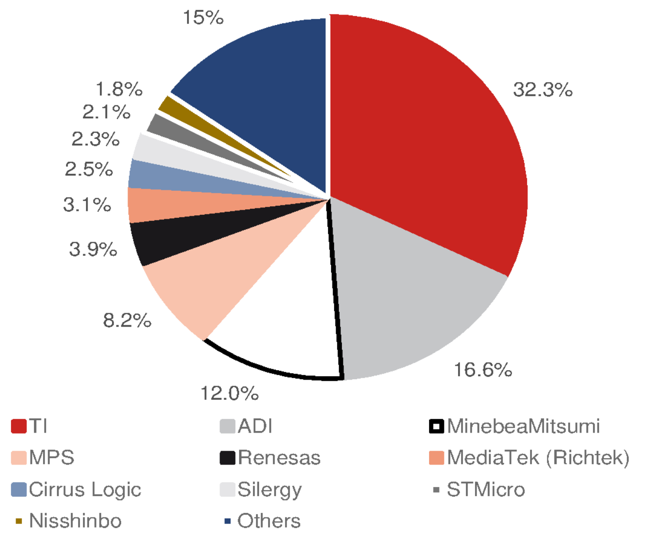
Source: Gartner, Nomura research

Fig. 125: Linear power IC market share by vendors' sales in 2022



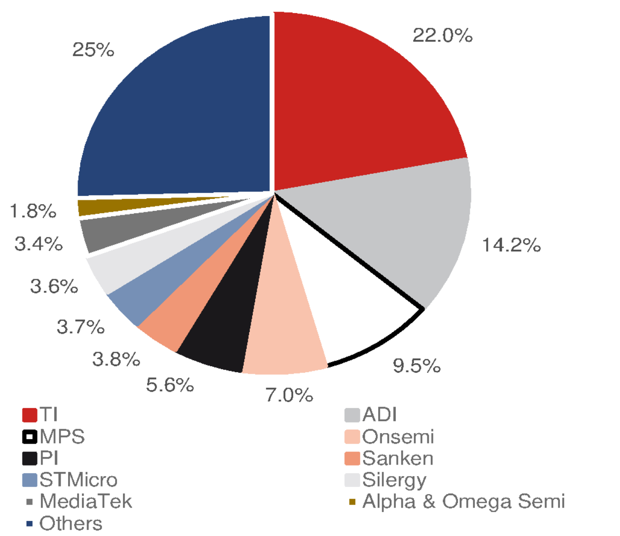
Source: Gartner, Nomura research

Fig. 126: Battery management semi market share by vendors' sales in 2022



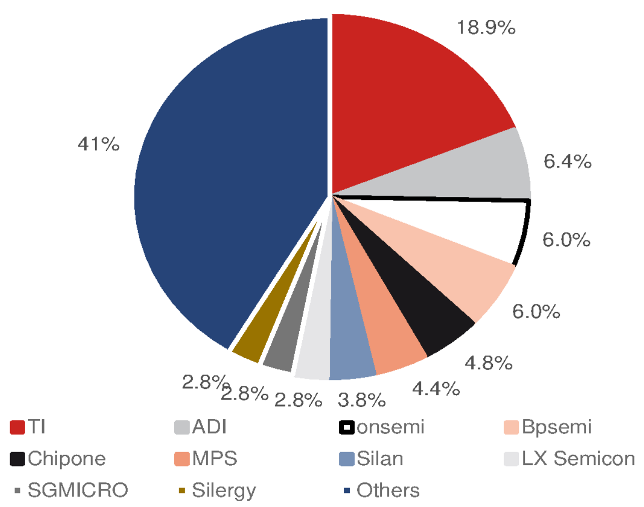
Source: Gartner, Nomura research

Fig. 127: Switching regulators power analog IC market share by vendors' sales in 2022



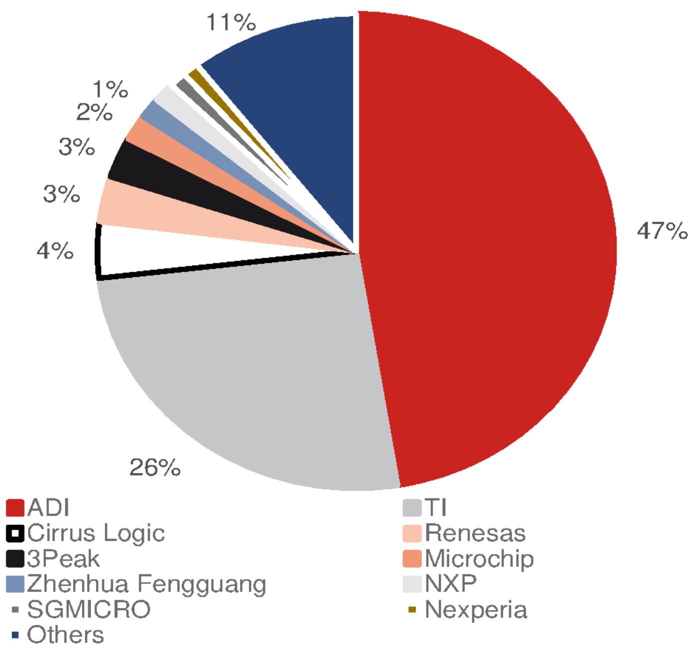
Source: Gartner, Nomura research

Fig. 128: LED and display power IC market share by vendors' sales in 2022



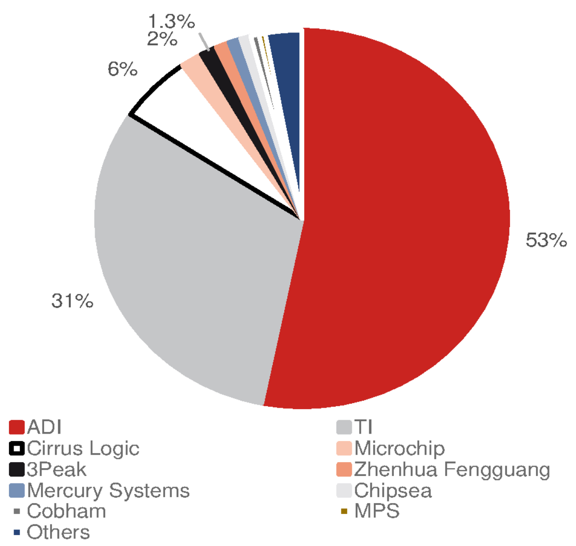
Source: Gartner, Nomura research

Fig. 129: Signal chain market share by vendors' sales in 2022



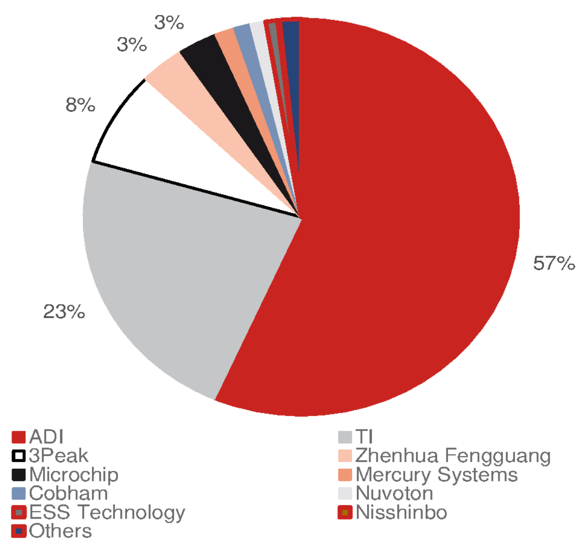
Source: Gartner, Nomura research

Fig. 130: ADC converters market share by vendors' sales in 2022



Source: Gartner, Nomura research

Fig. 131: DAC converters market share by vendors' sales in 2022



Source: Gartner, Nomura research

NOMURA

Silergy 6415.TW 6415 TT

Global Markets Research
12 April 2024

EQUITY: FABLESS

Best-in-class Asia analog fabless; initiate at Buy

Share gain potential ongoing, fab know-how critical, inventory normalizing, growth resuming

Rating Starts at	Buy
Target price Starts at	TWD 450.00
Closing price 9 April 2024	TWD 328.00
Implied upside	+37.2%
Market Cap (USD mn)	3,934.6
ADT (USD mn)	35.1

Initiate coverage at Buy with TP of TWD450 (37% implied upside)

We initiate coverage of Silergy with a Buy rating and TP of TWD450, based on 35x 2025F P/E (EPS of TWD12.9), supported by a 2023-25F net earnings CAGR of 186% (35x is at the mid-to-low-end of its P/E range of 20-60x in 2019-21; currently trading at 25x 2025F). From a market-share vs market-cap-share angle, we believe Silergy is also undervalued (*Fig. 221* - *Fig. 222*). Currently, “short term” visibility remains low (thus mainly rush orders) and TI’s growing capacity is still an investor concern, but its “long term” value could have emerged following the severe underperformance of the stock (down 74% 2022-YTD, vs TAIEX up 12%). Key downside risks include TI’s pricing and capacity plan, and end-demand strength.

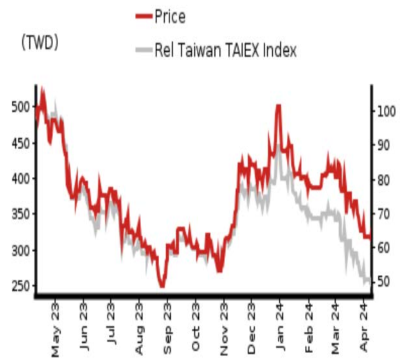
Share gain potential, fab know-how and 12” fab migration

Structurally, we like its: **1) further market share gain potential** from 1% currently in China, particularly following Silergy's fast product expansion beyond consumer applications, e.g., auto, and new energy, on top of our positive view on the ongoing China analog IC localization trend. Currently, the analog IC localization rate in China is likely at 15-20%, which is well below the 40-50% maturity level seen by Taiwan analog IC vendors in late 2010s; **2) fab process know-how:** Silergy is the only analog fabless company in Asia that has its own fab process know-how (like MPS in the US), which would be critical to its product design competitiveness, in our view; and **3) 12” fab migration for gen-4 chip design**: this should be helpful for Silergy’s cost competitiveness.

Aggressive inventory write-off, DOI back to pre-COVID level, GPM recovery, and likely easing price competition from TI

Silergy was one of the most aggressive analog IC vendors in Greater China in writing off its excess inventory value during the downturn in 2023-YTD (*Fig. 68*). Its inventory and DOI have both fallen back to even below pre-COVID levels by 4Q23 (*Fig. 67*) – which could make its recovery easier when demand returns. Also, with its inventory dollars peaking in 2H22, its write-off impact should peak out in 2H23-1H24 (*Fig. 68*). Its reported GPM should gradually approach product GPM toward end-2024. According to company disclosures, product GPM stabilized at the 50% level in 2023 after peaking at 56% in 3Q21 (*Fig. 69*), indicating that its cost-cut efforts should have started offsetting the impact of competitors’ pricing from 2H23. Of note, we believe TI’s pricing in 2024F can’t be as aggressive as it was in 2023.

Relative performance chart



Source: LSEG, Nomura

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Year-end 31 Dec	FY23	FY24F		FY25F		FY26F	
Currency (TWD)	Actual	Old	New	Old	New	Old	New
Revenue (mn)	15,427	0	18,572	0	23,449	0	27,577
Reported net profit (mn)	604	0	2,415	0	4,965	0	6,355
Normalised net profit (mn)	604	0	2,415	0	4,965	0	6,355
FD normalised EPS	1.57		6.28		12.92		16.54
FD norm. EPS growth (%)	-90.1		299.7		105.6		28.0
FD normalised P/E (x)	208.6	–	52.2	–	25.4	–	19.8
EV/EBITDA (x)	438.7	–	39.1	–	19.0	–	14.7
Price/book (x)	4.0	–	3.7	–	3.3	–	2.9
Dividend yield (%)	1.4	–	0.2	–	0.5	–	1.1
ROE (%)	1.9		7.4		13.8		15.7
Net debt/equity (%)	net cash		net cash		net cash		net cash

Source: Company data, Nomura estimates

Key data on Silergy

Performance

(%)	1M	3M	12M		
Absolute (TWD)	-14.6	-24.0	-31.5	M cap (USDmn)	3,934.6
Absolute (USD)	-16.4	-26.5	-35.1	Free float (%)	0.9
Rel to Taiwan TAIEX Index	-19.7	-42.6	-62.8	3-mth ADT (USDmn)	35.1

Income statement (TWDmn)

Year-end 31 Dec	FY22	FY23	FY24F	FY25F	FY26F
Revenue	23,511	15,427	18,572	23,449	27,577
Cost of goods sold	-11,152	-8,848	-9,700	-11,594	-13,568
Gross profit	12,359	6,579	8,872	11,855	14,009
SG&A	-1,970	-2,204	-2,124	-2,234	-2,523
Employee share expense	-4,460	-4,894	-4,656	-4,729	-5,205
Operating profit	5,929	-519	2,093	4,892	6,282
EBITDA	6,522	205	2,873	5,700	7,117
Depreciation	-389	-511	-561	-584	-608
Amortisation	-204	-213	-219	-223	-228
EBIT	5,929	-519	2,093	4,892	6,282
Net interest expense	332	470	682	798	1,005
Associates & JCEs	-53	55	0	0	0
Other income	450	584	-19	-23	-28
Earnings before tax	6,657	589	2,757	5,667	7,259
Income tax	-529	15	-342	-702	-904
Net profit after tax	6,128	604	2,415	4,965	6,355
Minority interests					
Other items	0	0	0	0	0
Preferred dividends					
Normalised NPAT	6,128	604	2,415	4,965	6,355
Extraordinary items					
Reported NPAT	6,128	604	2,415	4,965	6,355
Dividends	-1,692	-1,722	-192	-692	-1,383
Transfer to reserves	4,436	-1,118	2,222	4,274	4,972
Valuations and ratios					
Reported P/E (x)	20.6	208.6	52.2	25.4	19.8
Normalised P/E (x)	20.6	208.6	52.2	25.4	19.8
FD normalised P/E (x)	20.6	208.6	52.2	25.4	19.8
Dividend yield (%)	1.3	1.4	0.2	0.5	1.1
Price/cashflow (x)	18.0	32.5	49.5	21.6	19.1
Price/book (x)	4.0	4.0	3.7	3.3	2.9
EV/EBITDA (x)	17.2	438.7	39.1	19.0	14.7
EV/EBIT (x)	18.9	–	53.7	22.2	16.6
Gross margin (%)	52.6	42.6	47.8	50.6	50.8
EBITDA margin (%)	27.7	1.3	15.5	24.3	25.8
EBIT margin (%)	25.2	-3.4	11.3	20.9	22.8
Net margin (%)	26.1	3.9	13.0	21.2	23.0
Effective tax rate (%)	7.9	-2.5	12.4	12.4	12.5
Dividend payout (%)	27.6	285.0	8.0	13.9	21.8
ROE (%)	28.8	1.9	7.4	13.8	15.7
ROA (pretax %)	50.5	-2.2	9.4	21.4	26.7
Growth (%)					
Revenue		-34.4	20.4	26.3	17.6
EBITDA		-96.9	1,304.5	98.4	24.9
Normalised EPS		-90.1	299.7	105.6	28.0
Normalised FDEPS		-90.1	299.7	105.6	28.0

Source: Company data, Nomura estimates

Cashflow statement (TWDmn)

Year-end 31 Dec	FY22	FY23	FY24F	FY25F	FY26F
EBITDA	6,522	205	2,873	5,700	7,117
Change in working capital	2,115	-1,218	-501	-1,168	
Other operating cashflow	478	1,554	890	641	642
Cashflow from operations	7,000	3,873	2,545	5,840	6,592
Capital expenditure	-590	-1,001	-1,051	-1,103	-1,158
Free cashflow	6,410	2,873	1,494	4,737	5,433
Reduction in investments	-1,178	-3,303	0	0	0
Net acquisitions					
Dec in other LT assets	0	0	0	0	0
Inc in other LT liabilities	-239	-132	0	0	0
Adjustments	-1,237	-137	0	0	0
CF after investing acts	3,756	-700	1,494	4,737	5,433
Cash dividends	-1,692	-1,722	-192	-692	-1,383
Equity issue	0	0	0	0	0
Debt issue	0	160	0	0	0
Convertible debt issue	0	0	0	0	0
Others	1,266	12	0	0	0
CF from financial acts	-427	-1,549	-192	-692	-1,383
Net cashflow	3,329	-2,249	1,302	4,045	4,050
Beginning cash	11,614	14,944	12,694	13,996	18,042
Ending cash	14,943	12,694	13,996	18,042	22,092
Ending net debt	-14,944	-12,534	-13,836	-17,881	-21,932

Balance sheet (TWDmn)

As at 31 Dec	FY22	FY23	FY24F	FY25F	FY26F
Cash & equivalents	14,944	12,694	13,996	18,042	22,092
Marketable securities	2,425	5,729	5,729	5,729	5,729
Accounts receivable	1,618	1,785	2,159	2,445	3,035
Inventories	5,111	1,817	2,736	3,016	3,746
Other current assets	441	828	828	828	828
Total current assets	24,539	22,853	25,448	30,059	35,430
LT investments	5,275	5,602	5,034	4,466	3,898
Fixed assets	1,863	2,350	2,839	3,358	3,909
Goodwill	0	0	0	0	0
Other intangible assets	2,169	1,948	1,729	1,505	1,277
Other LT assets	1,428	1,670	1,670	1,670	1,670
Total assets	35,274	34,422	36,719	41,058	46,183
Short-term debt	0	160	160	160	160
Accounts payable	606	492	566	632	784
Other current liabilities	1,809	1,298	1,298	1,298	1,298
Total current liabilities	2,415	1,950	2,024	2,090	2,243
Long-term debt	0	0	0	0	0
Convertible debt	0	0	0	0	0
Other LT liabilities	1,367	931	931	931	931
Total liabilities	3,782	2,881	2,956	3,022	3,174
Minority interest					
Preferred stock	0	0	0	0	0
Common stock	954	960	960	960	960
Retained earnings	30,538	30,580	32,802	37,076	42,048
Proposed dividends					
Other equity and reserves					
Total shareholders' equity	31,492	31,541	33,763	38,036	43,009
Total equity & liabilities	35,274	34,422	36,719	41,058	46,183
Liquidity (x)					
Current ratio	10.16	11.72	12.57	14.38	15.80
Interest cover	–	–	–	–	–
Leverage					
Net debt/EBITDA (x)	net cash	net cash	net cash	net cash	net cash
Net debt/equity (%)	net cash	net cash	net cash	net cash	net cash
Per share					
Reported EPS (TWD)	15.95	1.57	6.28	12.92	16.54
Norm EPS (TWD)	15.95	1.57	6.28	12.92	16.54
FD norm EPS (TWD)	15.95	1.57	6.28	12.92	16.54
BVPS (TWD)	81.97	82.10	87.88	99.01	111.95
DPS (TWD)	4.40	4.48	0.50	1.80	3.60
Activity (days)					
Days receivable	40.3	38.9	35.8	36.3	
Days inventory	142.9	85.9	90.5	91.0	
Days payable	22.6	19.9	18.9	19.0	
Cash cycle	0.0	160.5	104.8	107.5	108.2

Source: Company data, Nomura estimates

Company profile

Silergy, founded in 2008, is a leading China analog IC fabless company that mainly provides PMIC products. Silergy's end-to-end analog, power, and signal chain solutions are widely used in automotive, industrial, consumer, computing, and telecom equipment.

Valuation Methodology

Our TP of TWD450 is based on a P/E of 35x on our 2025F EPS of TWD12.9. 35x P/E is at the mid-to-low-end of its P/E range of 20x-60x in 2019-2021 which earnings resumed growth from consecutive declining in 2017-2018. The benchmark index is TAIEX.

Risks that may impede the achievement of the target price

Downside risks include: 1) market share gain is slower than expected; 2) lower-than-expected auto sales growth; 3) higher than expected pricing competition; and 4) weaker than expected end-demand recover.

ESG

Silergy has focused on developing clean technology products and is committed to designing high efficiency power management chips with environmental protection and energy saving effects to provide customers with diverse and complete energy management solutions. Over the years, weSilergy has continued to refine IC design capabilities and wafer and packaging processes to increase the power density of our latest products to more than four times that of previous generation products, allowing consumer goods, communication, industrial and automotive products to be more compact and operate with the lowest power consumption, ultimately reducing carbon emissions.

Executive summary

Best-in-class analog fabless in Asia; initiating coverage with a Buy rating

We initiate coverage of Silergy with a Buy rating and TP of TWD450, based on a 35x target P/E multiple on 2025F EPS of TWD12.9, supprted by 2023-25 earnings CAGR of 186%. Our target P/E of 35x is at the mid-to-low-end of its P/E range of 20x-60x in 2019-21 when earnings growth resumed after declining consecutively in 2017-2018. From a market-share vs market-cap-share angle, we believe Silergy is also undervalued (*Fig. 221* - *Fig. 222*). Currently, "short term" visibility remains low and TI's growing capacity is still a competition concern. However, the severe share price underperformance (down 74% 2022-YTD, vs TAIEX's 12% appreciation) should have improved the risk-reward, too, in our view. The stock is currently trading at 25x 2025F EPS.

Structurally, we like its: **1) further market share gain potential** from 1% currently in China, particularly following Silergy's fast product expansion beyond consumer applications, e.g. auto, and new energy, on top of our positive view on ongoing China analog IC localization trend. Currently, the analog IC localization rate in China is likely at 15-20%, which is well below the 40-50% maturity level seen by Taiwan analog IC vendors in the late 2010s; **2) the ownership of fab process know-how:** Silergy is the only analog fabless company in Asia that has its own fab process know-how (like MPS in the US), which would be critical to its product design competitiveness, in our view; **3) 12" fab migration for gen-4 analog chip design:** this should help the company's cost competitiveness.

Silergy was one of the most aggressive analog IC vendors in Greater China in writing off its excess inventory value during the downturn (*Fig. 68*). Its inventory and DOI both came back to pre-COVID levels by 4Q23 (*Fig. 67*). While this indicates high pricing pressure and lengthy inventory digestion over the downturn, its aggressive approach on inventory has set itself up for an easier recovery when demand returns. Also, with its inventory dollar value peaking in 2H22, we expect its write-off impact will have peaked in 2H23-1H24 (*Fig. 68*). Its reported GPM should gradually approach product GPM toward end-2024. According to company disclosures, product GPM has stabilized at the 50% level after peaking at 56% in 3Q21 (*Fig. 69*), indicating that its cost- efforts should have started offsetting the impact of competitors' pricing from 2H23. Of note, we believe TI will not be able to be as aggressive with its pricing in 2024 as it was in 2023.

Silergy is among biggest analog IC vendors in Greater China

We believe Silergy is a leading analog fabless in terms of sales scale in China local fabless (*Fig. 132*) , and the ninth-largest general-purpose power IC vendor globally, as per Gartner's estimate.

According to management, Silergy is China's largest analog fabless (specializing in power management integrated circuits (PMIC), in terms of sales (*Fig. 137*); the #9 general-purpose analog supplier globally, and the #8 general power IC supplier globally, with estimated market share of 0.9%, ~2%, and 3% in 2022, respectively, as per Gartner. According to the company, it recorded power IC sales of c.USD670mn in 2022 vs a c.USD20bn general power analog IC total addressable market (TAM). We note the company's market share has expanded since 2019, when it reported USD350mn in analog IC sales (c.1.5% share vs market TAM of USD22.9bn in 2019, based on Gartner estimates). Besides mainly developing internally, Silergy acquired an LED driver business from NXP (NXPI US, Not rated) in 2016, and smart meter and energy analog business from Maxim (unlisted). Other than power IC, Silergy develops signal chain, sensor, and logic-function IC, which contributed c.USD120mn to 2022 sales. The sales contribution of power IC to Silergy for 2022 was 80-90%, and we expect power IC to remain a strong growth driver for the company over the next few years.

Of Silergy's analog business, consumer was the major application, at c.35-40% sales contribution in 2023, while the rest came from industrial, automotive, communication and computing. Among applications, in 2022, Silergy's auto sales grew by a significant c.300% y-y given its successful design in product mass production. On our forecast, auto sales accounted for 10% of Silergy's sales in 2023F, up from c.1% in 2021 (*Fig. 149*). High-end auto analog IC has a major role in its much stable pricing and profitability.

With automotive electrification mega trend, Silergy should maintain a dominant position in the China local market, in our view. As such, Silergy's auto design-in product mass production progress at auto analog remains the key factor to monitor. With still expanding product portfolio, we expect Silergy to regain market share from 2024, as we believe it would outrun China local peers by its Gen-4 platform product ramp-up, which can highly improve cost structure and product performance.

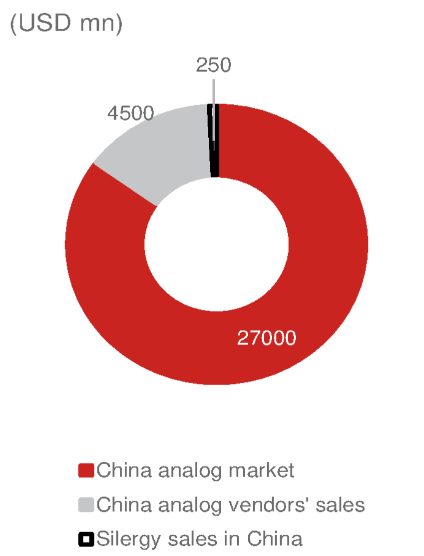
Beyond 2024, according to management, Silergy's auto sales likely to grow 100%+ in 2025F (from 2023 to 2025), as we believe that it should be able to capture the market share both from global and China customers. The growing market share in auto applications as well as server and industrial applications, and Silergy's focus on more high-end applications for the auto/server/communication markets to fulfill customers' localization requirements in China should help Silergy to deliver a 20-30% long-term sales CAGR over the next three years, in our view.

Fig. 132: Listed China analog fabless companies' ranking (by sales) – 2022

Rank	Company name	English name	Ticker	2022 analog sales (USD mn)
1	硅力杰	Silergy	6415 TT	767.0
2	圣邦	SGMicro	300661 CH	457.7
3	艾为电子	Awinic	688798 CH	300.0
4	思瑞浦	3 Peak	688536 CH	256.1
5	纳芯微	Novosense	688052 CH	237.6
6	杰华特	Joulwatt	688141 CH	201.6
7	上海贝岭	Belling	600171 CH	190.8
8	南芯	Southchip	688484 CH	186.8
9	韦尔	Will semi	603501 CH	173.0
10	晶丰明源	Bright Power	688368 CH	147.2
11	芯朋微	Chipown	688508 CH	103.3
12	英集芯	Injoinic	688209 CH	90.9
13	力芯微	Wuxi ETEK	688601 CH	90.8
14	富满微	Fine Made	300671 CH	88.9
15	帝奥	Dioo	688381 CH	79.3
16	必易微	Kiwi instruments	688045 CH	75.5
17	赛微	Cellwise	688325 CH	34.0

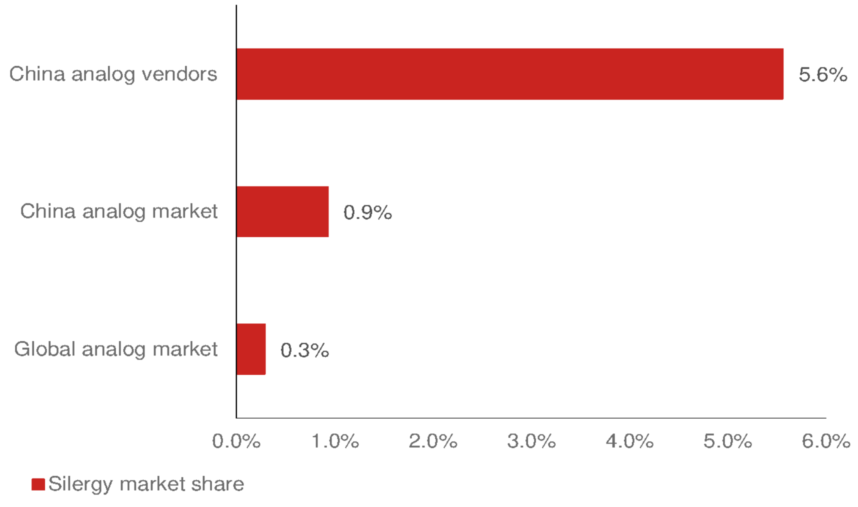
Source: Company data, Nomura research

Fig. 133: China analog market size vs China local analog vendors' sales and Silergy's local sales in 2023F



Source: Company data, Nomura estimates

Fig. 134: Silergy: market share in various markets in 2023F



Source: Nomura estimates

Fig. 135: Silergy: Product overview – 1

Applications	End-products	Products	Sales	GPM	Growth opportunity
Consumer	TWS, STB, Tablet, Smart watch, smart speaker, power bank	PMIC Signal chain	35-40%	40-50%	+
Industrial	Energy storage, building automation, factory automation, motor/ power control	PMIC Signal chain	30-35%	50-60%	+++
Computing	NB/PC, server, notebook docking station, SSD	PMIC Signal chain	10-15%	35-40%	Server ++++/ others +
Comm.	Smartphone, base station, optical module, enterprise swtich, small cell	PMIC Signal chain	8-10%	40-45%	+++
Auto	Telematics, lighting, infotainment, body control, ADAS sensor fusion	PMIC Signal chain	~10%	50-55%	+++++

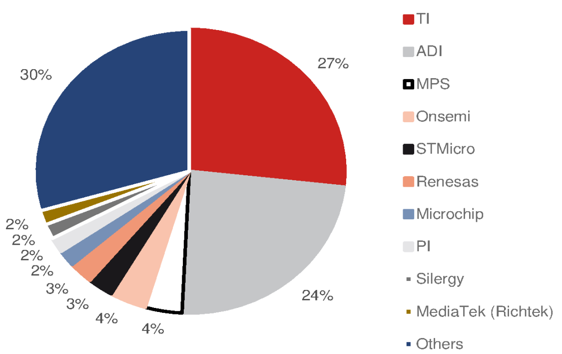
Source: Company data, Nomura estimates

Fig. 136: Silergy: Product overview – 2

Applications	End-products	Products	Great China competitors	Global competitors
Consumer	TWS, STB, Tablet, Smart watch, smart	PMIC Signal chain	SG Micro, Bright power, Silan, 3Peak SG Micro, GMT	PI, Sanken, MPS, TI ADI, TI, Cirrus logic
Industrial	Energy storage, building automation,	PMIC Signal chain	SG Micro, 3Peak, Bright power, Silan SG Micro, 3Peak, Novosense	ADI, TI, ON, MPS, PI, Microchip ADI, TI, Renesas
Computing	NB/PC, server, notebook docking	PMIC Signal chain	Mediatek, GMT, Anpec, Will semi, 3Peak SG Micro, 3Peak, GMT, Nuvoton	TI, MPS ADI, TI, MPS
Comm.	Smartphone, base station, optical	PMIC Signal chain	Awinic, Mediatek, SG Micro, Will semi 3Peak, SG Micro, Awinic	TI, ADI, MPS, ON, STMicro ADI, TI, Renesas
Auto	Telematics, lighting, infotainment, body	PMIC Signal chain	SG Micro, 3Peak SG Micro, Novosense	TI, ADI, infineon, Rohm, Sanken, ON, MPS ADI, TI, Renesas

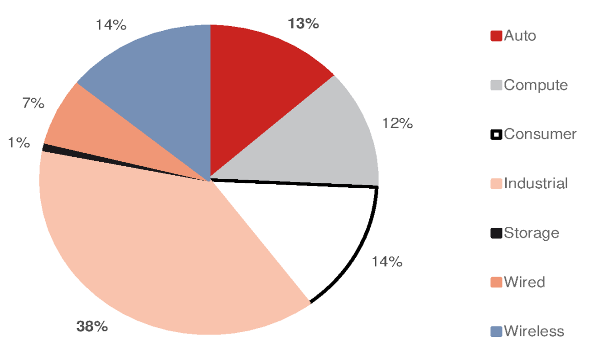
Source: Company data, Nomura estimates

Fig. 137: General purpose analog semi market share by vendors' sales in 2022



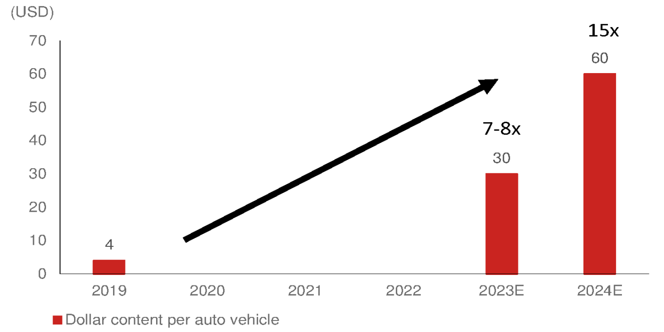
Source: Gartner, Nomura research

Fig. 138: General purpose analog semi sales mix by applications in 2022



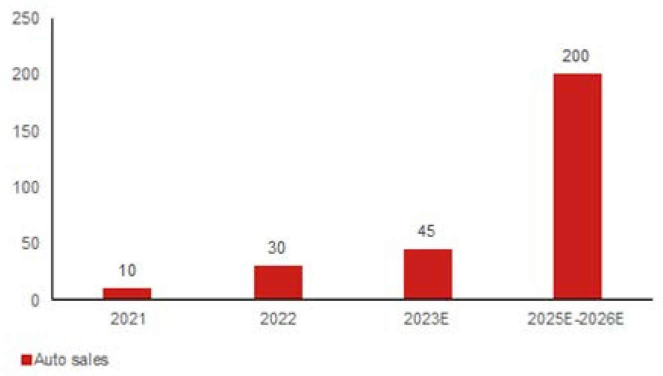
Source: Gartner, Nomura research

Fig. 139: Silergy: Auto products' dollar content per vehicle



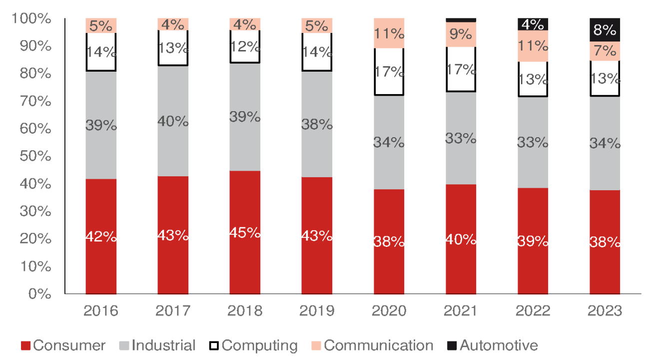
Source: Company data, Nomura research

Fig. 140: Silergy: Auto sales and management target



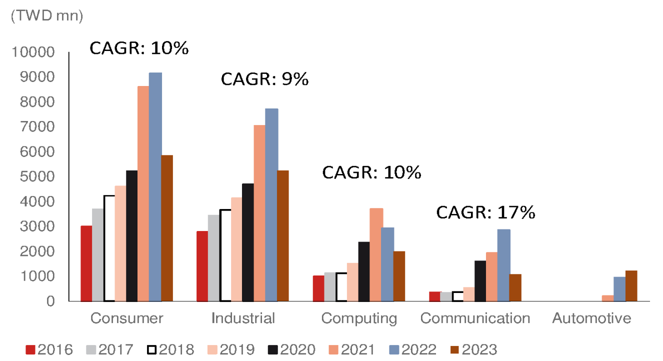
Source: Company data, Nomura research

Fig. 141: Silergy: Product mix (%)



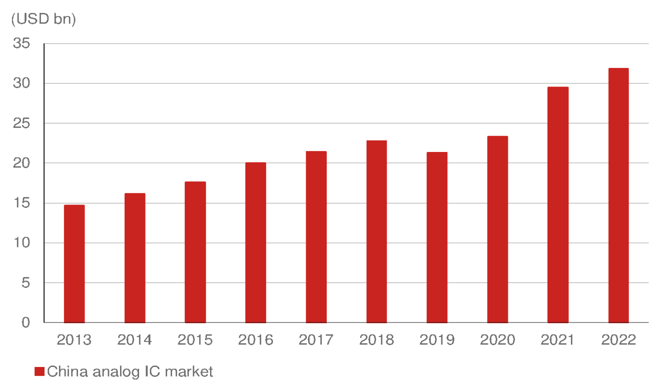
Source: Company data, Nomura research

Fig. 142: Silergy: Application sales CAGR, 2016-2023



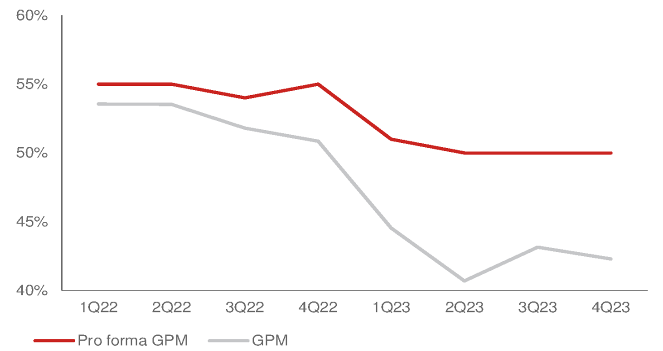
Source: Company data, Nomura research

Fig. 143: China analog market size



Source: WSTS, Nomura research

Fig. 144: Silergy: Reported GPM and pro-forma GPM



Source: Company data, Nomura research

Initiating coverage at Buy with TP of TWD450

We initiate coverage of Silergy, a leading PMIC vendor in the global analog market, with a Buy rating and a TP of TWD450 (based on 35x 2025F EPS of TWD12.9). Our target P/E of 35x is at the mid-to-low end of its P/E consensus trading range of 20-60x in 2019-2021 when earnings resumed growth after consecutive earnings declines y-y in 2017-2018. We think Silergy is well positioned to benefit from China analog market share gains through its healthy inventory level, as well as the growing auto analog market along with the significant dollar content growth in 2023-24F.

Silergy is one of the largest China analog vendors with a market share of 1% (declining from 1.4% in 2022F), according to our estimates. Silergy is a virtual IDM company and has its own manufacturing capabilities in different regions and foundry partners, and next generation-4 (Gen-4) platform products made by 300-mm fabs to better compete with other local vendors and global peers. We forecast Silergy's earnings will turn to growth of 300% in 2024F and 105% in 2025F, driven by: **1)** market share gains in the local China analog market; **2)** the easing of an inventory overhang, resulting in less pricing pressure on Silergy; and **3)** significant content growth in auto applications; and we believe the auto business could have long-term growth opportunities following consumer and smartphone demand recoveries in the near future.

Silergy products

Silergy is one of the few fabless companies that can produce high-voltage, high-current IC in small packages, and specializes in IC design and system design technologies. The company has a virtual IDM business, which is also equipped with wafer manufacturing and packaging technology, enabling Silergy to provide customers with highly integrated products of high performance and quality, according to management.

Silergy's end-user applications for its products can be divided into five major categories: consumer products, industrial products, computing products, communication products, and automotive products.

Fig. 145: Silergy: Product overview - 1

Applications	End-products	Products	Sales	GPM	Growth opportunity
Consumer	TWS, STB, Tablet, Smart watch, smart speaker, power bank	PMIC Signal chain	35-40%	40-50%	+
Industrial	Energy storage, building automation, factory automation, motor/ power control	PMIC Signal chain	30-35%	50-60%	+++
Computing	NB/PC, server, notebook docking station, SSD	PMIC Signal chain	10-15%	35-40%	Server ++++/ others +
Comm.	Smartphone, base station, optical module, enterprise swtich, small cell	PMIC Signal chain	8-10%	40-45%	+++
Auto	Telematics, lighting, infotainment, body control, ADAS sensor fusion	PMIC Signal chain	~10%	50-55%	+++++

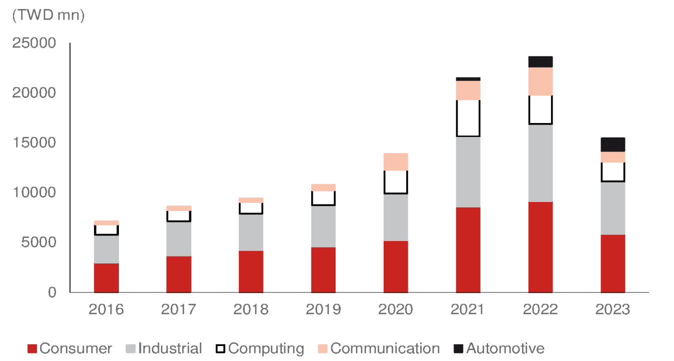
Source: Company data, Nomura estimates

Fig. 146: Silergy: Product overview - 2

Applications	End-products	Products	Great China competitors	Global competitors
Consumer	TWS, STB, Tablet, Smart watch, smart	PMIC Signal chain	SG Micro, Bright power, Silan, 3Peak SG Micro, GMT	PI, Sanken, MPS, TI ADI, TI, Cirrus logic
Industrial	Energy storage, building automation,	PMIC Signal chain	SG Micro, 3Peak, Bright power, Silan SG Micro, 3Peak, Novosense	ADI, TI, ON, MPS, PI, Microchip ADI, TI, Renesas
Computing	NB/PC, server, notebook docking	PMIC Signal chain	Mediatek, GMT, Anpec, Will semi, 3Peak SG Micro, 3Peak, GMT, Nuvoton	TI, MPS ADI, TI, MPS
Comm.	Smartphone, base station, optical	PMIC Signal chain	Awinic, Mediatek, SG Micro, Will semi 3Peak, SG Micro, Awinic	TI, ADI, MPS, ON, STMicro ADI, TI, Renesas
Auto	Telematics, lighting, infotainment, body	PMIC Signal chain	SG Micro, 3Peak SG Micro, Novosense	TI, ADI, infineon, Rohm, Sanken, ON, MPS ADI, TI, Renesas

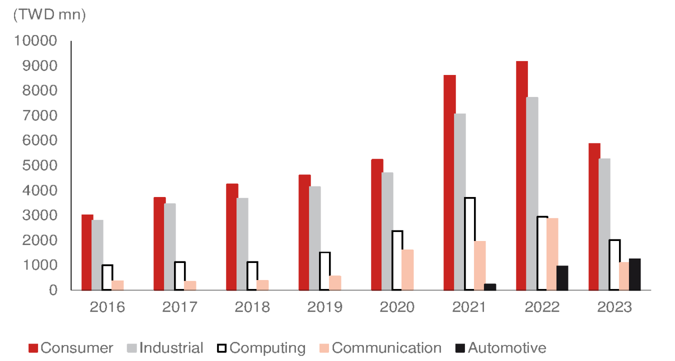
Source: Company data, Nomura estimates

Fig. 147: Silergy: Annual sales by application



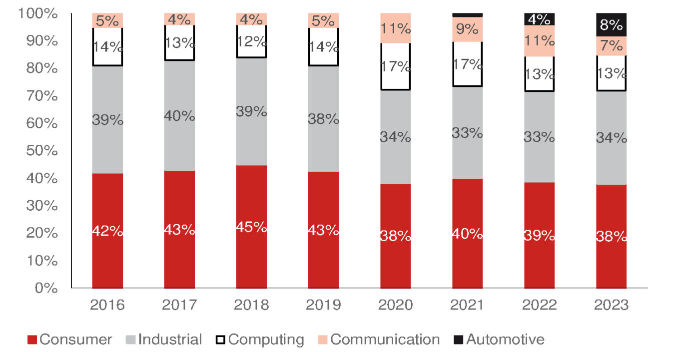
Source: Company data, Nomura research

Fig. 148: Silergy: Application sales



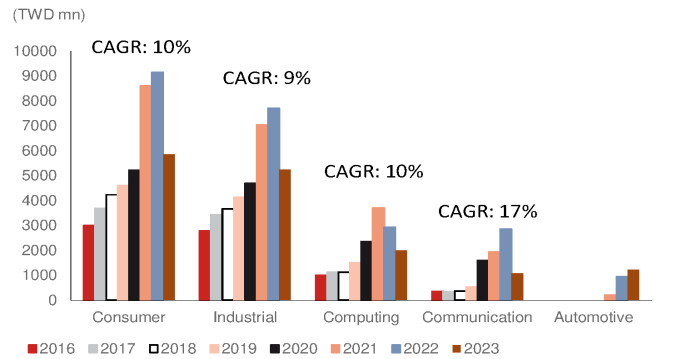
Source: Company data, Nomura research

Fig. 149: Silergy: Product mix (%)



Source: Company data, Nomura research

Fig. 150: Silergy: Application sales CAGR from 2016 to 2023



Source: Company data, Nomura research

Silergy's key Gen-4 product platform in 2024

New Gen-4 product platform to drive new product and better profitability

Silergy is a virtual IDM that uses its own manufacturing technology with its foundry partner to produce products, without high capex investment and depreciation cost. The self-owned manufacturing technology can cater to customers' unique requirements with tailor-made solutions, according to management. Currently, Silergy uses Gen-3, which accounted for 60% of 2023 wafer consumption, according to management estimates, as its major product platform. Silergy expects Gen-3 to account for 70% of 4Q23 sales, with Gen-4 starting to contribute sales from late-2024. Gen-4 is for larger-sized and higher-price products, mainly for HPC and auto.

Why is the Gen-4 process important for Silergy?

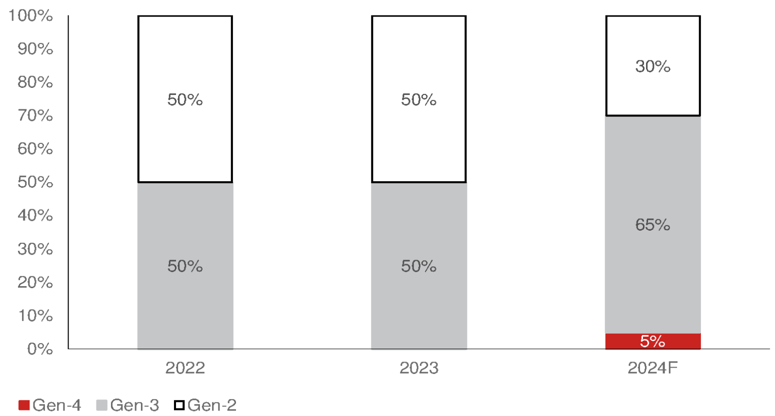
We think Silergy's Gen-4 has multiple advantages. **1) Cost:** 300-mm wafer can significantly reduce average cost in each die by a higher utilized area. Compared to other China analog peers mostly using a 200-mm wafer process, we think the Gen-4 process gives SIlergy a key advantage in providing competitive pricing for the company in the local China market; **2) More advanced process:** although traditional analog IC does not require an advanced node, we think that when Silergy develops products for higher-level computing applications or high-speed transmission, these products may have more digital functions along with analog IC; the IC may need to be manufactured under the 90nm process, and the process is usually made in 300-mm wafer fab, which is a Gen-4 product platform.

300-mm fab beneficiary in Gen-4

The major difference between Gen-4 and Gen-3 is using 300mm fab capacity (*Fig. 152*). In Gen-2, Silergy only uses 200mm and 0.35um processes, and Silergy started using Gen-3 in 2019-2020, with more advanced processes in 0.18um to 0.11um, but still mostly in 200mm. The first Gen-3 product applications are more focused on PC or consumer products, and further expands to industrial, communication and auto applications. From our discussions with Silergy and on TI's 300mm beneficiary commentary (*Fig. 153*), 300-mm can not only use larger wafer area to lower the front-end process cost, but also lower back-end cost through bumping or testing, thus improving profitability. By Silergy's estimate, the technology migration from Gen-3 to Gen-4 could reduce cost by 10%, thus lifting GPM by ~5pp, and it has better integration with logic IC, becoming more intelligent. Silergy has seen design wins with over 200 new clients, predominantly from auto and industrial applications. In foundry partners, we believe Silergy's Gen-4 Platform is working closely with key foundry partners in Greater China region, such as PSMC (6770 TT, Not rated) in Taiwan, and some fabs in China.

Fig. 151: Silergy: Wafer product platform mix

Gen-4 to start contributing to sales in 2024F



Source: Company data, Nomura estimates

Fig. 152: Silergy: Product platform comparison

	Gen-2	Gen-3	Gen-4
Launch year	2014	2018	End-2022
200-mm	O (0.35um)	O (0.11um-0.18um)	X
300-mm	X	X	O (90nm or below)
Detail		Higher precision, process migration, cost lowered 10%, expected Gen-3 product to accounts	Launched in end-22, strength in performance, current density, precision, and it increase the integration of MCU and DSP, expects sales mix increase in 2024E.
Foundry	UMC (Hejian)		PSMC and Others

Source: Company data, Nomura research

Fig. 153: TI estimates on 300-mm beneficiary

300-mm can reduce chip cost through better area utilization

TI		Build on 200-mm wafer	Build on 300-mm wafer
Sales price of example part		1	1
Cost of goods:	**Chip cost**	**0.2**	**0.12**
	Asembly, test, others	0.2	0.2
	Total	0.4	0.32
GPM (%)		**60%**	**68%**

Source: Company data, Nomura research

Auto business (8% of 2023 sales)

Auto sales is the major growth driver for Silergy

Silergy started developing auto analog products in 2019. Due to the long design cycle, long safety and verification process of auto semi products, Silergy's auto business began to contribute to sales in 2021 (though it has been co-working with auto customers since 2015-16), and recorded sales of around USD30mn in 2022. Silergy can provide various auto analog products, including PMIC and signal chain IC (mixed-signal IC), and both recorded mass production in 2022 for multiple tier-1 customers.

Silergy's auto products cover most of the major applications in vehicles, including infotainment, telematics (T-Box), ADAS sensor fusion, lighting, and heating, ventilation and air conditioning (HVAC) (*Fig. 156*). We think PMIC is still the major auto analog product tor Silergy.

Strong growth with significant dollar content increase in 2024

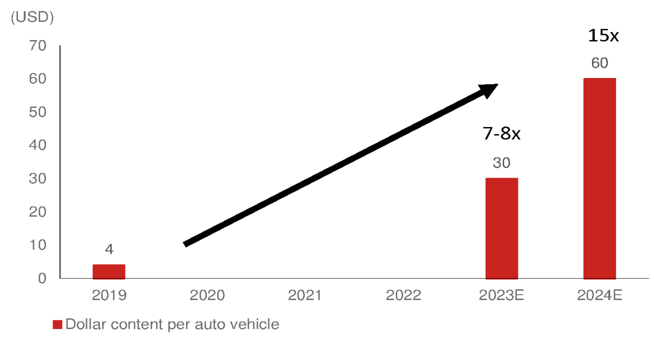
Silergy recorded auto sales of USD10mn/USD30mn in 2021/2022, and USD40-50mn in 2023, which suggests strong growth over the past three years in terms of dollar content growth and mass production of new products. We believe Silergy's auto sales can sustain growth higher than the corporate average.

Doubled dollar content in 2024 to boost future sales growth

According to management, Silergy had auto dollar content of USD3-5 in 2019, which increased to USD30 in 2023, and it expects to double this to USD60 in 2024 due to new product launches. Currently, Silergy's management estimates auto/EV average dollar content of analog products is USD200/USD600, due to increased analog IC consumption and ASP in motor, electricity controls, and battery. Silergy expects current global auto analog TAM of USD20bn, which is continuing to grow owing to likely higher EV penetration rates over the next three years.

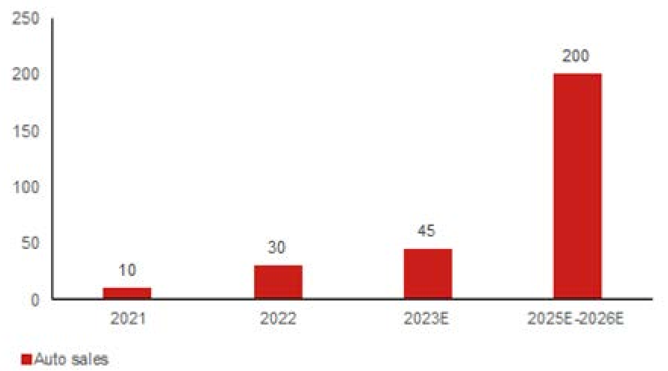
Based on the current auto product development schedule, Silergy expects its addressable market to be USD1-2bn between 2023 and 2026, and targets to record auto sales of USD200mn in 2026, which implies 10-20% market share. According to management, it was targeting to recorded auto sales of USD200mn within 2025E-26E, while management thinks the goal looks unachievable in 2025, and it will try to achieve the USD200mn goal in 2026.

Fig. 154: Silergy: Auto products' dollar content per vehicle



Source: Company data, Nomura research

Fig. 155: Silergy: Auto sales and management target



Source: Company data, Nomura research

Fig. 156: Silergy: Auto product portfolio

Applications	Analog IC	Products
ADAS sensor fusion ECU	PMIC	CAN ESD HV Buck/ LV Buck Buck-boost/ Pre-boost HV LDO
	Signal chain	n.a.
Telematics (T-BOX)	PMIC	LDO (LDO + charger/ 5V LDO) Gas gauge voltage detector Antenna detect/ CAN LIN ESD TVS DC-DC (Buck, 12V, 3.3/5V Buck/ LDO)
	Signal chain	eCall (audio amp.) Current sensing/ coulomb counter
Infotainment	PMIC	Linear charger LDO/ HV Buck ESD
	Signal chain	Audio amp./ MIC OP-Amp.
Lighting	PMIC	LDO ESD Multi-ch LED driver
	Signal chain	n.a.
HVAC	PMIC	LDO CAN/ LIN ESD High side switch Brushless motor drivers, Buck converters
	Signal chain	n.a.

Source: Company data, Nomura research

Auto customer exposure: ~50% came from overseas customers in 2023

Silergy's auto customers include global major tier-1 OEMs in the US, Japan and the EU. In 2022, Silergy received the "best semiconductor supplier of the year" award from Visteon (VC US, Not rated) (*link*), a recognition from a major auto tier-1 company in the market. According to management, Silergy's China auto customer sales contribution was around 50/50 in 1H23, and China auto customers' sales grew slighter more than that of overseas customers in 3Q23. Silergy expects overseas auto customers' sales to outgrow that of China local auto customers due to design-in new product mass production schedule in 2024E.

China EV customers are key long-term growth driver for Silergy

China EV OEMs have been growing quickly in recent years, and the entry barriers for component makers have also been reduced. The average verification cycle has been shortened to about 1.5 years from 2-year-plus. Also, China OEMs are more actively sourcing non-US vendors' components, which has also benefited Silergy. Overall, compared with other China local vendors, Silergy's auto business engages more with overseas customers, which diversifies its customer base to offset customer portfolio risks, in our view.

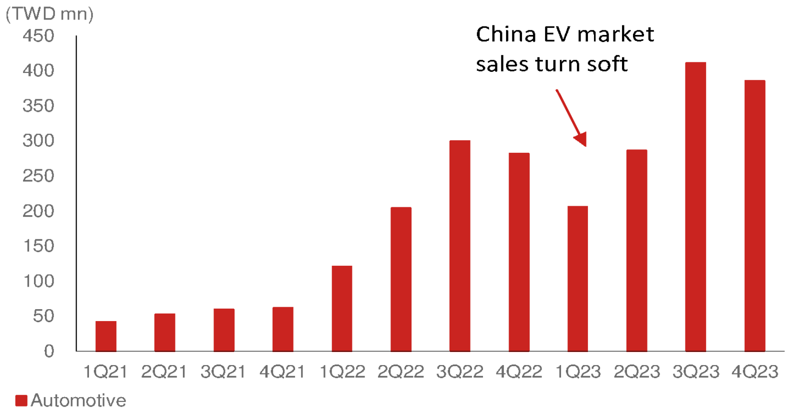
Limited competition with local analog vendors in China market

According to Silergy, the major competitors in the auto analog product market are mostly global IDMs, such as ADI (ADI US, Not rated), and TI (TXN US, Not rated). Based on our industry survey, other major auto analog players such as 3peak (688536 CH, Buy) and Novosense (688052 CH, Neutral) are more focused on signal chain or sensor products, and do not compete with Silergy in the auto analog market.

Sales slowed down in late-2022, but rebounded in 2Q23

Silergy's auto business started to contribute to sales in 2021, and continued to grow in 2022-2023. In 4Q22-1Q23, Silergy's auto business turned soft because of China EV market inventory adjustments due to an increase in ICE vehicle sales in 1H23. However, auto application sales recovered in 2Q-3Q23 when EV end-demand returned to a stable state.

Fig. 157: Silergy: Quarterly auto business sales



Source: Company data, Nomura research

Servers (1-3% of total sales): A key sales growth driver

Silergy expects HPC to be one of its sales growth drivers over the next few years in its new Gen-4 platform. Currently, Silergy provides power IC for communication, data transmitting, and storage applications in servers for the industrial and computing segments. A traditional server's power voltage is relatively lower than that of an AI server. AI server design requires very high power for multi-core and multi-processor designs that support cloud services or datacenters, like the power voltage is using 48V, instead of 12V in traditional server. Thus, Silergy integrates more components to sustain high current/ high voltage of power with MOSFET and drivers into a Vcore product, which give power to CPU core or GPU core in computing device. Based on Silergy's announcement, it has customers from China, Europe and US regions, and sales exposure to China and non-China is around 50/50.

Significant server dollar content growth by 4x in Gen-4 platform

In a traditional server product, Silergy thinks the analog product dollar content per device is around USD50-100, and currently Silergy has USD8-9 dollar content by current Gen-3 products portfolio, while Silergy expects the USD50-100 in each server motherboard will continue to grow due to AI server upgrades. The server sales mix accounted for 3-5% total 2020 sales, 10% less than 2021 sales, by Silergy estimates.

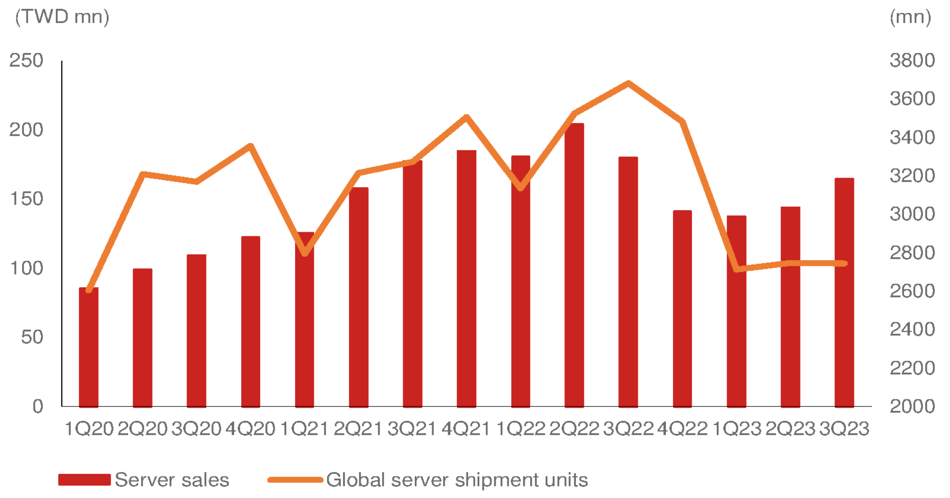
Due to strong customer design in, Silergy expects its dollar content in data center can significantly increase from Gen-3 of **USD8-9** to **USD30-40** in Gen-4, which has higher ASP, higher integration, and better product performance.

Fig. 158: Silergy: server product portfolio

Applications	Analog IC	Products
Server	PMIC	DC/DC POL
		E-Fuse
		Power Stage DrMOS
		USB Power Switch
		ESD protection
		VDD regulator
		VTT regulator
		LDO (under development)
	Signal Chain	Current sensing

Source: Company data, Nomura research

Fig. 159: Silergy server sales vs global server shipments



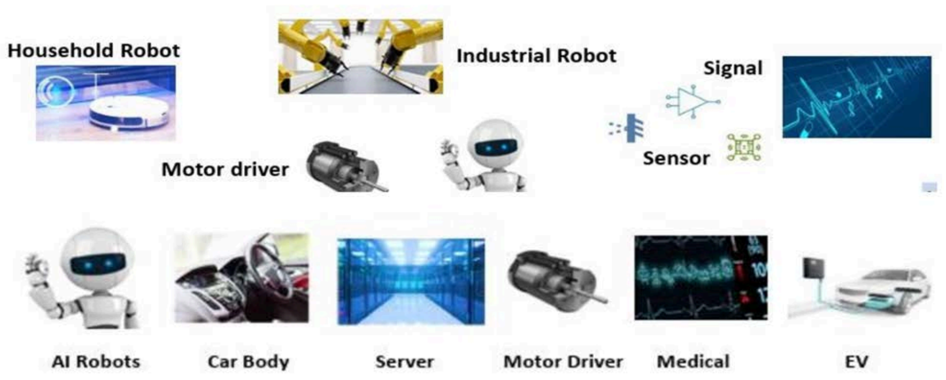
Source: IDC, Nomura estimates

Industrial business (34% of 2023 sales)

Silergy's industrial business has two major categories: **1)** traditional industrial, and **2)** datacenter related, which cover various industrial end-markets.

- **Traditional industrial:** products include smart meters, smart lighting, video monitors, mobile payment end devices, medical and factory testing equipment, mid-high power LED, and SSD.
- **Datacenter related:** server motherboards and SSD products used in datacenters.

Fig. 160: Silergy: Industrial application products

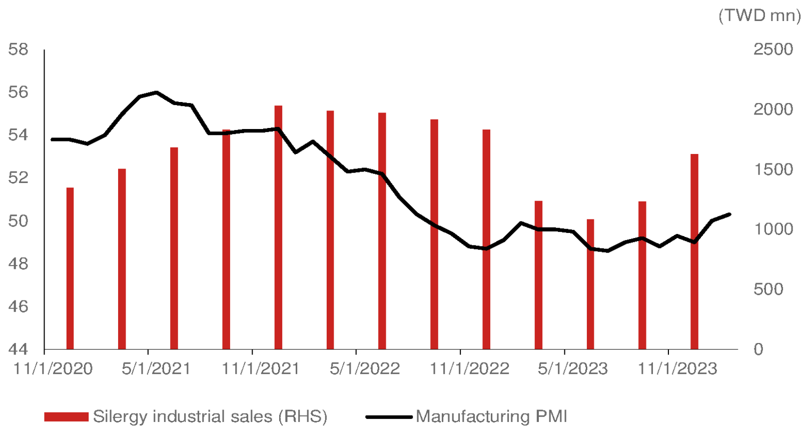


Source: Company data, Nomura research

Traditional industrial applications

Silergy's traditional industrial business has various end-applications, e.g., energy storage, electric power, motor control, and automation in factories and buildings. The business is highly correlated with global economy and industrial expenditure. Thus, we think industrial sales will rebound as the global economy recovers.

Fig. 161: Silergy industrial sales vs global manufacturing PMI



Source: Bloomberg Financial LP., Company data, Nomura research

Fig. 162: Silergy: Industrial product portfolio

Major Applications		Analog IC	Major products
Energy storage system	Battery Monitor Unit	PMIC	ADC, HV buck, ESD TVS, fan controller, battery monitor AFE
		Signal chain	Hall sensor, Amplifier
	PCS	PMIC	Gate driver, reset, ESD, DC/DC LDO, E-fuse
		Signal chain	V/I sense, temp. sense
Power automation	smart meter, solar, inverter..etc	PMIC	AC/DC, DC/DC, reset, LDO, gate driver,
		Signal chain	Current sensor, voltage sensing
Motor/ Power control	Servo	PMIC	AC/DC LDO, isolator
		Signal chain	n.a.
	Inverter	PMIC	AC/DC LDO, isolator
		Signal chain	n.a.
	EV charger	PMIC	AC/DC, DC/DC, LDO, isolator, low-side driver
		Signal chain	Temp. sensor
	UPS	PMIC	AC/DC, DC/DC, LDO, isolator
		Signal chain	Temp. sensor, Hall sensor
Factory automation	Human-machine interface	PMIC	LDO, load switch, LED driver, HV/LV buck, LCD bias
		Signal chain	Optical sensor
	Industrial computer	PMIC	HV/ MV/ LV buck, LDO, LED driver, LCD bias, load switch
		Signal chain	n.a.
	Distribution control system	PMIC	DC/DC, LDO, E-fuse, reset, half-bridge driver
		Signal chain	Temp. sensor
Building automation	Video Matrix	PMIC	LDO, DDR LDO, DC-DC module, ESD
		Signal chain	
	IP camera	PMIC	LED driver, LDO, HV/ LV Buck
		Signal chain	POE PA(interface)
	Elevator control system	PMIC	LCD power, LDO, Buck
		Signal chain	n.a.

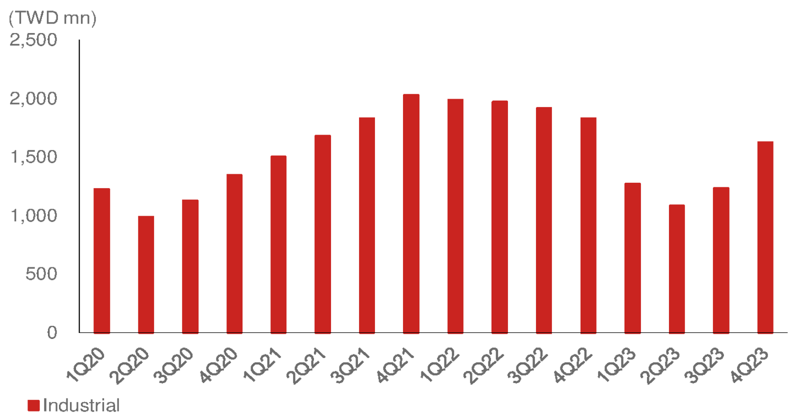
Source: Company data, Nomura research

China sales turned soft in 2H22, non-China dropped in 1H23

Silergy's industrial sales portfolio is more impacted by EU/US customers than China customers, while server sales were around 50:50 for China /non-China customers in 2023.

Silergy's 2H22 industrial sales turned soft due to demand weakness from China, while non-China sales remained stable. This impacted Silergy's China industrial and server market sales in 2H22 (*Fig. 163*). Silergy attributed this to large China internet companies lowering server investments to reserve more cash in this high-interest-rate environment. On the other hand, demand from non-China region customers declined owing to an inventory overhang that significantly impacted Silergy's 1H23 sales despite the recovery in China industrial demand, according to management.

Fig. 163: Silergy: Quarterly industrial sales



Source: Company data, Nomura research

Consumer (38% of 2023 sales)

Consumer applications was Silergy's main market in its early stage of development. The main reasons are that the technology entry barriers are low, the product verification period is short, and the major customers are closer to Asia. Besides self-development, Silergy acquired a lighting business from NXP (NXPI US, Not rated) in 2016.

Fig. 164: Silergy: consumer application products

Source: Company data, Nomura research

Silergy has a strong product portfolio for most key consumers' electronics needs

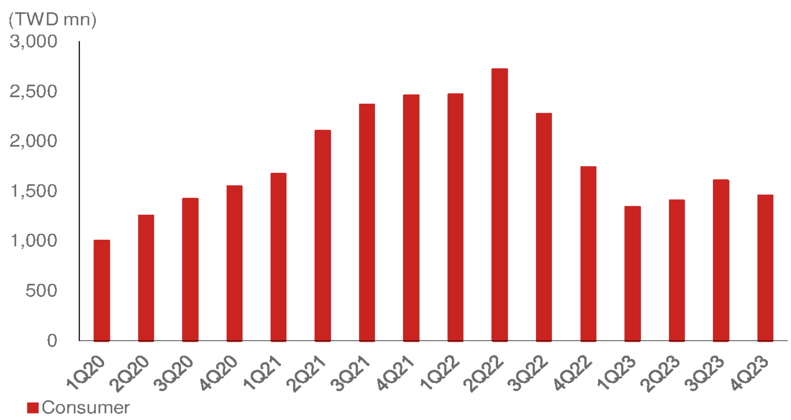
Consumer applications have several end-markets, and most are fragmented. The major applications before 2018 were mobile payment, wearables, and POS machines, while TVs, panels, tablets, and smartphones became top products in recent years (*Fig. 165*). By Silergy's estimate, TV, smartphones, and notebook computers (NB; although smartphone and NB are not categorized as consumer applications) have usually accounted for 20% of total annual sales.

Although consumer applications are not a focus area for Silergy now, we think Silergy is still one of the largest consumer analog vendors in Greater China, and we believe it will continue to grow owing to its broad product portfolio and localization demand from China customers.

Sales impacted by China customers' inventory overhang in 2H22, but sales recovered in 2H23F

We think Silergy's consumer sales are predominantly from China, and the China consumer electronics industry began to adjust inventory in 2022. Thus, Silergy's consumer sales were impacted by customers' inventory overhang in mid-2022 due to overbooking during the previous chip shortage period. However, the weakness in consumer sales improved in 2H23, when inventory returned close to healthy levels after over four quarters of digestion. Silergy management has stated that the consumer product inventory adjustment has begun earlier than for industrial and server applications, resulting in a recovery that has also come earlier than for other applications.

Fig. 165: Silergy: quarterly consumer sales



Source: Company data, Nomura research

Fig. 166: Silergy: consumer product portfolio

Major Applications		Analog IC	Major products
Electronic payment	Point of Sale termainal	PMIC	HV/MV/LV buck, LDO, Load switch, ESD, motor drive, Charger
		Signal chain	n.a.
	Barcode Scanner	PMIC	Load switch, boost, LDO, charger, TVS protection
		Signal chain	n.a.
Consumer Electronics	True Wireless Stereo(TWS)	PMIC	Linear Charger, Boost, LDO, load switch, ESD, buck, OVP
		Signal chain	Proximity sensor
	Set-top-box	PMIC	AC/DC, MV buck, USB protection, ESD
		Signal chain	audio power amplifier, ethernet
	Tablet PC	PMIC	LED driver, power switch, core buck, LV buck, charger, OVP
		Signal chain	Optical sensor, PXS+ALS, audio power amplifier
	Power Bank	PMIC	Charger, OVP, current limit
		Signal chain	n.a.
	Smart watch	PMIC	Charger, LDO, load switch, boost, LV buck
		Signal chain	Optical sensor, ALS
	Smart Speaker	PMIC	LDO, Boost, charger, OVP, ethernet ESD
		Signal chain	audio power amplifier, current sensor
Home appliance	Smart TV	PMIC	AC/DC, HV LV buck, power switch, PD PMIC, TVS, LED driver
		Signal chain	Audio power amplifier, current sense amp.
	Vacuum robot	PMIC	AC/DC charger, HV/MV/LV buck, load switch, laser/ BDC driver
		Signal chain	n.a.
	Air conditioner	PMIC	AC/DC, DC/DC, LDO, BDC driver, motor drive, DCDC converter
		Signal chain	current sensor
	Refrigerator	PMIC	AC/DC, DC/DC, LDO, motor drive, LED driver, LCD bias, reset
		Signal chain	current sensor
	Washing Machine	PMIC	AC/DC, DC/DC, LDO, LED driver, LCD bias, reset, E-fuse
		Signal chain	current sensor
LED lighting	AC/DC lighting	PMIC	Buck-boost, flyback controller
		Signal chain	n.a.
	Intelligent lighting	PMIC	Buck, Buck-boost, Flyback, controller
		Signal chain	
	Stage light	PMIC	Fan driver, controller, LDO, HV buck
		Signal chain	Current sensor
AC-DC power supply	AC/DC general charger	PMIC	Side regulation
		Signal chain	n.a.
	AC/DC fast charger	PMIC	PFC controller, side regulation
		Signal chain	n.a.
	Small appliance adapter	PMIC	Isolated PSR/ SSR, Buck regulator
		Signal chain	n.a.

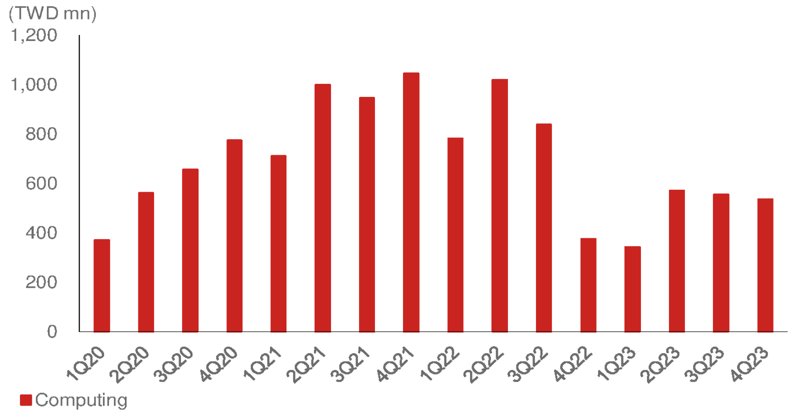
Source: Company data, Nomura research

Computing (13% of 2023 sales)

In the computing business, the major product is NB/PC, while Silergy also provides computing peripheral components in notebook docking stations, and client and enterprise SSD products (*Fig. 170*). Silergy offers complete, high-performance solutions for a variety of functions in NB/PC products, as well as a USB interface protection solution. Silergy's computing sales have been highly correlated with global NB/PC shipments (*Fig. 168*, *Fig. 169*).

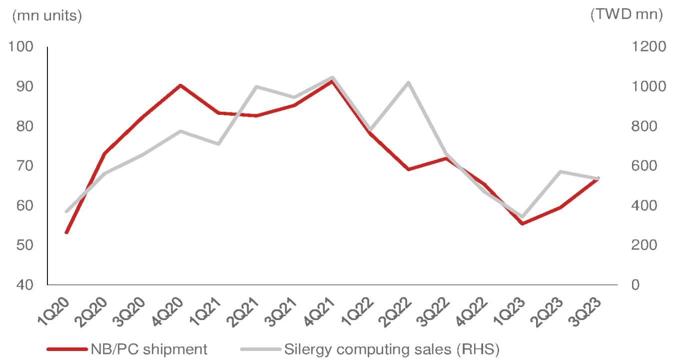
Therefore, when the global NB/PC markets turned weak, Silergy's computing application sales significantly declined in 2H22 due to customers' inventory corrections. However, Silergy thinks customers' inventory has returned to normal levels, and they began to restock in late-2Q23 after four quarters of digestion.

Fig. 167: Silergy: quarterly computing sales



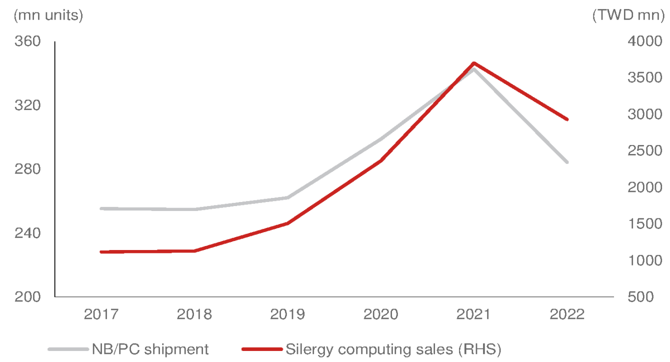
Source: Company data, Nomura research

Fig. 168: Quarterly global NB/PC shipments vs Silergy computing sales



Source: IDC, company data, Nomura research

Fig. 169: Annual global NB/PC shipments vs Silergy computing sales



Source: IDC, company data, Nomura research

Fig. 170: Silergy: computing product portfolio

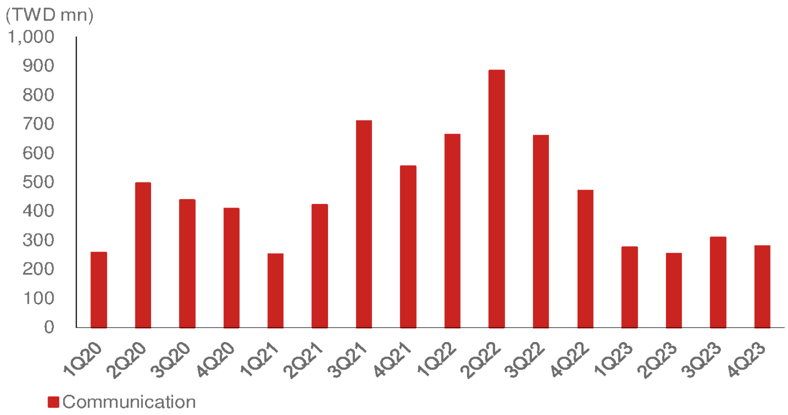
Applications	products	Analog IC	Products
Computing	NB	PMIC	Load switch, LED driver, DDR buck, MV/LV buck, ESD, power switch
		Signal Chain	Current sensor
	PC	PMIC	Adaptor, LDO, MV/LV buck, LED driver, LCD bias, ESD, load switch
		Signal Chain	Audio power amp.
	Client SSD	PMIC	Protection Switch, PMIC
		Signal Chain	n.a.
	Enterprise SSD	PMIC	Protection Switch, MV/LV buck, power loss protection, PMIC
		Signal Chain	n.a.
	Docking station	PMIC	OVP, Boost, LDO, MV/LV buck, load switch, ESD protection, USB PD
		Signal Chain	n.a.

Source: Company data, Nomura research

Communications business (7% of 2023 sales)

Silergy's communication business includes smartphone, base stations, optical modules, enterprise switches, and other networking products (*Fig. 176*). We think smartphones accounted for 50%+ of communication sales in 2023.

Fig. 171: Silergy: Quarterly communication sales



Source: Company data, Nomura research

Dollar content growth owing to the launch of new products drove sales growth in 2020-2022

Silergy started developing communication products in 2019, and sales gradually materialized in 2020. Communication sales experienced significant growth in 2021-2022 due to dollar content growth and the mass production of base stations and networking products' new design-in. The sales growth also highly benefited from Huawei's (unlisted) product shipments, but Silergy stopped shipping to Huawei after Huawei was banned by the US government. According to the company, its dollar content per base station was USD10 before 2020, and it increased to USD20-30 in 2020, thus contributing to strong growth in 2021-2022. Silergy targets a dollar content of USD50 as a long-term goal. On the other hand, Silergy expects its communication products to continue to gain share from global peers due to localization requirements from customers, and expects to outgrow the corporate average given the strong design-in activity in its customers' product road map as well as strong design-in activities with China local customers.

2023 is a year of tough environment for communication

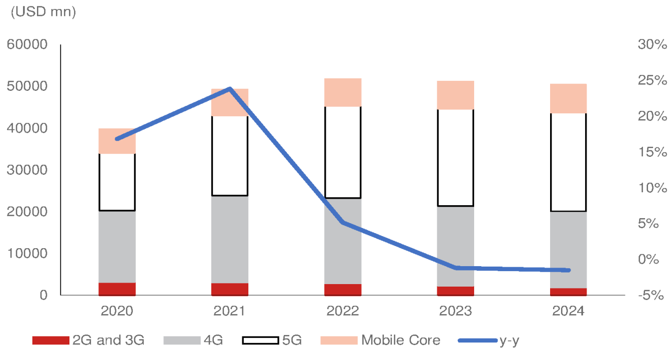
Smartphone: The global smartphone market turned soft in 2022, and the smartphone market bottomed in 1H23 (*report*). We think smartphone sales, which are highly impacted by global smartphone demand, especially for China smartphone brands. accounted for over 50% of communication sales in 2023. Along with the smartphone sell-in recovery of China and some emerging markets in 3Q23, Silergy's communication sales improved in 3Q23, growing 30% q-q.

Communication infrastructure: According to Gartner, major North American CSPs have started to reduce their capex on 5G coverage and China 5G investment with 2mn deployed base stations in 2023, given the limited demand for new sites. 4G remains the major access technology in use in many parts of the world, and 5G deployment is expected to continue unless a new successful application emerges. Therefore, the communication market slowed down in 2022, and Gartner expected a decline in 2023E. The inventory or DOI of leading global base station equipment vendors Ericsson and Nokia is still being adjusted, thus impacting communication components providers' sales in late-2022 and 2023.

Overall, Silergy thinks its communication business has now bottomed, and expects to see demand improvement in the near term owing to healthier inventory levels for customers and better smartphone sell-in.

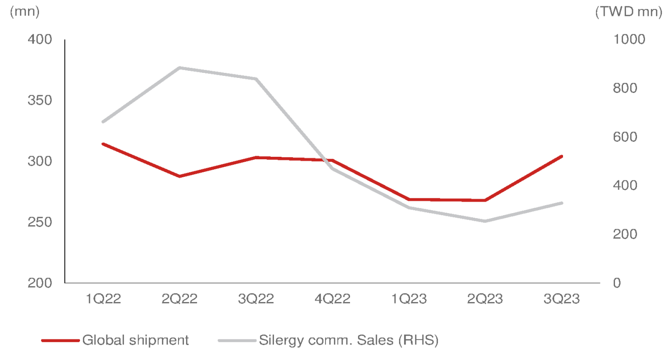
Fig. 172: Global communications service provider market value

Slow market growth in 2023



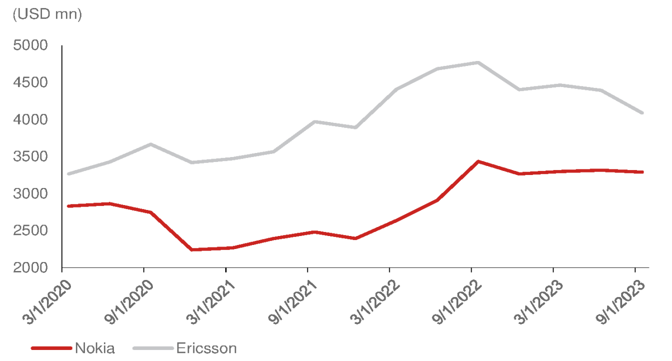
Source: Gartner, Nomura research

Fig. 173: Global smartphone shipments and Silergy's comm. sales



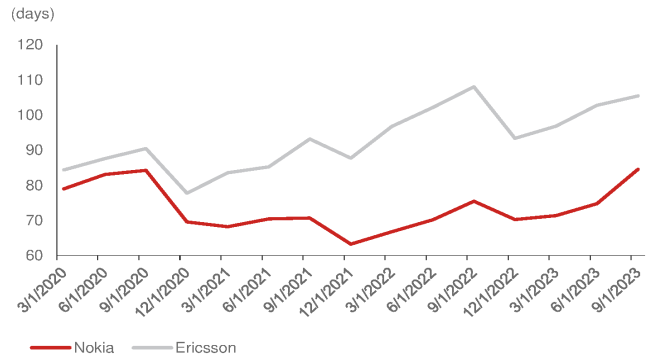
Source: Company data, Nomura estimates

Fig. 174: Inventory of Nokia and Ericsson



Source: Company data, Nomura research

Fig. 175: Inventory days of Nokia and Ericsson



Source: Company data, Nomura research

Fig. 176: Silergy: Communication product portfolio

Applications	Analog IC	Products
Smartphone	PMIC	OVP Boost Load switch LED driver OLED bias LDO LV buck
	Signal chain	Optical sensor, PXS+ALS
Optical module	PMIC	Load Switch (protector) Buck-Boost Negative Charge Pump LV buck/ LV buck module Buck for PIN/ Boost for APD
	Signal chain	n.a.
Enterprise Switch	PMIC	Current/ Power monitor OR-ing controller HV/ MV/ LV buck DDR regulator PWM converter
	Signal chain	n.a.
Communication power	PMIC	Isolator DC/DC LDO AC/DC Digital isolator Isolated driver
	Signal chain	Current sensor Temp sensor Hall sensor
Video conference power	PMIC	DC/DC, DC/DC module LDO Load Switch E-fuse DDR termination ESD Current limit
	Signal chain	Current sensing Audio-power amp.
Small Cell	PMIC	DC/DC, DC/DC module Buck module DDR termination POE PD Ethernet PHY ESD ESD/ Surge protection LDO PWM controller Boost controller
	Signal chain	Current sense amp.
WLAN AP	PMIC	LV/ MV Buck
	Signal chain	Interface controller

Source: Company data, Nomura research

Competition: globally and locally

More local competition in consumer applications

We believe most industrial and NB/PC customers are based abroad, so Silergy is not facing much competition from China vendors for localization requirements, in our view. However, smartphone and TV application markets will be challenging, in our view, due to new suppliers using aggressive pricing in the market. We think Silergy will continue to face local competition in the consumer application market, while management believes the company can provide a better portfolio of products, performance in high-end consumer applications, and a new Gen-4 technology platform to protect its profitability. Thus, management believes the impact of local competition should not be very significant

Better product portfolio through high RD expense

Analog products is a fragmented market, and customers might be better served with total solutions. Thus, we believe a company's product portfolio is one of the key factors to increase share in the analog market. Silergy's number of products increased to 5,000+ in 2022 from 2,000+ in 2019. Silergy has spent the most in R&D in the China analog industry (*Fig. 180*), and it has continued to raise its R&D expenses to speed up product development (*Fig. 178*, *Fig. 179*).

Cost structure through Gen-4 in 300-mm fab

Fabless companies bear more foundry/OSAT costs than IDM vendors, so cost structure is one of the key considerations for fabless to compete with IDM. We think that as long as Silergy adopts more products using the Gen-4 platform, this could make up for its weakness on cost structure when competing with global peers and other fabless.

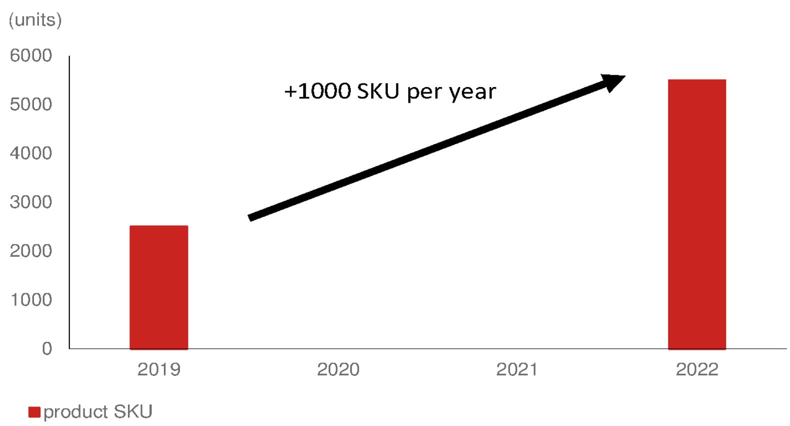
Own manufacturing technology through Virtual IDM model

Silergy's virtual IDM model is to develop its own manufacturing process and save repeated development cost and time in foundry partners. As well, virtual IDM can reduce the risk of competitor plagiarism. Also, analog IC may require special process technology for co-work in the layout design and manufacturing process to improve performance and reduce costs. Thus, we think having its own manufacturing technology is a key strength that should allow the company to grow further and compete not only in China against its local peers, but also against global vendors.

Global partnerships in China and non-China

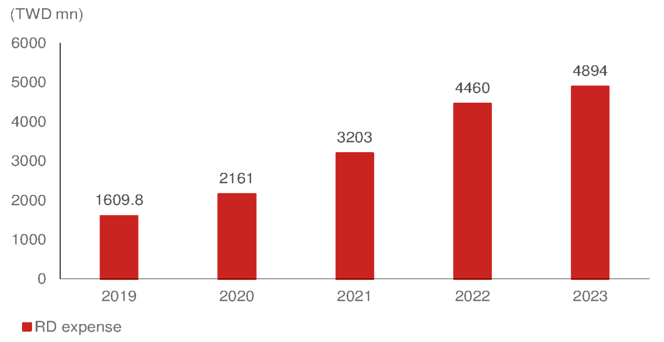
Silergy's major foundry partners are UMC (2303 TT, Buy) (HeJiang fabs) in China. In order to diversify its manufacturing sites to cater to customer requirements or to improve supply chain stability, Silergy started working with its Taiwan foundry partner, such as new Gen-4 platform with PSMC [6770 TT, Not rated], and is also looking for other regional partners, according to management. Therefore, we think Silergy has better geographic supply chain partners than its China peers.

Fig. 177: Silergy: number of products



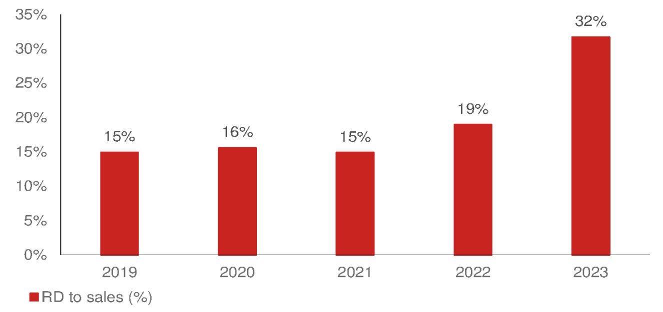
Source: Company data, Nomura research

Fig. 178: Silergy: RD expense



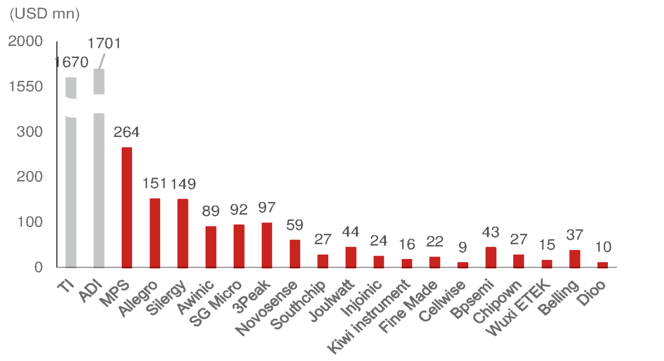
Source: Company data, Nomura research

Fig. 179: Silergy: RD expense to sales (%)



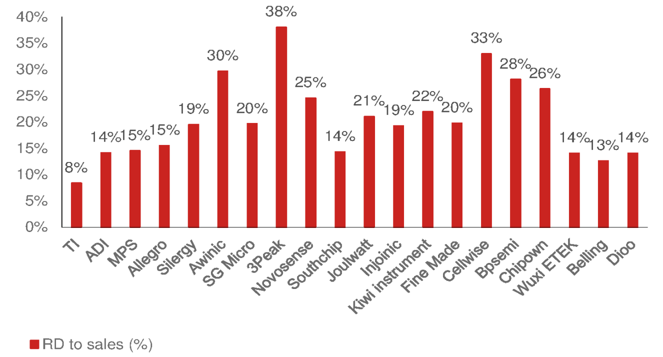
Source: Company data, Nomura research

Fig. 180: Global and China analog IC companies' RD expense in 2022



Source: Company data, Nomura research

Fig. 181: Global and China analog companies' RD expense to sales (%)



Source: Company data, Nomura research

Major customers and suppliers

Silergy is a fabless company, so its major cost is mostly related to foundry/OSAT. Over the past five years, its largest supplier (which we believe was the HeJian fab under UMC [2303 TT, Buy]), accounted for 56% of total sales, and we believe the percentage has gradually decreased each year due to the company's strategy to diversify its supply chain. Besides the major foundry partners in China, Silergy works with TSMC (2330 TT, Buy), Vanguard (5347 TT, Neutral), SilTerra (unlisted), and we think PSMC (6770 TT, Not rated) will become a major foundry partner in Gen-4 platform from 2024. On the other hand, Silergy usually sells products through distributors, and thus its major customers are channel partners or distributors, in which distributors' sales account for 70% of total sales; while Silergy still has direct customers, they are not significant sales contributors. According to Silergy, the largest customer J is a distributor partner and accounted for 16% of 2022 sales, and also works with direct customers such as Samsung [005930 KS, Buy] for SSD and TV products.

Fig. 182: Silergy: Major suppliers

	2018		2019		2020		2021		2022	
(TWD '000)	Amount	(%)	Amount	(%)	Amount	(%)	Amount	(%)	Amount	(%)
Supplier R	2,037,598	71%	2,526,756	76%	2,997,690	71%	3,850,168	61%	5,092,634	56%
Supplier T							1,180,148	19%	1,831,533	20%
Supplier L									916,847	10%
Others	826,425	29%	801,329	24%	1,230,902	29%	1,311,867	21%	1,256,830	14%
Total	2,864,023	100%	3,328,085	100%	4,228,592	100%	6,342,183	100%	9,097,844	100%

Source: Company data, Nomura research

Fig. 183: Silergy: Major customers

(TWD '000)	2018 Amount	2018 (%)	2019 Amount	2019 (%)	2020 Amount	2020 (%)	2021 Amount	2021 (%)	2022 Amount	2022 (%)
Customer J	1,264,774	13%	1,596,376	15%	1,906,993	14%	2,403,903	11%	3,788,580	16%
Customer R			1,090,125	10%	1,402,482	10%				
Others	7,194,488	87%	8,091,280	75%	10,556,970	76%	19,102,163	89%	19,722,506	84%
Total	8,459,262	100%	10,777,781	100%	13,866,445	100%	21,506,066	100%	23,511,086	100%

Source: Company data, Nomura research

Cost and ASP have moved along with semi cycle

Silergy's cost and ASP declined in 2018-2019 when the semi industry was in a downcycle, with both bottoming in 2020. They then started to see a significant rebound due to the IC shortage period that followed. We think both cost and ASP likely declined in 2023F because of the semi downcycle, which we expect will continue to weigh on Silergy in some low-end or consumer application products in 1H24F. Cost and ASP have followed similar trends over the past five years (*Fig. 186*). We think Silergy suffered more from the ASP decline in 2023 as foundry price did not decline as significantly, and resulting a y-y decline in GPM in 2023. However, Silergy expects that further ASP declines will not continue to significantly impact GPM due to further reduced foundry costs.

Fig. 184: Silergy: Production unit and value amount

(TWD '000; '000 units)	2018 Amount	2018 Value	2019 Amount	2019 Value	2020 Amount	2020 Value	2021 Amount	2021 Value	2022 Amount	2022 Value
	3,224,657	5,022,442	3,878,492	5,583,280	5,434,201	7,381,730	6,827,452	10,123,607	5,133,849	11,979,398
Cost per unit		1.56		1.44		1.36		1.48		2.33
y-y				-8%		-6%		9%		57%

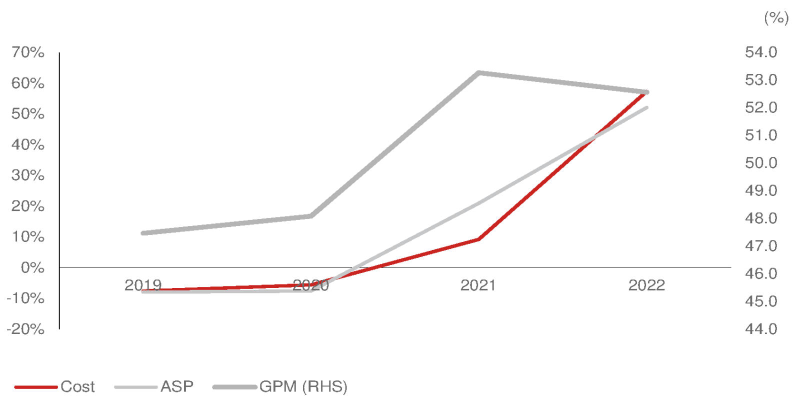
Source: Company data, Nomura research

Fig. 185: Silergy: Sales volume and value

(TWD '000; '000 units)	2018 Amount	2018 Value	2019 Amount	2019 Value	2020 Amount	2020 Value	2021 Amount	2021 Value	2022 Amount	2022 Value
	3,088,143	9,414,159	3,840,684	10,777,781	5,350,301	13,876,445	6,855,326	21,506,066	4,928,294	23,511,086
ASP per unit		3.05		2.81		2.59		3.14		4.77
y-y				-8%		-8%		21%		52%

Source: Company data, Nomura research

Fig. 186: Silergy: Cost and ASP y-y per unit vs GPM



Source: Company data, Nomura research

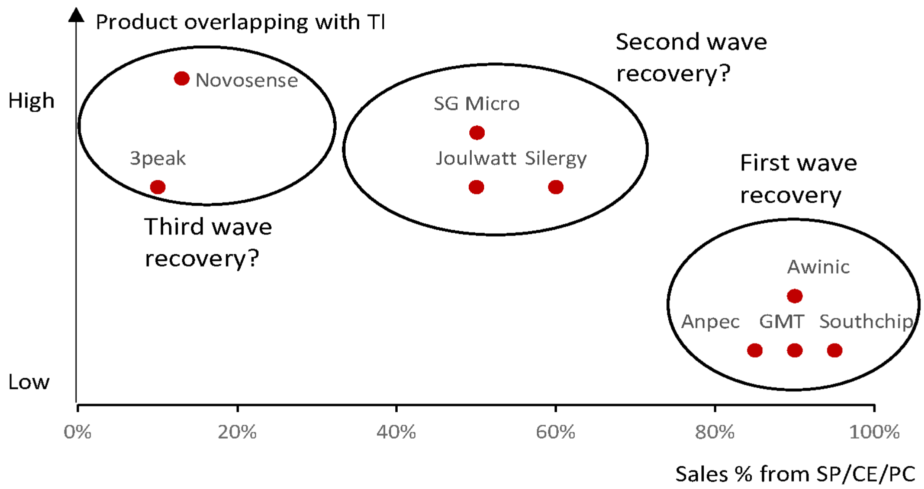
Silergy: Sooner recovery through analog cycle bottom than global peers

In our Nov 2023 analog IC *report*, we noted our view that companies' recovery depended on product coverage and percentage overlap with TI (*Fig. 187*). Silergy has five major product lines, and different products have had different cycle patterns over the past two years. Silergy's sales turned soft from mid-2022 for most of its applications except auto applications, and inventory piled up entering 2023.

Better inventory level in second wave recovery group

However, we think Silergy managed its inventory aggressively in late-2022, and inventory levels continued to decline in 2023. Compared to other local China peers, Silergy's DOI returned to a healthy level of 70 days in 4Q23, which is healthier than other China local peers that have high sales exposure to TI, and it has similar inventory days to the first-wave recovery group.

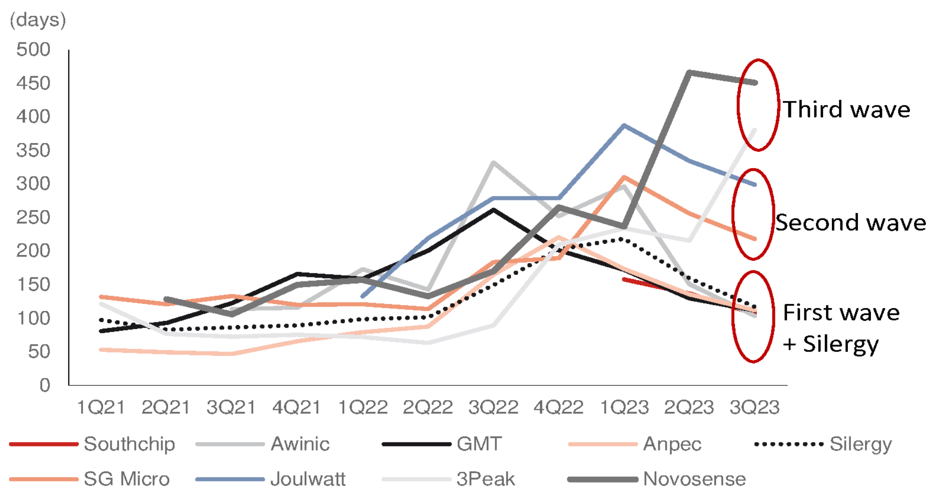
Fig. 187: Greater China analog IC companies: product overlapping with TI vs sales % from SP/CE/PC



Source: Company data, Nomura estimates

Fig. 188: Greater China analog IC companies' DOI

DOI declines also follow the sales % of TI exposure



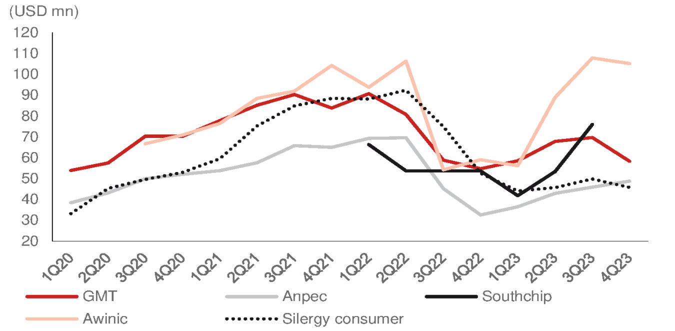
Source: Company data, Nomura research

Consumer applications

In our July 2023 *report* on Awinic, we noted that we had seen consumer application demand recovering from a low base since 2H22. Other fabless also saw a similar sales trend, with demand gradually improving in mid-2023.

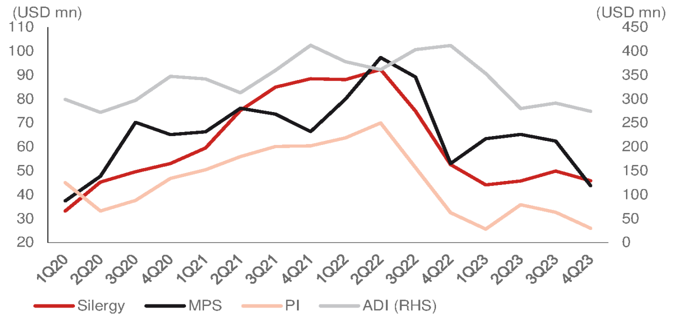
Compared to the sales of other greater China consumer analog IC companies, the Silergy's consumer sales recovery was slightly behind (*Fig. 189*), more similar to the trend from other global peers (*Fig. 189*). We think it is also supported our view that the sales exposure to TI and product mix from SP/CE/PC will drive a different pattern of recovery. Also, we think more of the greater China analog IC vendors' customers are from China, which started an inventory correction earlier than in other regions, resulting in an earlier recovery. Therefore, Silergy and other global peers have higher sales to other regions to see slightly behind rebound in 2H23.

Fig. 189: Greater China consumer analog IC companies sales



Source: Company data, Nomura research

Fig. 190: Global analog IC vendors' consumer sales



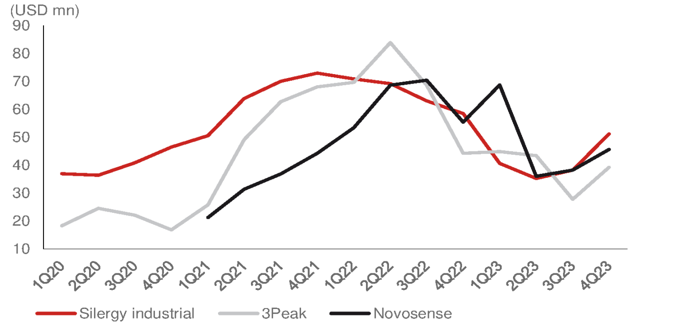
Source: Company data, Nomura research

Industrial applications

Server exposure and geographic sales impact on recovery trend

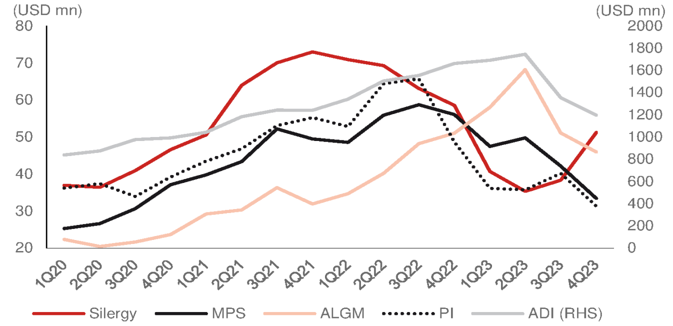
Industrial is the application that is currently seeing a recovery in the market, in our view. Silergy's industrial sales turned soft in 2Q22, and earlier than China peers' in 2H22. We think this was due to higher server exposure than other local peers, as server demand declined from 1Q22. Furthermore, Silergy's industrial sales softness was also earlier than global peers in 2H22-1H23, which we think was due to higher China sales exposure than others.

Fig. 191: Greater China industrial analog IC companies sales



Note: 3Peak and Novosense are total sales, and Silergy shows industrial sales only.
Source: Company data, Nomura research

Fig. 192: Global analog IC vendors' industrial sales



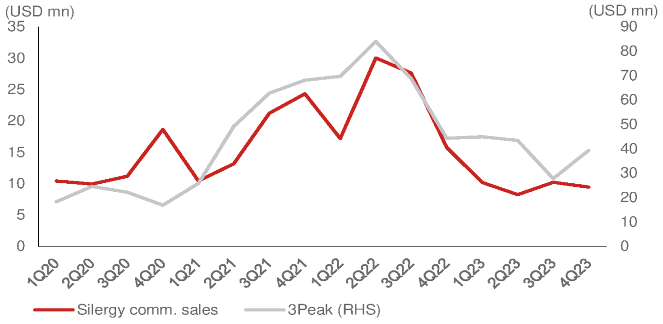
Source: Company data, Nomura research

Communication applications

Impact of China exposure reflected predominantly in Silergy sales pattern

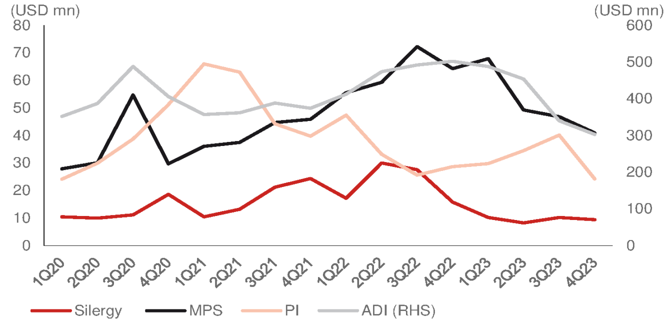
Silergy's communication sales have declined significantly from 3Q22, and the trend is similar for local peers. However, global peers' communication sales turned soft in 1H23, which we think was due to customer exposure, as Silergy has higher communication sales from China.

Fig. 193: Greater China communication analog IC companies sales



Source: Company data, Nomura research

Fig. 194: Global analog IC vendors' communication sales



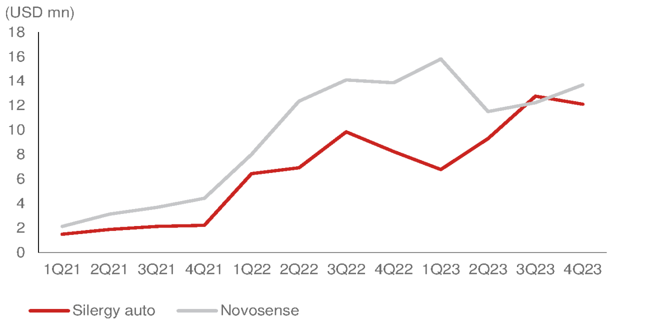
Source: Company data, Nomura research

Auto applications

Dollar content growth to outperform other peers

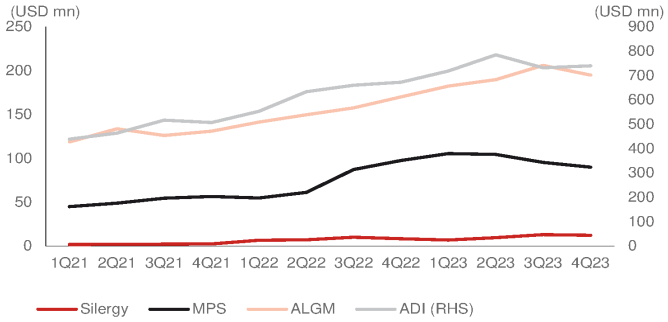
Silergy's auto application sales have experienced significant growth since 2021, and have also seen a trend similar to that for Novosense (688052 CH, Neutral), another major auto analog vendor in China. Both companies' auto applications sales also saw a slowdown in 2H22-1H23 due to inventory adjustments by China auto customers, but demand stabilized or rebounded in 3Q23. However, global peers' auto application sales remained strong until 2H23, which we think was due to global auto OEMs having a slower inventory adjustment period. Thus, we think Silergy can outperform global peers through its high dollar content growth in 2023-2024F and because of the earlier inventory correction for China auto OEM customers.

Fig. 195: Greater China auto analog IC companies sales



Source: Company data, Nomura research

Fig. 196: Global analog IC vendors' auto sales



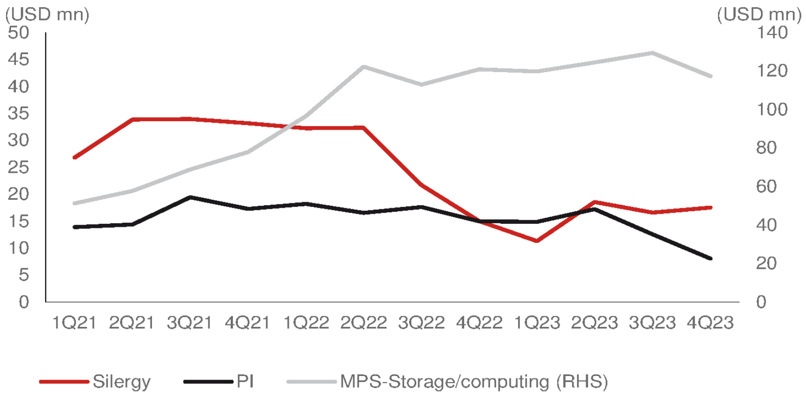
Source: Company data, Nomura research

Computing application

Dependency on NB/PC market demand recovery

In computing applications, Silergy turned soft in 2H22 when NB/PC demand declined, similar to PI's computing sales slowdown in 4Q22, while MPS sales remained stable and continued to grow in 2023 due to market share gains. There is no vendor with high sales exposure to computing applications in China. We think Silergy's computing sales could recover when NB/PC market demand recovers.

Fig. 197: Silergy: Computing sales vs global analog IC vendors



Source: Company data, Nomura research

Summary of financials

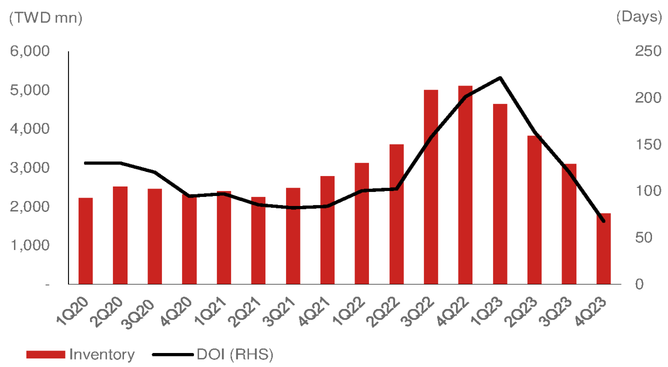
Silergy to record a 23% sales CAGR over 2024-25F, driven by cyclical recovery and market share gains

We forecast Silergy's sales to grow 20%/26% in 2024F/25F, driven by auto sales' dollar content growth and a cyclical recovery for industrial applications. We assume consumer application demand recovery and auto sales to grow by a mid-to-high double-digit CAGR over 2023-2025F, driven by the mass production of new products. We also forecast earnings will grow in 2024F/25F by 300%/105% on higher GPM owing to lower inventory provision costs and lower foundry costs. In terms of other business, we think industrial will remain weak in 1H24 due to weak global macro conditions and continued excess inventory overhang in customers.

GPM could return to normal level in 2025F

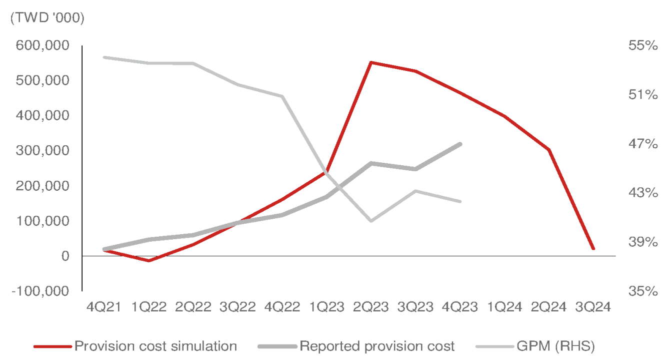
We forecast Silergy's GPM can continue to improve from the 2Q23 trough into 2024F owing to a reduction in inventory provision cost recognized in COGS. According to Silergy accounting policy, inventory provision costs begin to be recognized after more than three quarters on balance sheet, and 20% of products' inventory value will be recognized in next five quarters. The inventory write-off impact to GPM shall peak out in 2H23-1H24, since inventory peaked at end-2022 (*Fig. 198*). Thus, the low reported GPM level was in 2H23, which is three quarters' after the inventory peak level in 4Q22. We calculate in our simulation (*Fig. 199*) that the provision cost for excess inventory peaked in 2Q23. Our provision cost simulation is based on no inventory depleted by Silergy's customer, which resulted a higher number than company-reported provision cost when inventory was digested by customers. On the other hand, the provision cost in COGS needs to be substracted from actual GPM to reflect the actual product GPM. In addition to provision cost, depreciation costs in COGS need to be subtracted from pro-forma GPM thus pro-forma GPM would be higher than actual GPM. Therefore, as long as inventory provision costs ease in 2H24, we expect Silergy's reported GPM to return to normal 50% levels by 2025F.

Fig. 198: Silergy: inventories and DOI



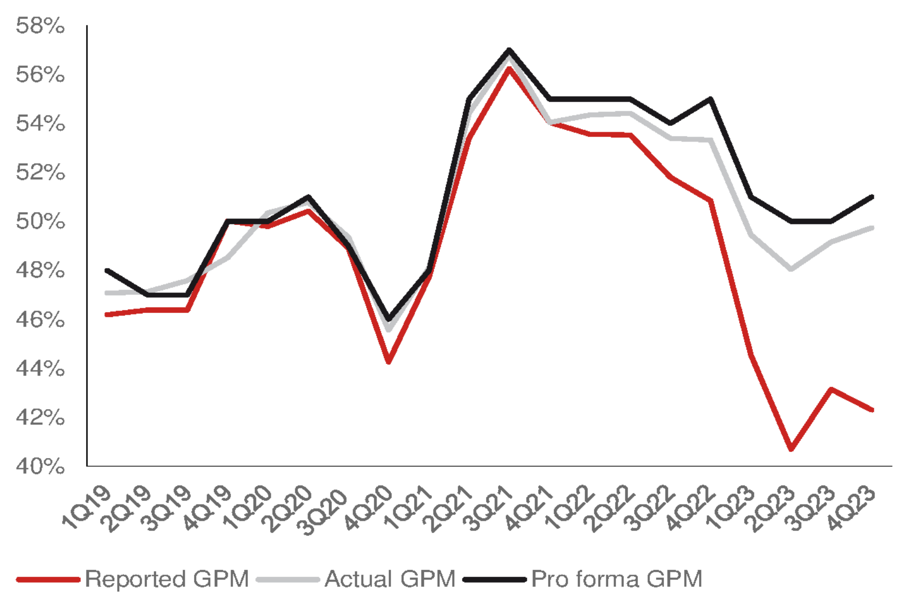
Source: Company data, Nomura research

Fig. 199: Silergy: Inventory provision cost simulation vs GPM



Source: Company data, Nomura estimates

Fig. 200: Silergy: Reported GPM, actual GPM, pro-forma GPM



Note: Actual GPM is based on reported GPM add inventory provision cost to reflect Actual product GPM
Pro-forma GPM is based on GAAP policy, which doesn't includes depreciation and amortization expense.
Source: Company data, Nomura research

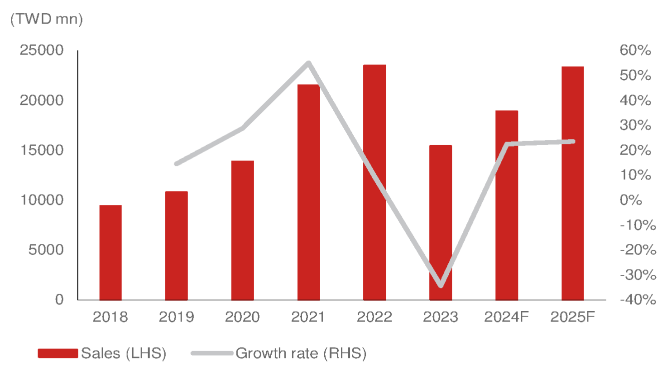
We view the Gen-4 platform as one of the key drivers of improved GPM in 2024F-25F for Silergy. Thus, we estimate Silergy's GPM recovery sensitivity analysis to earnings by different level in 2024-25F.

Fig. 201: Sensitivity analysis of Silergy's GPM recovery since using Gen-4 platform

	2024F			2025F		
Gen-4 Platform benefits GPM recovery	43%	48%	50%	46%	51%	56%
		Base case			Base case	
Revenues	18,572	18,572	18,572	23,449	23,449	23,449
Gross profit	7,986	8,872	9,286	10,786	11,855	13,131
Operating profit	1,207	2,093	2,507	3,823	4,892	6,168
Pre-tax profit	1,871	2,757	3,171	4,598	5,667	6,943
Net profit	1,638	2,415	2,777	4,029	4,965	6,083
EPS (TWD)	4.26	6.28	7.23	10.49	12.9	15.83
Margins (%)						
Gross margin	43%	48%	50%	46%	51%	56%
Operating margin	6%	11%	13%	16%	21%	26%
Pre-tax margin	10%	15%	17%	20%	24%	30%
Net margin	9%	13%	15%	17%	21%	26%

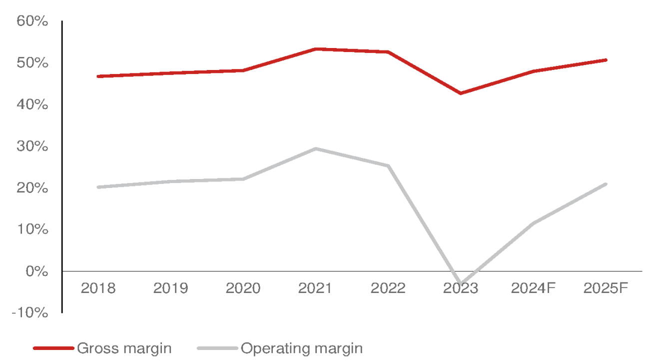
Source: Nomura estimates

Fig. 202: Silergy: Annual sales and y-y growth rate



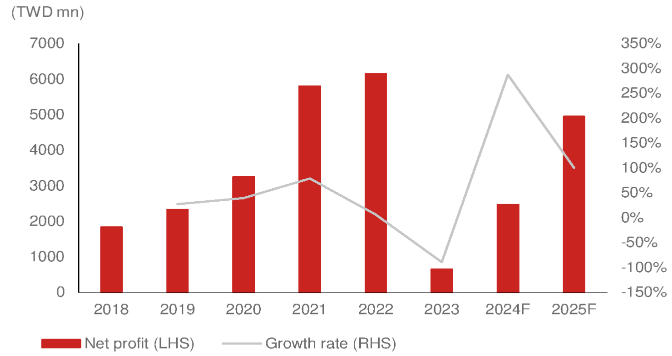
Source: Company data, Nomura estimates

Fig. 203: Silergy: GPM and OPM



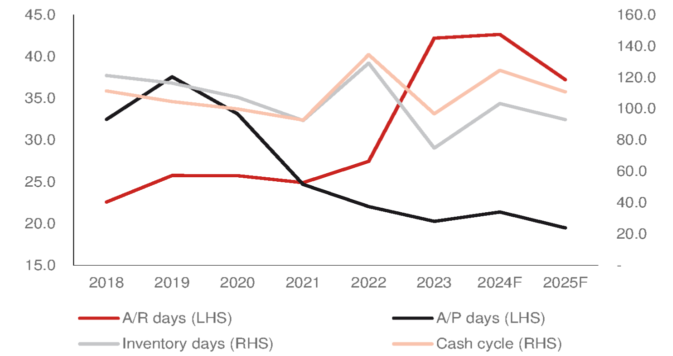
Source: Company data, Nomura estimates

Fig. 204: Silergy: Net income and y-y growth rate



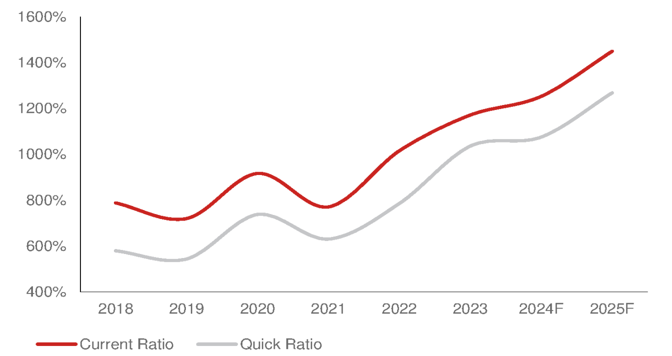
Source: Company data, Nomura estimates

Fig. 205: Silergy: CCC analysis



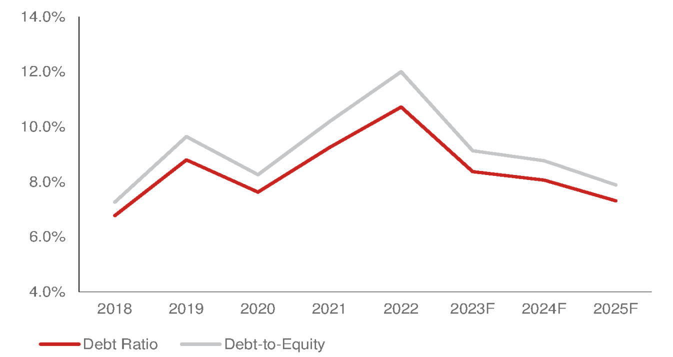
Source: Company data, Nomura estimates

Fig. 206: Silergy: Current ratio and quick ratio



Source: Company data, Nomura estimates

Fig. 207: Silergy: Net debt / net equity and debt ratio



Source: Company data, Nomura estimates

Fig. 208: Silergy: P&L

(TWD mn)	1Q23	2Q23	3Q23	4Q23	1Q24F	2Q24F	3Q24F	4Q24F	1Q25F	2Q25F	3Q25F	4Q25F	2023	2024F	2025F	2026F
Net sales	3,435	3,598	4,111	4,284	3,842	4,314	5,047	5,369	5,172	5,728	6,435	6,113	15,427	18,572	23,449	27,577
- COGS	1,905	2,134	2,338	2,472	2,110	2,280	2,604	2,706	2,576	2,830	3,177	3,011	8,848	9,700	11,594	13,568
Gross profit	1,530	1,464	1,774	1,812	1,732	2,035	2,443	2,663	2,596	2,898	3,259	3,102	6,579	8,872	11,855	14,009
- SG&A	(582)	(536)	(559)	(529)	(461)	(544)	(555)	(564)	(517)	(544)	(592)	(581)	(2,204)	(2,124)	(2,234)	(2,523)
- R&D expenses	(1,168)	(1,240)	(1,230)	(1,256)	(1,037)	(1,122)	(1,262)	(1,235)	(1,138)	(1,146)	(1,223)	(1,223)	(4,894)	(4,656)	(4,729)	(5,205)
Operating profit	(220)	(312)	(15)	28	234	369	626	864	941	1,209	1,444	1,299	(519)	2,093	4,892	6,282
Net non-op income	363	151	469	126	161	168	166	168	177	194	194	209	1,109	664	775	978
Pretax profit	143	(162)	454	153	395	538	792	1,032	1,118	1,403	1,638	1,508	589	2,757	5,667	7,259
Net profit	171	(139)	446	126	329	448	779	859	931	1,168	1,611	1,256	604	2,415	4,965	6,355
EPS (TWD)	0.46	(0.35)	1.20	0.50	0.86	1.17	2.03	2.24	2.42	3.04	4.19	3.27	1.57	6.28	12.9	16.5
Profitability (%)	**1Q23**	**2Q23**	**3Q23**	**4Q23**	**1Q24F**	**2Q24F**	**3Q24F**	**4Q24F**	**1Q25F**	**2Q25F**	**3Q25F**	**4Q25F**	**2023**	**2024F**	**2025F**	**2026F**
Gross margin	44.5%	40.7%	43.1%	42.3%	45.1%	47.2%	48.4%	49.6%	50.2%	50.6%	50.6%	50.7%	42.6%	47.8%	50.6%	50.8%
- OPEX ratio	(51.0%)	(49.4%)	(43.5%)	(41.7%)	(39.0%)	(38.6%)	(36.0%)	(33.5%)	(32.0%)	(29.5%)	(28.2%)	(29.5%)	(46.0%)	(36.5%)	(29.7%)	(28.0%)
Operating margin	(6.4%)	(8.7%)	(0.4%)	0.6%	6.1%	8.6%	12.4%	16.1%	18.2%	21.1%	22.4%	21.2%	(3.4%)	11.3%	20.9%	22.8%
Net margin	5.0%	(3.9%)	10.9%	2.9%	8.6%	10.4%	15.4%	16.0%	18.0%	20.4%	25.0%	20.5%	3.9%	13.0%	21.2%	23.0%
QoQ (%)	**1Q23**	**2Q23**	**3Q23**	**4Q23**	**1Q24F**	**2Q24F**	**3Q24F**	**4Q24F**	**1Q25F**	**2Q25F**	**3Q25F**	**4Q25F**	**2023**	**2024F**	**2025F**	**2026F**
Net sales	(26.9%)	4.7%	14.3%	4.2%	(10.3%)	12.3%	17.0%	6.4%	(3.7%)	10.8%	12.3%	(5.0%)				
Gross profit	(36.0%)	(4.3%)	21.2%	2.2%	(4.4%)	17.5%	20.1%	9.0%	(2.5%)	11.7%	12.4%	(4.8%)				
Operating profit	(133.2%)	41.9%	(95.3%)	(287.5%)	740.7%	58.1%	69.5%	38.0%	8.9%	28.5%	19.5%	(10.0%)				
Net profit	(79.8%)	(181.3%)	(422.0%)	(71.8%)	161.5%	36.2%	74.0%	10.3%	8.4%	25.5%	37.9%	(22.1%)				
YoY (%)	**1Q23**	**2Q23**	**3Q23**	**4Q23**	**1Q24F**	**2Q24F**	**3Q24F**	**4Q24F**	**1Q25F**	**2Q25F**	**3Q25F**	**4Q25F**	**2023**	**2024F**	**2025F**	**2026F**
Sales	(43.0)	(47.1)	(31.4)	(8.8)	11.9	19.9	22.8	25.3	34.6	32.8	27.5	13.9	(34.4)	20.4	26.3	17.6
Gross profit	(52.6)	(59.8)	(42.9)	(24.2)	13.2	39.0	37.7	46.9	49.9	42.5	33.4	16.5	(46.8)	34.8	33.6	18.2
Op profit	(112.4)	(115.3)	(101.0)	(95.8)	(206.1)	(218.3)	(4,324.9)	3,010.0	302.8	227.2	130.7	50.3	(108.8)	(503.0)	133.8	28.4
Net profit	(89.6)	(106.4)	(70.0)	(85.1)	92.8	(423.0)	74.5	583.1	183.1	160.8	106.8	46.2	(90.1)	299.7	105.6	28.0

Source: Company data, Nomura estimates

Fig. 209: Nomura forecasts vs Bloomberg consensus estimates

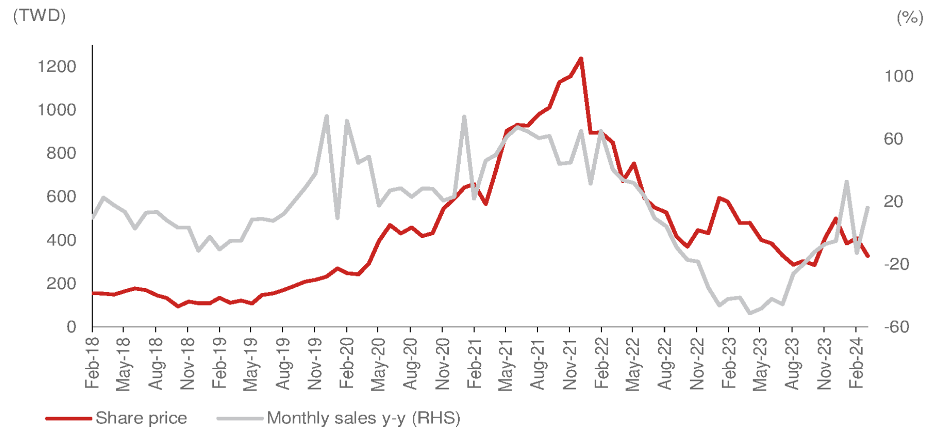
	2024F			2025F			2026F		
(TWD mn)	NMR	BBG	Diff (%)	NMR	BBG	Diff (%)	NMR	BBG	Diff (%)
Sales	18,569	19,532	(4.9)	23,271	24,974	(6.8)	27,402	30,463	(10.0)
Gross profit	8,876	9,163	(3.1)	11,771	12,506	(5.9)	13,926	15,256	(8.7)
Operating profit	2,093	1,913	9.4	4,861	4,576	6.2	6,247	6,768	(7.7)
Net profit	2,414	2,364	2.1	4,939	4,782	3.3	6,326	6,666	(5.1)
EPS (TWD)	6.28	6.15	2.1	12.86	12.45	3.3	16.46	17.35	(5.1)
Margin	NMR	BBG	Diff (ppts)	NMR	BBG	Diff (ppts)	NMR	BBG	Diff (ppts)
Gross margin (%)	47.8	46.9	0.9	50.6	50.1	0.5	50.8	50.1	0.7
Operating margin (%)	11.3	9.8	1.5	20.9	18.3	2.6	22.8	22.2	0.6
Net margin (%)	13.0	12.1	0.9	21.2	19.1	2.1	23.1	21.9	1.2

Source: Bloomberg Financials L.P., Nomura estimates

Share price drivers

In our view, y-y sales and earnings are the company's major share price drivers. Silergy's sales grew significantly owing to an IC shortage that drove a share price rally in 2021. The share price peaked in late-2021 to reflect a decrease in sales in 1Q22 when the semi cycle downturn began (*Fig. 210*). We also believe that the uptrend in consensus earnings revisions continued to drive the share price higher in 2020-2021, while downward earnings estimate revisions put the share price under pressure in 2022-2023 (*Fig. 211*, *Fig. 212*).

Fig. 210: Silergy: Share price vs monthly sales y-y

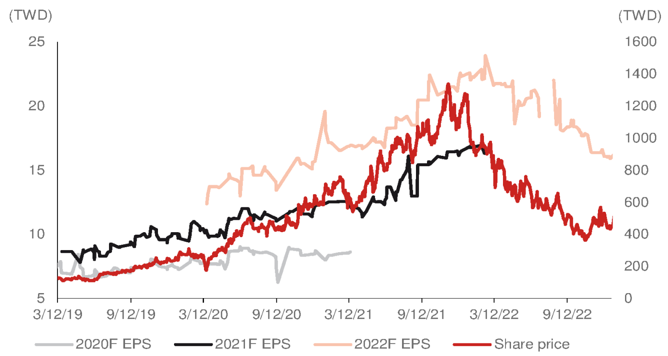


Source: Company data, Nomura research

Consensus earnings forecast revisions

Fig. 211: Silergy: Consensus earnings forecast revisions - I

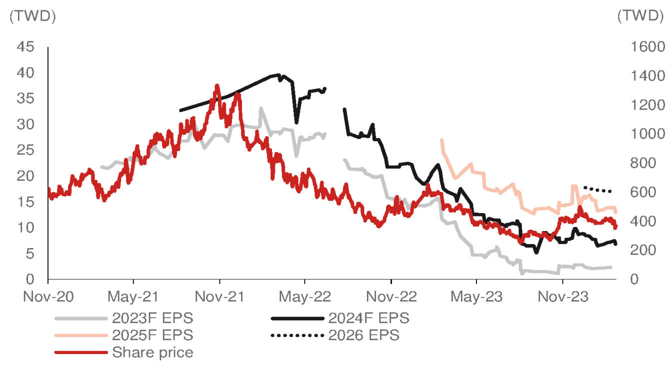
Earnings upward revision to drive share price strength



Source: Bloomberg Finance LP., Nomura research

Fig. 212: Silergy: Consensus earnings forecast revisions - II

Earnings downward revisions to put share price under pressure



Source: Bloomberg Finance LP., Nomura research

Fig. 213: Share price event chart

Source: Company data,TEJ, Nomura research

Valuation and risks

Our TP of TWD450 is based on 35x P/E and 2025F EPS of TWD12.9. Our 35x target P/E is at the mid-to-low-end of its consensus P/E trading range of 20-60x in 2019-2021 when net income resumed growth from net income declines y-y in 2017-2018, which in our view reflects the similar expected earnings growth resumption of 300%/105% y-y in 2024F/25F. We think Silergy will maintain its leading position in China's analog market, gaining share from local and global peers, and benefiting from growing content in the auto analog IC market.

Fig. 214: Analog vendors' valuation comparison

Ticker	Company	Nomura Analyst	Nomura Rating	Price(Local CCY) 4/9/2024	Mkt cap USDmn	P/E (x)		P/S(x)		P/B (x)		ROE (%)		EPS (Local CCY)		Div Yield (%)	
						FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F
Global analog																	
TXN US	TI		Not Rated	173.5	152,306	31.4	24.9	8.7	9.7	8.9	8.3	33.4	30.6	5.3	6.7	3.3	3.5
ADI US	ADI		Not Rated	204.1	96,583	32.5	25.3	7.9	10.6	2.8	2.7	11.4	13.7	6.0	7.7	2.0	2.2
MPWR US	MPS		Not Rated	682.2	31,839	51.1	40.5	17.5	15.6	13.7	11.6	28.9	29.8	12.8	16.1	0.8	0.9
ALGM US	Allegro		Not Rated	28.0	5,091	29.8	19.6	4.9	5.4	4.0	3.4	13.5	17.7	0.9	1.3	n.a.	n.a.
POWI US	Power Integration		Not Rated	71.7	3,917	57.5	33.2	8.8	8.7	5.1	4.7	10.3	16.8	1.2	2.1	1.2	1.2
Global Avg						40.5	28.7	9.5	10.0	6.9	6.2	19.5	21.7			1.8	2.0
Taiwan analog																	
2454 TT	MediaTek	Aaron Jeng	Neutral	1,160.0	57,896	23.4	21.0	3.7	3.3	4.6	4.4	19.2	21.1	49.6	55.2	3.6	4.0
6415 TT	Silergy	Aaron Jeng	Buy	328.0	3,946	51.0	25.5	6.7	5.4	3.7	3.3	7.6	13.8	6.4	12.9	0.2	0.6
8081 TT equity	GMT		Not Rated	271.5	728	14.8	14.9	2.8	2.6	2.6	n.a	18.1	n.a.	20.4	18.0	n.a.	n.a.
6719 TT equity	uPI		Not Rated	288.5	717	42.8	26.7	7.6	5.8	3.1	3.1	11.6	n.a.	6.0	11.8	1.8	2.9
TW Avg						33.0	22.0	5.2	4.3	3.5	3.6	14.1	17.4			1.8	2.5
A-share analog																	
688052 CH	Novosense	Aaron Jeng	Neutral	94.4	1,859	(0.0)	0.0	8.9	6.9	2.1	2.2	(2.5)	(3.0)	(1.4)	0.0	0.0	0.0
688536 CH	3Peak	Aaron Jeng	Buy	96.2	1,763	64.4	36.0	8.4	6.4	2.4	2.3	6.5	7.6	1.5	2.7	0.2	0.2
688798 CH	Awinic	Donnie Teng	Buy	53.0	1,699	142.4	38.9	3.8	3.2	3.4	3.2	2.3	8.1	0.4	1.4	0.1	0.1
300661 CH	SG Micro	Donnie Teng	Neutral	61.9	3,999	67.6	40.0	8.8	7.2	7.2	6.3	10.9	16.5	0.9	1.6	0.4	0.4
688484 CH	Southchip		Not Rated	30.0	1,689	35.5	26.0	7.4	5.5	3.4	0.4	10.9	12.5	0.8	1.2	0.8	1.5
688141 CH	Joulwatt		Not Rated	16.0	934	(50.0)	50.8	5.2	4.0	2.8	2.7	(5.7)	5.0	(0.3)	0.3	n.a.	n.a.
A share Avg						43.3	32.0	10.0	7.5	3.6	2.8	3.8	7.8			0.3	0.4

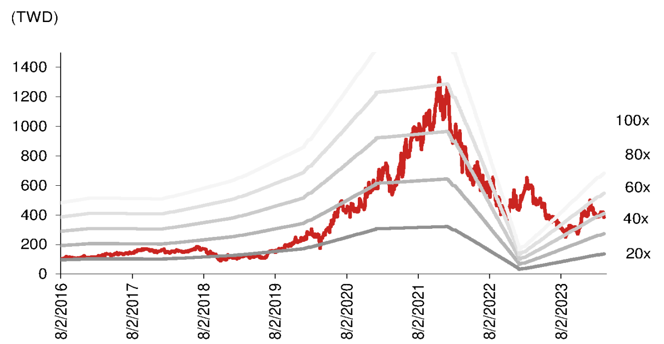
Source: Bloomberg Finance L.P., Nomura estimates for coverage companies

Fig. 215: Global power semi peers valuation

Ticker	Company	Nomura Analyst	Nomura Rating	Price(Local CCY) 4/9/2024	Mkt cap USDmn	P/E (x)		P/S(x)		P/B (x)		ROE (%)		EPS (Local CCY)		Div Yield (%)	
						FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F	FY24F	FY25F
Global Power semi																	
NXPI US	NXP		Not Rated	252	61,827	18.7	16.4	4.9	4.9	7.3	6.7	35.8	33.7	13.5	15.4	1.8	2.0
STM US	STM		Not Rated	44	38,178	13.8	10.8	2.5	2.3	2.4	2.1	19.1	18.6	3.1	4.0	0.7	0.6
IFNNY US	Infineon		Not Rated	37	47,849	17.9	14.2	3.4	2.7	2.8	2.6	17.5	18.6	2.0	2.6	1.2	1.3
6723 JP	Renesas	Masaya Yamasaki	Buy	2,761	34,048	15.4	13.1	3.0	3.5	2.5	2.2	15.5	15.6	183.0	215.0	0.0	0.0
ON US	ON Semi		Not Rated	71	19,758	16.7	13.8	3.7	3.7	3.8	3.2	22.7	25.5	4.2	5.2	-	-
TXN US	TI		Not Rated	173	153,596	32.5	25.8	7.9	9.0	9.4	9.2	34.6	31.7	5.3	6.7	3.2	3.4
ADI US	ADI		Not Rated	204	97,361	34.1	26.5	8.5	8.2	2.9	2.9	12.0	14.4	6.0	7.7	1.9	2.1
VSH US	Vishay		Not Rated	23	2,990	17.7	11.5	0.9	0.9	n.a.	n.a.	n.a.	n.a.	1.3	2.0	1.9	n.a.
DIOD US	Diode		Not Rated	72	3,210	31.8	18.9	1.7	2.0	1.9	1.8	n.a.	n.a.	2.3	3.8	n.a.	n.a.
Average						22.1	16.8	4.0	4.1	4.1	3.8	22.4	22.6			1.3	1.4

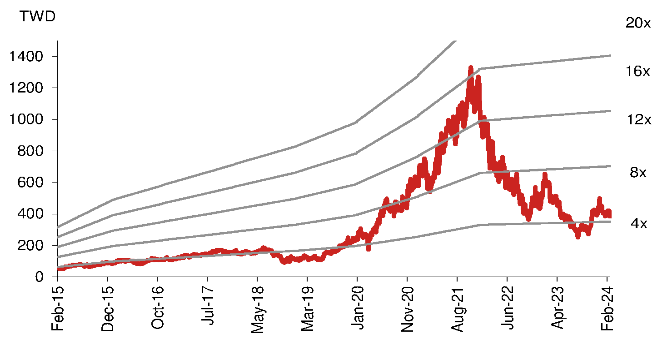
Source: Bloomberg Finance L.P., Nomura estimates for coverage companies

Fig. 216: Silergy: One-year-forward P/E



Source: Company data, Nomura estimates

Fig. 217: Silergy: One-year-forward P/B

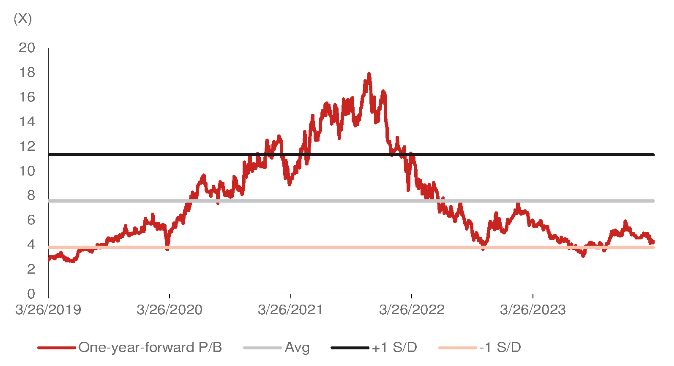


Source: Company data, Nomura estimates

Fig. 218: Silergy: One-year-forward consensus P/E

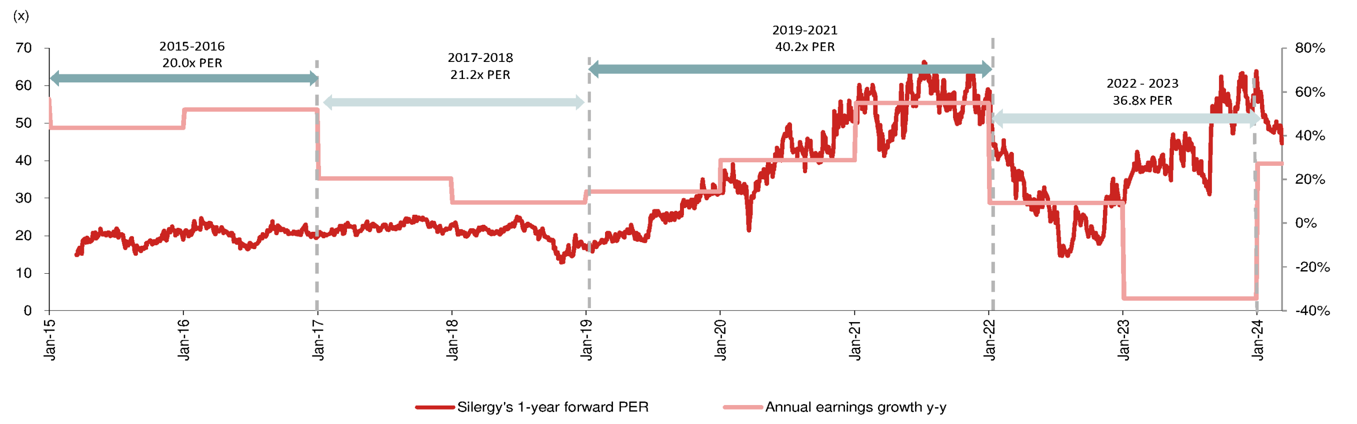
Source: Bloomberg Finance LP., Nomura research

Fig. 219: Silergy: One-year-forward consensus P/B



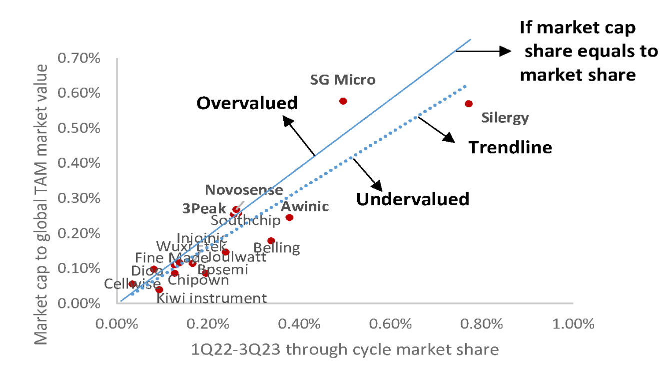
Source: Bloomberg Finance LP., Nomura research

Fig. 220: Silergy: One-year-forward consensus P/E and net income y-y



Source: Bloomberg Finance LP., Nomura research

Fig. 221: China analog through cycle sales' market share vs market cap share



Source: Company data, Nomura estimates

Fig. 222: Valuation analysis of major China analog vendors

	Market cap (USD mn)	1Q22-3Q23 sales (USD mn)	1Q22-3Q23 market share	Market cap to global TAM market value
Silergy	3,946	1,152.9	0.77%	0.57%
SG Micro	3,999	741.4	0.50%	0.58%
AWINIC	1,702	565.3	0.38%	0.25%
Southchip	1,782	397.6	0.27%	0.26%
Joulwatt	1,019	356.3	0.24%	0.15%
Injoinic	795	248.2	0.17%	0.11%
Kiwi instrument	274	139.0	0.09%	0.04%
Fine Made	755	189.9	0.13%	0.11%
Cellwise	388	51.1	0.03%	0.06%
Bpsemi	598	291.3	0.19%	0.09%
Chipown	600	189.6	0.13%	0.09%
Wuxi ETEK	808	205.4	0.14%	0.12%
Belling	1,240	505.6	0.34%	0.18%
3Peak	1,764	382.8	0.26%	0.25%
Novosense	1,860	391.1	0.26%	0.27%
Dioo	678	121.3	0.08%	0.10%
Total	22,209		4.0%	3.2%

Source: Bloomberg Finance L.P., Nomura research

Risks to our investment view

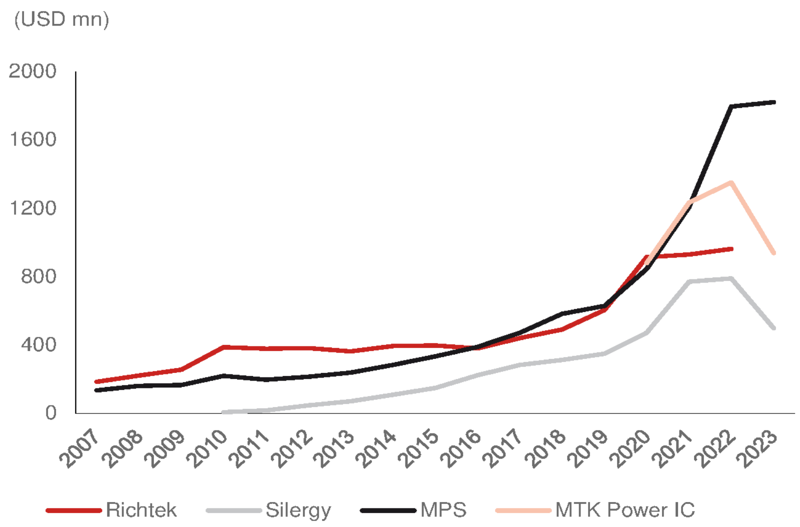
1) Softer-than-expected end-demand recovery; 2) slower-than expected market share gains in China and oversea market; 3) fiercer-than-expected pricing competition from global IDM and local peers; 4) higher-than-expected foundry price; 5) slower-than-expected product transition in Gen-4 platform; and 6) slower-than-expected GPM recovery.

Appendix

Major power IC fabless sales performance

From the history of the power IC fabless industry, we think Richtek (unlisted, acquired by MTK [2454 TT, Neutral]) and MPS (MPWR US, Not rated) are the comparable peers of Silergy. Richtek's sales of USD350mn-400mn were at similar levels with Silergy between 2010 and 2016, which we think was due to Richtek's mature market for its computing applications. MPS's sales grew y-y for 12 consecutive years from 2009, and surpassed Richtek's sales from 2016. Between 2022 and 2023, the power IC industry was in a downturn, but MPS maintained its sales growth momentum led by auto and AI server application sales that offset the decline in consumer and comm. applications.

Fig. 223: Global power IC fabless sales



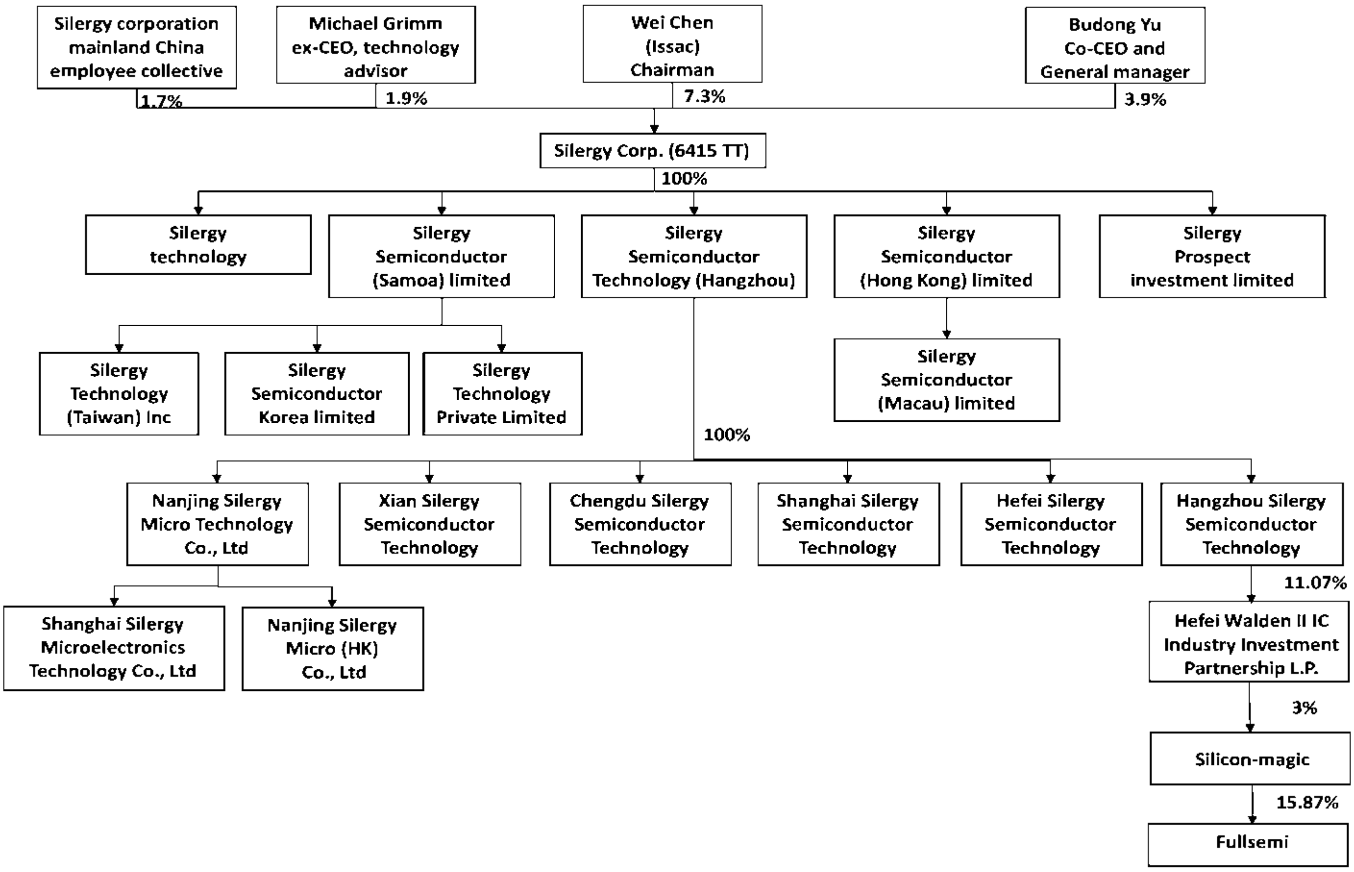
Source: Company data, Nomura research

Company background

Established in Silicon Valley, USA and Hangzhou, China in 2008, Silergy is one of the largest analog companies in China. Silergy Hangzhou is the headquarters, and is responsible for product design, operation, and technical support; the US office is responsible for advanced technology development. Silergy is one of the few fabless companies in the world capable of producing high-voltage, high-current IC in small packages in the greater China area.

Silergy's management team has years of analog IC experience. With a team that specializes in key technologies, Silergy is able to quickly respond to the latest specifications for products and provide customers with a self-owned development platform equipped with comprehensive specifications, designs, and real-time customer service during the product development stage. With the rapid development of information products, Silergy has demonstrated it is a professional IC design house with competitive advantages, in our view.

Fig. 224: Silergy: Company structure



Source: Company data, Nomura research

Fig. 225: Silergy: Management team

Title	Name	Nationality	Main experience and academic background
Chairman	Wei Chen (Issac)	USA	PhD, Department of Electrical Engineering, Virginia Polytechnic Institute and State University, USA; Technical Manager, Linear Technology; Deputy Chief System and Applications Technology Officer, Monolithic Power Systems, Inc.
Co-CEO and General Manager	Budong You	USA	PhD, Department of Electrical Engineering, Virginia Polytechnic Institute and State University, USA; Deputy Technology Manager, Volterra Semiconductor
VP of Asia Sales	Chih-chung Lu	ROC	Deputy Marketing Manager, Magnachip Semiconductor Corp; General Manager of Taiwan Region, ON Semiconductor Marketing Manager of Taiwan Region, Fairchild Semiconductor
Fellow	Jaime Tseng	USA	Electrical Engineering, University of California, Berkeley; Design Manager, Linear Technology
Chief Financial Officer	Shao-wei Chen	ROC	Department of Accounting, Chung Yuan Christian University; Financial manager, BenQ Materials Corp.

Source: Company data, Nomura research

Fig. 226: Silergy: Board of directors

Title	Name	Nationality	Main experience and academic background
Chairman	Wei Chen (Issac)	USA	PhD, Department of Electrical Engineering, Virginia Polytechnic Institute and State University, USA; Technical Manager, Linear Technology; Deputy Chief System and Applications Technology Officer, Monolithic Power Systems, Inc.
Director	Budong You	USA	PhD, Department of Electrical Engineering, Virginia Polytechnic Institute and State University, USA; Deputy Technology Manager, Volterra Semiconductor
Director	Jiun-Huei Shih	ROC	JD, Stanford University Law school; BS, United States Military Academy; Managing Director and Partner at JP Morgan/One Equity Partners Industry banker at both Merrill Lynch and Deutsche; Founding Partner, Hudson Highland Partners Advisory Partner, eJnnn Phecda Partners
Director	Sophia Tong	ROC	Department of foreign languages, National Taiwan University; President, Test Rite International Co., Ltd.; GM of IBM Taiwan; CEO, HonTai Group Chairman, ZHAN TENG Ltd.;Independent Director, Acer; Cyber Security Inc. Director, Dotmore Media
Independent Director	Yong-Song Tsai	ROC	Masters, International Business, National Taiwan University (NTU); Partner, APP Capital Limited Deputy General Manager; Walden International Taiwan Co., Ltd.; Independent director, WAFER WORKS Corporation
Independent Director	Henry King	ROC	EMBA Enterprise Class, National Cheng-Chi University, TAIWAN; MBA in Finance, Loyola University of Chicago, USA; BS in Electrical Engineering (minor in BA), National Central University, TAIWAN; Managing Director, Co-head of Asia Technology team, Head of Taiwan research, Goldman Sachs Asia; Senior analyst, Credit Suisse; Chairman, Kashman Investment Co., Ltd; Director, GOLDEN BRIDGE ELECTECH INC.; Independent Director, CHIP HOPE CO., LTD; Independent Director, PANRAM INTERNATIONAL CORP.
Independent Director	Jet Tsai	ROC	MBA, National Taipei University; Electrophysics / Chiao Tung University; CPA, Jianda Lianhe Accounting Firm; Accounting Firm Independent Director, GLOBAL VIEW CO., LTD.; Chairman, JIE DENG CO.,LTD.; Chairman, DA SHU ENERGY CO.,LTD

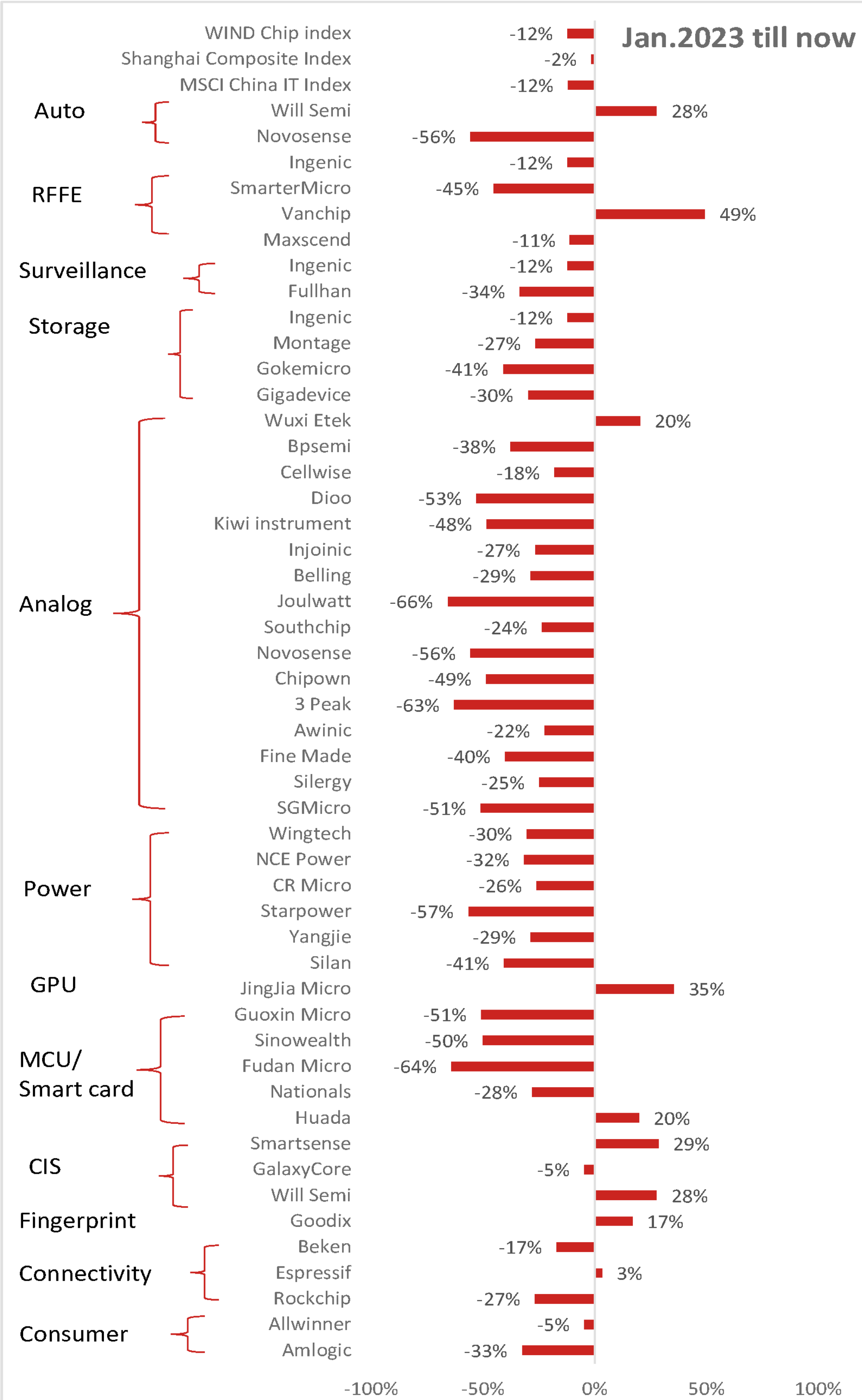
Source: Company data, Nomura research

Fig. 227: Silergy: Major shareholders

Major share holder	Ownership (%)
Issac Chen	7.31%
Budong You	3.92%
Michael Grimm	1.91%
Fubon Life insurance	1.71%
Silergy Corporation Mainland China area employee collective	1.69%
Standard Charted in custody for invesco emerging market equity fund	1.62%
Standard Charted in custody for Fidelity fund	1.55%
JP Morgan Chase Bank N.A. Taipei Branch in custody for New World Fund Inc.	1.53%
JP Morgan Chase Bank N.A. Taipei Branch in custody for JP Morgan emerging market equity fund Inc.	1.45%
JP Morgan Chase Bank N.A. Taipei Branch in custody for JP Morgan asset management	1.43%

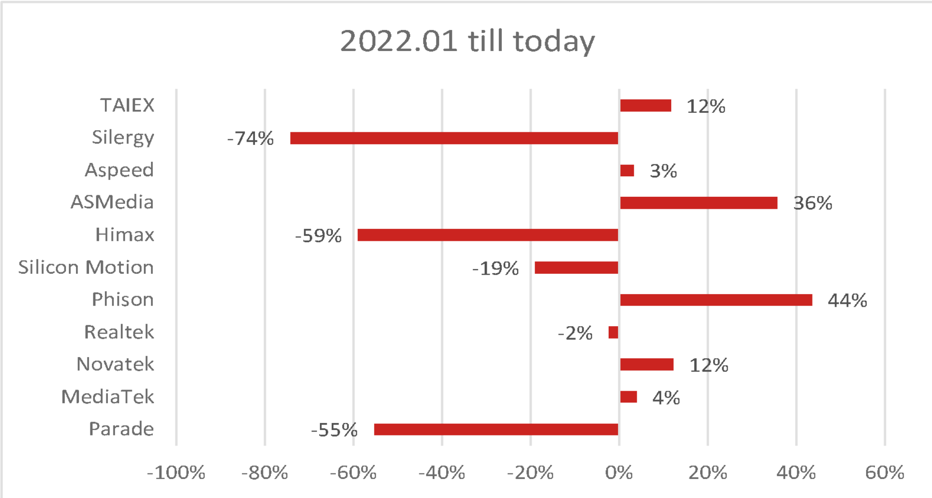
Source: Company data, Nomura research

Fig. 228: China semi companies' share price performance since Jan-2023



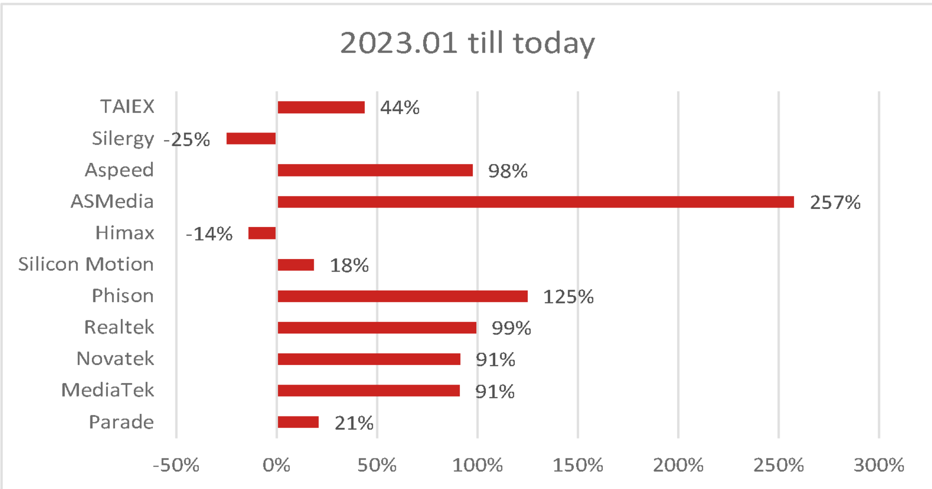
Source: Bloomberg Finance L.P., Nomura research

Fig. 229: Taiwan fabless share price performance since Jan-22



Source: Company data, Nomura research

Fig. 230: Taiwan fabless share price performance since Jan-2023



Source: Company data, Nomura research

Global Markets Research
12 April 2024

Suzhou Novosense Microelectronics 688052.SS 688052 CH

EQUITY: FABLESS

Pace of recovery lags behind peers

Recovering from 2023 trough, but earnings may not reach breakeven until 2025; initiate coverage at Neutral

Rating Starts at	Neutral
Target price Starts at	CNY 105.00
Closing price 9 April 2024	CNY 94.37
Implied upside	+11.3%
Market Cap (USD mn)	1,859.4
ADT (USD mn)	29.9

Action: Initiating coverage at Neutral with a TP of CNY105

We initiate coverage of Novosense, one of the leading signal chain analog IC fabless firms in China (next to 3Peak [688536 CH, Buy] in terms of sales and market cap). Novosense has high sales exposure to industrial and auto applications, which contributed 80-90% of its total sales in 2023F. We expect the company to return to the growth path in 2024, after it experienced considerable inventory adjustment by customers in 2023. We however assign Novosense a Neutral rating as we envisage: **(1) a breakeven is only likely in 2025F –** we still see Novosense incurring a small loss at the operating level in 2025F due to a slower GPM recovery and higher option cost despite a sequential decline in its operating loss over 2024-25F; **(2) higher GPM pressure –** due to higher price pressure from auto customers during the 2023 industry downturn (*Fig. 237*); and **(3) higher inventory overhang –** given stronger inventory pressure than peers (*Fig. 238*, *Fig. 239*). Our TP of CNY105 is based on 2.5x 2025F BVPS of CNY43.1. The 2.5x target P/B is at the mid- to low-end of the P/B range (2x-5x) between Jan 2023 and Apr 2024 when its earnings were at loss-making or breakeven levels. The stock is trading at 2.2x of 2025F BVPS.

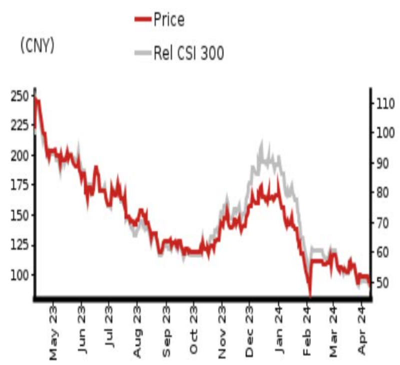
Novosense mainly has exposure to industrial and auto applications

Industrial analog market demand is highly influenced by the macro economy. The recent US PMI data seems to suggest that the macro cycle is bottoming (*Fig. 240*), and China's manufacturing PMI in March has recovered back above 50 (*Fig. 241*) after remaining below 50 for most of 2023 – possibly a reason why ADI (ADI US, Not rated), the largest industrial signal chain vendor globally by sales, expects a recovery for industrial applications in 2H24. Novosense has 50-55% sales exposure to industrial applications, which should benefit it if such a recovery materializes. However, a potential recovery for Novosense may trail that of 3Peak owing to the former facing greater inventory overhang and higher pricing pressure (from its auto exposure – accounted for 30-35% of Novosense sales in 2023F; since late 2022, Novosense has faced significant pricing pressure from Texas Instruments (TI) [TXN US, Not rated]). Thus, we prefer 3Peak in the China industrial analog IC sector. For Novosense, we project a 22% sales CAGR over 2023F-25F owing to its end-demand recovery, market share gains in local customers and a recovery of GPM to the low-mid-40% level, but we believe its earnings will remain negative in 2024F and should break even in 2025F. Downside risks: (1) slower-than-expected China automotive demand growth; (2) stronger-than-expected price competition; and (3) slower-than-expected China analog semi localization rate growth.

Year-end 31 Dec	FY22	FY23F		FY24F		FY25F	
Currency (CNY)	Actual	Old	New	Old	New	Old	New
Revenue (mn)	1,670	0	1,311	0	1,500	0	1,940
Reported net profit (mn)	251	0	-236	0	-166	0	9
Normalised net profit (mn)	251	0	-236	0	-166	0	9
FD normalised EPS	1.77		-1.67		-1.18		6.67c
FD norm. EPS growth (%)	12.0		-194.1		–		–
FD normalised P/E (x)	53.3	–	–	–	–	–	1,414.7
EV/EBITDA (x)	51.0	–	–	–	–	–	378.3
Price/book (x)	2.1	–	2.1	–	2.2	–	2.2
Dividend yield (%)	0.7	–	–	–	–	–	–
ROE (%)	7.1		-3.7		-2.7		0.2
Net debt/equity (%)	net cash		net cash		net cash		net cash

Source: Company data, Nomura estimates

Relative performance chart



Source: LSEG, Nomura

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Key data on Suzhou Novosense Microelectronics

Performance

(%)	1M	3M	12M		
Absolute (CNY)	-11.3	-34.1	-56.9	M cap (USDmn)	1,859.4
Absolute (USD)	-11.9	-34.8	-59.1	Free float (%)	22.6
Rel to CSI 300	-11.0	-41.4	-42.6	3-mth ADT (USDmn)	29.9

Income statement (CNYmn)

Year-end 31 Dec	FY21	FY22	FY23F	FY24F	FY25F
Revenue	862	1,670	1,311	1,500	1,940
Cost of goods sold	-401	-835	-782	-865	-1,066
Gross profit	461	835	529	634	874
SG&A	-100	-239	-368	-361	-404
Employee share expense	-107	-404	-656	-525	-514
Operating profit	254	193	-495	-252	-43
EBITDA	276	240	-436	-199	35
Depreciation	-19	-39	-59	-53	-78
Amortisation	-4	-8	0	0	0
EBIT	254	193	-495	-252	-43
Net interest expense	-3	24	13	7	5
Associates & JCEs					
Other income	-2	36	179	49	49
Earnings before tax	248	253	-303	-196	11
Income tax	-25	-3	67	29	-2
Net profit after tax	224	250	-236	-166	9
Minority interests	0	0	0	0	0
Other items	0	0	0	0	0
Preferred dividends					
Normalised NPAT	224	251	-236	-166	9
Extraordinary items					
Reported NPAT	224	251	-236	-166	9
Dividends	-2	-87	0	0	0
Transfer to reserves	221	163	-236	-166	9
Valuations and ratios					
Reported P/E (x)	59.7	53.3	–	–	1,414.7
Normalised P/E (x)	59.7	53.3	-56.6	-80.2	1,414.7
FD normalised P/E (x)	59.7	53.3	–	–	1,414.7
Dividend yield (%)	0.0	0.7	–	–	–
Price/cashflow (x)	132.7	–	–	523.6	710.1
Price/book (x)	24.3	2.1	2.1	2.2	2.2
EV/EBITDA (x)	48.8	51.0	–	–	378.3
EV/EBIT (x)	53.1	63.4	–	–	–
Gross margin (%)	53.5	50.0	40.4	42.3	45.1
EBITDA margin (%)	32.0	14.3	-33.3	-13.3	1.8
EBIT margin (%)	29.4	11.5	-37.7	-16.8	-2.2
Net margin (%)	26.0	15.0	-18.0	-11.1	0.5
Effective tax rate (%)	9.9	1.2	–	–	15.0
Dividend payout (%)	1.1	34.8	–	–	0.0
ROE (%)	–	7.1	-3.7	-2.7	0.2
ROA (pretax %)	–	6.1	-8.6	-4.2	-0.7
Growth (%)					
Revenue	256.0	93.8	-21.5	14.4	29.4
EBITDA	384.0	-13.2	-282.1	–	–
Normalised EPS	323.0	12.0	-194.1	–	–
Normalised FDEPS	323.0	12.0	-194.1	–	–

Source: Company data, Nomura estimates

Cashflow statement (CNYmn)

Year-end 31 Dec	FY21	FY22	FY23F	FY24F	FY25F
EBITDA	276	240	-436	-199	35
Change in working capital	253	-420	-218	139	-69
Other operating cashflow	-428	-49	259	85	53
Cashflow from operations	101	-229	-395	25	19
Capital expenditure	-187	-398	-200	-200	-199
Free cashflow	-87	-626	-595	-175	-180
Reduction in investments	1	-3,508	0	0	0
Net acquisitions					
Dec in other LT assets					
Inc in other LT liabilities					
Adjustments	0	-66	0	0	0
CF after investing acts	-86	-4,200	-595	-175	-180
Cash dividends	-2	-87	0	0	0
Equity issue	0	5,812	0	0	0
Debt issue	45	-66	0	0	0
Convertible debt issue					
Others	-3	-271	-1	0	0
CF from financial acts	39	5,388	-1	0	0
Net cashflow	-46	1,187	-596	-175	-180
Beginning cash	124	78	1,265	670	495
Ending cash	78	1,265	669	495	315
Ending net debt	16	-1,238	-642	-468	-288

Balance sheet (CNYmn)

As at 31 Dec	FY21	FY22	FY23F	FY24F	FY25F
Cash & equivalents	78	1,265	670	495	315
Marketable securities	0	3,464	3,464	3,464	3,464
Accounts receivable	111	198	54	234	149
Inventories	224	605	894	671	789
Other current assets	107	192	192	192	192
Total current assets	520	5,724	5,273	5,055	4,909
LT investments	0	44	44	44	44
Fixed assets	179	344	486	633	754
Goodwill	38	0	0	0	0
Other intangible assets	22	32	32	32	32
Other LT assets	82	716	716	716	716
Total assets	841	6,861	6,551	6,481	6,455
Short-term debt	94	20	20	20	20
Accounts payable	74	144	71	166	131
Other current liabilities	98	163	163	163	163
Total current liabilities	266	327	254	350	314
Long-term debt	0	7	7	7	7
Convertible debt					
Other LT liabilities	25	29	28	28	28
Total liabilities	291	363	289	385	350
Minority interest					
Preferred stock					
Common stock	76	101	101	101	101
Retained earnings	247	387	151	-15	-6
Proposed dividends					
Other equity and reserves	227	6,010	6,010	6,010	6,010
Total shareholders' equity	550	6,498	6,262	6,095	6,105
Total equity & liabilities	841	6,861	6,551	6,480	6,454
Liquidity (x)					
Current ratio	1.95	17.52	20.78	14.46	15.61
Interest cover	83.4	–	–	–	–
Leverage					
Net debt/EBITDA (x)	0.06	net cash	–	–	net cash
Net debt/equity (%)	2.9	net cash	net cash	net cash	net cash
Per share					
Reported EPS (CNY)	1.58	1.77	-1.67	-1.18	6.67c
Norm EPS (CNY)	1.58	1.77	-1.67	-1.18	6.67c
FD norm EPS (CNY)	1.58	1.77	-1.67	-1.18	6.67c
BVPS (CNY)	3.89	45.92	44.26	43.08	43.15
DPS (CNY)	0.02	0.62	0.00	0.00	0.00
Activity (days)					
Days receivable	32.5	33.7	35.0	35.1	36.0
Days inventory	140.6	181.2	350.0	330.9	250.0
Days payable	47.5	47.6	50.0	50.1	51.0
Cash cycle	125.6	167.3	335.0	315.9	235.0

Source: Company data, Nomura estimates

Company profile

Novosense is one of China's leading analog IC vendors, which ranked fifth largest fabless vendors in China listed analog market. It specializes in industrial and automotive products. Novosense focused on signal chain, power management IC (PMIC) and sensor products to provide customers total solution in analog products.

Valuation Methodology

Our TP of CNY105 is based on 2.5x 2025F BVPS of 43.1, at the mid-to-low-end of the 2x-5x PE range between Jan 2023-Mar 2024 when it was loss-making or at breakeven point. The benchmark index for this stock is SSE Composite Index.

Risks that may impede the achievement of the target price

Downside risks: 1) Slower than expected China automotive demand growth; 2) stronger than expected price competition; and 3) slower than expected China analog semi localization rate growth. Upside risks: 1) Faster inventory depletion than our expectation, 2) fewer pricing pressure than our expectation, and 3) lower wafer cost than our expected

ESG

Novosense continue to monitor its foundry partners to set reduction objectives for energy management greenhouse gas emissions, air pollution, water pollution, and waste management to minimize the environment impact of the production process. Novosense is also actively to engage in environment protection, shareholders and employee interest, and social responsibilities in operation.

Executive summary

We initiate coverage of Novosense with a **Neutral rating** and a TP of **CNY105**. Novosense is one of China's leading analog fabless companies, a vendor highly focused on the automotive and industrial markets. We believe Novosense can return to growth and gaining share in 2024F with its auto sales likely to nearly double from 2023F to 2025F. The company is the second-largest China analog signal chain focused fabless (next to 3Peak [688536 CH, Buy]) in terms of 2022 sales, and No #5 China analog IC fabless supplier, with a market share that we estimate at ~2.3% in 2022 (*Fig. 231*).

According to the company, it recorded c.USD152mn in analog sales in 2022 vs a c.USD6.5bn general purpose signal chain total addressable market (TAM). We note the company's market share has expanded since 2020, when it reported USD37mn of signal chain sales (c. 0.6% share vs market TAM of USD4.4bn in 2020, according to Gartner estimates). Other than signal chain products, Novosense also recorded around USD74mn in PMIC sales and USD17mn in sensor sales in 2022.

Among Novosense's business segments, industrial and energy are the major applications, at c.70% of sales combined in 2022, vs automotive at 23% and consumer at 7%. In 2022, Novosense grew auto sales by a significant +350% y-y, led by its successful product design-in for mass production for China EV customers, and we estimate it grew by 10%+ y-y in 2023F despite China auto semiconductor market's inventory adjustments. We expect auto accounted for 33% of total sales in 2023F, up from c.10% in 2021.

With the auto electrification megatrend continuing, Novosense should maintain its dominant position in auto isolator products among China analog IC vendors, in our view. As such, we think Novosense's progress in ramping up sales of automotive products remains the most important application to monitor. We expect Novosense to sustain its share gains in 2024-2025F, as we believe it will outpace that of global competitors in the China local analog market.

We prefer 3Peak over Novosense in the China industrial analog IC sector. The two major industrial/auto application focused analog IC companies in China had high industrial application sales exposure of 47% and 55% in 2023F, respectively, by our estimate. The two companies had similar sales growth patterns between 2018 and 2023 (*Fig. 236*), as both benefited from analog IC localization.

However, we think 3Peak managed inventory better by decreasing inventory level faster and reducing inventory days, and thereby protecting profitability better than Novosense despite substantial price cuts by global and local peers since 4Q22 (*Fig. 237* , *Fig. 238* , *Fig. 239*). We believe Novosense faced higher pricing pressure which resulted in a lower GPM from China auto customers given its high auto sales exposure (30%+ vs. 3Peak at 10-20%).

We project Novosense's auto sales will grow 60% from 2023F to 2025F, as we believe it will be able to expand its market share across local auto OEM customers. The growing market share in auto applications as well as industrial and energy applications, and Novosense's ambitions in acquiring more relevant analog businesses, such as its acquisition of KT Micro (unlisted) in 2023, will support Novosense in delivering a 20-30%+ sales CAGR over the next three years, in our view.

We initiate coverage of Novosense with a Neutral rating and a TP of CNY105, based on 2.5x our 2025F BVPS of CNY43.1. The 2.5x target P/B is at the mid- to low-end of Novosense's **2-5**x P/B band between Jan 2023 to Mar. 2023, when its earnings were at loss-making or breakeven levels.

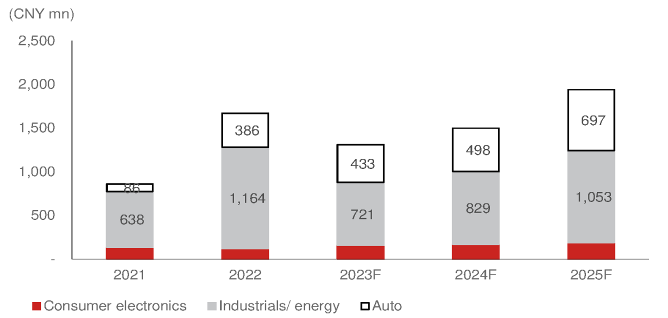
Downside risks: (1) Slower-than-expected China automotive demand growth; (2) stronger-than-expected price competition; and (3) slower-than-expected China analog semi localization rate growth. **Upside risks: (**1) Faster inventory depletion than our expectation, (2) fewer pricing pressure than our expectation, and (3) lower wafer cost than our expected.

Fig. 231: Listed China analog fabless companies ranking (by sales) - 2022

Rank	Company name	English name	Ticker	2022 analog sales (USD mn)
1	硅力杰	Silergy	6415 TT	767.0
2	圣邦	SGMicro	300661 CH	457.7
3	艾为电子	Awinic	688798 CH	300.0
4	思瑞浦	3 Peak	688536 CH	256.1
5	纳芯微	Novosense	688052 CH	237.6
6	杰华特	Joulwatt	688141 CH	201.6
7	上海贝岭	Belling	600171 CH	190.8
8	南芯	Southchip	688484 CH	186.8
9	韦尔	Will semi	603501 CH	173.0
10	晶丰明源	Bright Power	688368 CH	147.2
11	芯朋微	Chipown	688508 CH	103.3
12	英集芯	Injoinic	688209 CH	90.9
13	力芯微	Wuxi ETEK	688601 CH	90.8
14	富满微	Fine Made	300671 CH	88.9
15	帝奥	Dioo	688381 CH	79.3
16	必易微	Kiwi instruments	688045 CH	75.5
17	赛微	Cellwise	688325 CH	34.0

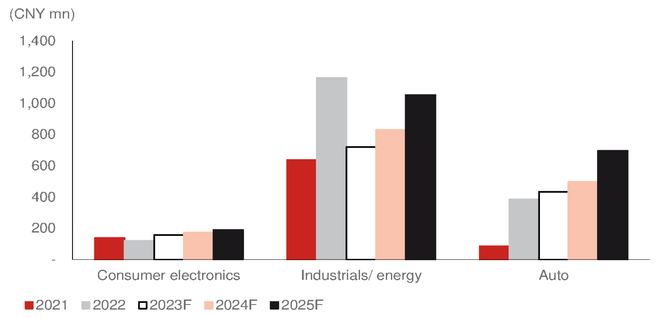
Source: Company data, Nomura research

Fig. 232: Novosense: Application sales



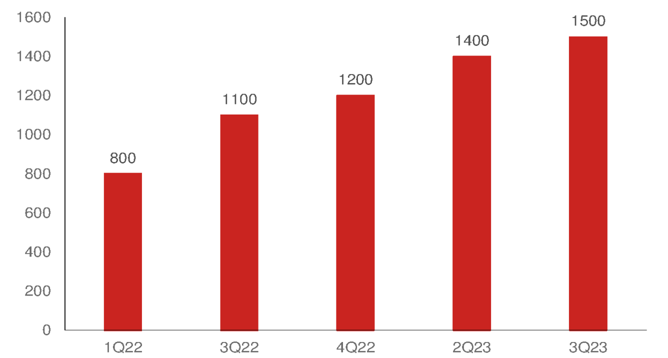
Source: Company data, Nomura estimates

Fig. 233: Novosense: Applications sales by year



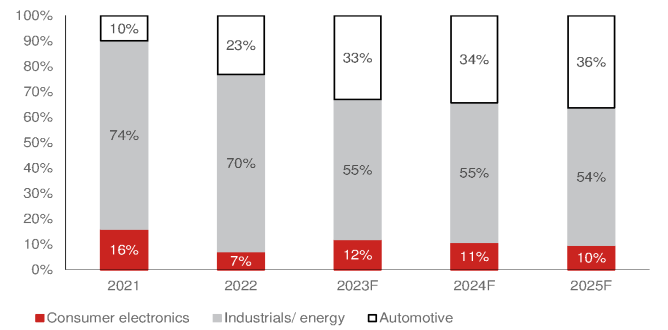
Source: Company data, Nomura estimates

Fig. 234: Novosense: growth in number of products



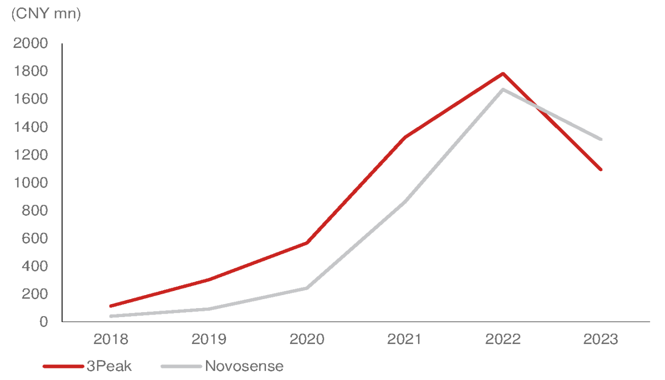
Source: Company data, Nomura research

Fig. 235: Novosense: application sales mix (%) in 2023F



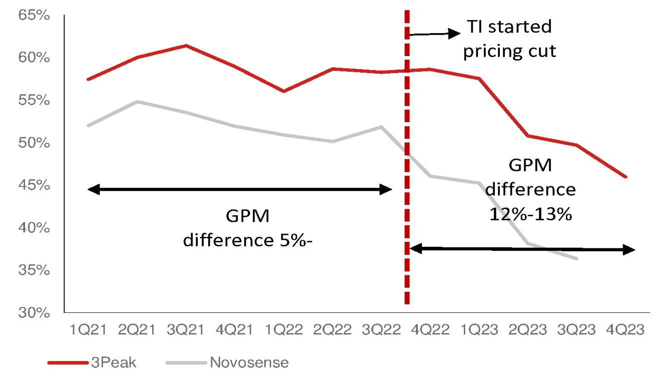
Source: Company data, Nomura estimates

Fig. 236: 3Peak and Novosense sales



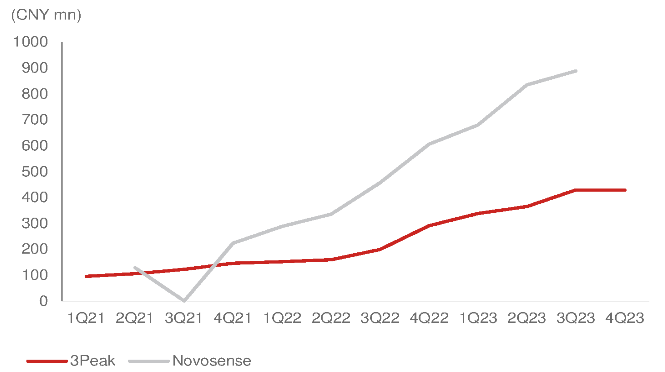
Source: Company data, Nomura research

Fig. 237: 3Peak and Novosense GPM



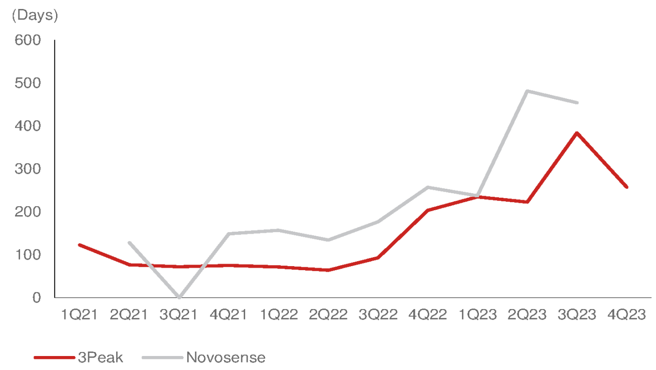
Source: Company data, Nomura research

Fig. 238: 3Peak and Novosense's inventories



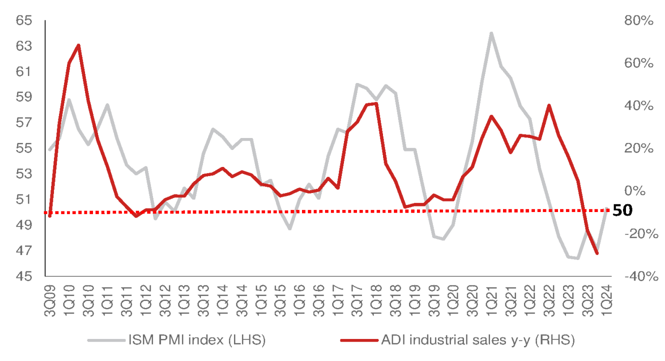
Source: Company data, Nomura research

Fig. 239: 3Peak and Novosense inventory days



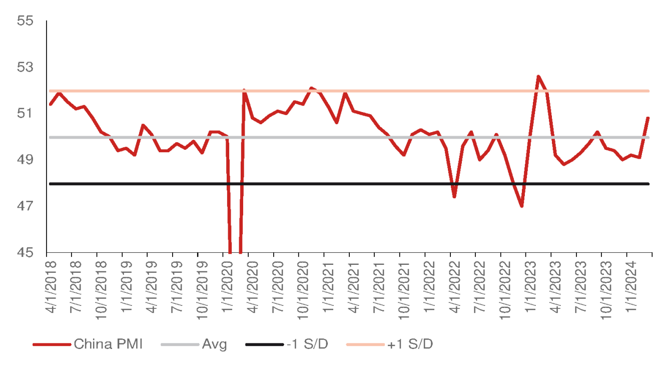
Source: Company data, Nomura research

Fig. 240: ADI industrial sales y-y vs. ISM PMI index



Source: Company data, Nomura research

Fig. 241: China manufacturing PMI



Source: Nomura research

Company background

We initiate coverage of Suzhou Novosense Microelectronics (Novosense) with a Neutral rating and a CNY105 TP. Novosense is one of the leading analog IC vendors in China with high sales exposure to high-end analog IC of industrial and automotive applications. We expect Novosense to record strong sales growth, as the industry cycle is bottoming out, and to sustain its market share gains in 2024-2025F.

Novosense, established in 2013, is one of the leading analog integrated circuit (IC) vendors in China, which started with signal conditioning sensor ASIC (application-specific integrated circuits); in recent years, the company has expanded its product portfolio to include signal chain IC and Power Management Integrated Circuits (PMIC). Novosense has a 5% market share in global isolator shipments, according to the company.

It has developed analog IC in high-end products within industrial and automotive applications, and currently has three major product lines: (1) signal chain IC; (2) PMIC; and (3) sensor products. It has more than 1,400 product models for the three product lines. The performance of some of its products is close to or even surpasses that of global peers, according to the company.

Industrial and automotive applications accounted for approximately 90% of total sales in 2022. In the industrial sector, the products are mostly related to energy systems, from power generation, power transmission, to power distribution, and finally to power consumption. Novosense has a broad spectrum of products for these applications, which are well recognized in terms of brand and product quality by its China customers. The sales mix by application has grown the most in the automotive segment. Novosense is one of the few companies in China that can supply isolators for vehicles. In recent years, owing to its strong performance, development capabilities and product diversification, the company has managed to enter the supply chain of several mainstream China car OEMs and achieved mass production.

KT Micro acquisition

Novosense in July 2023 announced the acquisition of a 33.6% stake in KT Micro (unlisted) for a total consideration of CNY504.5mn, at an implied valuation of less than CNY1.5bn. According to Novosense, management is in discussions with other shareholders to take control of the company.

Novosense product line: (1) Signal chain; (2) PMIC; and (3) Sensors

Novosense previously had three major product lines (*Fig. 244*): (1) driver and sampling; (2) isolator and interface; and (3) signal and sensors. In order to present information more accurately and clearly, the company re-categorised its products starting from 2023. Novosense integrated signal sensors into **sensor** products; moved signal/sensor ASIC, isolator/ interface, and sampling IC into **signal chain**; and moved driver IC into **PMIC**. Therefore, Novosense's current product lines are: **(1) Signal chain, (2) PMIC, and (3) Sensors.**

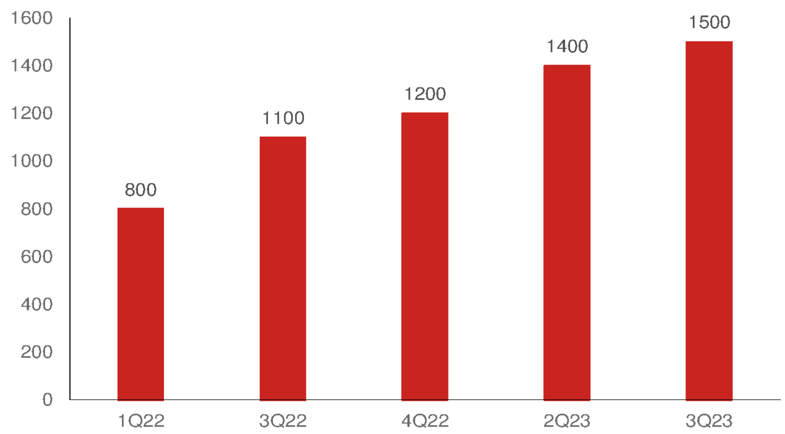
Novosense's products cover various of analog functions, and they have now grown to over 1,500 products (*Fig. 243*), nearly doubling within a span of one-and-a-half year since its listing in April 2022.

Fig. 242: Novosense: Three major product lines

Sensor Products	Signal Chain Products	PMIC Products
• Magnetic Sensor	• Sensor signal ASIC	• Gate Driver
• Pressure Sensor	• Isolator	• LED driver
• Thermal/Humidity Sensor	• General interface	• Power supply
	• Amplifier	• Motor driver
	• Data converter	• Power path protection

Source: Company data, Nomura research

Fig. 243: Novosense: growth in number of products



Source: Company data, Nomura research

KT Micro (昆腾微) products

KT Micro is an analog and mixed-signal fabless company, and its products include audio system on a chip (SoC) for consumer markets, such as wireless audio transceiver IC, which accounted for 82% of SoC sales and 69% of total sales in 2022, FM/AM receiver IC, and USB audio IC. On the other hand, KT Micro also has signal chain products for communication and industrial control applications – including analog-to-digital converters (ADC), digital-to-analog converters (DAC), and integrated ADC & DAC ICs – which accounted for 16% of total sales in 2022.

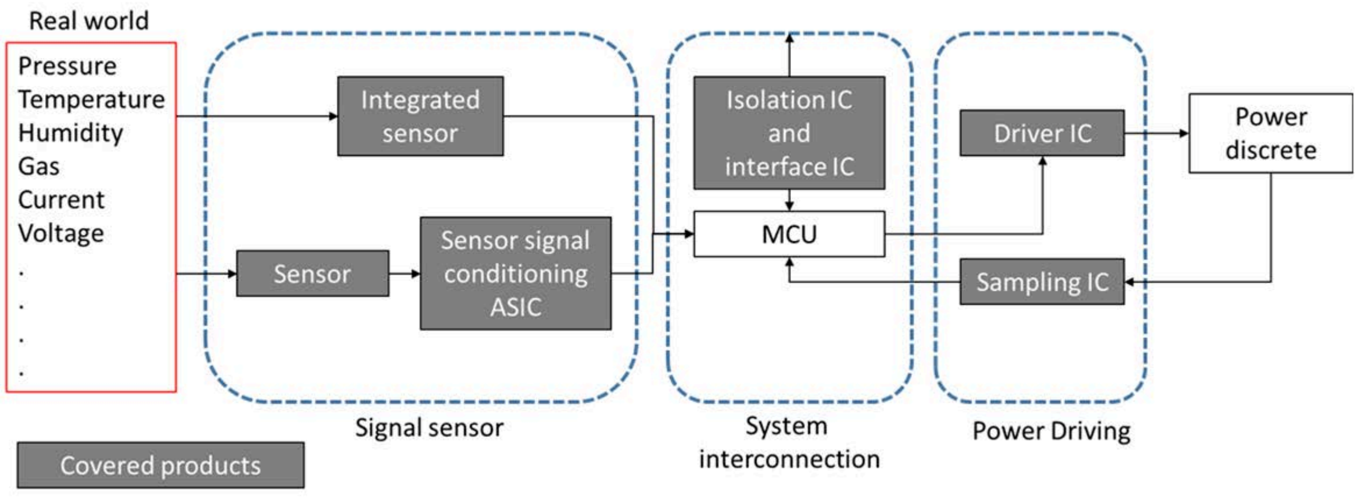
According to KT Micro, it has shipped over 200 mn SoC units since its establishment in 2006, and is one of the leading audio SoC companies in China. Furthermore, KT Micro is one of the dominant players in high-speed and high-precision DAC and ADC products in China. KT Micro recorded annual sales of CNY217mn/ CNY315mn/ CNY305mn, and GPM of 58.2%/ 60.0%/ 61.2% in 2020/2021/2022.

Fig. 244: Novosense: Product portfolio

Products (sales % in 2022)	Previous product category	Introduction	Products	Global and China Peers
Signal Chain (63%)	Digital isolation IC	Converts the signal input and outputs it to achieve electrical isolation between the input and output ends	 Standard digital isolation IC; Digital isolation IC with integrated power supply	Global: ADI, TI, Melexis, Infineon, Renesas, China: 3Peak
	Interface IC	The IC for signal transmission between different electronic systems	 Isolated interface IC; Non-isolated interface IC	
	Sampling IC	The IC to understand high precision signal acquisition and transmission	 Sampling IC	
	Sensor signal conditioning ASIC	Highly integrated IC for sampling and processing the signal output of the sensor sensitive element	 MEMS Pressure sensor; MEMS microphone/ Acceleration sensor/ current sensor	
PMIC (31%)	Driver IC	IC for power conversion, testing and other power management responsibilities	 Driver IC	Global: TI, ADI, Infineon, Renesas, China: Silergy, SG Micro
Sensor (7%)	Integrated sensor	The sensor integrated with the sensitive components	 Integrated temperature sensor; Integrated pressure sensor; Magnetic sensor	Sensata, BOSCH, Denso, Delphi
KTmicro(昆腾微)	Analog and Mix-signal IC	Specialized in audio SoC and signal chain products, including wireless IC, USB audio IC, ADC-DAC IC	 Audio IC; Wireless IC; Signal Chain	Global: ADI, TI, Cirrus Logic China: Audio SoC (Beken, Bestechnic), Signal chain (SG Micro, 3Peak)

Source: Company data, Nomura research

Fig. 245: Novosense: Product coverage



Source: Company data, Nomura research

Signal chain: (64% sales in 2023F)

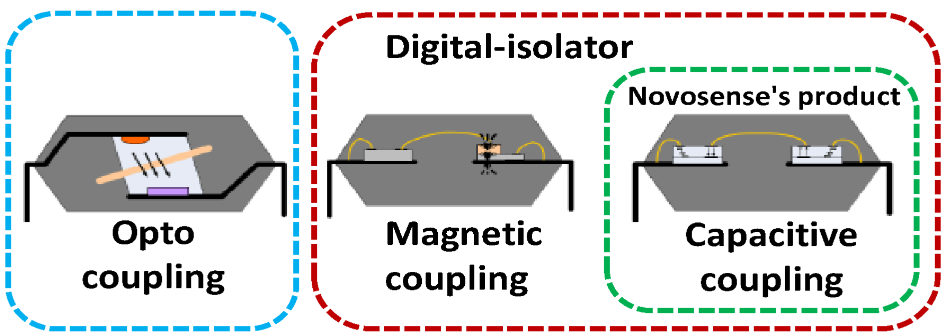
Novosense combines isolation/interface/sampling/sensor signal conditioning ASIC in its signal chain product line

1. **Isolation IC**: It is the component that isolates the electric signal output and input, protects the signal transmission between the strong and the weak current circuit, and makes the electronic system to achieve higher robustness and longer life. When involving the connection between high- and low-voltage signal transmission, the device requires isolation ICs for current isolation. It is widely used in industrial, automotive, communications, and power electronics.

The isolation products are of three different types, and opto coupling is the oldest and the current mainstream isolation solution. According to Omdia, opto-coupling accounted for over 70% share in isolation in 2022. However, more and more applications are starting to adopt isolators in devices; internal combustion engines (ICE) rarely use isolators in vehicles due to their distribution electronic system architecture. After vehicle electrification and electrics/electronics (E/E) architecture transition to domain and zonal system, isolators have become a necessary component to ensure the safety and signal transmission in different voltages in the vehicle. Considering the higher product requirements of temperature, signal transmission speed, and power consumption in industrial and auto applications, we believe digital isolator will continue to drive digital isolator to gain share from opto solutions. And Novosense is the largest local digital isolation vendor in China.

Fig. 246: Isolator structure comparison

Novosense focuses on capacitive coupling isolator



Source: Company data, Nomura research

Fig. 247: Isolator product comparison

	Opto	Magnetic	Capacitive
Signal	Optical	Magnetic	Electric
Input	LED	CMOS+coil structure	CMOS
Output	Phototransistor	CMOS+coil structure	CMOS
Isolation layer materials	Polyimide	Polyimide	SiO2
Life	Short	Long	Long
Speed	Slow	Fast	Fast
Size	Big	Small	Small
Temperature range	Low	High	High
Cost	Low	High	High
Power consumption	High	Low	Low
Note			Can't pin-to-pin to opto-coupling product

Source: Nomura research

Isolator: Automotive and industrial are key applications

According to Novosense, in 2020, industrial and auto applications accounted for 44% of the total digital isolation IC market, and in 2026, industrial and auto should account for 46%. This market is dominated by global analog leaders, including TI (TXN US, Not rated), Silicon Labs (SLAB US, Not rated), ADI (ADI US, Not rated), Broadcom (AVGO US, Not rated), and Infineon (IFX GY, Not rated), which account for 40-50% of the global market. The other vendors include Rohm (6963 JP, Buy), Maxim (unlisted), Vicor (VICR US, Not rated), and ON Semi (ON US, Not rated). According to the company, the isolation TAM is around USD1bn globally, and USD500mn+ in China.

- **In Auto: more signal connection and higher voltage to drive market growth**

The EV trend provides a strong growth opportunity for isolation ICs, because in EV high-wattage electronic systems, including OBC, Battery Management System, DC/DC converter, Motor driver, CAN/LIN communication system, etc., isolation ICs become the key electronic components in EV's power and signal transmission.

In addition, the EV voltage system will also be upgraded from 400V to 800V. With the voltage increase, the power charging time can be decreased, heat loss can be reduced, the weight of the vehicle can also be reduced, and the power structure can be strengthened. The voltage isolation requirement will also have higher standard, and the consumption of isolation IC will also increase.

Other than that, the latest Tesla (TSLA US, Not rated) Cybertruck upgraded its electric system into **48V** from current EV mainstream of 12V. The upgrade has several advantages. **1) Demand requirement**: higher voltage can meet the requirement to the increasing demand in vehicle control system. **2) Power consumption:** the 4x increase in voltage makes the current passing through smaller, which could reduce the loss during energy transmission, and can reduce the temperature. **3) Cost:** It could reduce the wire thickness in vehicle and save cost.

- **In industrial: more device connection and safety standard to drive demand**

In industrial automation, human-computer interaction is becoming more frequent. In order to ensure the safety of operators, the signal transmission between high and low voltage must be isolated to avoid machine failure leading to any injury. In addition, different machine has different operating voltage. When machine connecting with each other, isolation IC are also needed for transmission protection. Thus, the higher the degree of automation, the higher the requirements for precise control, the higher demand for isolation ICs when automation trend continues to increase.

2. **Interface IC**: interface IC is for communication functions in general or specific protocols, which are widely used in signal transmission between different electronic systems. For example, the CAN and LIN interface are both for vehicles. When signal transmission, data transmission must meet the corresponding protocol standard, while CAN is suitable for high-reliability, high-common-mode voltage devices such as auto, and LIN bus communication is largely used in auto pressure sensing, air-conditioning, and thermal management systems. According to IC Insight estimates, the global interface market size was about USD2.8bn in 2021.

3. **Sampling IC**: Sampling IC captures the continuous analog signal, and then translates it

into a digital signal, which is mainly used in a closed-loop control system. Analog to Digital converter (ADC) is the most common sampling IC, which converts a continuous signal in analog form into a discrete signal in digital form for electronics to read the signal. Sampling rate (how often the input analog signal is converted; the higher the sampling rate, the more continuous the digital signal obtained) and resolution (the higher the resolution, the more accurate the signal obtained) are the key to the ADC product. The major competitors are all global peers, including ADI and TI.

4. **Sensor signal conditioning ASIC**: The product is a highly integrated ASIC for sampling and processing the output signal of sensor sensitive component. It can measure various environment data (pressure, sound, temperature, humidity, magnetic field, light intensity), can amplify the signal, and realize the conversion of analog and digital signals. In addition, this product needs to be calibrated to remove the output deviation caused by environmental factors. Novosense's ASIC products are mostly used with its micro-electro-mechanical systems (MEMS) components. According to Novosense, the global industrial MEMS sensor market will grow from USD18.2bn in 2020 to USD29bn in 2025 with a CAGR of 9.8%.

In the auto MEMS market, currently it is mainly adapted with acceleration sensors, pressure sensors, and gyro sensors. However, for the precision and safety of auto system control, the MEMS sensors equipped in vehicles continue to increase. According to Novosense, it had a 18.7% share in the China sensor signal ASIC market in 2020.

PMIC: (27% sales in 2023F)

Previously as Driver IC: Amplifies the signal of the control circuit to drive the circuit of the power discrete. By controlling the current signal of the logic IC, including amplifying the voltage range and enhancing the current output capability, it can quickly turn on and off the power device. Novosense products are widely used in energy industry and auto applications, such as power supply and power protection.

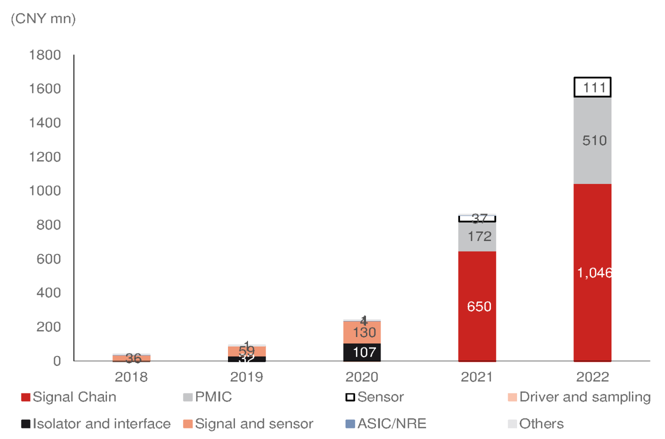
Sensor: (10% sales in 2023F)

Previously as Integrated Sensor: Sensor products include magnetic sensors, pressure sensors, and temperature/humidity sensors. In recent years, Novosense has expanded into integrated products from ASIC products previously, launching pressure sensors, temperature sensors, and magnetic sensors. Novosense continues to increase its integrated sensor sales mix, from 3% in 2018 to 10% in 1H21. Currently, the global magnetic sensor market is dominated by Allegro (ALGM US, Not Rated) in the US. According to Novosense prospectus, the market size was about USD2.6bn in 2021, and it is expected to grow to USD4.5bn by 2027 (CAGR 9%), and auto is the main application market, accounting for a 65% share in 2021.

Product sales

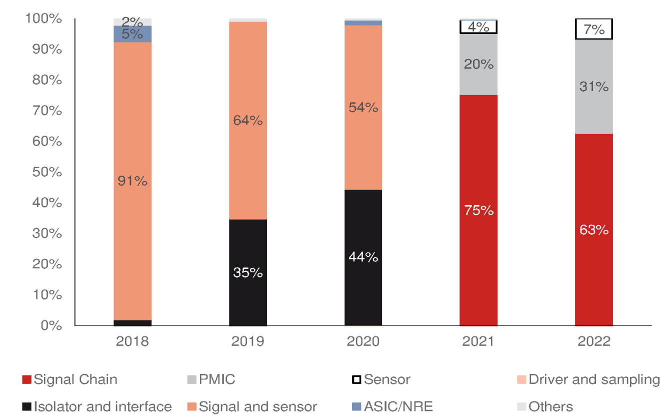
The three major products have maintained strong growth in the past few years, and the revenue between 2018 and 2022 achieved a CAGR of 154%. The current product revenue breakdown is mainly due to Signal chain (68%), followed by PMIC (31%), and Sensors (7%) and the three product lines recorded 61%, 197% and 202% y-y sales growth, respectively, in 2022. We believe the strong growth came from the application in industrial and auto new product enter mass production and market share gains.

Fig. 248: Novosense: Product sales



Source: Company data, Nomura research

Fig. 249: Novosense: Product mix

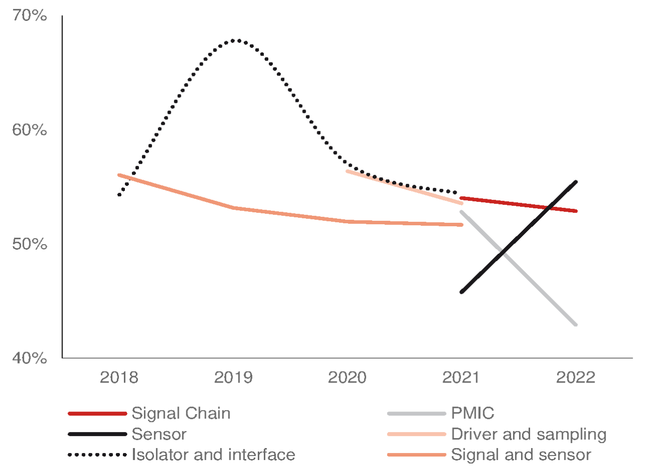


Note: Novosense change its product portfolio catagories in 2022
Source: Company data, Nomura research

GPM: PMIC faced pricing pressure in 2022

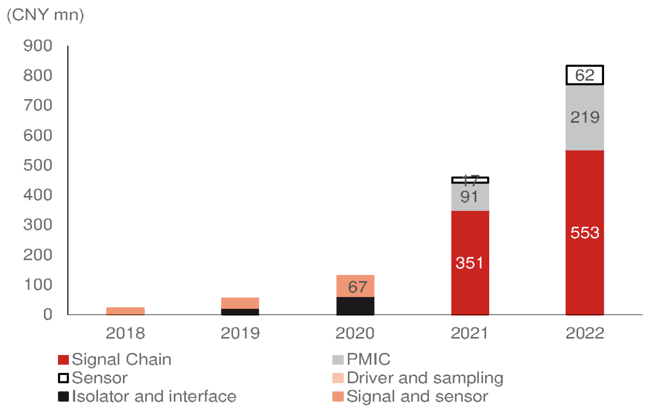
Novosense products' GPM were high in the early stage when Signal chain and Sensor's GPM in 2022 remained stable y-y, but PMIC GPM declined significantly, from 53% in 2021 to 43% in 2022. We believe Novosense has more pin-to-pin in its PMIC product line, and pin-to-pin products are easy for customers replacing suppliers. Therefore, Novosense faced severe pricing pressure in PMIC from competitors in 2022, resulting in lower GPM. However, Novosense's high wafer price by LTA ended in 2H23, and the company expects to consume lower wafer cost in 2024. According to management's estimate, 2024 GPM can gradually be improved to above the 40% level.

Fig. 250: Novosense: Product portfolio GPM trend



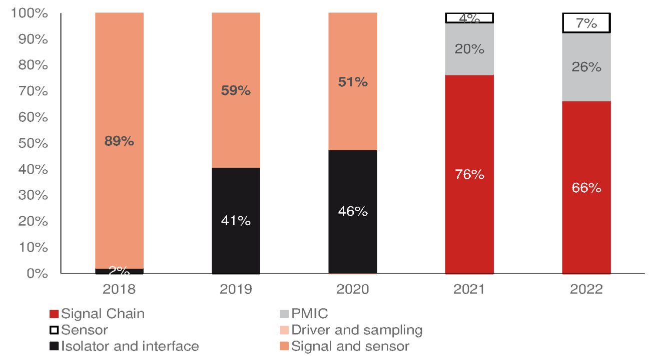
Source: Company data, Nomura research

Fig. 251: Novosense: Gross profit by product



Source: Company data, Nomura research

Fig. 252: Novosense: Gross profit mix (%)



Source: Company data, Nomura research

Application sales: Auto sales continue to outgrow others

Novosense's application sales were mainly driven by industrial/energy (70%) and automotive (23%) in 2022, which grew 82% and 349% y-y (*Fig. 253*), respectively. The strong auto sales growth in the past two years was largely due to: **(1)** mass production of new products – Novosense's new isolation products have undergone design-in to customers and entered the production stage, and the magnetic current sensor has also gone through continuous design-in to new customers; **(2)** auto LED headlight driver and power supply products are gradually being mass-produced as well; **(3)** the share in the customers has also been increased.

The strong industrial application sales growth mainly came from expanding customer coverage and increasing market share of its existing customers. In addition, new products are being continuously adopted by the leading local customers, in which Novosense has continued to increase market share. Besides, in order to address its global customers' supply chain localization needs in China, Novosense continues to expand its overseas customer coverage in the China market.

Auto products

Sensors: Magnetic sensors and pressure sensors are its major products, and the company is developing angle sensors and speed sensors, too.

Signal chain: Auto-grade sensor signal conditioning ASIC, and general signal chain product (ADC), and specialized application-specific standard parts (ASSP) and SoC products. Interface: mass production CAN/LIN networking products.

PMIC/ Driver: Isolated gate driver, motor driver, lighting driver in auto application, and in power-related: auto power supply IC, low-dropout regulators (LDO), and highly-integrated

PMIC. According to Novosense, current auto application products are pin-to-pin product, which may face higher pricing pressure, but Novosense also collaborates with customers to develop custom-made products to offset the pricing impact.

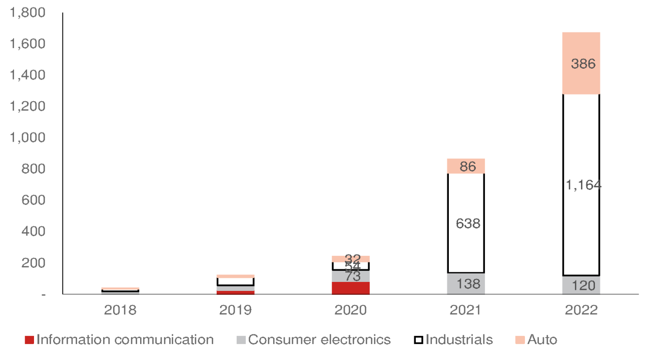
Novosense can supply CNY400 (~USD50-60) in total chip value in one vehicle, according to management estimates.

Industrial products

Novosense's industrial application product covers many end-markets and electronic products. In industrial/energy applications, Novosense is one of the few suppliers to provide total solutions including general isolator, isolation interface, driver, current sensor, and isolator sampling. In 2022, industrial applications accounted for 70% of its total sales. Among industrial products, solar energy storage accounted for 20% of total industrial application sales, while digital power IC in PC/server, base station, and charging pile accounted for another 20%of total sales. Industrial electric driver inverter and industrial server system, and power grid and electrical applications accounted for the remaining 30% of total sales.

Fig. 253: Novosense: Application sales

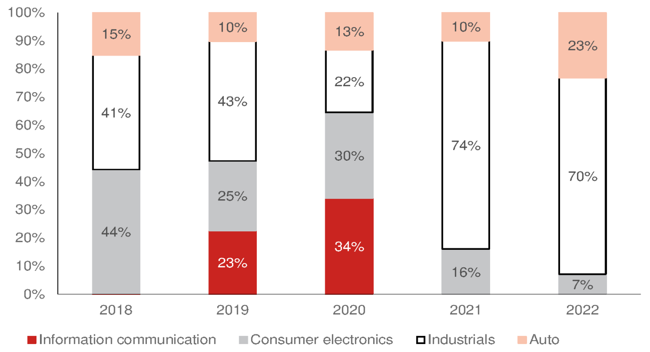
Auto sales increased significantly in 2021-2022



Source: Company data, Nomura research

Fig. 254: Novosense: Application sales mix(%)

Auto sales outgrew corporate average to a significant sales mix of 23% in 2022.



Source: Company data, Nomura research

Key supply-chain partners

Novosense enjoys rapid growth through its fabless model by relying on the support of its supply-chain partners. Its foundry partners include SMIC (981 HK, Neutral), Dongbu HiTek (000990 KS, Not rated), TSMC (2330 TT, Buy), X-fab (unlisted) and other large local and oversea foundries. The wafer process is mainly using 0.13mm/0.18mm BCD processes.

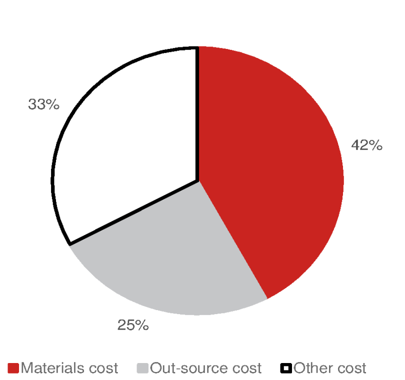
According to Novosense, SMIC was the largest foundry supplier (30% of total cost) in 1H21, followed by Dongbu Hitek (000990 KS, Not Rated). Among OSAT suppliers, ASE (3711 TT, Neutral) is the main OSAT supplier, followed by JCET (600584 CH, Buy). However, Novosense also has its own testing capacity and its only outsources 60-70% to other OSAT suppliers.

Fig. 255: Novosense: Top 5 suppliers list and percentage

Suppliers	2018	2019	2020	1H21	2022
1	Dongbu Hitek	Dongbu Hitek	Dongbu Hitek	ASE	Supplier A
2	TSMC	ASE	ASE	SMIC	Supplier B
3	ASE	TSMC	SMIC	Dongbu Hitek	Supplier C
4	SMIC	SMIC	TSMC	TSMC	Supplier D
5	Xiangyang Chengxin	Xiangyang Chengxin	JCET	JCET	Supplier E
Amount					
1	9.6	16.5	60.6	71.1	464.6
2	5.7	10.5	41.0	58.8	395.1
3	1.9	6.6	27.8	40.3	138.3
4	1.7	5.8	10.5	4.2	27.4
5	1.1	2.2	3.9	3.4	23.6
Mix					
1	42%	34%	34%	36%	40%
2	25%	22%	23%	30%	34%
3	9%	14%	16%	20%	12%
4	8%	12%	6%	2%	2%
5	5%	5%	2%	2%	2%

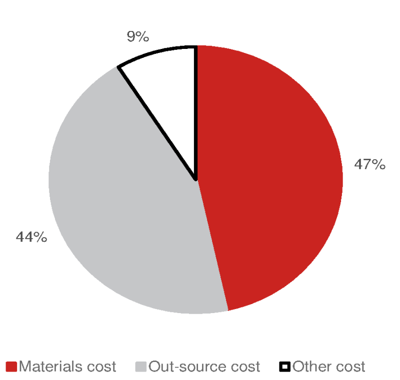
Note: Novosense did not disclose 2H21 suppliers detail
Source: Company data, Nomura research

Fig. 256: Novosense: sensor product cost structure



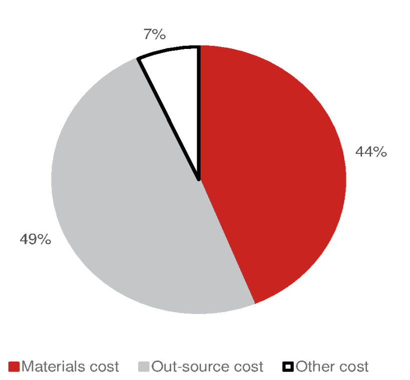
Source: Company data, Nomura research

Fig. 257: Novosense: signal chain product cost structure



Source: Company data, Nomura research

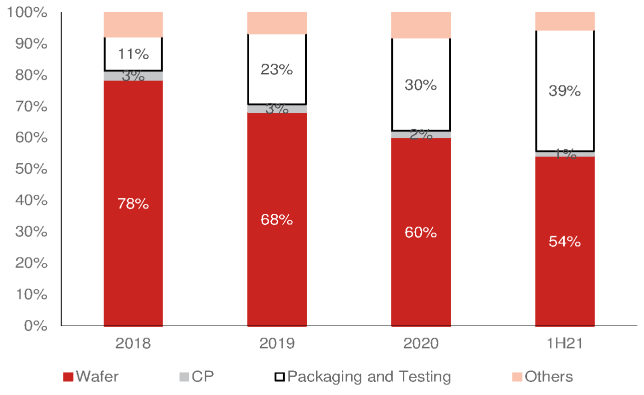
Fig. 258: Novosense: PMIC product cost structure



Source: Company data, Nomura research

Fig. 259: Novosense: Cost structure

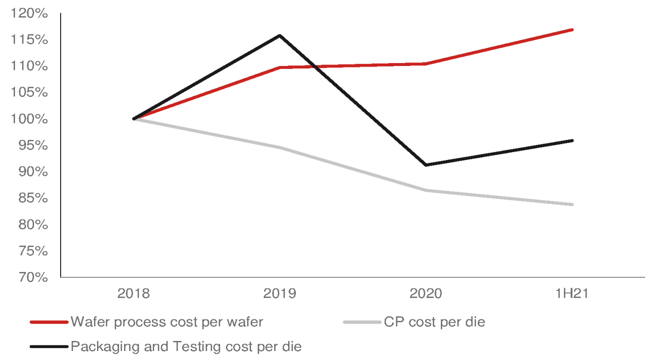
The packaging and testing cost % increase due to product mix change by higher sales in PMIC and automotive application.



Source: Company data, Nomura research

Fig. 260: Front-end wafer cost continue to increase

... back-end OSAT cost declined



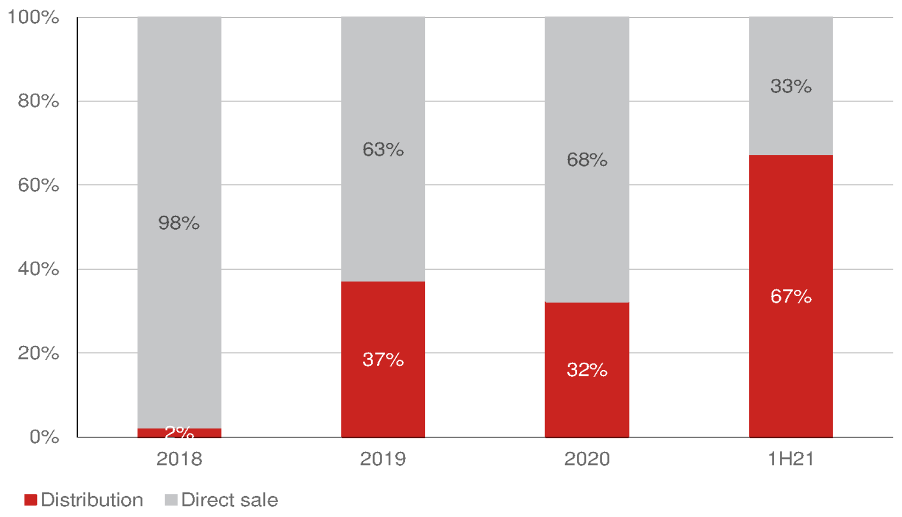
Note: Set 2018 as 100%

Source: Company data, Nomura research

Major customers

At the beginning of a company's business development stage, direct sales are key. Direct sales are a beneficial way for a company to catch customer needs in the first place, and can allow them to accurately control the direction of product research and development, and better serve customers. After 2019, there has been a surge in number of distribution selling models, driven by: **1)** some customers designating companies to sell products to them through distributors; and **2)** expanded product categories, and the expansion of application fields require the assistance of distributors to achieve fast and increased sales scale, and distributors can also provide better service and management experience to customers. Therefore, some small customers and orders have been handed over to distributors. With the continuous launch of new products, distributors can also help expand market share and raise brand awareness.

Fig. 261: Novosense: distribution and direct sales



Source: Company data, Nomura research

Therefore, the list of the top five customers has also changed due to its changed selling model. The largest customer in 1H21 was Nanjing grnuo (unlisted) distributor, which accounted for 43% of the company’s total sales. The distributor is responsible for the expansion of isolator/interface, driver/sampling business. The other top 2-5 customers of Novosense are all direct sales customer groups. However, the top five customers in 2022 accounted for only 13% of Novosense's total sales. Novosense has significantly reduced its customer concentration risk.

Among its end-customers, the industrial energy category includes Sungrow (阳光电源

,300274 CH, Buy), Inovance Technology (汇川技术, 300124 CH, Buy), Zhixinwei (智芯微, unlisted) and ZTE Communications (中兴通讯,763 HK, Buy). Its automotive customers include BYD (1211 HK, Buy), Dongfeng Motor (东风汽车, unlisted), Wuling Motors (五菱汽车, unlisted), Great Wall Motors (长城汽车, unlisted), and SAIC (上汽大众, unlisted).

Fig. 262: Novosense: Major customers

	2018		2019		2020		1H21		2022	
1	MiraMEMS (苏州明皜) (signal and sensor)	24%	Ameston(亚美斯通) (isolator and interface)	26%	Customer X (isolator and interface)	17%	Nanging grnuo Distributor(南京基尔诺) (driver and sampling, isolator and interface)	43%	Customer A	13%
2	Will MEMS(无锡韦感, previous name上海盘巨) (signal and sensor)	9%	MiraMEMS (苏州明皜) (signal and sensor)	17%	Nanging grnuo Distributor(南京基尔诺) (driver and sampling, isolator and interface)	10%	MiraMEMS (苏州明皜) (signal and sensor)	4%	Customer B	9%
3	OULD(深圳欧利德) (signal and sensor)	6%	Sinomags(宁波希磁) (signal and sensor, NRE)	5%	MiraMEMS (苏州明皜) (signal and sensor)	9%	Smartchip(智芯) (isolator and interface)	4%	Customer C	7%
4	GSR sensor (NRE)	5%	Avnet(distributors) (isolator and interface)	4%	Ameston(亚美斯通) (isolator and interface)	6%	Teampo(深圳霆宝) (isolator and interface, driver and sampling)	3%	Customer D	6%
5	Sinomags(宁波希磁) (signal and sensor, NRE)	4%	GMEMS(通用微) (signal and sensor)	3%	Sinomags(宁波希磁) (signal and sensor, NRE)	4%	Will MEMS(无锡韦感) (signal and sensor)	3%	Customer E	5%
Top 5		49%		55%		46%		57%		40%

Source: Company data, Nomura research

Financials

Novosense recorded strong sales and profit growth after its IPO in April 2022, but growth rates have slowed down due to increased sales scale, weak macroeconomic environment/demand, customers' inventory adjustments, and price competition in 2023. However, we expect Novosense's to record a sales CAGRs of 22% over 2023-25F, supported by industrial and auto market recoveries and China market-share gains, but we expect earnings to remain negative in 2024F and to only reach breakeven in 2025F.

Novosense's GPM is also affected by the downturn in the global semi cycle and has faced pricing competition since late-2022. The price cut caused Novosense's GPM to decline from 3Q22 of 52% to 3Q23's 36%. Factoring in likely declining GPM and higher operating expense, we expect Novosense's 2023F results to be negative/loss-making, with EPS of -CNY1.7. However, we expect that Novosense will only be able to return to low- to mid-40s level GPM in 2024-25F (vs low- to mid-50s between 1Q21 and 3Q22) on higher pricing pressure from auto customers and inventory levels.

Operating ratio

We estimate inventory days and cash cycle days increased in 2023 due to inventory corrections in industrial and auto products. But we expect inventory to gradually improve in 2024F-25F as inventory continues to be digested by customers.

Leverage and coverage ratio

Novosense has announced that it will invest in packaging and testing capacity for custom-made and pressure sensor products, which could reduce cost, as they tend to have low production volume. The company still has a strong cash position on hand, according to management; hence, we believe it does not have to raise debt to fulfil its capital needs. We believe Novosense can sustain healthy debt and debt-to-equity ratios during 2023-25F, and we project both its current ratio and quick ratio can be sustained above 1.0, in view of its few fixed-asset investment requirement, sales growth momentum, and strong profitability.

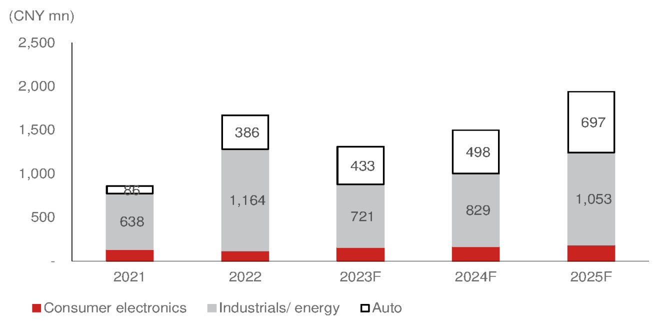
Sensitivity analysis

Fig. 263: Sensitivity analysis of Novosense's GPM recovery in 2024F-25F

	2024F			2025F		
Lower wafer cost and eased pricing pressure benefit GPM recovery	38%	42%	45%	40%	45%	50%
		Base case			Base case	
Revenues	1,500	1,500	1,500	1,940	1,940	1,940
Gross profit	570	634	675	776	874	970
Operating profit	(316)	(252)	(211)	(142)	(43)	52
Pre-tax profit	(260)	(196)	(155)	(87)	11	107
Net profit	(221)	(166)	(132)	(74)	9	91
EPS (TWD)	(2.2)	(1.2)	(1.3)	(0.7)	0.1	0.9
Margins (%)						
Gross margin	38%	42%	45%	40%	45%	50%
Operating margin	-21%	-17%	-14%	-7%	-2%	3%
Pre-tax margin	-17%	-13%	-10%	-5%	1%	5%
Net margin	-15%	-11%	-9%	-4%	0%	5%

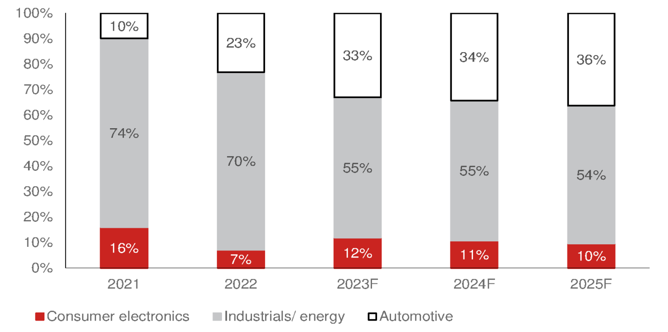
Source: Nomura estimates

Fig. 264: Novosense: Application sales



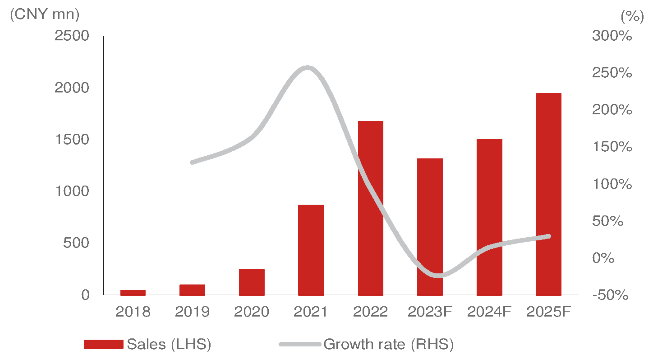
Source: Company data, Nomura estimates

Fig. 265: Novosense: Application sales mix (%)



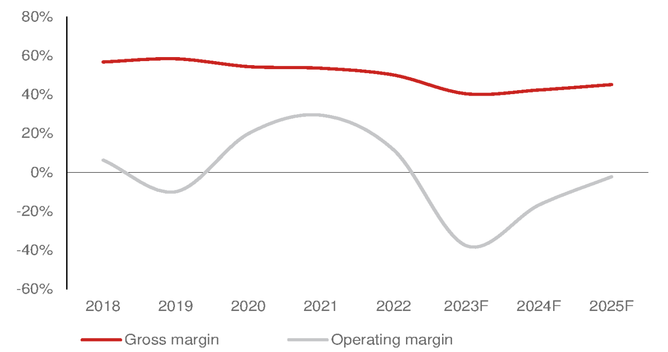
Source: Company data, Nomura estimates

Fig. 266: Novosense: Annual sales and y-y growth rate



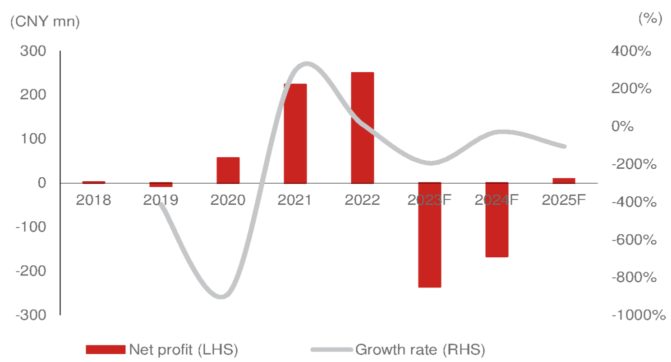
Source: Company data, Nomura estimates

Fig. 267: Novosense: GPM and OPM



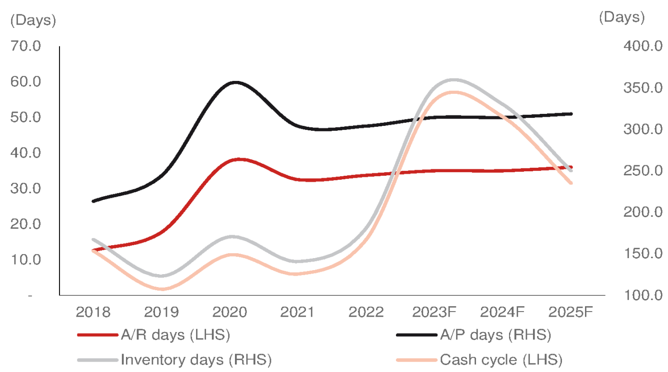
Source: Company data, Nomura estimates

Fig. 268: Novosense: Net income and y-y growth rate



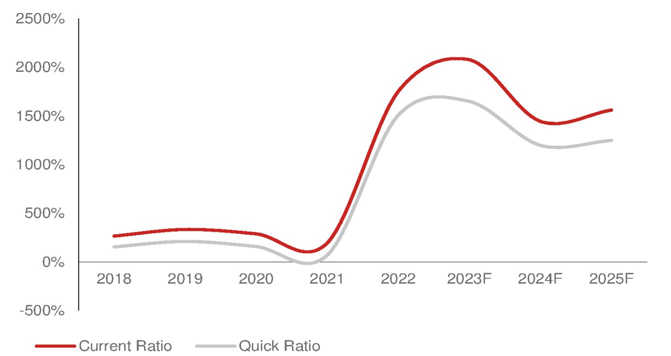
Source: Company data, Nomura estimates

Fig. 269: Novosense: CCC analysis



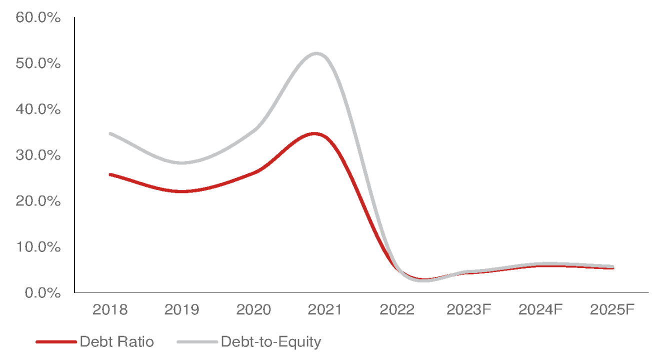
Source: Company data, Nomura estimates

Fig. 270: Novosense: Current ratio and quick ratio



Source: Company data, Nomura estimates

Fig. 271: Novosense: Net debt/net equity and debt ratios



Source: Company data, Nomura estimates

ESOP: Significant opex impact before 2023; management expects a sequential gradual decline from 2024

Novosense is a fabless design house, and employees are the most valuable asset of the company. So the company has an employee stock option plan (ESOP) incentive to retain and attract talent, and this will affect the company's financial performance in 2022-2026F. In 2022, Novosense recorded a stock option cost of CNY183.56mn (implying 52% of 2022 net income). Management estimates ESOP expense reached ~CNY280mn in 2023, and believes ESOP cost peaked in 2023.

Fig. 272: Novosense: Stock option cost

Stock option cost peaked in 2023E and decline gradually y-y in 2026E

CNY mn	2022	2023E	2024E	2025E	2026E
Stock option cost	196.7	279.9	149.1	75.3	22.7
RD expense	404				
OPEX	643				
Normalized RD expense	207				

Source: Company data, Nomura research

Fig. 273: Novosense: 2022 ESOP goal

Stage	Goal	Y-Y	Result
1st stage	2022's sales not less than CNY1300mn	51%	Yes
2nd stage	2023's sales not less than CNY1800mn	38%	No
3rd stage	2024's sales not less than CNY2300mn	28%	
4th stage	2025's sales not less than CNY2800mn	22%	

Source: Company data, Nomura research

Fig. 274: Novosense: 2023 ESOP

Stage	Goal	Y-Y	Result
1st stage	2023's sales not less than CNY1300mn	-22%	Yes
2nd stage	2024's sales not less than CNY2300mn	77%	
3rd stage	2025's sales not less than CNY2800mn	22%	

Source: Company data, Nomura research

Fig. 275: Novosense: P&L

(CNY mn)	2018	2019	2020	2021	2022	2023F	2024F	2025F
Sales	40	92	242	862	1,670	1,311	1,500	1,940
Gross profit	23	54	131	461	835	529	634	874
- OPEX	(20)	(63)	(83)	(208)	(643)	(1,024)	(886)	(918)
Operating income	3	(9)	48	254	193	(495)	(252)	(43)
Pretax income	2	(7)	60	248	253	(303)	(196)	11
Net income	2	(7)	56	224	250	(236)	(166)	9
EPS	0.0	(0.2)	0.4	1.6	1.8	(1.7)	(1.2)	0.1
Profitability								
Gross margin	57%	58%	54%	54%	50%	40%	42%	45%
- OPEX ratio	-50%	-68%	-34%	-24%	-38%	-78%	-59%	-47%
Operating margin	6%	-10%	20%	29%	12%	-38%	-17%	-2%
Net margin	6%	-8%	23%	26%	15%	-18%	-11%	0%
YoY								
Sales		129%	163%	256%	94%	-22%	14%	29%
Gross profit		136%	145%	251%	81%	-37%	20%	38%
- OPEX		210%	32%	150%	209%	59%	-13%	4%
Operating income		-461%	-628%	425%	-24%	-357%	-49%	-83%
Pretax income		-411%	-940%	316%	2%	-220%	-35%	-106%
Net income		-412%	-888%	297%	12%	-194%	-29%	-106%

Source: Company data, Nomura estimates

Fig. 276: Novosense: P&L

(CNY mn)	1Q21	2Q21	3Q21	4Q21	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23
Sales	138	203	239	283	339	454	483	394	471	253	277
Gross profit	72	111	128	147	173	228	250	182	213	96	101
- OPEX	(33)	(46)	(53)	(72)	(63)	(120)	(212)	(244)	(233)	(268)	(262)
Operating income	39	65	75	75	109	108	38	(63)	(20)	(172)	(161)
Pretax income	38	64	72	74	94	125	64	(29)	(3)	(147)	(140)
Net income	34	56	64	70	84	111	47	8	2	(133)	(119)
EPS	0.45	0.74	0.84	0.92	1.11	1.10	0.46	0.08	0.02	(0.93)	(0.84)
Profitability											
Gross margin	52%	55%	54%	52%	51%	50%	52%	46%	45%	38%	36%
- OPEX ratio	-24%	-23%	-22%	-26%	-19%	-26%	-44%	-62%	-50%	-106%	-94%
Operating margin	28%	32%	31%	26%	32%	24%	8%	-16%	-4%	-68%	-58%
Net margin	25%	28%	27%	25%	25%	24%	10%	2%	0%	-53%	-43%
YoY											
Sales					146%	124%	102%	39%	39%	-44%	-43%
Gross profit					141%	105%	96%	24%	23%	-58%	-60%
- OPEX					95%	160%	301%	237%	267%	124%	23%
Operating income					179%	66%	-49%	-184%	-118%	-259%	-523%
Pretax income					150%	94%	-11%	-139%	-103%	-218%	-318%
Net income					148%	97%	-27%	-88%	-98%	-220%	-354%
EPS					149%	48%	-45%	-91%	-99%	-185%	-280%
QoQ											
Sales		47%	18%	19%	20%	34%	6%	-18%	19%	-46%	10%
Gross profit		55%	15%	15%	18%	32%	10%	-27%	17%	-55%	5%
Operating income		42%	14%	37%	-12%	89%	77%	15%	-4%	15%	-2%
Pretax income		66%	15%	0%	47%	-1%	-65%	-265%	-68%	752%	-6%
Net income		71%	12%	3%	26%	33%	-49%	-145%	-91%	5553%	-5%
EPS		65%	14%	9%	21%	31%	-58%	-82%	-81%	-8582%	-10%

Source: Company data, Nomura research

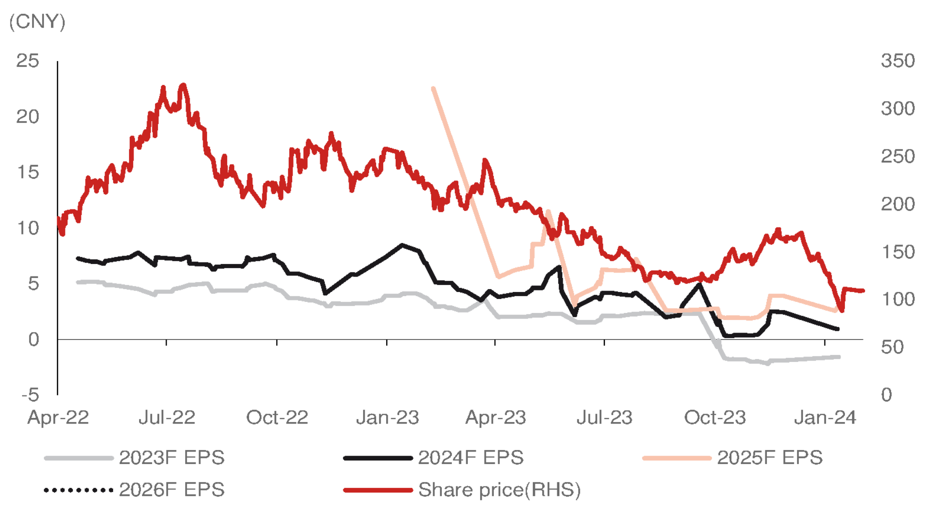
Valuation methodology and risks

Our TP of CNY105 is based on 2.5x P/B and 2025F BVPS of CNY43.1. Our 2.5x P/B multiple is at its mid to low end of the range of 2-5x between January 2023 and March. 2024, when its earnings were at loss-making or breakeven levels.

Novosense's share price declined significantly in 2Q23 when consensus earnings estimates were downgraded. Novosense's P/E valuation decreased alongside the share price decline.

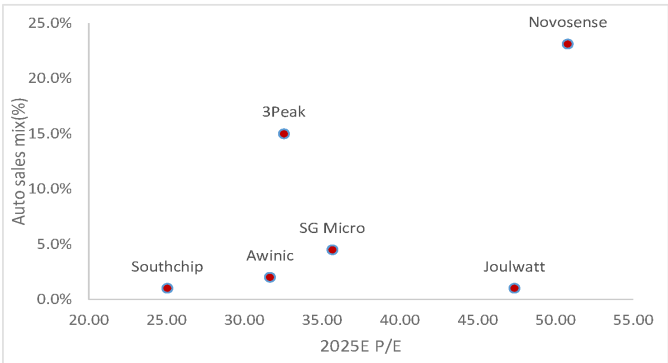
Our analysis of analog IC vendors with auto sales exposure suggests that among China A-share analog peers, Novosense has high sales exposure to auto applications, vs others with less than 10% exposure. We also detect little correlation between valuation and auto sales exposure (*Fig. 278* - *Fig. 280*). Likewise, among non-A share analog peers, we detect little correlation between auto sales and P/E multiples. In our view, auto sales growth is more critical than consumer and energy/industrial sales for P/E valuation.

Fig. 277: Novosense: Consensus earnings forecast revisions



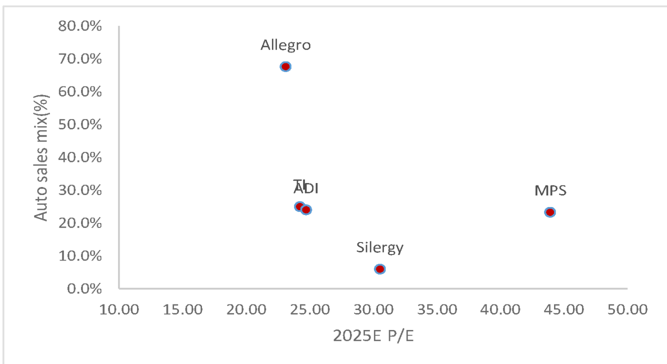
Source: Bloomberg Finance LP., Nomura research

Fig. 278: China A-share analog auto sales vs 2025E P/E multiples



Source: Bloomberg Finance LP., Nomura research

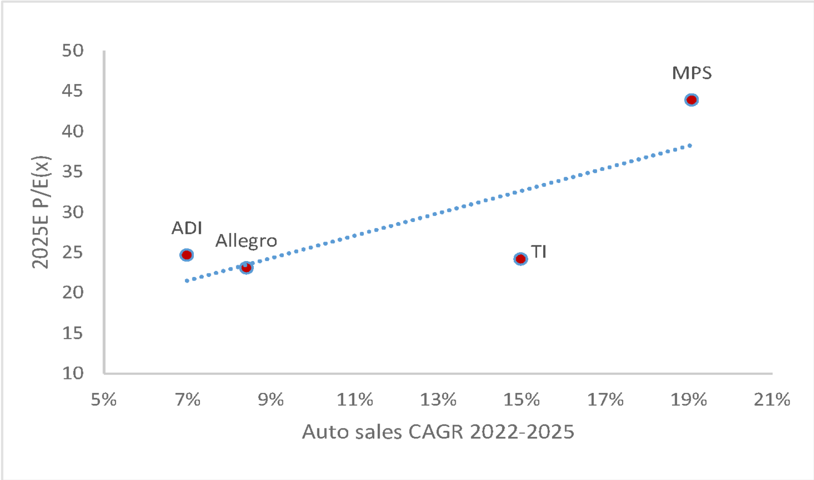
Fig. 279: Non-A share analog peers auto sales vs 2025E P/E multiples



Source: Bloomberg Financel LP., Nomura research

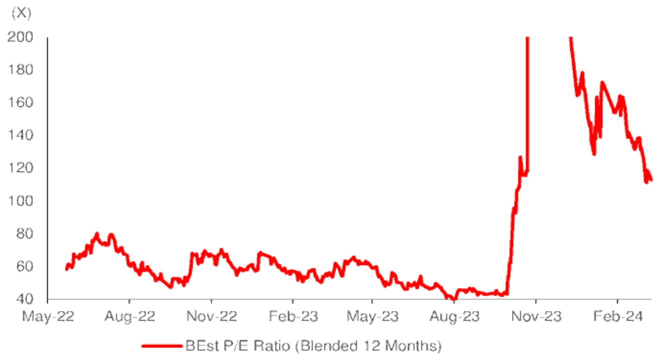
Fig. 280: Analog semi auto sales CAGR vs 2025E P/E

Higher auto sales CAGR supports higher valuation.



Source: Bloomberg Finance LP., Nomura research

Fig. 281: Novosense: 5-year P/E

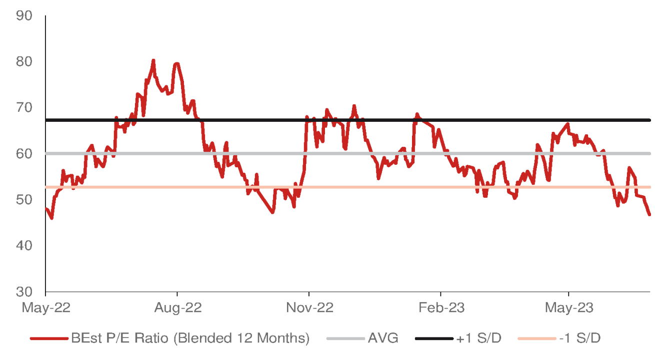


Source: Bloomberg Finance LP., Nomura research

Fig. 282: Novosense: 5-year P/B

Source: Bloomberg Finance LP., Nomura research

Fig. 283: Novosense: consensus P/E April 2022- June 2023



Source: Bloomberg Finance LP., Nomura research

Risks to our investment view

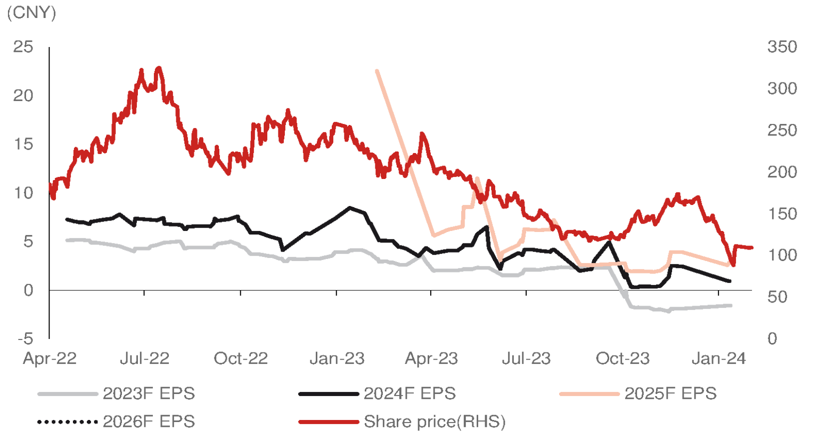
Downside risks: 1) softer-than-expected China auto demand; 2) slower-than expected adoption of local analog IC; and 3) fiercer-than-expected competition from global and local peers. Upside risks: 1) Faster inventory depletion than our expectation, 2) fewer pricing pressure than our expectation, and 3) lower wafer cost than our expected.

Share price drivers

In our view, earnings are Novosense's major share price driver. Novosense was publicly listed in April 2022, and the share price started to decline significantly thereafter amid consensus earnings forecast cuts. In our view, these cuts were due to an industry inventory correction. We believe Novosense's competitive edge remains solid, and the analog industry will continue to grow. However, we forecast Novosense will only be able to resume earnings at a breakeven level in 2025F on higher inventory pressure and pricing pressure from auto customers. Thus, it will take time for Novosense to see a significant share price catalyst.

On the other hand, we think one-year-forward consensus sales is not a powerful indicator of share prices (*Fig. 285*). Other than that, quarterly EPS have been correlated with the share price (*Fig. 286*, *Fig. 287*, *Fig. 288*). Therefore, we forecast that with negative or breakeven EPS, it will take longer for EPS to become a meaningful share price catalyst.

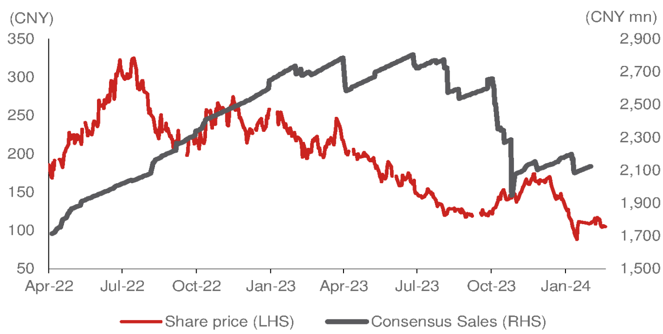
Fig. 284: Novosense: consensus earnings revisions



Source: Bloomberg Finance LP., Nomura research

Fig. 285: Novosense: one-year-forward sales vs share price

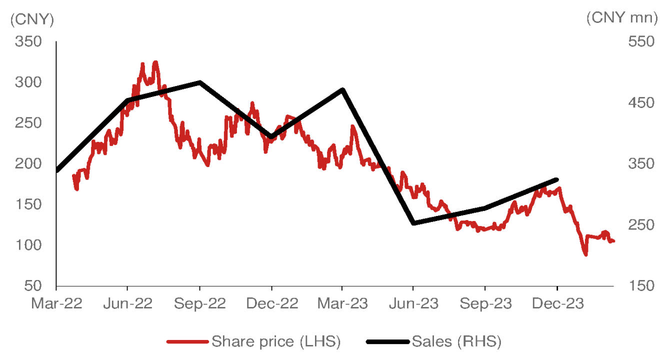
Sales consensus is not correlated to share price



Source: Bloomberg Finance LP., Nomura research

Fig. 286: Novosense: quarterly sales vs share price

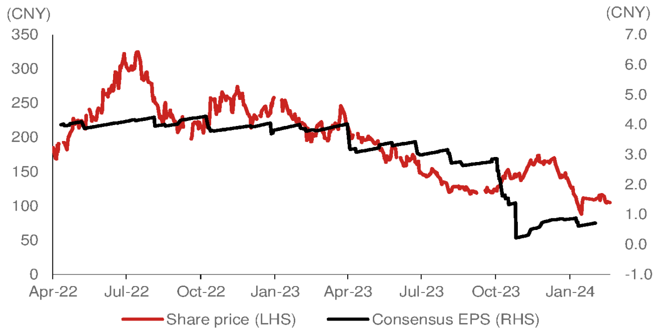
Quarterly sales is correlated to share price



Source: Company data, Nomura research

Fig. 287: Novosense: one-year-forward EPS vs share price

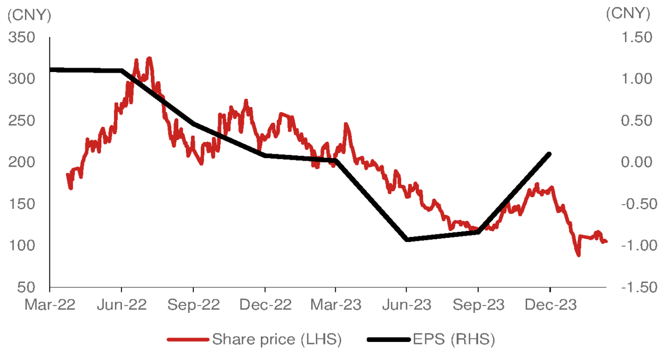
Earnings is correlated to share price



Source: Bloomberg Finance LP., Nomura research

Fig. 288: Novosense: quarterly EPS vs share price

Quarterly EPS is correlated to share price



Source: Company data, Nomura research

Fig. 289: Novosense: 5-year P/E

Source: Bloomberg Finance LP., Nomura research

Fig. 290: Novosense: 5-year P/B

Source: Bloomberg Finance LP., Nomura research

Appendix

Fig. 291: Novosense: Management team

English Name	Chinese Name	Title	Experience
Wang Shengyang	王升杨	Chairman/ President	Mr. Wang worked as a design engineer in ADI (Shanghai) from June 2009 to March 2012, and as a R&D manager of Wuxi Naxun Microelectronics Co. Ltd from March 2012 to May 2013. Mr. Wang has been the Chairman and President from September 2013 to now.
Zhu Ling	朱玲	Contoller	Ms. Zhu worked as cost accountant of Suzhou Zhaoke Electronics Co., Ltd from August 2010 to March 2012; served as cost accountant and chief accountant of Xinyin Electronics (China) from April 2012 to In July 2014, and she has served as the company's chief financial officer from November 2014 to now.
Sheng Yun	盛云	VP/ head of R&D	Mr. Sheng worked as a senior design engineer in ADI (Shanghai) from June 2008 to September 2011, and as a R&D director of Wuxi Naxun Microelectronics Co. Ltd from Oct 2011 to May 2013. Mr. Sheng has been the board member, R&D director, and vice president from September 2013 to now.
Wang Yifeng	王一峰	Vice President	Mr. Wang worked as a product manager in Wuxi Ruiwei Optoelectronics Technology from Sep 2009 to August 2013, and has been the marketing director, Vice President, board member, and secretary of the board of Novosense from September 2013 to now.
Jiang Chaoshang	姜超尚	Secretary/ IR	Mr. Jiang holds BS degree, and served as project manager and business director of Soochow Securities Co., Ltd from July 2011 to January 2020.; Starting from February 2020 to July 2020, Mr. Jiang served as the director in the Office of the Secretary of the Board of Directors of Novosense; from August 2020 to now, he served as the secretary of the company's board of directors.

Source: Company data, Nomura research

Fig. 292: Novosense: Board members

English Name	Chinese Name	Title	Experience
Wang Shengyang	王升杨	Chairman	Mr. Wang worked as a design engineer in ADI (Shanghai) from June 2009 to March 2012, and as a R&D manager of Wuxi Naxun Microelectronics Co. Ltd from March 2012 to May 2013. Mr. Wang has been the Chairman and President from September 2013 to now.
Sheng Yun	盛云	Director	Mr. Sheng worked as a senior design engineer in ADI (Shanghai) from June 2008 to September 2011, and as a R&D director of Wuxi Naxun Microelectronics Co. Ltd from Oct 2011 to May 2013. Mr. Sheng has been the board member, R&D director, and vice president from September 2013 to now.
Wang Yifeng	王一峰	Director	Mr. Wang worked as a product manager in Wuxi Ruiwei Optoelectronics Technology from Sep 2009 to August 2013, and has been the marketing director, Vice President, board member, and secretary of the board of Novosense from September 2013 to now.
Wu Jie	吴杰	Director	Mr. Wu worked as a director of Midea Group from Midea from July 2005 to January 2013, worked as the deputy general manager of Guangdong Haiwu Technology from January 2013 to September 2017, and served as the rotating general manager of Shenzhen Hengxin Huaye Equity Investment Fund Management from September 2017 to now. From August 2020 to now, he has served as the director of Novosense.
Yin Yifeng	殷亦峰	Director	Mr. Yin worked as director of Beijing Nengtai Tongda Technology from December 2011 to February 2018, and served as executive director and manager of the company; he has served as director and manager of Guorun Asset Management (Beijing) since April 2018, and Mr. Yin is the director of the company since August 2020.
Jiang Chaoshang	姜超尚	Director	Mr. Jiang holds BS degree, and served as project manager and business director of Soochow Securities Co., Ltd from July 2011 to January 2020.; Starting from February 2020 to July 2020, Mr. Jiang served as the director in the Office of the Secretary of the Board of Directors of Novosense; from August 2020 to now, he served as the secretary of the company's board of directors.
Chen Xichan	陈西婵	Independent Director	Ms. Chen hold a commerce PhD degree, and serve as an accounting professor in Soochow University. Mr. Chen has been as independent director of Novosense since August 2020.
Hong Zhiliang	洪志良	Independent Director	Mr.Hong holds PhD degree from Zurich institute of technology Switzerland. He worked as a professor at Hannover University from March 1993 to August 1994. He worked as a professor at Fudan University since January 1988. Mr. Jiang has been as independent director of Novosense since August 2020.
Wang Ruwei	王如伟	Independent Director	Mr. Wang has been a lawyer at Beijing Yingke (Suzhou) law firm since November 2018. Mr. Wang has been as independent director of Novosense since August 2020.

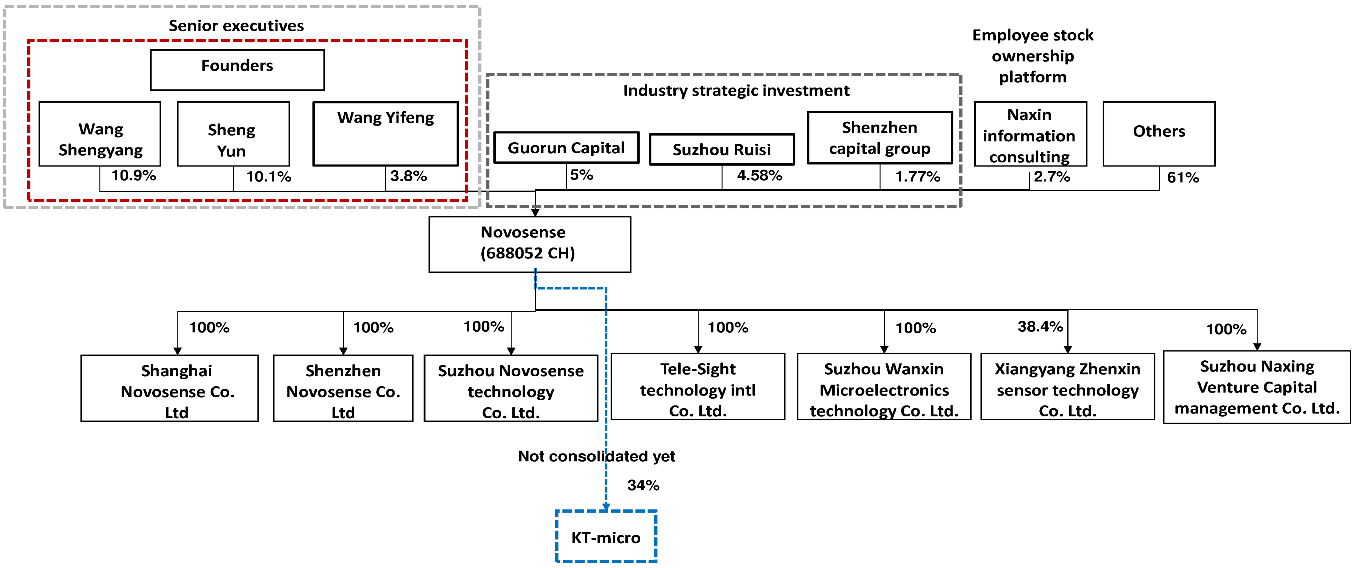
Source: Company data, Nomura research

Fig. 293: Novosense: Major shareholders

Major share holder	Ownership (%)
Wang Shengyang	10.87%
Sheng Yun	10.13%
Guorun Capital	5.00%
Suzhou Ruisi information consulting	4.58%
Wang Yifeng	3.80%
Huiyue Growth fund	3.65%
Huaxia Shanghai 50 ETF	3.37%
Suzhou Naxin information consulting	2.73%
Shangyun sensor investment	1.84%
Shenzhen Capital group (SCGC)	1.77%

Source: Company data, Nomura research

Fig. 294: Novosense: Company structure



Source: Company data, Nomura research

NOMURA

Global Markets Research
12 April 2024

3Peak 688536.SS 688536 CH

EQUITY: FABLESS

Cycle appears to be bottoming out; initiate at Buy

A high quality industrial analog IC vendor in China with long term growth potential and cycle bottoming

Rating Starts at	**Buy**
Target price Starts at	**CNY 120.00**
Closing price 9 April 2024	**CNY 96.17**
Implied upside	**+24.8%**
Market Cap (USD mn)	1,762.9
ADT (USD mn)	19.5

Initiate coverage with a Buy rating and TP of CNY120, implying 25% upside

We initiate coverage of 3Peak, the largest analog signal chain fabless vendor in China in terms of sales and market cap, we estimate (about 3% global market share in the signal chain market). We think 3Peak's fundamentals bottomed in 2H23, and expect it to turn profitable in 2024F-25F, driven by: **1) industrial applications (45-50% of 23F sales) likely bottoming** following several quarters of correction and recent rebound of US and China PMI; **2) GPM bottoming out** with TI's pricing easing into 2024 and the end of high-price foundry long term agreements (LTAs) in 2H23; **3) communication applicatons (30-35% of 23F sales) inventory digestion completed** by 1H24, and; **4) new MCU product lines to make contribution from 2025F.** 3Peak expects the new MCU business to grow as fast as its PMIC in 2021-22. Therefore, we initiate coverage of 3Peak with a Buy rating and TP of CNY120 based on 45x 2025F P/E (EPS: CNY2.7), which is at the mid- to low-end of its 40x-80x range during the 2020-22 profit-making period. We prefer 3Peak to Novosense, another local signal chain vendor in China, on the former's stronger margins resilience (*Fig. 306*), better inventory control (*Fig. 307*, *Fig. 308*), and stronger profitability (*Fig. 304*). 3Peak currently trades at 36x 2025F EPS of CNY2.7.

3Peak a major beneficiary of industrial analog IC applications recovery

Industrial applications account for 50% of global signal chain market demand, according to Gartner. ADI (ADI US, Not rated; the largest industrial analog vendor) guides for global industrial demand to bottom in 2H23-1H24E, and resume growth in 2H24E – which seems supported by our observation of the US PMI cycle (*Fig. 310*). China PMI also recovered back to the above-50 level in March 2024, following a slow 2023 (*Fig. 309*). Other than the industrial recovery, we expect the inventory of 3Peak's communication customers to return to normal levels in 2Q24F, and shipments to gradually recover in 2H24F.

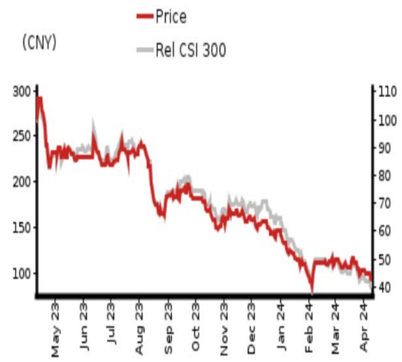
GPM bottoming out in 2024F-25F

3Peak suffered significant GPM pressure due to a worsening product mix of less communication sales, price cuts, and high wafer cost since late-2022 (*Fig. 306*). However, we think 3Peak's communication sales can gradually recover from 2Q24F, and the pricing pressure has started to ease as TI (TXN US, Not rated) lowered its internal fabs' utilization rate in 2H23, and 3Peak's wafer LTA ended in 3Q23. Thus, we expect 3Peak's GPM to moderately recover from 2H23's mid-to-high-40s% to low-to-mid 50s% in 2024F-25F.

Year-end 31 Dec	FY23	FY24F		FY25F		FY26F	
Currency (CNY)	Actual	Old	New	Old	New	Old	New
Revenue (mn)	1,094	0	1,518	0	1,981	0	2,413
Reported net profit (mn)	-40	0	190	0	352	0	487
Normalised net profit (mn)	-40	0	190	0	352	0	487
FD normalised EPS	-29.91c		1.44		2.66		3.68
FD norm. EPS growth (%)	-113.5		–		85.0		38.4
FD normalised P/E (x)	–	–	67.0	–	36.2	–	26.2
EV/EBITDA (x)	–	–	55.4	–	27.2	–	17.0
Price/book (x)	2.3	–	2.2	–	2.1	–	1.9
Dividend yield (%)	–	–	–	–	–	–	–
ROE (%)	–		–		–		–
Net debt/equity (%)	net cash		net cash		net cash		net cash

Source: Company data, Nomura estimates

Relative performance chart



Source: LSEG, Nomura

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Key data on 3Peak

Performance

(%)	1M	3M	12M		
Absolute (CNY)	-9.2	-22.0	-65.9	M cap (USDmn)	1,762.9
Absolute (USD)	-9.8	-22.8	-67.6	Free float (%)	44.0
Rel to CSI 300	-8.8	-29.3	-51.6	3-mth ADT (USDmn)	19.5

Income statement (CNYmn)

Year-end 31 Dec	FY22	FY23	FY24F	FY25F	FY26F
Revenue	1,783	1,094	1,518	1,981	2,413
Cost of goods sold	-738	-527	-720	-939	-1,140
Gross profit	1,045	566	798	1,041	1,273
SG&A	-212	-191	-184	-200	-234
Employee share expense	-656	-554	-455	-495	-531
Operating profit	177	-179	159	346	508
EBITDA	208	-120	197	415	619
Depreciation	-23	-45	-38	-69	-110
Amortisation	-8	-14	0	0	-1
EBIT	177	-179	159	346	508
Net interest expense	19	25	15	18	14
Associates & JCEs					
Other income	71	72	50	50	51
Earnings before tax	267	-82	224	414	573
Income tax	0	48	-34	-62	-86
Net profit after tax	267	-35	190	352	487
Minority interests	0	0	0	0	0
Other items	-1	-5	0	0	0
Preferred dividends					
Normalised NPAT	266	-40	190	352	487
Extraordinary items					
Reported NPAT	266	-40	190	352	487
Dividends					
Transfer to reserves	266	-40	190	352	487
Valuations and ratios					
Reported P/E (x)	43.5	–	67.0	36.2	26.2
Normalised P/E (x)	43.5	-321.5	67.0	36.2	26.2
FD normalised P/E (x)	43.5	–	67.0	36.2	26.2
Dividend yield (%)	–	–	–	–	–
Price/cashflow (x)	21.8	–	20.3	–	12.8
Price/book (x)	3.1	2.3	2.2	2.1	1.9
EV/EBITDA (x)	51.1	–	55.4	27.2	17.0
EV/EBIT (x)	59.9	–	68.8	32.7	20.7
Gross margin (%)	58.6	51.8	52.6	52.6	52.8
EBITDA margin (%)	11.7	-11.0	13.0	21.0	25.7
EBIT margin (%)	9.9	-16.4	10.5	17.5	21.1
Net margin (%)	14.9	-3.6	12.5	17.8	20.2
Effective tax rate (%)	0.1	–	15.0	15.0	15.0
Dividend payout (%)	0.0	–	0.0	0.0	0.0
ROE (%)	–	–	–	–	–
ROA (pretax %)	–	-5.6	3.7	7.4	10.4
Growth (%)					
Revenue		-38.7	38.8	30.5	21.8
EBITDA		-157.7	–	110.5	49.1
Normalised EPS		-113.5	–	85.0	38.4
Normalised FDEPS		-113.5	–	85.0	38.4

Source: Company data, Nomura estimates

Cashflow statement (CNYmn)

Year-end 31 Dec	FY22	FY23	FY24F	FY25F	FY26F
EBITDA	208	-120	197	415	619
Change in working capital	3,217	-142	400	-604	401
Other operating cashflow	-2,895	97	31	6	-21
Cashflow from operations	530	-165	628	-182	1,000
Capital expenditure	-184	-234	-200	-199	-198
Free cashflow	346	-399	428	-381	802
Reduction in investments	1,495	-2,077	0	0	2
Net acquisitions					
Dec in other LT assets					
Inc in other LT liabilities					
Adjustments	-34	-36	0	0	1
CF after investing acts	1,807	-2,511	428	-381	805
Cash dividends	-49	-25	0	0	-1
Equity issue	0	1,784	0	0	1
Debt issue	0	0	0	0	2
Convertible debt issue					
Others	79	107	-105	0	1
CF from financial acts	30	1,867	-105	0	3
Net cashflow	1,837	-644	323	-381	808
Beginning cash	297	2,134	1,490	1,814	1,432
Ending cash	2,134	1,490	1,813	1,432	2,240
Ending net debt	-2,134	-1,490	-1,814	-1,432	-2,238

Balance sheet (CNYmn)

As at 31 Dec	FY22	FY23	FY24F	FY25F	FY26F
Cash & equivalents	2,134	1,490	1,814	1,432	2,240
Marketable securities	802	2,880	2,880	2,880	2,879
Accounts receivable	189	208	167	322	286
Inventories	291	428	45	572	183
Other current assets	127	79	79	79	79
Total current assets	3,543	5,085	4,984	5,285	5,668
LT investments	105	104	104	104	103
Fixed assets	84	112	274	403	491
Goodwill	0	0	0	0	0
Other intangible assets	70	62	62	62	61
Other LT assets	350	545	650	650	647
Total assets	4,151	5,908	6,074	6,505	6,970
Short-term debt	0	0	0	0	1
Accounts payable	95	101	76	155	132
Other current liabilities	230	191	191	191	191
Total current liabilities	325	292	267	346	324
Long-term debt	0	0	0	0	1
Convertible debt					
Other LT liabilities	40	37	37	37	37
Total liabilities	366	329	304	383	362
Minority interest					
Preferred stock					
Common stock	120	133	133	133	133
Retained earnings	846	786	976	1,328	1,814
Proposed dividends					
Other equity and reserves	2,819	4,661	4,661	4,661	4,661
Total shareholders' equity	3,786	5,579	5,769	6,121	6,608
Total equity & liabilities	4,151	5,908	6,074	6,504	6,970
Liquidity (x)					
Current ratio	10.89	17.43	18.66	15.28	17.50
Interest cover	–	–	–	–	–
Leverage					
Net debt/EBITDA (x)	net cash	–	net cash	net cash	net cash
Net debt/equity (%)	net cash	net cash	net cash	net cash	net cash
Per share					
Reported EPS (CNY)	2.21	-29.91c	1.44	2.66	3.68
Norm EPS (CNY)	2.21	-29.91c	1.44	2.66	3.68
FD norm EPS (CNY)	2.21	-29.91c	1.44	2.66	3.68
BVPS (CNY)	31.49	42.07	43.51	46.16	49.83
DPS (CNY)	0.00	0.00	0.00	0.00	0.00
Activity (days)					
Days receivable	66.2	45.1	45.0	46.0	
Days inventory	248.9	120.3	120.0	121.0	
Days payable	68.0	45.1	45.0	46.0	
Cash cycle	0.0	247.1	120.3	120.0	121.0

Source: Company data, Nomura estimates

Company profile

Founded in 2012, 3peak is a China leading analog company and based in Shanghai city. The company is focused on high-performance, high-quality, and highly reliable IC products, including signal chains, power analog chips, mixed-signal front-ends, and embedded processors to provide customers with all-around solutions. The applications of the products cover many fields such as communication, industrial, security monitoring, medical and health, instrumentation, new energy, and automotive industry.

Valuation Methodology

Our TP of TWD120 is based on a P/E of 45x on our 2025F EPS of CNY2.7. 50x P/E is at the mid-low-end of its past five-year P/E range of 40x-80x in profi-making period of 2020-2022. The benchmark index is CSI300.

Risks that may impede the achievement of the target price

Downside risks include: 1) market recovery is slower than expected; 2) lower-than-expected auto sales growth; and 3) higher than expected pricing competition. Upside risks include 1) Higher-than-expected market share gain; 2) less pricing competition than our expected.

ESG

3peak continue to monitor its foundry partners to set reduction objectives for energy management greenhouse gas emissions, air pollution, water pollution, and waste management to minimize the environment impact of the production process. 3peak is also actively to engage in environment protection, shareholders and employee interest, and social responsibilities in operation.

Executive summary

We initiate coverage of 3Peak with a Buy rating, with TP of CNY120. It is the No.1 analog signal chain fabless (*Fig. 302*) in China, and No.4 general purpose analog IC vendor in 2022. We expect 3Peak to steadily gain market share and grow sales by 39%/30% in 2024F-25F.

3Peak is the largest China analog signal chain fabless. It is the No.4 general purpose signal chain IC vendor globally, with an estimated market share of ~3% (*Fig. 295*) in 2022 as per Gartner. According to the company, it recorded c.USD183mn in signal chain sales in 2022 vs a c.USD6.5bn signal chain total addressable market (TAM). We note the company's market share has expanded since 2019, when it reported USD44mn of signal chain sales (c. 1% share vs a TAM of USD4.3bn in 2019, according to Gartner estimates). Other than signal chain, 3Peak's PMIC started to contribute meaningfully at USD75mn (29% of 2022 sales), and 3Peak announced to acquire ICE-semi in 2023 (the deal is to be approved by shareholders in 2024E, according to management), and its products mainly comprise PMIC. We think 3Peak can leverage ICE-semi's customer portfolio in consumer applications and product development experience to accelerate its PMIC product development schedule. The sales contribution of analog IC to 3Peak was almost c.100% in 2023, and we expect MCU, its new product line, to be another major growth driver for the company over the next few years

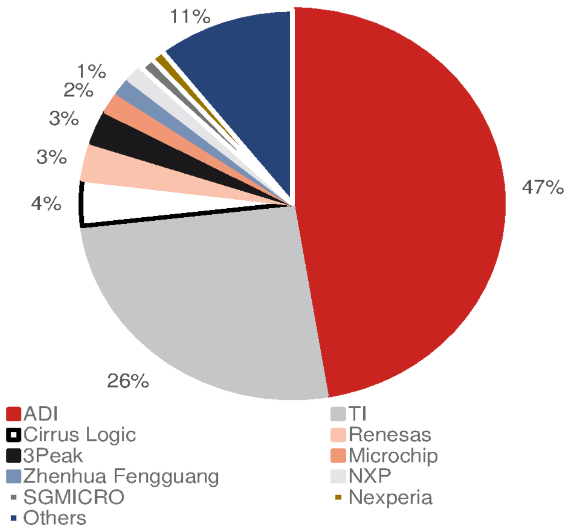
For 3Peak's analog business, communication was the major application, at c.50% of sales in 2022, while the rest was contributed by consumer, industrial, and automotive. In 2023, 3Peak's sales declined by a significant c.40% y-y given: **1)** analog industry cycle downturn; **2)** aggressive pricing strategy; and **3)** communication (comm.) customer inventory overhang. On our forecast, comm. accounted for 32% of 3Peak's comm. sales in 2023F, down from 55% in 2022. Comm. sales matter as it has a much higher ASP and GPM.

With the large scale of China's analog market (30-40% of global analog sales of c.USD85bn) and rising demand for localization, 3Peak should maintain its dominant position in the China local analog market, in our view. As such, 3Peak's localization progress at auto and industrial remains the most important factor to watch in the long term. We expect 3Peak to steadily gain market share in 2024F-25F, as we believe it would outgrow global IDMs in the local analog market as 3Peak's sales suffered order correction earlier in 1H23, and expect it to recover in 2Q24F.

Beyond 2024, we forecast 3Peak's auto sales to double in 2025F, as we believe that it should be able to capture market share driven by its China local EV customers. The growing market share at auto analog, and industrial/comm. applications, as well as 3Peak's ambition to acquire more relevant businesses (e.g., MCU) should support the company in delivering a 30% sales CAGR over 2023-25F, in our view.

With 3Peak's position as a China local analog supplier and improving product portfolio, we initiate coverage of 3Peak with a Buy rating and TP of CNY120, based on 45x our 2025F EPS, supported by a 35% sales CAGR over 2023 to 2025F. 45x is the mid- to low-end range of 40x-80x during the 2020-22 profit-making period.

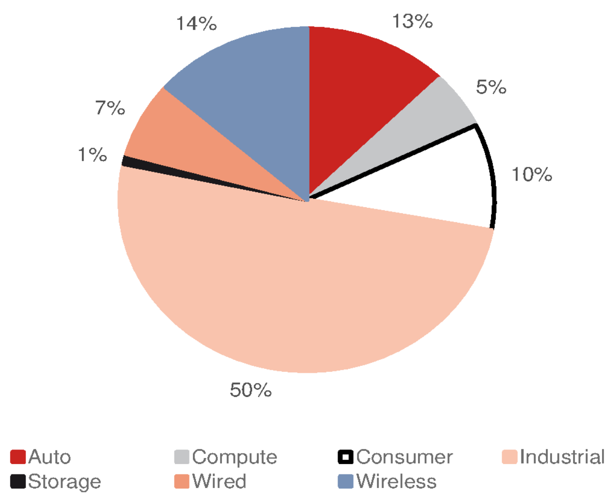
Fig. 295: Signal chain market share by vendors' sales in 2022



Source: Gartner, Nomura research

Fig. 296: Industrial accounts for the majority of general purpose signal chain market

General purpose signal chain market revenue mix – 2022



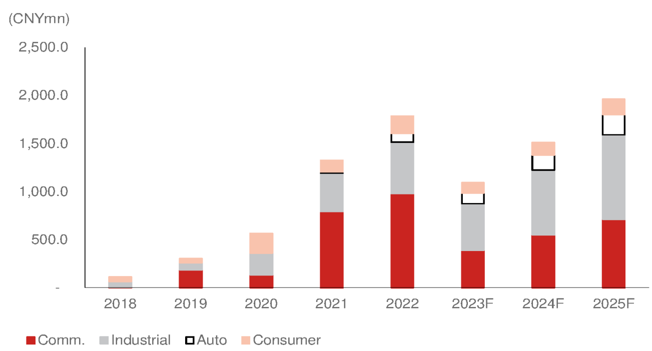
Source: Gartner, Nomura research

Fig. 297: 3Peak: Summary of major product lines

Applications	Sales % in 2023F	2024F outlook	Gross Margin	China competitors	Global competitors	Key customers
Industrial	47%	++	Higher than corporate average	Novosense, Zhenhua Fengguang, SG Micro	ADI, TI, Microchip, Mercury systems, Cobham, Nuvoton	Inovance, Haier, Ecovacs(603486 CH)
Communication	32%	++	Highest	Silergy, Chipsea	ADI, TI, Microchip	ZTE, Luxshare, Shiner Fiber
Consumer	9%	+	Lower than corporate	SG Micro, Nexperia, Bpsemi, Giantec	ADI, TI, Infineon, Onsemi, Cirrus logic	iFlytek, Changhong(長虹)
Automotive	12%	+++	Similar to corporate average	Novosense, Silergy	ADI, TI, Infineon, Onsemi, Renesas, Microchip	CATL, Auto-star(澳仕達), HZSTI(科島微)

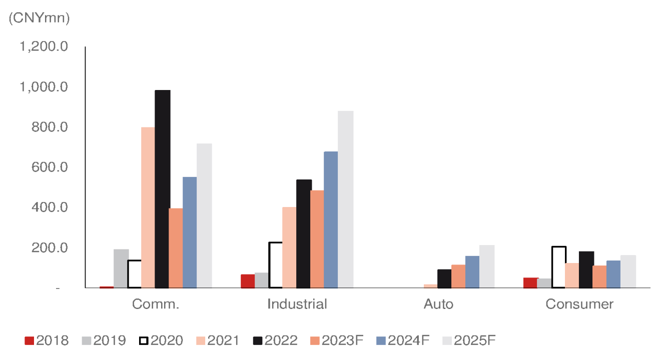
Source: Company data, Nomura estimates

Fig. 298: 3Peak: Application sales



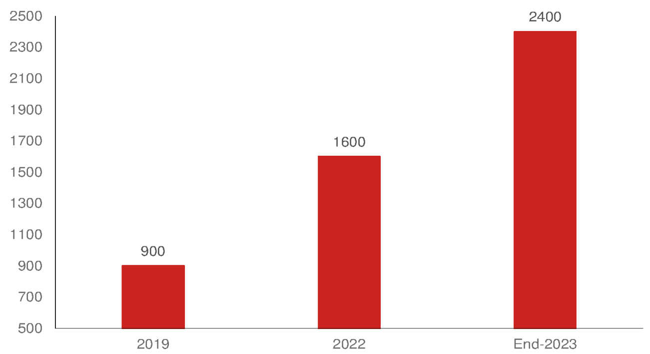
Source: Company data, Nomura estimates

Fig. 299: 3Peak: Application sales by year



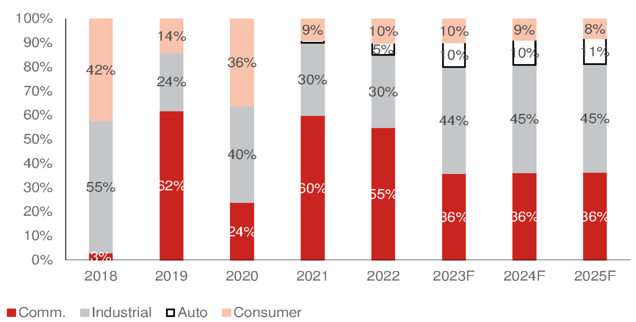
Source: Company data, Nomura estimates

Fig. 300: 3Peak: Number of products



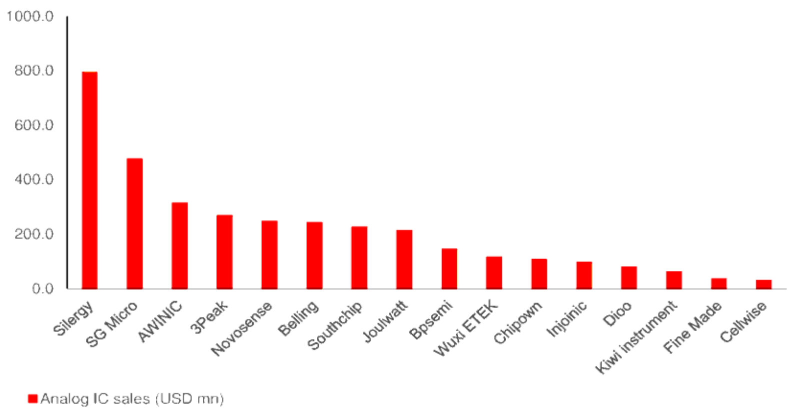
Source: Company data, Nomura research

Fig. 301: 3Peak: Application mix (%)



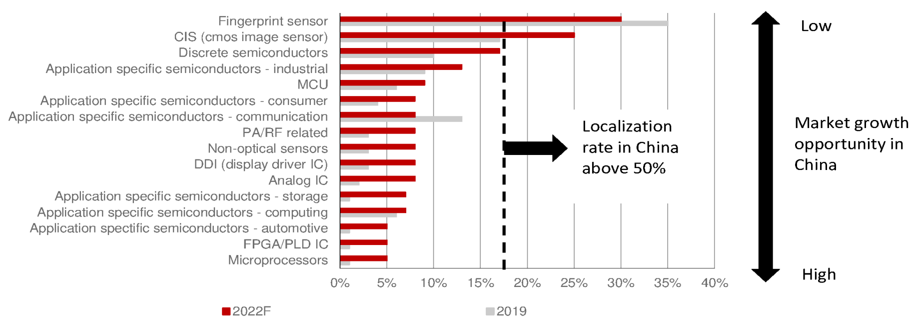
Source: Company data, Nomura estimates

Fig. 302: China local vendors' analog IC sales in 2022



Source: Company data, Nomura research

Fig. 303: China local vendors' market share globally in 2019 vs 2022



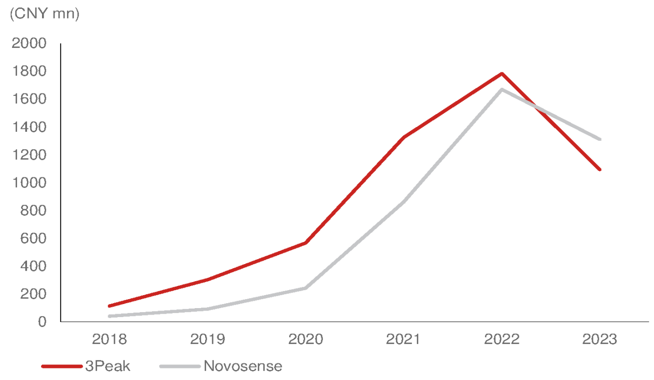
Source: Nomura estimates

Fig. 304: Industrial/ auto analog vendors' comparison

	3Peak	Novosense
Founded	2012	2013
2023 Sales (CNYmn)	1,094	1,311
2023 GPM (%)	52%	40%
Product SKU	2,400	1,500
2022 Employee	714	645
RD employee (%)	71.2%	50.5%
2023 RD expense (CNYmn)	554	650
2023 RD to sales (%)	51%	50%
1Q-3Q23 Application sales mix (%)		
Auto	12%	33%
Industrial	47%	55%
Consumer	9%	12%
Product sales mix (%)		
Signal chain	79%	65%
PMIC	20%	27%
Supply chian		
Key foundry partner	Tower, SMIC	SMIC, DB Hitek

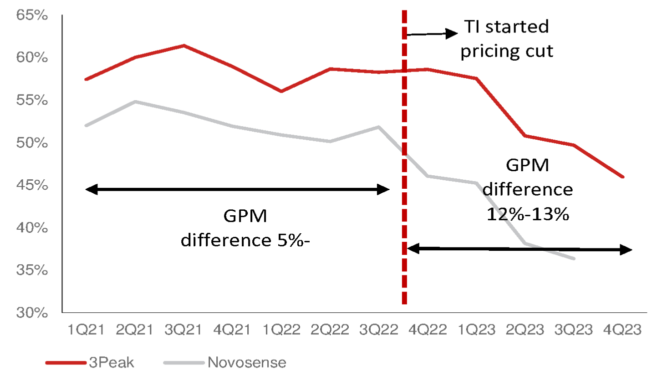
Source: Nomura research

Fig. 305: 3Peak and Novosense sales



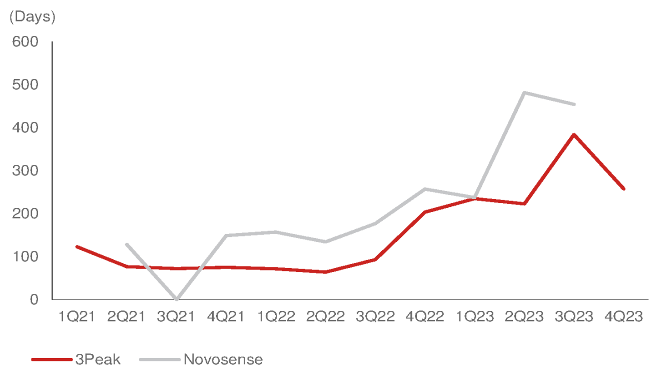
Source: Company data, Nomura research

Fig. 306: 3Peak and Novosense GPM



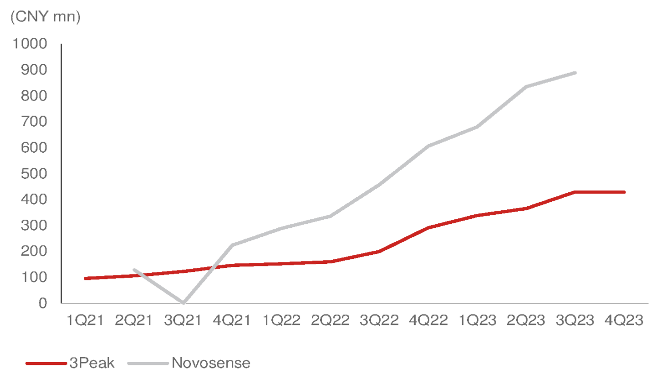
Source: Company data, Nomura research

Fig. 307: 3Peak and Novosense inventory days



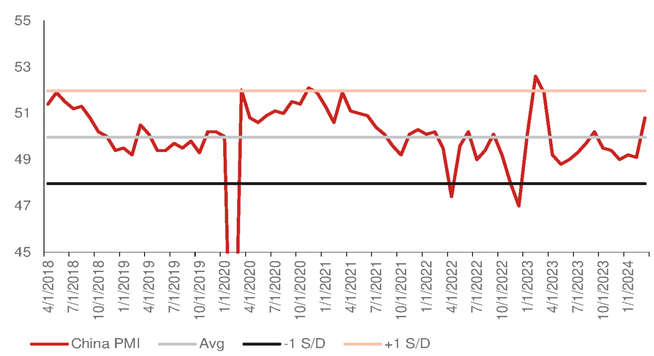
Source: Company data, Nomura research

Fig. 308: 3Peak and Novosense's inventories



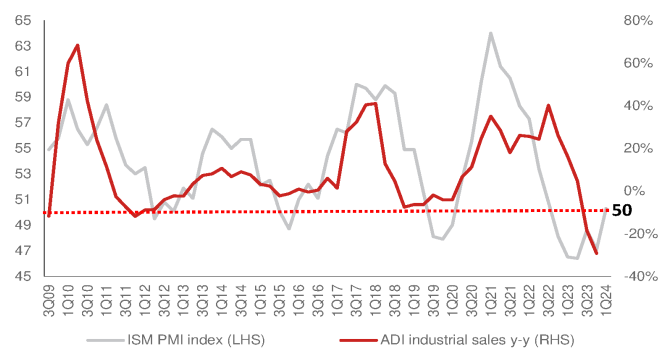
Source: Company data, Nomura research

Fig. 309: China manufacturing PMI



Source: Nomura research

Fig. 310: ADI industrial sales y-y vs. ISM PMI index



Source: Company data, Nomura research

3Peak: Key signal chain vendor benefiting from China analog localization trends

Established in 2012, 3Peak is a China analog fabless company focused on signal chain products, and it recently expanded its product portfolio to PMIC. 3Peak's over 2,400 products cover broad end-applications of networking, industrial, and automotive industries, and are supplied to more than 3,700 customers. In 2021, 3Peak set up a new business of embedded process group, which designs MCU products and had its first tape-out in 2023.

2012-19: 3Peak launched linear, interface, and converter signal chain, and power IC products in 2018

3Peak launched several new products with performance close to global peers, such as general converters for local leading communication equipment customers, and video amplifiers for leading surveillance companies. Both products are a breakthrough for local analog vendors.

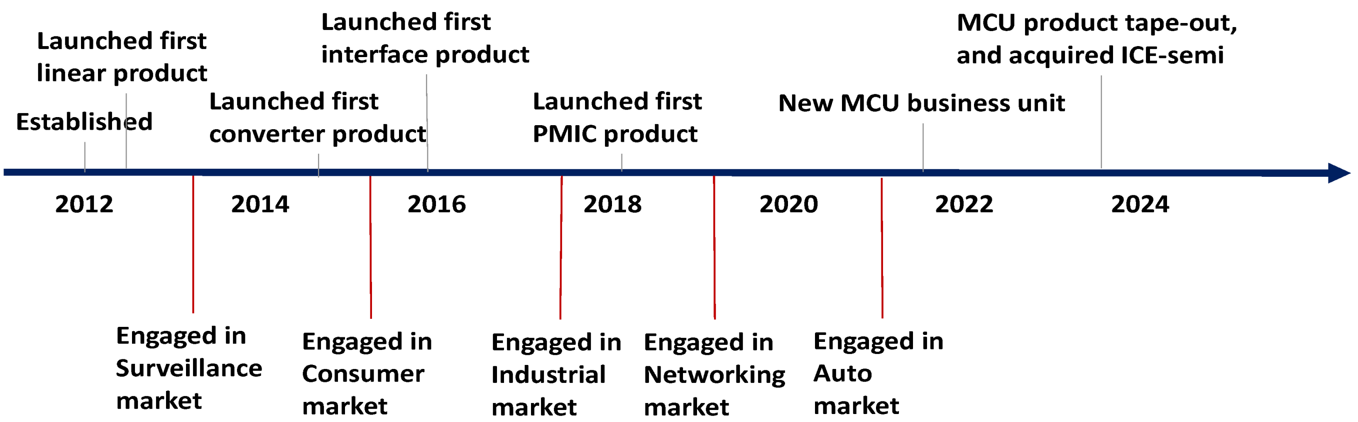
2020-22: PMIC sales to customers materialize

PMIC became another sales driver for 3Peak in this period. PMIC sales (which included low-dropout regulator (LDO), power supervisor IC, and motor drivers) to customers materialized during the semi shortage period. Thus, the sales mix of PMIC increased from 2% in 2019 to 29% in 2022.

2023-now (transition and new focus): New sales growth driven by auto, new MCU products, and M&A growth

3Peak started to engage with the auto industry in 2021, and it had over 100 auto analog products in 2023, from nearly 0 in early 2021. The auto sales mix also significantly increased to 15-20% in 3Q23, vs 1% in 2021. Also, 3Peak taped out its first industrial MCU product in 2023, and provided its customers better solutions overall by offering more embedded products, thereby increasing customer loyalty. On the other hand, 3Peak announced to acquire 100% shares of ICM-semi (创芯微, unlisted) in June 2023, which is a PMIC company specialized in battery management IC and AC/DC products in consumer applications.

Fig. 311: 3Peak: Company history and milestones



Source: Company data, Nomura research

3Peak: Product portfolio

Strong signal chain products and growing PMIC

3Peak has a strong signal chain product portfolio, including linear products, converters, and interface products. Signal chain IC is an analog chip to transmit, receive, convert, amplify, and filter analog signals. Some of its products have reached global peers' standards in terms of product performance and reliability, and have achieved localization requirement for similar global vendors' products, such as amplifiers, high-voltage comparators, and high-precision digital-to-analog converters.

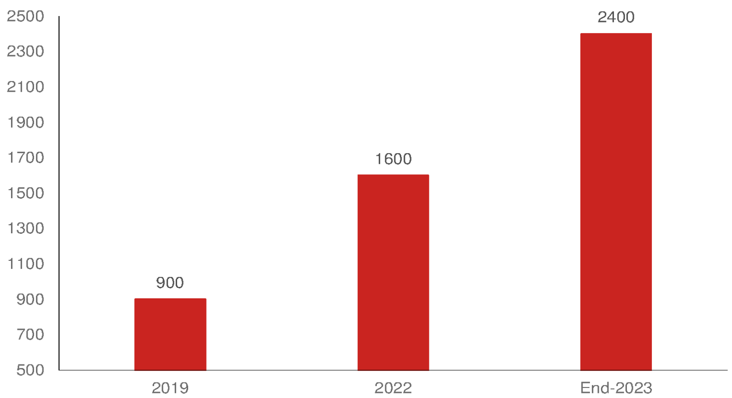
3Peak's products mainly focus on non-consumer electronics industrial systems, with a high proportion of applications in high-barrier fields such as communications, industrial, and automotive. From its latest disclosures, 3Peak has more than 2,400 products, a significant jump from 900 in 2019 (*Fig. 313*).

Fig. 312: 3Peak: Product portfolio

Products (sales % in 2022)	Previous product category	Introduction	Major applications	Global and China Peers
Signal Chain (71%)	Amplifiers: (Operational amplifiers, Special function amplifiers, Comparators, sensor interface)	A component that can amplify the voltage or power of an input signal, or compare the size, and select the signal is on or off by turning the control on or not, or select the needed signal from multiple sources, thereby connecting or disconnecting the signal link.	Industrial, auto, audio, IoT, meters	Global: ADI, TI, STM, NXP, Renesas, Infineon China: Silergy, Novosense, SG Micro, Chipsea
	Isolation: (Ditial isolator, Isolated Amplifier, Modulator)	A safety function responsible for converting input signals and outputting them to achieve electrical isolation between input and output ends. Electrical isolation can ensure the safety of signal transmission between strong current circuits and weak current circuits	Industrial, power electricity, medical	
	Interface: (Analog Switch, CAN, LVDS)	For digital signal transmission between electronic systems	New energy, auto, industrial, communication, surveillance, power electricity	
	Data converter: (Precision converters, High speed converters, Special converters)	Convert analog signals to digital signals	Industrial, communicatin, medical, auto, LiDAR	
PMIC (29%)	LDO/ Voltage Reference	Using a transistor or FET operating within its linear region to produce a regulated output voltage through applied input voltage.	Industrial meter, base station, image sensor, auto camera and radar	Global: TI, ADI, Infineon, Renesas, MPS China: Silergy, SG Micro, Beken, BP semi, Giantec, Silan, Chipone
	Supervisors: (Power sequencer, Supervisors & Reset ICs)	Control the sequence of powering on and off different power supplies during startup or shutdown, and regularly check the internal conditions of the IC and send a restart signal to the IC if an error occurs; or monitor the power supply voltage.	Industrial, smart device	
	Drivers: (Motor drivers, Gate Drivers)	Used to control the rotation of mechanical motors	Smart door, surveillance	
	DCDC: (Buck Converters, Boost converter)	To achieve the purpose of voltage conversion and improve the efficiency of power conversion.	Communication, industrial, medical	
MCU		Integrating memory, clock, timer/counter, display interface, etc. on a single chip to control systems with software control	Industrial	Global: Infineon, NXP, STM, TI China: Gigadevice, ingenic, sinowealth, Espressif
ICM-semi (创芯微)	Battery management IC/ AC-DC	3Peak acquired ICM-semi in June-2023, current product mostly in linear products	Communication, industrial, consum	Global: TI, ADI China: Silergy, SG Micro, Awinic

Source: Company data, Nomura research

Fig. 313: 3Peak: Number of products



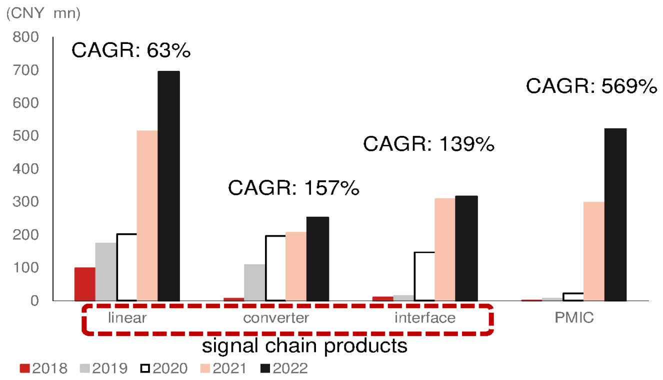
Source: Company data, Nomura research

Mass production of new products driving sales growth since 2018

Signal chain is the major product of 3Peak and sales have recorded strong growth since 2018 due to industrial and communication applications, and PMIC products have started to contribute to sales from 2021-22 and have become another growth driver. 3Peak was also a semi shortage beneficiary during 2020-22. Signal chain products accounted for 71% of total sales in 2022, while PMIC accounted for 29%.

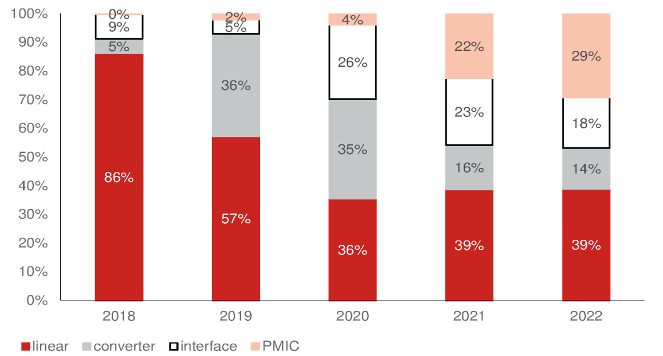
Also, PMIC profitability improved to c.50% in 2021 due to mass production of new products and price hikes, but it declined from late-2022 due to inventory adjustment and lowering of prices because of higher competition. We think profitability can be gradually improved by a better product mix and lower wafer cost in 2024F-25F.

Fig. 314: 3Peak: Product sales and CAGR



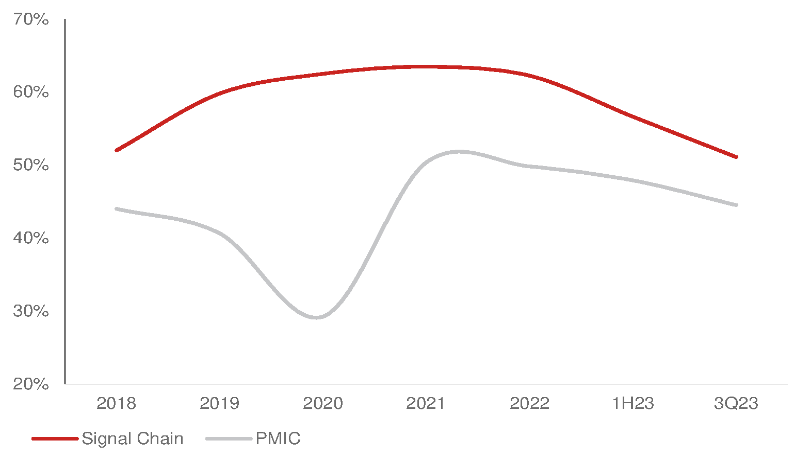
Source: Company data, Nomura research

Fig. 315: 3Peak: Product mix (%)



Source: Company data, Nomura research

Fig. 316: 3Peak: Product line GPM



Source: Company data, Nomura research

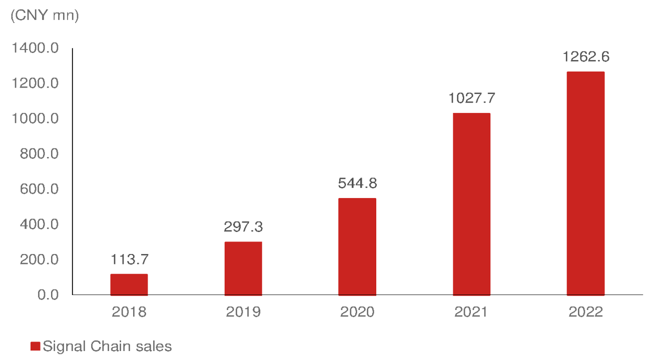
Signal chain (75-80% of 2023F sales)

3Peak has a strong signal chain product portfolio and higher global market share than other China analog competitors

We think 3Peak owns the most complete signal chain product portfolio across China analog vendors. Its portfolio covers three major products: amplifiers, converters, and interface (*Fig. 321*). According to our observation, there are few signal chain-focused companies in China in the analog market, which suggests 3Peak is competing more with global analog peers, such as TI, ADI, etc.

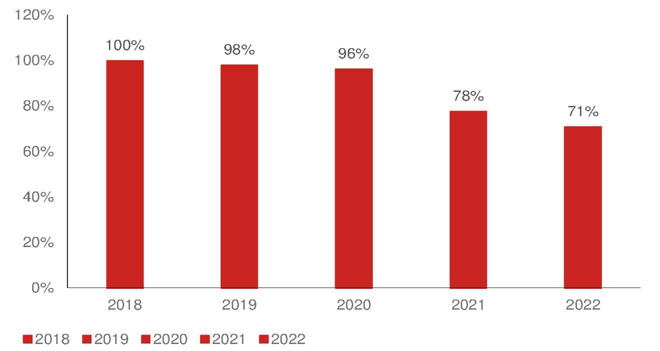
3Peak started its signal chain business with linear products in 2018 (86% of 2018's signal chain sales), and all three signal chain products have recorded strong growth in the past five years driven by mass production of new products, localization, and price hikes during the semi IC shortage period.

Fig. 317: 3Peak: Signal chain sales



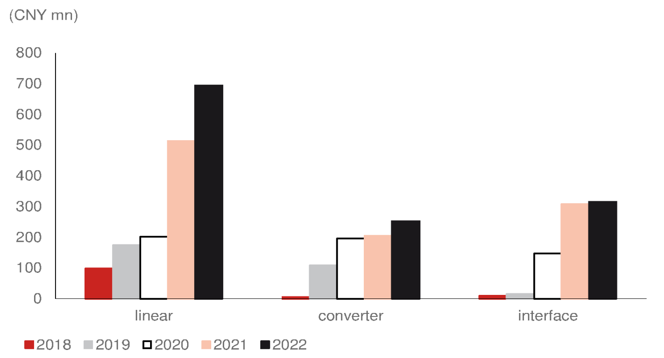
Source: Company data, Nomura research

Fig. 318: 3Peak: Signal chain sales mix (%)



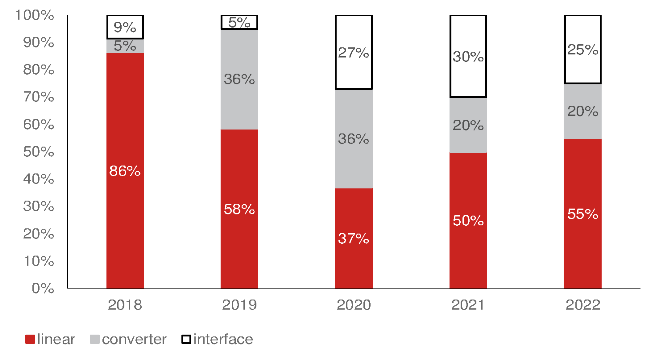
Source: Company data, Nomura research

Fig. 319: 3Peak: Signal chain sales by product



Source: Company data, Nomura research

Fig. 320: 3Peak: Signal chain sales breakdown (%) by product



Source: Company data, Nomura research

Fig. 321: Major China analog IC companies' signal chain product portfolios

	Amplifier			Converter			Driver		Analog Switch	Interface circuits	Isolator
	Operational Amp	Audio/ Video Amp	Comparator	ADC	DAC	AFE	Audio	Video			
Sliergy	√	√			√				√		√
SG Micro	√	√	√	√			√	√	√	√	√
Awinic		√					√	√	√	√	√
3Peak	√	√	√	√	√	√	√		√	√	√
Novosense		KTmicro		KTmicro			KTmicro			√	√
Southchip											
Joulwatt						√				√	
Injoinic											
Chipown											
Wuxi ETEK		√		√					√		
Dioo	√	√	√				√	√	√		
Cellwise											
Fine Made											
Belling	√	√		√	√				√	√	√
Kiwi instrument						√					
Bpsemi											

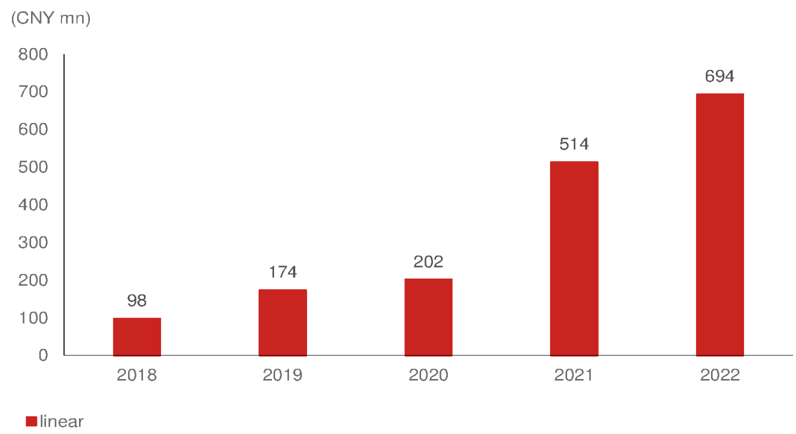
Source: Company data, Nomura research

1) Linear products (55% of signal chain sales)

Operational amplifiers are the major linear products, and the core technology is high precision, high speed, and low power consumption. In high-end applications, these three technologies are often needed simultaneously. As per Gartner's estimate, the amplifier/comparators market size was USD4.5bn in 2022, and industrial was the major application, which accounted for 47% of the market.

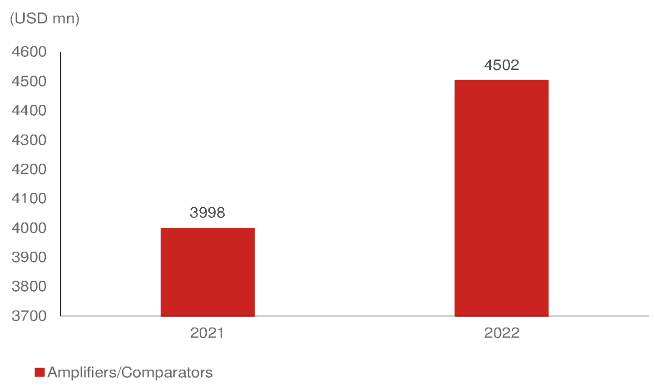
Linear products are 3Peak's major contributors by sales, and also the most technologically advanced product line. Compared with local and global products, 3Peak's operational amplifier products' low power consumption and high precision puts the company in a leading position. According to Gartner, 3Peak had a 2.1% global market share in 2022.

Fig. 322: 3Peak: Annual linear sales



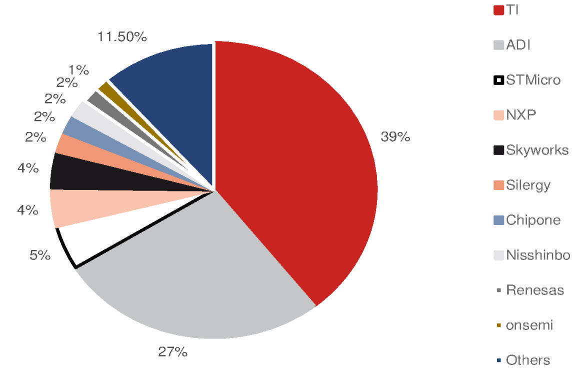
Source: Company data, Nomura research

Fig. 323: Amplifier/comparators market size in 2021-22



Source: Gartner, Nomura research

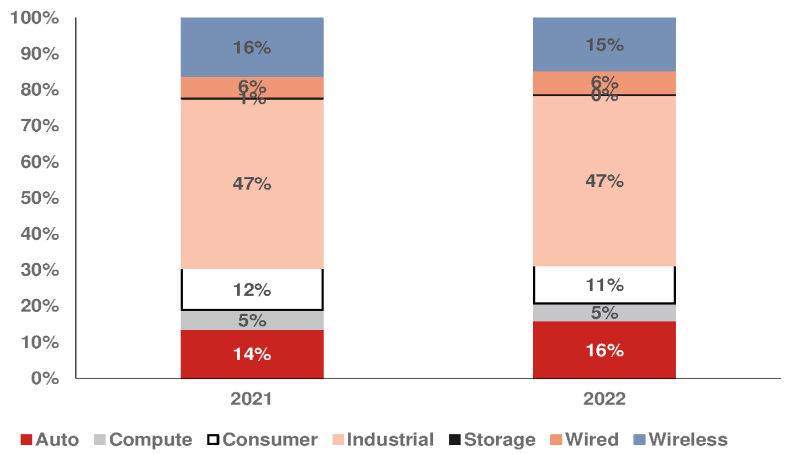
Fig. 324: Amplifier/comparator market share by vendor in 2022



Source: Gartner, Nomura research

Fig. 325: Amplifier/comparators sales mix (%) by application in 2021-22

Industrial is the major application.



Source: Gartner, Nomura research

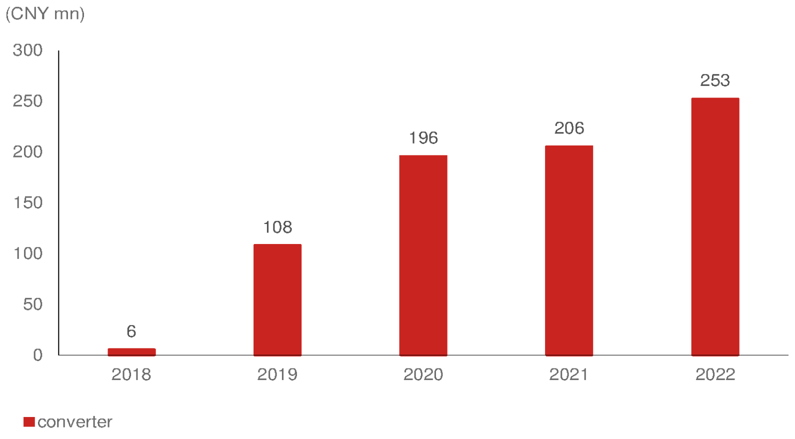
2) Converters (20% of signal chain sales)

Data converters are divided into two major products: digital-to-analog converters (DACs) and analog-to-digital converters (ADCs), both of which are mainly used for conversion between analog signals and digital signals. According to Gartner data, ADC accounts for 74% of the data converter market value, while DAC accounts for another 26%. The market is dominated by ADI and TI, while 3Peak also plays an important role and ranked as the fifth largest vendor in 2022, according to Gartner estimates.

Technical difficulties involved in manufacturing high-speed and high-precision ADCs make it one of the most complex signal chain products. **1) Technology:** ADCs' resolution and sampling rate affect each other. The higher the resolution, the more parameters are sampled and compared, resulting in a more complex chip architecture, which may slow down the operational speed. Also, the higher the resolution, the more sensitive it is to external environment signals, which affects the sampling data, and thereby increases the time required to simulate the performance repeatedly. **2) Patent:** Patents of major global IDMs are highly protected, making it increasingly difficult for late entrants to break through as global IDMs' products have already advanced into high-end ADCs. **3) Process technology**: High-end ADCs are mainly produced by BiCMOS, for which foundry partners do not have significant capacity.

3Peak is a leading data converter vendor in China with a global market share of 3% (*Fig. 34*) and has launched several products with similar performance as global peers. Other local vendors are: Chipsea (芯海, 688595 CH, Not rated), Zhenhua Fengguang (振华风光, 688439 CH, Not rated), SG Micro (300661 CH, Neutral), and KT-micro (unlisted, acquired by Novosense).

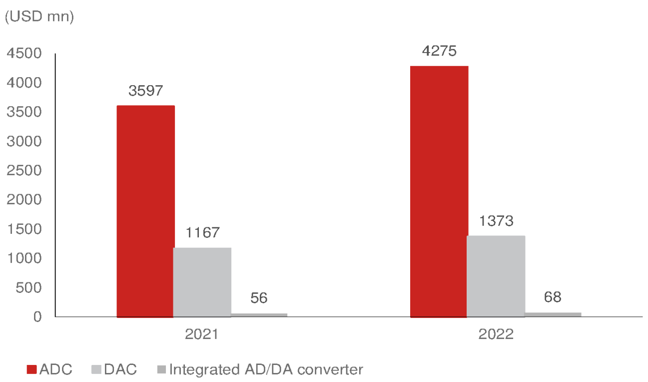
Fig. 326: 3Peak: Annual converter sales



Source: Company data, Nomura research

Fig. 327: Data converter market size in 2022

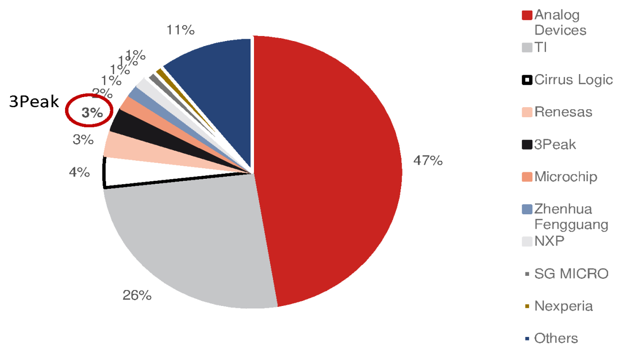
ADC is the major product



Source: Gartner, Nomura research

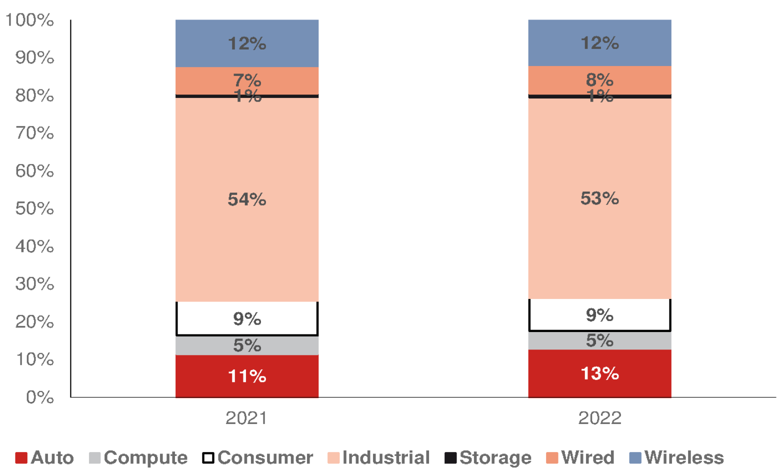
Fig. 328: Data converter market share by vendor in 2022

3Peak was ranked fifth largest in 2022



Source: Gartner, Nomura research

Fig. 329: Data converter sales mix (%) by application in 2021-22



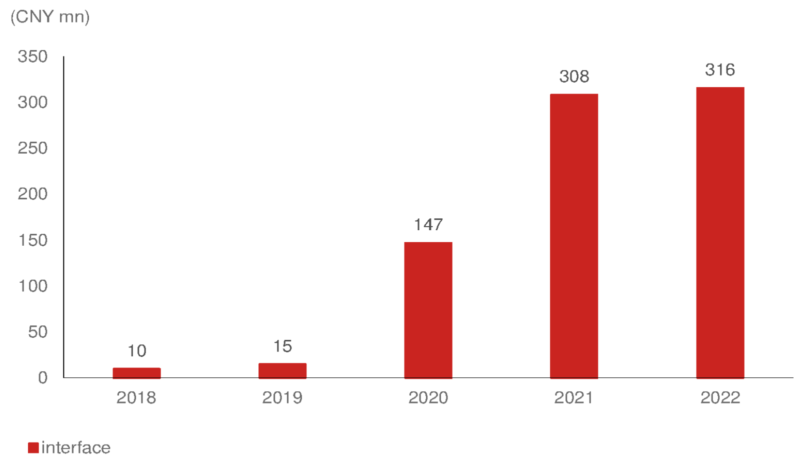
Source: Gartner, Nomura research

3) Interface (25% of signal chain sales)

The interface IC is a chip with communication functions based on general and specific protocols. It is widely used in signal transmission between electronic systems to improve system performance and reliability. As per Gartner's estimates, the global interface market size was USD3.2bn in 2022, and industrial is the major end-application.

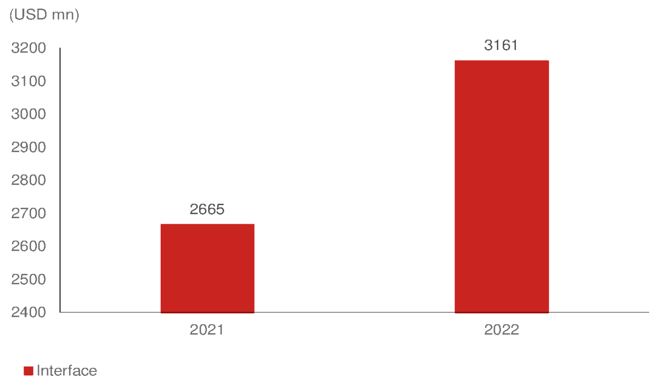
The interface IC market is still dominated by global vendors (*Fig. 332*), in which 3Peak has 1.4% share. Interface IC is the latest product line where 3Peak has invested, and it has already started seeing meaningful sales contribution – CNY300mn in 2022 (23% of total sales). 3Peak's interface sales maintained strong growth and surpassed converter sales in 2021 due to strong demand in communication and industrial applications.

Fig. 330: 3Peak - annual interface sales



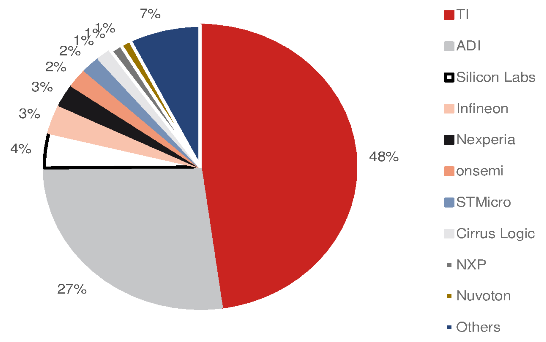
Source: Company data, Nomura research

Fig. 331: Interface market size in 2021-22



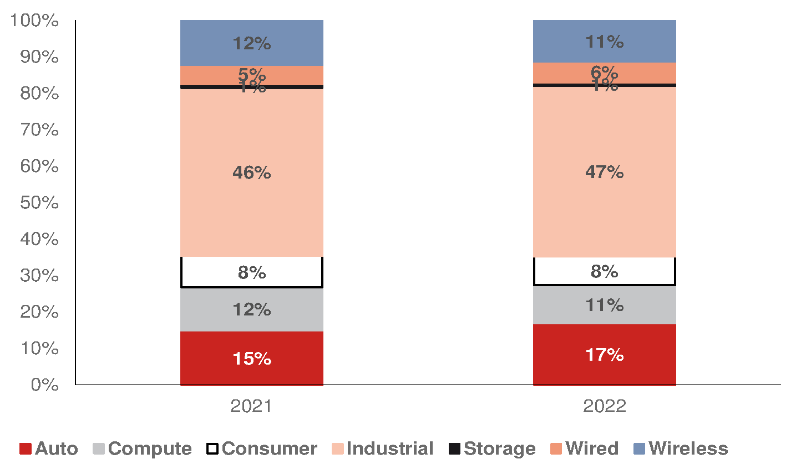
Source: Gartner, Nomura research

Fig. 332: Interface market share by vendor in 2022



Source: Gartner, Nomura research

Fig. 333: Interface sales mix (%) by application in 2021-22



Source: Gartner, Nomura research

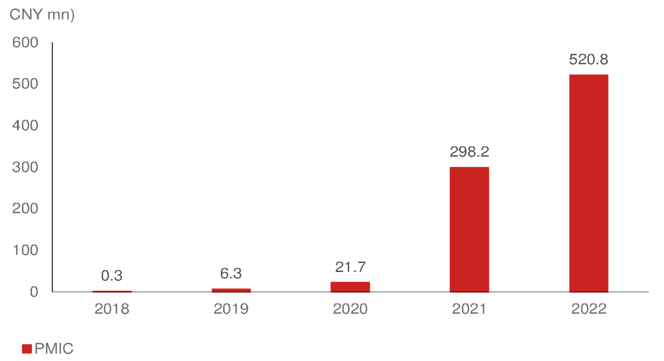
PMIC (20-25% of 2023F sales): LDO a major sales growth driver in 2021-22

Strong PMIC sales growth through mass production for customers; more complete analog product portfolio to provide better customer service

3Peak started providing PMIC products to customers in 2021, which mainly are the linear products of low-dropout regulator (LDO). However, 3Peak also provides customers with other PMIC products, including supervisors, drivers, and DC-DC products. However, the PMIC market is more competitive than signal chain, which we believe will face more pricing pressure and fewer localization opportunity in customers in the China market.

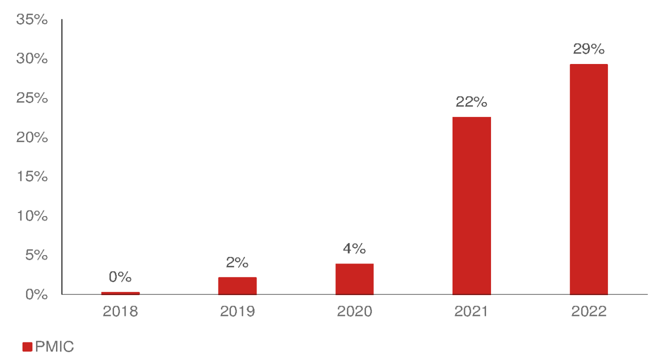
Despite being a late entrant in the China PMIC market, 3Peak merged ICM-semi with itself in June-2023 to strengthen its PMIC products portfolio, which has AC-DC and battery management IC product lines, and alleviate the weakness of fewer PMIC products.

Fig. 334: 3Peak: PMIC sales over 2018-22



Source: Company data, Nomura research

Fig. 335: 3Peak: PMIC sales mix (%)



Source: Company data, Nomura research

Fig. 336: The major China analog IC companies' PMIC product portfolios

	Driver			Non-driver							
	LED	MOSFET	Motor	LDO	DC-DC	AC-DC	Battery management: Charge management	Battery management: Power Monitoring	Battery management: Overvoltage protection	Load Switch	Supervisors/ Reset
Sliergy	√	√	√	√	√	√	√	√	√	√	√
SG Micro	√	√		√	√		√	√	√	√	√
Awinic	√		√	√	√		√	√	√	√	
3Peak		√	√	√	√	ICM-semi	ICM-semi	ICM-semi	ICM-semi		√
Novosense	√	√	√	√			√		√	√	
Southchip					√	√	√		√	√	
Joulwatt				√	√	√	√		√	√	
Injoinic					√	√	√				
Chipown	√	√		√	√	√					√
Wuxi ETEK	√			√	√		√		√	√	
Dioo	√	√	√	√	√	√	√			√	√
Cellwise			√	√	√		√	√	√	√	
Fine Made	√			√	√	√	√				
Belling	√	√	√	√	√	√	√			√	√
Kiwi instrument	√	√	√	√	√	√	√	√		√	
Bpsemi	√		√		√	√					

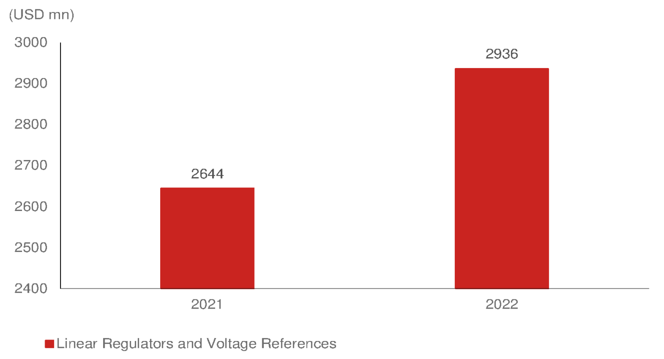
Source: Company data, Nomura research

LDO/voltage reference (80%+ of PMIC sales)

LDO is now 3Peak's major PMIC product, and is the second-largest product line, contributing c.25% of total sales in 2022. 3Peak's LDO product sales had significant growth in 2020-21 due to 5G infrastructure deployment and mass production of new products for telecom customers, which surpassed converter and interface products sales.

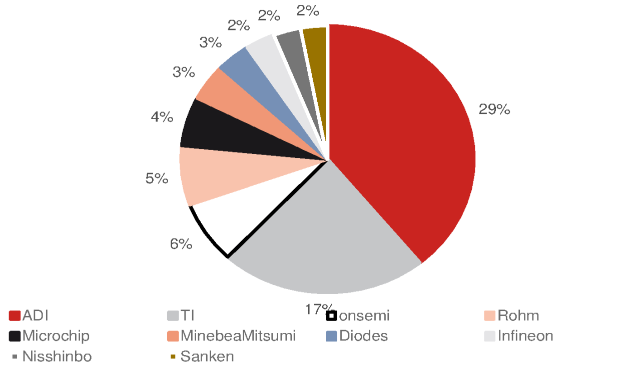
LDO is a lower-end product in the PMIC market, which is a DC linear voltage regulator that can only set down voltage, but can't step up. As per Gartner's estimates, the LDO/voltage reference market size is USD2.94bn, and current major vendors are ADI and TI. The end-applications are more diversified than signal chain product's market. Considering Gartner's market size data, we think 3Peak's market share is around 2.1%, which is close to that of the global top 10 vendors.

Fig. 337: Linear regulator and voltage references market size in 2021-22



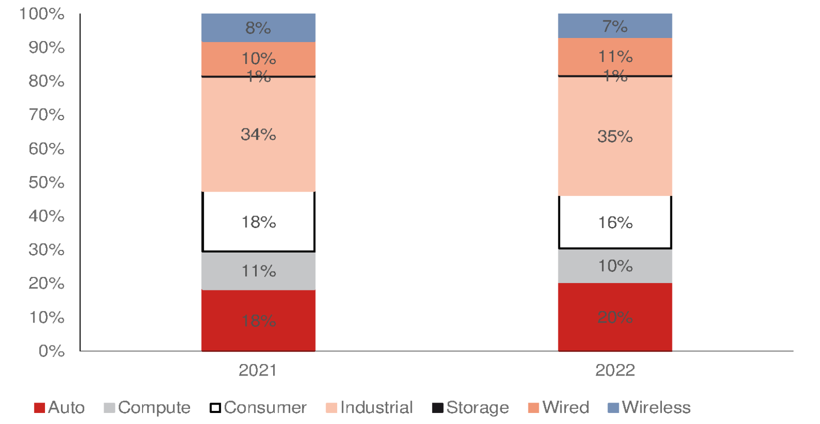
Source: Gartner, Nomura research

Fig. 338: Linear regulator and voltage references market share in 2022



Source: Gartner, Nomura research

Fig. 339: Linear regulators and voltage references sales mix (%) by application



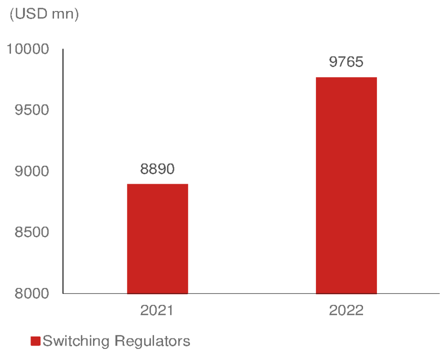
Source: Gartner, Nomura research

Other PMIC (DC-DC, drivers, and supervisors)

Although LDO is the only product that has meaningful sales contribution, 3Peak continues to expand its PMIC product portfolio. Switching voltage regulators and drivers are new PMIC products launched by 3Peak in 2021, including DC/DC buck/boost/and flyback, which also has a far higher market size than that of LDO. Moreover, 3Peak's planned merger with ICE-semi to acquires the latter's AC-DC and battery management IC product lines in 2024 would also strengthen 3Peak's product portfolio.

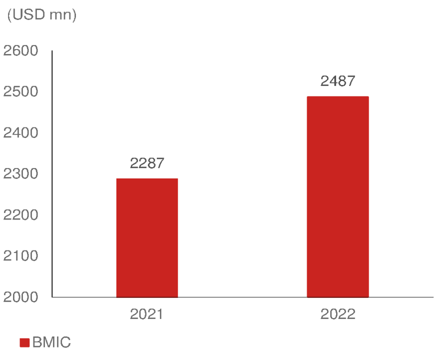
Similar to other PMIC products, the major vendors are global peers, but some China vendors are also gaining a meaningful share in the market, such as Bright Power [688368 CH, Not rated] and Chipone [unlisted] in LED/display power, and Silergy [6415 TT, Buy] in BMIC and switching regulators markets. We believe 3Peak will introduce customers to new PMIC with a strong signal chain product portfolio in order to better meet their demand. However, we think the PMIC market faces intense competition and is also characterized by higher localization than the signal chain market, which will make it more difficult for new entrants to gain a foothold in the market.

Fig. 340: Switching regulator market size in 2021-22



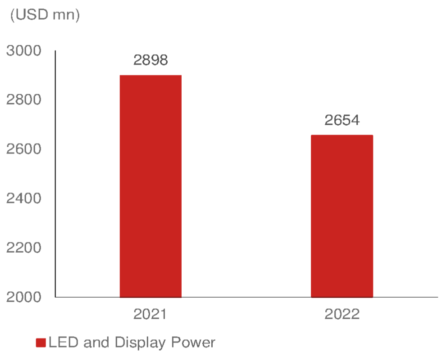
Source: Gartner, Nomura research

Fig. 341: Battery management IC market size in 2021-22



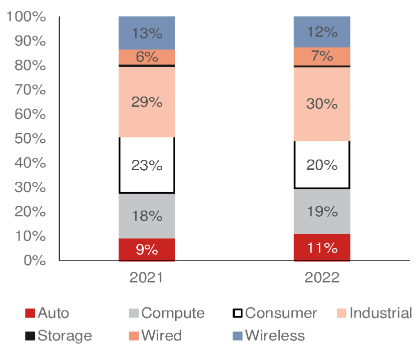
Source: Gartner, Nomura research

Fig. 342: LED and display power market size in 2021-22



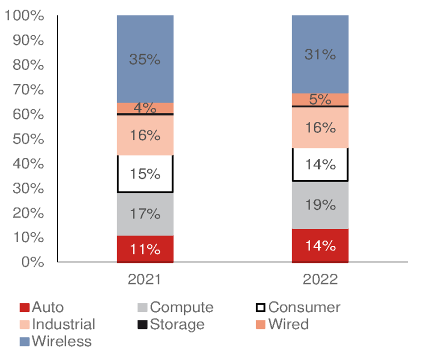
Source: Gartner, Nomura research

Fig. 343: Switching regulators sales mix (%) by application



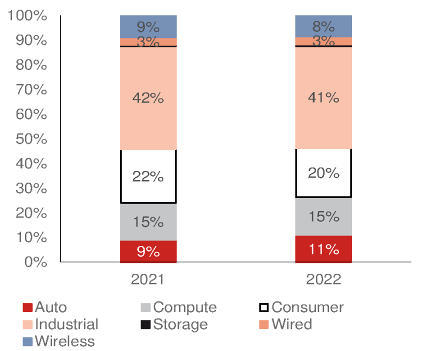
Source: Gartner, Nomura research

Fig. 344: Battery management IC sales mix (%) by application



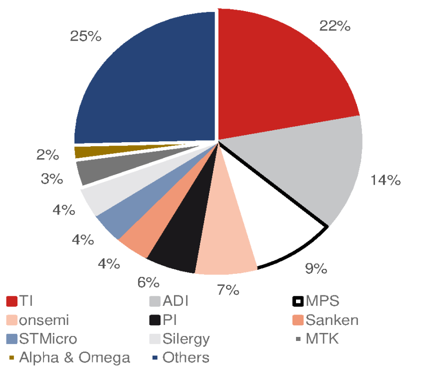
Source: Gartner, Nomura research

Fig. 345: LED and display power sales mix (%) by application



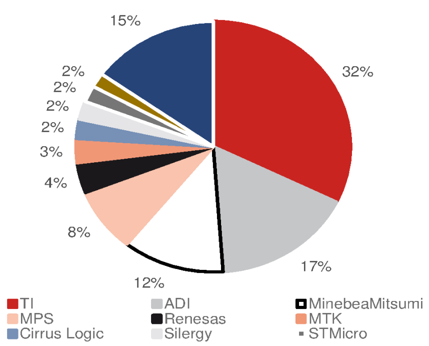
Source: Gartner, Nomura research

Fig. 346: Switching regulator market share in 2022



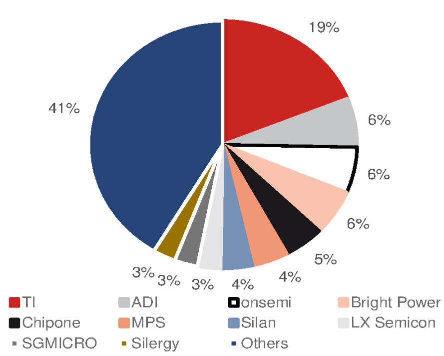
Source: Gartner, Nomura research

Fig. 347: Battery management IC market share in 2022



Source: Gartner, Nomura research

Fig. 348: LED and display power market share in 2022



Source: Gartner, Nomura research

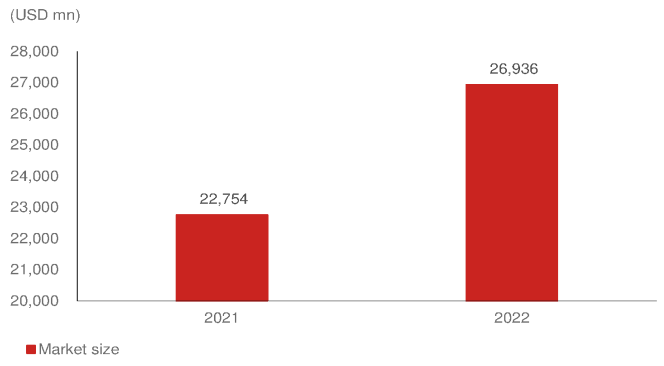
MCU: New business focused on industrial and auto

The first MCU (microcontroller unit) product was taped out in a foundry in Oct-2022, and 3Peak started providing samples to customers in late 2023. An MCU focuses first on industrial applications, and then will further expand to auto. The intention behind developing MCU products is that many analog IC product design definitions came from MCU products, helping 3Peak to identify product trends quickly. Furthermore, 3Peak believed logic IC has higher-integration architecture, and may consume less analog ICs. Consequently, 3Peak decided to build MCU products. 3Peak's MCU will try to differentiate from peers in terms of performance and die area by integrating its own analog technology IP, and engaging more in products with low localization.

By integrating with analog products, 3Peak will provide a one-stop solution to customers to meet their requirement. Currently, MCU is under customer sampling and is yet to contribute sales. In addition to the first MCU product, 3Peak has another three in R&D stage.

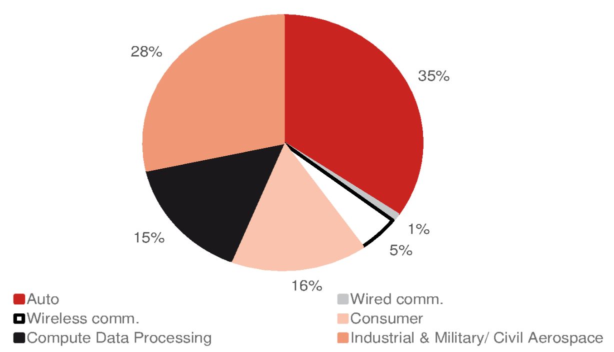
As per Gartner estimates, the MCU market size was USD26.9bn in 2022. The major vendors are global players and the top five have over 76% market share, while some China local MCU vendors already have meaningful sales, such as Gigadevice (603986 CH, Not rated), Fudan (688385 CH, Not rated), or Sinowealth (300327 CH, Not rated) with 1~2% of global share.

Fig. 349: MCU market size in 2021-22



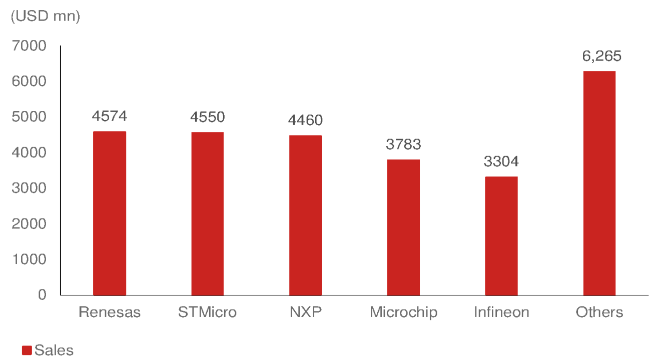
Source: Gartner, Nomura research

Fig. 350: MCU sales mix (%) by application



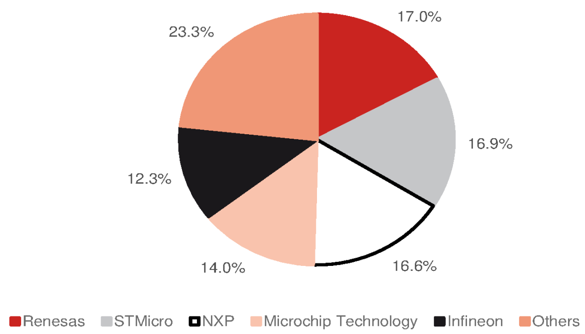
Source: Gartner, Nomura research

Fig. 351: MCU vendor sales in 2022



Source: Gartner, Nomura research

Fig. 352: MCU market share (%) by vendor



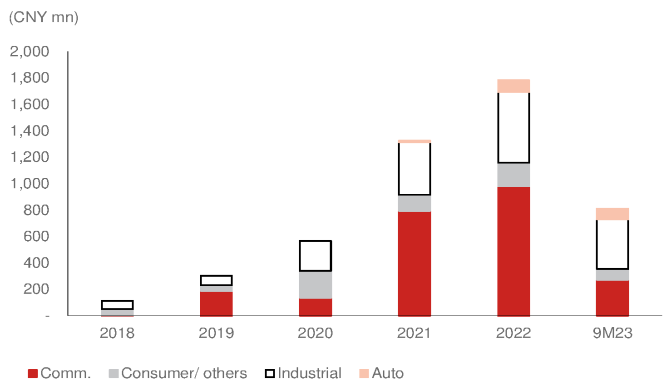
Source: Gartner, Nomura research

3Peak's product applications

3Peak is one of the few fabless companies focusing on non-consumer applications and providing customers with high-end analog products. 3Peak's end-user application of products can be divided into four major categories: consumer, industrial, communication, and automotive.

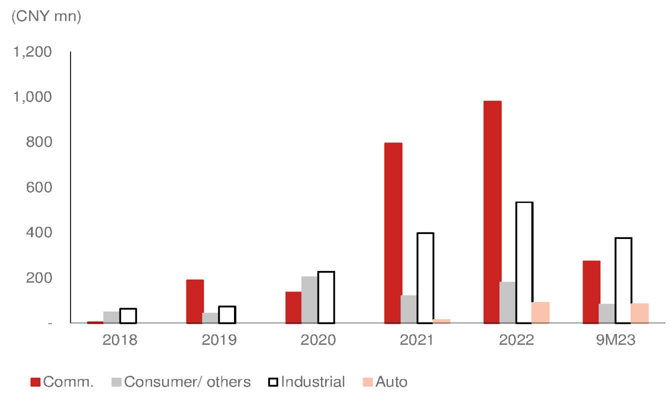
In 3Peak's product application sales mix, communication sales are witnessing significant growth -- had the highest sales mix in 2021-22 due to strong communication equipment demand along with semi shortage, while consumer remained stable at 10% of total sales. Industrial application sales growth has also remained stable since 2018, with auto sales showing meaningful contribution from 2022.

Fig. 353: 3Peak: Annual sales by application



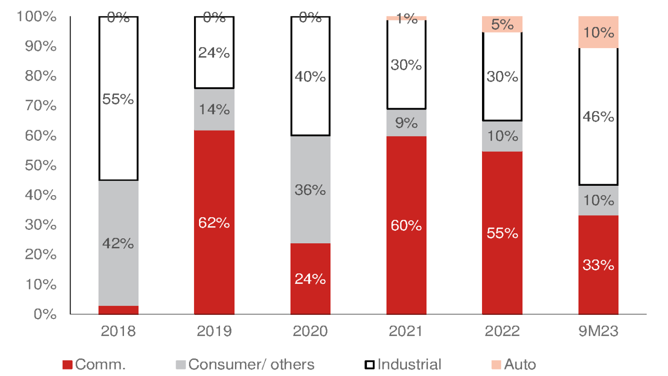
Source: Company data, Nomura research

Fig. 354: 3Peak: Application sales



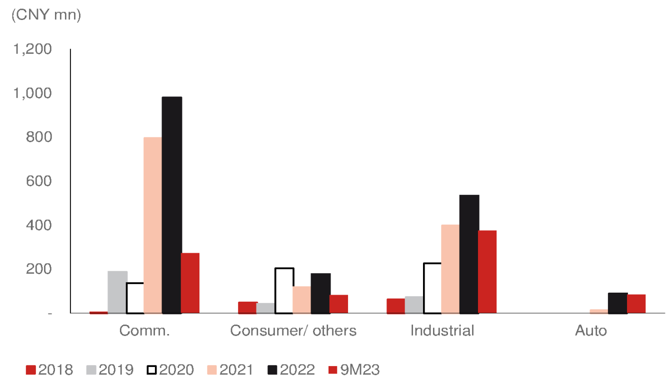
Source: Company data, Nomura research

Fig. 355: 3Peak: Product application mix (%)



Source: Company data, Nomura research

Fig. 356: 3Peak: Product application sales



Source: Company data, Nomura research

Communication (30-35% of 2023F sales)

3Peak's communication applications cover most communication devices and can provide a cost-effective way to support a wide range of communication methods. It has been a major sales contributor since 2021 owing to new products being mass-produced and strong end-demand in the China telecom market. Besides the beneficiary from China 5G market growth, 3Peak has also benefited from localization demand during 2021-22. 5G equipment requires dollar content of up to USD100 from USD50 in 4G for analog IC in base station, as per our industry survey.

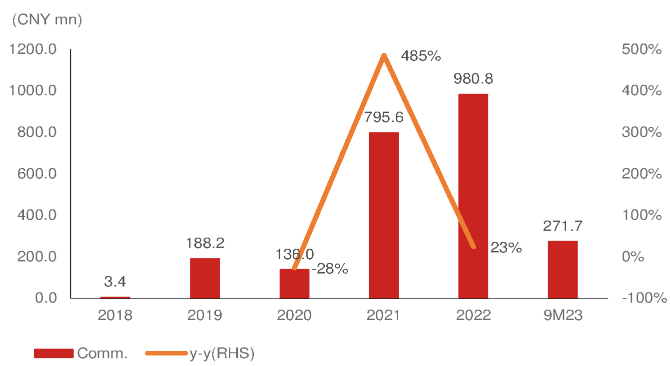
2023, a challenging year

However, the industry has been weakening since late 2022, when telecom end-customers were digesting inventory and also lowering their equipment capex. Similar to other communication semi players, 3Peak suffered a significant sales decline in 2023, and 3Peak has also been facing high pricing pressure in both signal chain and PMIC products due to competition and excess inventory overhang. From 3Peak's observation, overseas customers are more stable, while China customers remain muted due to a lowered 5G base station installations target number resulting in weaker demand. We think 3Peak's communication sales is more exposed to the China market, and can start to recover once China telecom demand rebounds after digesting inventories.

End of excess inventory digestion and recovery likely in mid-24F

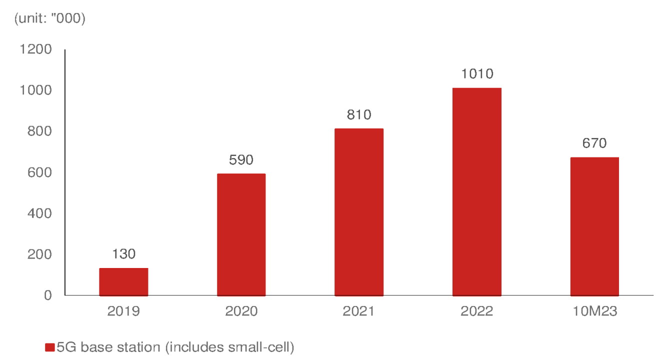
Although local 5G infrastructure deployment remains at a similar level y-y, we think 3Peak's communication sales should start recovering in mid-24F after significant inventory digestion during 2023-1H24. We forecast communication sales to record 40-45% y-y growth in 2024F.

Fig. 357: 3Peak: Communication sales



Source: Company data, Nomura research

Fig. 358: China 5G base station installations



Source: China miit, Nomura research

Fig. 359: 3Peak: Communication products portfolio

Applications		Analog IC	Products
Wired Infrastructure	Switch	PMIC	LDO, DC-DC, DDR power, prower detection, power sequence, reference
		Signal chain	I2C Level shift/ Switch/ Expander/ Hot swap
	Erbium-doped fiber amplifier (EDFA)	PMIC	Reference, LDO, reset, Log Amplifier
		Signal chain	OPA, ADC, DAC
Wireless Infrastructure	Base station	PMIC	Reset, LDO, DC-DC, power sequence, monitor
		Signal chain	ADC, DAC, current sensor, digital isolator, switch, I2C level shift/ expander
Enterprise Systems	Server	PMIC	LDO, power sequencer, power monitor
		Signal chain	I2C level shift/ expande, current sensing

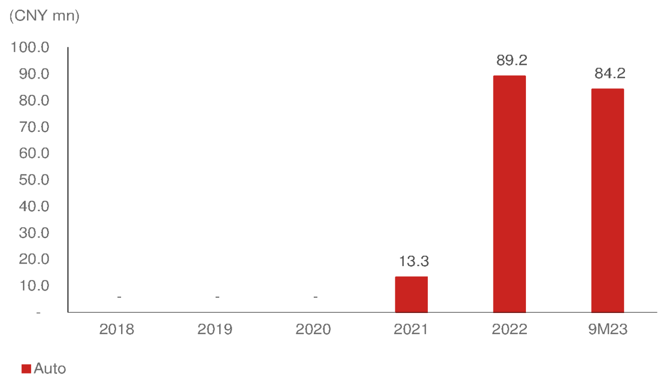
Source: Company data, Nomura research

Auto (10-15% of 2023F sales)

3Peak began its auto product development in 2020, and now provides a comprehensive product portfolio of full signal chain and PMIC solution. 3Peak's auto business strategy is to broaden its customer portfolio to cover most Tier-1 or auto OEM brands, on which 3Peak has made significant progress in 2023. In 2024, 3Peak will focus on gaining more market share by providing more comprehensive solutions on analog and MCU products. 3Peak's current auto dollar content per vehicle is in the USD40-50 range, and it targets to provide up to USD150 per vehicle.

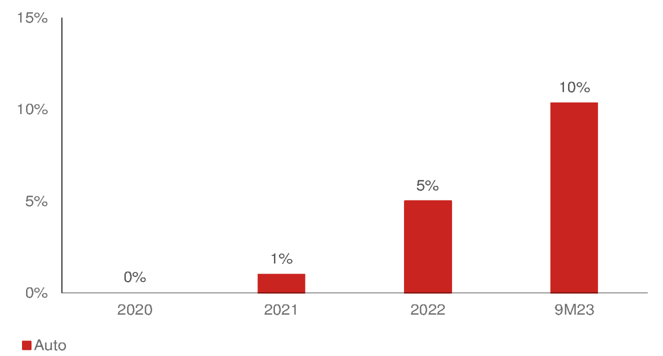
Auto sales started to see meaningful contribution in 2022-23, which accounted for 10% of total sales in 9M23, though the increased mix is also due to a decline in other application sales. However, 3Peak's auto sales remained strong in 2023, and management thinks auto sales will account for 10-15% of total sales, meaning 4Q23 auto sales continued to outperform other applications in 4Q23 due to strong growth among local EV customers. Management expects auto sales to remain strong in 2024E driven by higher market share and mass production of new products.

Fig. 360: 3Peak: Auto sales



Source: Company data, Nomura research

Fig. 361: 3Peak: Auto sales mix



Source: Company data, Nomura research

Fig. 362: 3Peak: Auto product portfolio (signal chain)

Applications	CAN/ LIN/ SBC	Half-bridge drivers	Low-side drivers	Isolated drivers	Digital isolator	Special amplifiers	General operatinal amplifiers	ADC	Transformer drivers
Body control	√	√	√	√	√	√	√		
ADAS	√	√	√			√	√	√	
Powertrain system	√	√	√	√	√	√	√	√	√
Infotainment and cluster	√	√	√	√		√	√	√	

Source: Company data, Nomura research

Fig. 363: 3Peak: Auto product portfolio (PMIC)

Applications	Supervisors and reset	Voltage reference	PMIC	BUCK	LDO	Comparators
Body control	√	√	√	√	√	√
ADAS	√	√	√	√	√	√
Powertrain system	√	√	√	√	√	√
Infotainment and cluster	√	√	√	√	√	√

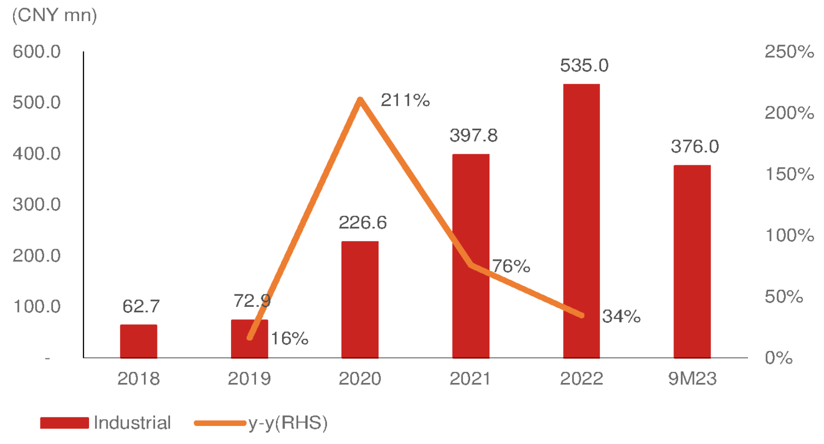
Source: Company data, Nomura research

Industrial (45-50% of 2023F sales)

Analog IC plays a key role in industrial applications in either performance, electrification, reliability, or securities in different and extreme environments. Current analog IC has increasing product performance requirements. As per ICinsights estimates, the global industrial analog IC market size is USD4.1bn, and is expected to record a CAGR of 6% over 2021-26.

3Peak's industrial products' applications are surveillance, smart meters, instruments, scanners, electric automation, and drones. And 3Peak has also covered most of China industrial customers in each application, such as Inovance (300124 CH, Buy), Ecovacs (603486 CH, Not rated), and Haier (600690 CH, Not Rated) in the industrial control market; Hikvision (002415 CH, Not rated), and Dahua (002415 CH, Not rated) in surveillance; and Neware technology (unlisted), Newland (unlisted), and MCGS (unlisted) in the instrument/ meters industry.

Fig. 364: 3Peak: Industrial sales



Source: Company data, Nomura research

Fig. 365: 3Peak: Industrial products portfolio

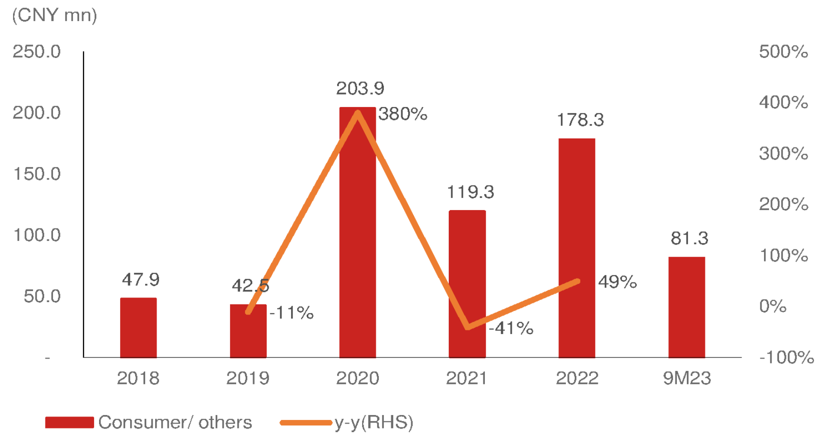
Applications		Analog IC	Products
Industrial	Encoder	PMIC Signal chain	LDO, DC-DC, reference OPA, ADC, comparators

Source: Company data, Nomura research

Consumer and others (5-10% of 2023F sales)

In 3Peak's consumer/other applications, it covers industries such as data storage, home entertainment, medical, and electronic cigarettes. Although the consumer and others application is not a major focus area for 3Peak, it still contributed around 10% of its 2023F total sales. According to management, the company's consumer sales likely accounted for 5-10% of its total 2023E sales. 3Peak does not have Tier-1 consumer application customers so far. However, ICE-semi focused on consumer applications prior to the planned acquisition by 3Peak. We think 3Peak can further leverage ICE-semi's channels to broaden its consumer application clients list in the future.

Fig. 366: 3Peak: Consumer/others sales



Source: Company data, Nomura research

Fig. 367: 3Peak: Consumer and others product portfolio

Applications		Analog IC	Products
Data storage	SSD	PMIC Signal chain	LDO, reset n.a.
Home Theater& Entertainment	Set-top box(STB), Over-the-top(OTT)	PMIC Signal chain Others	DC-DC, LDO, reset n.a. MCU
Medical	Electronic thermometer/ Blood pressure monitor/ Blood glucose meter	PMIC Signal chain	LDO, DC-DC, reference, reset OPA, current sense, switch
Others	Electronics cigarettes	PMIC Signal chain	Linear charger, LDO, drivers Current sensor amplifier, comparators

Source: Company data, Nomura research

Customers are likely to end inventory adjustment in 2024E

Communication

3Peak expects China local 5G base station infrastructure deployment to remain at low levels of 400K-500K (vs. 2023's 500K or 2022's 800K). However, after digesting excess inventory over a year since early-2023, its major local communication customer's inventory level will return to normal by 2Q24E, according to management, which estimates the customer's sales contribution accounted for 50-70% of peak comm. sales in 2021-2022. We believe 3Peak's comm. sales will improve in 2024F as local customers' inventories return to healthy levels, and new overseas customers' sales contribution begins in 2024F.

Auto

3Peak's auto application sales accounted for 5% of total sales in 2022, which is still not a meaningful sales mix to us. Thus, 3Peak's auto application sales were not significantly impacted by auto customers' inventory adjustment in 2023 due to its low sales base and new product mass production. By 3Peak's discussion with auto application customers, inventory remains at normal levels. 3Peak expects auto sales to continue to outgrow the company average due to new designs in product mass production in local and overseas customers.

Industrial

Industrial application, another key application, accounted for 40-50% of 3Peak's 2023F sales. Customers of the industrial segment started inventory adjustment in 2H22-1H23, and industrial customers continued to digest inventory in 1Q24E. Meanwhile, 3Peak is seeing industrial customers' demand gradually improving in new energy and industrial control applications. 3Peak expects its industrial application sales to outgrow global vendors' as its customers adjust orders earlier.

Consumer

3Peak's consumer application sales continued to improve in 2023 sequentially q-q. Consumer applications is not 3Peak's major focused industry now. However, ICE-Semi is a consumer-applications-focused company, which has a different customer base to 3Peak. Once 3Peak completes the acquisition, we believe 3Peak and ICE-Semi can increase sales through leveraging each other's customer portfolios.

Summary of financials

2024F/25F sales to grow by 39%/30%, driven by demand recovery in industrial and communication applications and auto sales growth

We forecast 3Peak to turn profitable at the operating level in 2024F/2025F supported by auto sales growth, less inventory provision cost, and industrial/communication demand recovery. We expect 3Peak to gain market share in auto and industrials. Thus, we forecast 3Peak's sales to record a 35% CAGR over 2023-2025F, increasing to CNY2.0bn, from the low base of CNY1.1bn in 2023F.

35% sales CAGR over 2023F-25F

We forecast 3Peak's sales to grow 39%/30% in 2024F/25F, driven by dollar content growth in auto, demand recovery from communication customers' normalized inventory, and resuming growth q-q in mid-2024F. We assume auto sales to record a 37% sales CAGR over 2023-25F driven by new products' mass production, and a recovery of industrial demand.

Gross margin pressure stabilized

We expect 3Peak's gross margin to improve after the pricing pressure eases along with a fall in foundry cost. We also expect this trend to continue, given increasing mix of high margin comm./auto/industrial analog sales. Therefore, we forecast GM to bottom in 2H23F at high-40s%, and gradually improve to low-50s% in 2024F driven by lower wafer cost as it had entered into an LTA with its foundry partner in 3Q23, and a better product mix along with a recovery of communication from 2Q24F.

Earnings rebound in 2024F/25F

Due to an improvement in both sales and blended GPM, driven by a better product mix and reasonable opex control, we expect the company to become profitable at the operating level in 2024F/2025F. We expect the high GPM for the communication and auto segments to not only accelerate earnings growth but also improve the company's ROE.

Sensitivity analysis by GPM improvement in 2024F-25F

Fig. 368: Sensitivity analysis of 3Peak's GPM improvement in 2024F-25F

	2024F			2025F		
Lower wafer cost and eased pricing pressure benefit GPM recovery	51%	53%	55%	51%	53%	55%
		Base case			Base case	
Revenues	1,518	1,518	1,518	1,981	1,981	1,981
Gross profit	774	798	835	1,010	1,041	1,089
Operating profit	135	159	196	315	346	394
Pre-tax profit	200	224	261	383	414	462
Net profit	170	190	222	326	352	393
EPS (TWD)	1.28	1.4	1.7	2.4	2.7	3.0
Margins (%)						
Gross margin	51%	53%	55%	51%	53%	55%
Operating margin	9%	10%	13%	16%	17%	20%
Pre-tax margin	13%	15%	17%	19%	21%	23%
Net margin	11%	13%	15%	16%	18%	20%

Source: Nomura estimates

Fig. 369: 3Peak: P&L

(CNY mn)	2018	2019	2020	2021	2022	2023	2024F	2025F	2026F
Sales	114	304	566	1,326	1,783	1,094	1,518	1,981	2,413
Gross profit	59	180	347	803	1,045	566	798	1,041	1,273
- OPEX	(70)	(112)	(184)	(433)	(868)	(746)	(639)	(695)	(765)
Operating income	(10)	68	163	370	177	(179)	159	346	508
Pretax income	(6)	71	181	445	267	(82)	224	414	573
Net income	(6)	71	183	444	267	(35)	190	352	487
EPS	(0.3)	1.7	2.3	5.5	2.2	(0.3)	1.4	2.7	3.7
Profitability									
Gross margin	52%	59%	61%	61%	59%	52%	53%	53%	53%
- OPEX ratio	-61%	-37%	-32%	-33%	-49%	-68%	-42%	-35%	-32%
Operating margin	-9%	22%	29%	28%	10%	-16%	10%	17%	21%
Net margin	-6%	23%	32%	33%	15%	-3%	13%	18%	20%
YoY									
Sales		166%	87%	134%	34%	-39%	39%	30%	22%
Gross profit		204%	92%	131%	30%	-46%	41%	30%	22%
- OPEX		61%	64%	136%	101%	-14%	-14%	9%	10%
Operating income		-764%	139%	127%	-52%	-201%	-189%	118%	47%
Pretax income		-1220%	156%	146%	-40%	-131%	-372%	85%	38%
Net income		-1227%	158%	142%	-40%	-113%	-649%	85%	38%

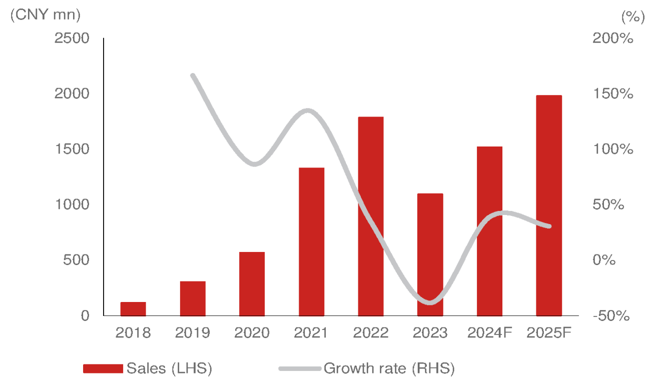
Source: Company data, Nomura estimates

Fig. 370: Nomura forecasts vs Bloomberg consensus estimates

	2024F			2025F			2026F		
(TWD mn)	NMR	BBG	Diff (%)	NMR	BBG	Diff (%)	NMR	BBG	Diff (%)
Sales	1,518	1,661	(8.6)	1,981	2,675	(26.0)	2,413	2,889	(16.5)
Gross profit	798	912	(12.5)	1,041	1,497	(30.5)	1,273	1,473	(13.6)
Operating profit	159	198	(19.7)	346	359	(3.7)	508	420	21.0
Net profit	190	197	(3.6)	352	351	0.4	487	391	24.8
EPS (CNY)	1.44	1.49	(3.7)	2.66	2.66	(0.2)	3.68	2.95	24.6
Margin	**NMR**	**BBG**	**Diff (ppts)**	**NMR**	**BBG**	**Diff (ppts)**	**NMR**	**BBG**	**Diff (ppts)**
Gross margin (%)	52.6	54.9	(2.4)	52.6	56.0	(3.4)	52.8	51.0	1.8
Operating margin (%)	10.5	11.9	(1.4)	17.5	13.4	4.0	21.1	14.5	6.5
Net margin (%)	12.5	11.9	0.7	17.8	13.1	4.7	20.2	13.5	6.7

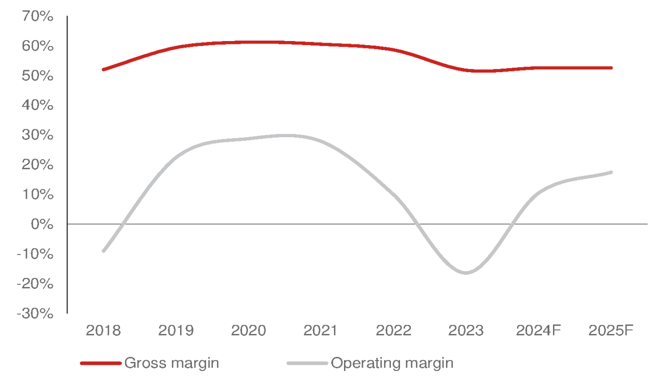
Source: Bloomberg Financials L.P., Nomura estimates

Fig. 371: 3Peak: Sales and growth rate



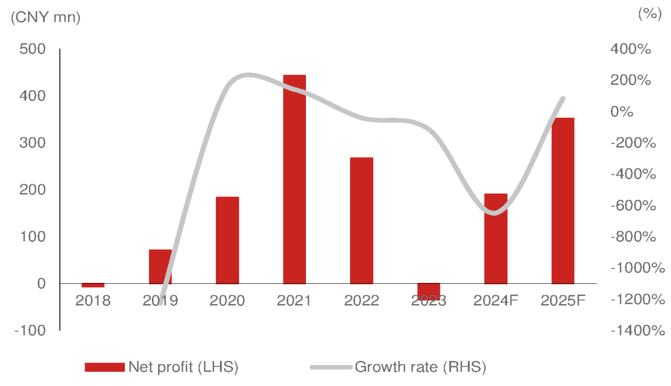
Source: Company data, Nomura estimates

Fig. 372: 3Peak: Gross margin and OP margin



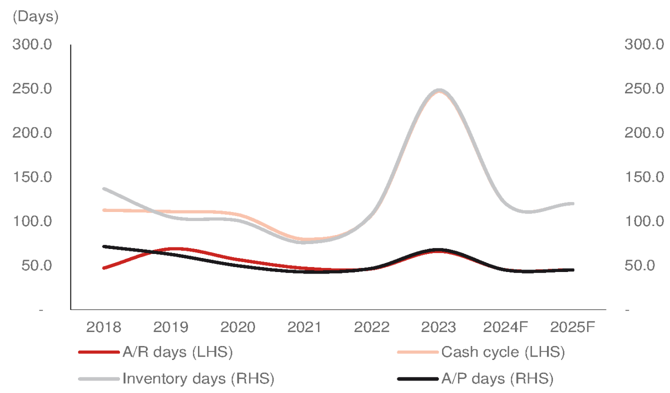
Source: Company data, Nomura estimates

Fig. 373: 3Peak: Net profit and growth rate



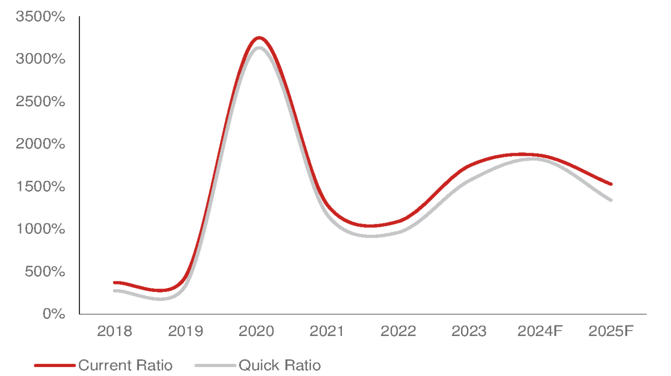
Source: Company data, Nomura estimates

Fig. 374: 3Peak: CCC analysis



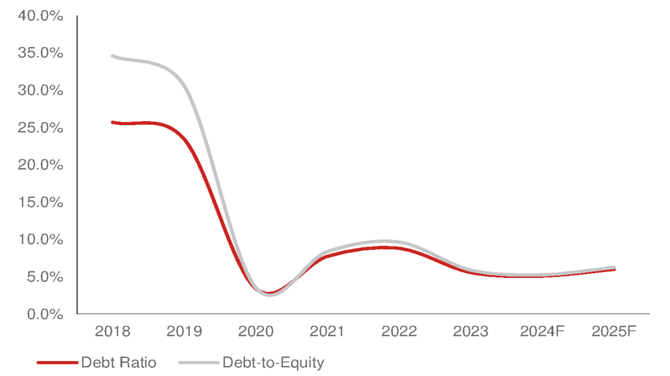
Source: Company data, Nomura estimates

Fig. 375: 3Peak: Current ratio and quick ratio



Source: Company data, Nomura estimates

Fig. 376: 3Peak: Debt ratio and debt-to-equity



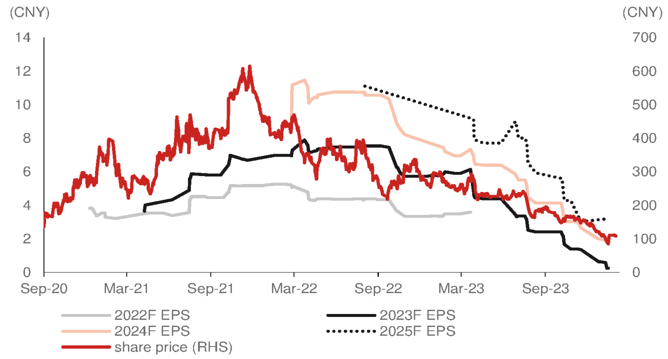
Source: Company data, Nomura estimates

Valuation and risks

Our TP of CNY120 is based on 45x 2025F P/E (EPS: CNY2.7). 3Peak's one-year forward P/E ranged between 40x and 80x during the profit-making period of 2020-22 (it was publicly listed in 2020). The stock is currently trading at 36x 2025F P/E.

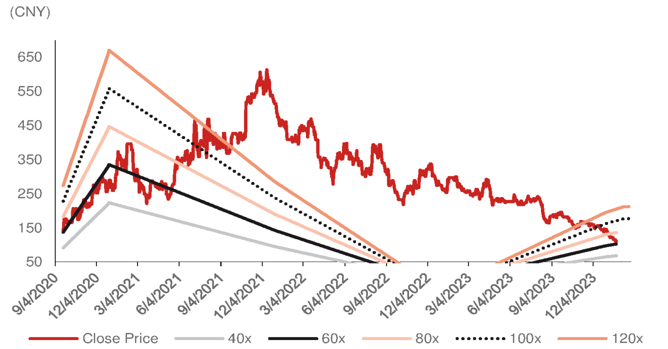
Downside risks to our call include: 1) a softer-than-expected end-demand recovery in the industrial and communication segments; 2) slower-than expected gross margin recovery, and; 3) fiercer-than-expected price competition from global and local peers.

Fig. 377: 3Peak: Consensus earnings revisions



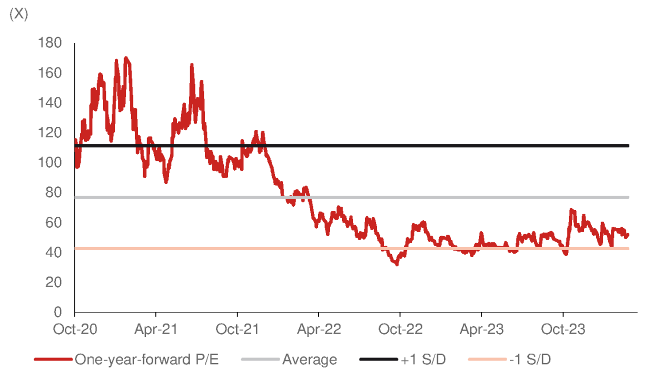
Source: Bloomberg Finance L.P., Nomura research

Fig. 378: 3Peak: One-year forward P/E



Source: Company data, Nomura estimates

Fig. 379: 3Peak: One-year forward consensus P/E



Source: Bloomberg Finance L.P., Nomura research

Fig. 380: 3Peak: One-year forward consensus P/B

Source: Bloomberg Finance L.P., Nomura research

Share price drivers

In our view, 3Peak's analog business growth is the company's major share price driver. Since its public listing in 2020, 3Peak's share price has factored in the semiconductor shortage and benefits of localization, driving sales growth significantly in 2020-21. After the market entered a downturn and customers started digesting inventory, we think 3Peak's quarterly sales have become a powerful share price driver.

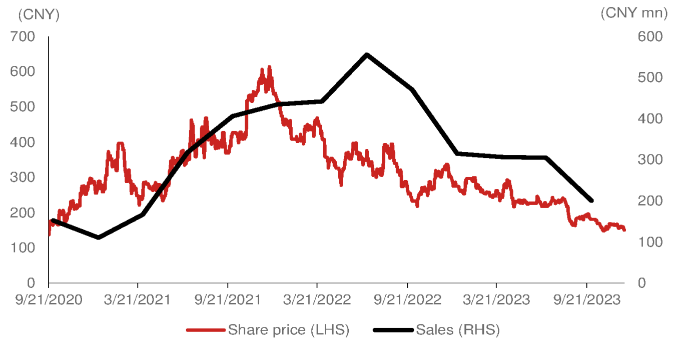
Fig. 381: 3Peak: One-year forward consensus sales

Source: Bloomberg Finance LP., Nomura research

Fig. 382: 3Peak: One-year forward consensus net income

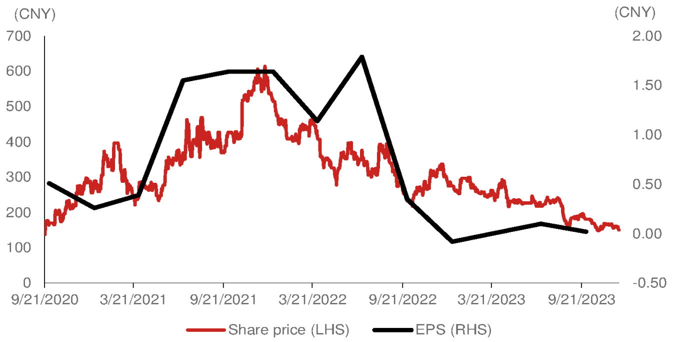
Source: Bloomberg Finance LP., Nomura research

Fig. 383: 3Peak: Share price vs quarterly sales



Source: Company data, Nomura research

Fig. 384: 3Peak: Share price vs quarterly EPS

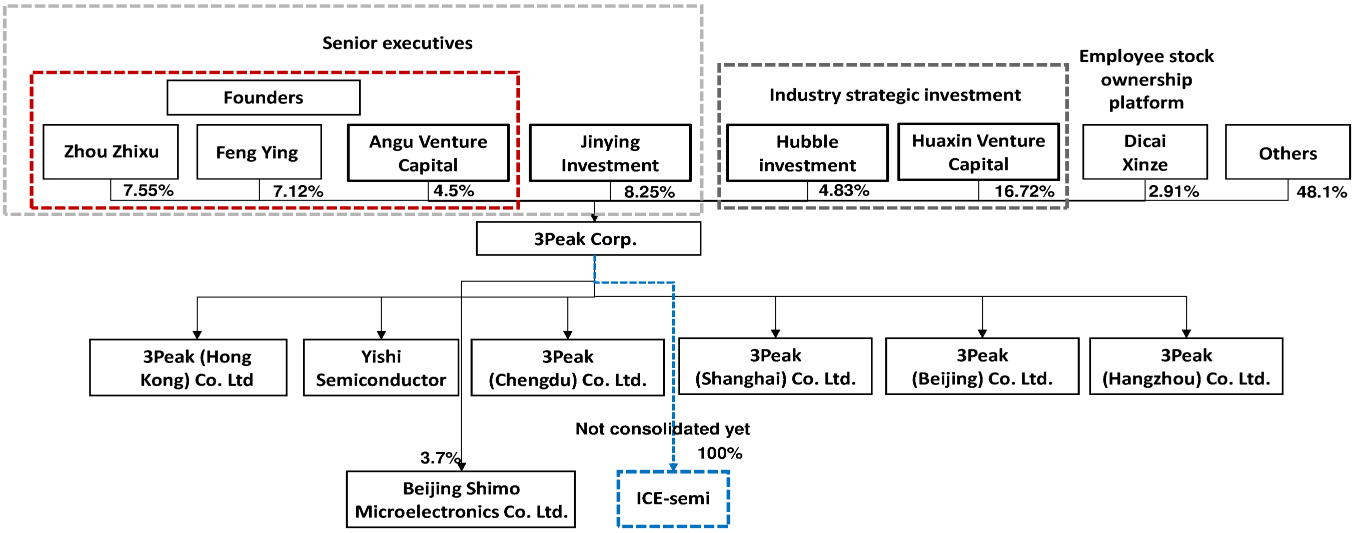


Source: Company data, Nomura research

Company background

Established in 2012, 3Peak is committed to making high-performance, high-quality, and highly reliable IC products, including signal chains, power analog chips, mixed-signal front-ends, and embedded processors (MCU) to provide customers with a one-stop solution. The applications of the products cover many fields such as communication, industrial, security monitoring, medical and health, instrumentation, new energy, and automotive industry. On 21 September 2020, 3Peak was listed on the Shanghai Science and Technology Innovation Board. In May 2023, 3Peak announced to acquire ICE-semi, a consumer PMIC analog company, to broaden 3Peak's customer base and product portfolio. The above deal is required to be approved by shareholders in 2024E.

Fig. 385: 3Peak: Company structure



Source: Company data, Nomura research

Fig. 386: 3Peak: Board members

English Name	Chinese Name	Title	Experience
Zhou Zhixu		Chairman	Mr. Zhou has a B.S., M.S., and Ph.D. in electrical engineering from Arizona State University, US, and 27 years of experience in analog chips at global companies at China and oversea. From 1994 to 2007, he served as a device and process R&D engineer, chief researcher, chief engineer of analog circuit design, and senior member of the committee on science and technology at Motorola. He has been at 3Peak since 2008, and is the chairman
Ying Feng		Director, deputy general manager, head of R&D	Mr. Ying has a B.S. in physics from Zhejiang University, an M.S. in physics from the Institute of Physics, Chinese Academy of Sciences, an M.S. in electrical engineering from the University of Michigan, a Ph.D. in microelectronics from the University of Texas at Dallas and 23 years of experience in analog chips at global companies at China and oversea. From 1998 to 2007, he served as the technical manager of the hybrid digital IC design department at Texas Instruments. From 2007 to 2009, he served as the design director at C2 Microsystems Inc. He has been at 3Peak since 2009, and is director, deputy general manager, and head of R&D.
Wu Jiangkang	吴建刚	Director	From 2007 to 2017, Mr. Wu served as a analog design vice director in Spreadtrum, and design director, vice president, and general manager in 3Peak since 2017.
Wong Hing		Director	From 1990 to 1997, Mr. Wong served as a engineer in IBM, and Asia Pacific business vice president in Silicon Access Networks from 1997 to 2003, and senior consultant in Silicon Federation from 2005 till now, and Managing Director in Waldon investment since 2012.
Zhang Xiaojun	章晓军	Director	Mr. Zhang has 29 years of experience in the power industry and has been a director at 3Peak since 2021.
Wang Lin	王林	Director	Mr. Wang has a B.S. from Hangzhou University of Electronic Science and Technology, an M.S. from Zhejiang University, and 17 years of experience in well-known semi companies at China and abroad. From 2004 to 2012, he served as an engineer, senior engineer, and technical planning manager at Samsung Semiconductor (China) Research and Development Co., Ltd. He has served as a director of 3Peak since 2019.
Hong Zhiliang	洪志良	Independent Director	From 1970 to 1980, Mr. Hong served as a lecturer in Shenyang University of Technology, and served as a post-doc from 1985 to 1987 and as a professor from 1988 to now in Fudan University, and has been a independent director in 3Peak since 2019.
Lo Yan	罗研	Independent Director	From 2010 till now, Ms. Lo served as a professor at finance department in Fudan University, and has been a independent director in 3Peak since 2019.
Chu Kuangwei	朱光伟	Independent Director	From 1993 to 1997, Mr. Chu served as a marketing director and product manager in Heraeus, and from 1997 to 2013, Mr. Chu served as a director and vice president in Continental, and has served as a vice president in Zongmutech from 2022 till now, and has been a independent director in 3Peak since 2022.

Source: Company data, Nomura research

Fig. 387: 3Peak: Management team

English Name	Chinese Name	Title	Experience
Wu Jiangkang	吴建刚	Director	From 2007 to 2017, Mr. Wu served as a analog design vice director in Spreadtrum, and design director, vice president, and general manager in 3Peak since 2017.
Ying Feng		Director, deputy general manager, head of R&D	Mr. Ying has a B.S. in physics from Zhejiang University, an M.S. in physics from the Institute of Physics, Chinese Academy of Sciences, an M.S. in electrical engineering from the University of Michigan, a Ph.D. in microelectronics from the University of Texas at Dallas and 23 years of experience in analog chips at global companies at China and oversea. From 1998 to 2007, he served as the technical manager of the hybrid digital IC design department at Texas Instruments. From 2007 to 2009, he served as the design director at C2 Microsystems Inc. He has been at 3Peak since 2009, and is director, deputy general manager, and head of R&D.
Leng Aikuo	冷爱国	Vice President	From 2005 to 2006, Mr. Leng served as a senior engineer in ZTE communication, and from 2006 to 2008 served as a marketing manager and senior FAE in Maxim, and from 2008 to 2016, served as a senior FAE, system engineer in Texas Instrument, and has been a product manager, sales director, and vice president in 3Peak.
Li Shuhuang	李淑环	Board Secretary	From 2002 to 2012, Ms. Li served as a vice president in Shanghai Aijia electronics, and served as a marketing manager in east China region, central China sales general manager and assistant to general manager in 3Peak since 2013.
Wang Wenping	王文平	CFO	From 2000 to 2013, Mr. Wang served as audit manager in Deloitte(Shanghai), and served as a Chief finance officer in Something Big technology from 2014 to 2017, and served as a executive director, CFO, vice president in Foliday(01992 HK, NR) from 2017 to 2022, and has been a CFO in 3Peak since 2022.
Chu Yiping	朱一平	Director of quality engineering	From 2008 to 2009, Mr. Chu served as post-doc position in University of Florida, and served as a Researcher of post-doc in University of California, Berkeley from 2009 to 2012, served as a vice researcher in East China Normal University from 2012 to 2018, and has been a director of quality engineering in 3Peak since 2018.
He Dejun	何德军	Director, new technical director	Mr. He has a B.S. in information and electronic technology from Zhejiang University and 22 years of experience at global semi companies at China and oversea. From 1999 to 2004, he served as an engineer at Suzhou Huaxin Micro-electronics Co., Ltd. and at Kaiming Information Technology Co., Ltd. From 2004 to 2008, he worked as an engineer at ATMEL Corporation. He has been at 3Peak since 2008, and is a new technical director.

Source: Company data, Nomura research

Fig. 388: 3Peak: Major shareholders

Major share holder	Ownership (%)
Shanghai Huaxin Venture Capital	16.68%
Zhou Zhixu	7.53%
Suzhou Golden Sakura Venture Capital Partnership	7.48%
Ying Feng	7.06%
Galaxy innovation growth investment fund	4.85%
Huaxia Shanghai STAR board 50 ETF	4.44%
Hubble Technology investment	4.38%
Suzhou Angu Venture Capital Co. Ltd	3.67%
Wanjia industry selective investment fund	3.02%
Jiaxing Xunxinze Enterprise Management Partnership	2.46%

Source: Company data, Nomura research

Appendix A-1

Analyst Certification

We, Aaron Jeng, Donnie Teng, Albert Lin and Vivian Yang, hereby certify (1) that the views expressed in this Research report accurately reflect our personal views about any or all of the subject securities or issuers referred to in this Research report, (2) no part of our compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this Research report and (3) no part of our compensation is tied to any specific investment banking transactions performed by Nomura Securities International, Inc., Nomura International plc or any other Nomura Group company.

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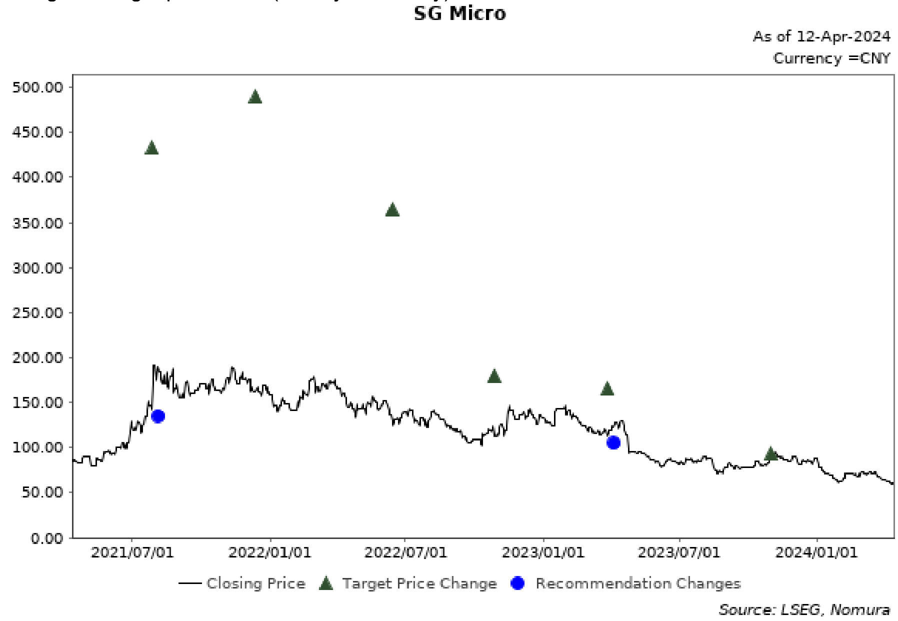
Materially mentioned issuers

Issuer	Ticker	Price	Price date	Stock rating	Sector rating	Disclosures
SG Micro	300661 CH	CNY 59.58	11-Apr-2024	Neutral	N/A	
Silergy	6415 TT	TWD 328	09-Apr-2024	Buy	N/A	
Suzhou Novosense Microelectronics	688052 CH	CNY 94.37	09-Apr-2024	Neutral	N/A	
3Peak	688536 CH	CNY 96.17	09-Apr-2024	Buy	N/A	
Awinic	688798 CH	CNY 50.90	11-Apr-2024	Buy	N/A	

SG Micro (300661 CH) **CNY 59.58 (11-Apr-2024)** Neutral (Sector rating: N/A)

Rating and target price chart (three year history)



Date	Rating	Target price	Closing price
01-Nov-23		93.00	88.19
28-Mar-23	Neutral		111.88
28-Mar-23		166.00	111.88
27-Oct-22		180.00	114.35
13-Jun-22		365.00	128.97
13-Dec-21		490.00	161.95
28-Jul-21	Buy		142.09
28-Jul-21		433.00	142.09

For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of CNY93 is based on 60x 2025F EPS of CNY1.55. Our target P/E of 60x is SG Micro's mid to low level of its historical P/E range of 40x-120x. We continue to use the PE valuation method for fabless companies, given their product cycle-driven earnings growth momentum. The benchmark index is SCI 300.
Risks that may impede the achievement of the target price Major upside/downside risks include: 1) better/weaker-than-expected analog IC demand; 2) easing/rising competition causing higher ASP or gross margin erosion; and 3) faster progress/delays in new product launches and new customers' project wins due to better/worse execution, supply chain over supply/shortage or easing/escalation of China-US tech tensions.

Silergy (6415 TT) **TWD 328 (09-Apr-2024)** Buy (Sector rating: N/A)

Rating and target price chart (three year history)

Date	Rating	Target price	Closing price

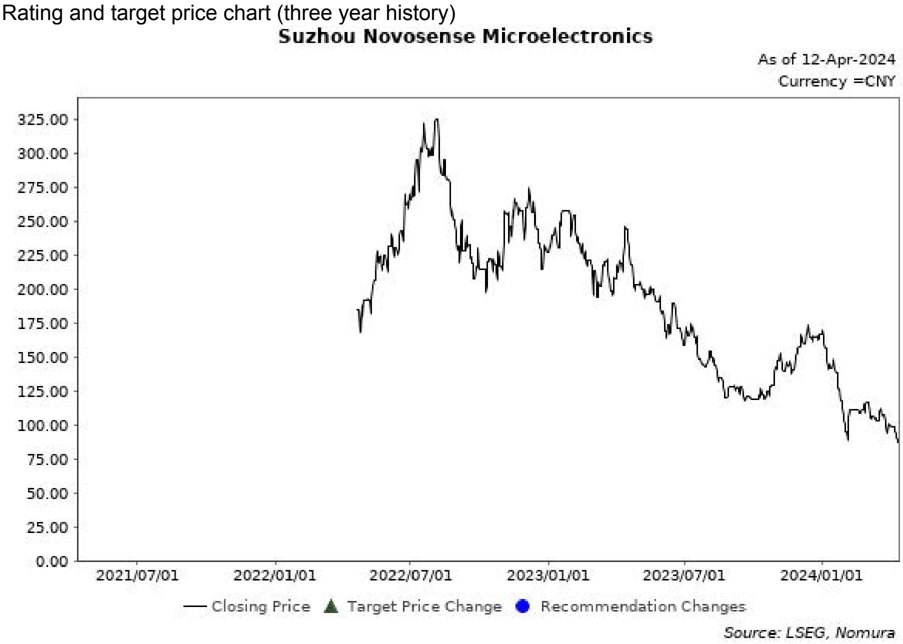
For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of TWD450 is based on a P/E of 35x on our 2025F EPS of TWD12.9. 35x P/E is at the mid-to-low-end of its P/E range of 20x-60x in 2019-2021 which earnings resumed growth from consecutive declining in 2017-2018. The benchmark index is TAIEX.

Risks that may impede the achievement of the target price Downside risks include: 1) market share gain is slower than expected; 2) lower-than-expected auto sales growth; 3) higher than expected pricing competition; and 4) weaker than expected end-demand recover.

Suzhou Novosense Microelectronics (688052 CH) **CNY 94.37 (09-Apr-2024)** Neutral (Sector rating: N/A)

Rating and target price chart (three year history)



Date	Rating	Target price	Closing price

For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of CNY105 is based on 2.5x 2025F BVPS of 43.1, at the mid-to-low-end of the 2x-5x PE range between Jan 2023-Mar 2024 when it was loss-making or at breakeven point. The benchmark index for this stock is SSE Composite Index.

Risks that may impede the achievement of the target price Downside risks: 1) Slower than expected China automotive demand growth; 2) stronger than expected price competition; and 3) slower than expected China analog semi localization rate growth. Upside risks: 1) Faster inventory depletion than our expectation, 2) fewer pricing pressure than our expectation, and 3) lower wafer cost than our expected

3Peak (688536 CH) **CNY 96.17 (09-Apr-2024)** Buy (Sector rating: N/A)

Rating and target price chart (three year history)

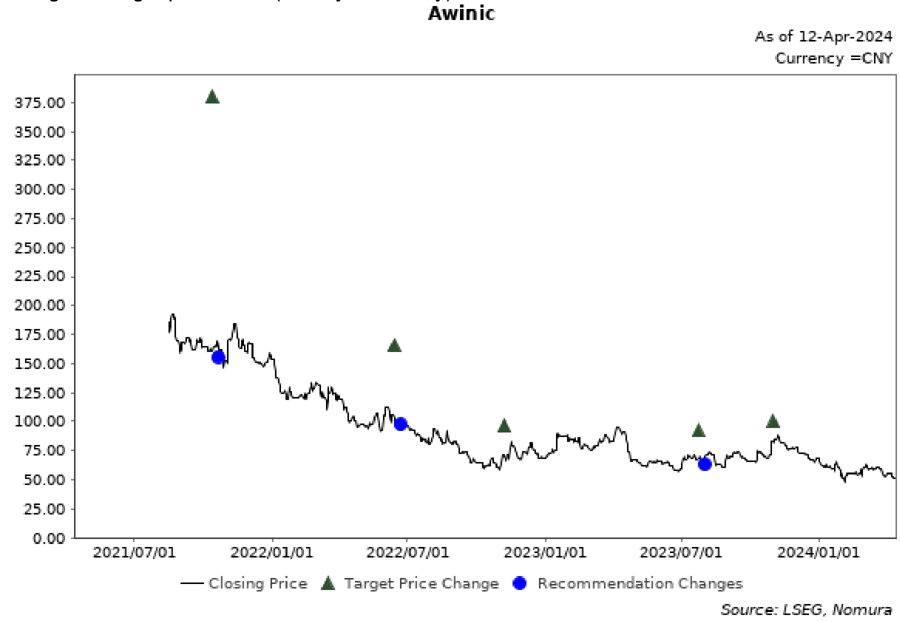
Date	Rating	Target price	Closing price

For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of TWD120 is based on a P/E of 45x on our 2025F EPS of CNY2.7. 50x P/E is at the mid-low-end of its past five-year P/E range of 40x-80x in profi-making period of 2020-2022. The benchmark index is CSI300.
Risks that may impede the achievement of the target price Downside risks include: 1) market recovery is slower than expected; 2) lower-than-expected auto sales growth; and 3) higher than expected pricing competition. Upside risks include 1) Higher-than-expected market share gain; 2) less pricing competition than our expected.

Awinic (688798 CH) **CNY 50.90 (11-Apr-2024)** Buy (Sector rating: N/A)

Rating and target price chart (three year history)



Date	Rating	Target price	Closing price
01-Nov-23		100.00	83.00
24-Jul-23	Buy		68.60
24-Jul-23		92.00	68.60
07-Nov-22		97.00	70.05
13-Jun-22	Neutral		102.87
13-Jun-22		166.00	102.87
13-Oct-21	Buy		161.09
13-Oct-21		380.00	161.09

For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of CNY100 is based on 6x 2024F BVPS of CNY16.8. Our target P/B of 6x is at the lower-range of Awinic's historical P/B multiple range of 4-8x. The benchmark index is CSI300.
Risks that may impede the achievement of the target price The major downside risks are: 1) weaker-than-expected analog IC demand; 2) rising foundry cost and competition causing gross margin erosions; 3) worsening competition landscape; and 3) the slower in new product launches and customers' project wins due to worse execution, supply shortage or escalation of China-US tensions.

Rating and target price changes

Issuer	Ticker	Old Stock Rating	New Stock Rating	Old Target Price	New Target Price
Silergy	6415 TT	Not Rated	Buy	N/A	TWD 450
Suzhou Novosense Microelectronics	688052 CH	Not Rated	Neutral	N/A	CNY 105
3Peak	688536 CH	Not Rated	Buy	N/A	CNY 120

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